

North Carolina State University Mechanical and Aerospace
Engineering Department

MAE 481



Final Report

S.M.A.R.T.

Soil Moisture And Response Tracking CubeSat

Table of Contents

1. Information	14
1.1 Project Customers	14
1.2 S.M.A.R.T. Squadron CubeSat Personnel	15
1.3 S.M.A.R.T. Squadron Project Organizational Structure	16
2. Project Description	16
2.1 Project Overview	16
2.2 Purpose & Need	17
2.3 Objectives	17
2.4 CONOPS	17
2.5 Design Solution	18
2.6 Functional Block Diagram	20
3. Functional Requirements	20
4. Key Design Alternatives	25
4.1 Electrical Power System	25
4.1.1 Battery	25
4.1.1.1 Lithium-Ion	26
4.1.1.2 Lithium-Polymer	27
4.1.1.3 Nickel Cadmium	28
4.1.1.4 Battery Comparison	29
4.1.2 Electronic Power System Architecture	29
4.1.2.1 Standalone Commercial EPS Architectures	30
4.1.2.3 Custom EPS	32
4.1.2.4 EPS Architecture Comparison	33
4.1.3 Solar Panels	33
4.1.3.1 Fixed Solar Panels	34
4.1.3.2 Deployable Solar Panels	35
4.1.3.4 Solar Panel Comparison	38
4.2 Payload Alternatives	38
4.2.1 Antennas	39
4.2.1.1 RainCube Adjacent Antenna/Dish Antenna	39
4.2.1.2 RADARSAT-1 Adjacent Antenna/Panel Antenna	40
4.2.1.3 Sentinel-1 Adjacent Antenna/Array Antenna	40
4.2.1.4 Capella Space Synthetic Aperture Radar Adjacent Antenna/Mesh Antenna	41
4.2.1.5 Horn Antenna	41
4.2.2 Radar Bands	41
4.2.2.1 P Band: 250 to 500 MHz	42
4.2.2.2 L Band: 0.5 to 1.5 GHz	42
4.2.2.3 S Band: 2 to 4 GHz	42
4.2.2.4 C Band: 4 to 8 GHz	42
4.2.2.5 X Band: 8 to 12 GHz	42

4.2.2.6 K Band: 12 to 26 GHz	42
4.3 Guidance, Navigation and Control	43
4.3.1 Attitude Control Sensors	43
4.3.1.1 Magnetometer	43
4.3.1.2 Sun Sensor	43
4.3.1.3 Earth Sensor/Horizon Sensor	44
4.3.1.4 Star Tracker	44
4.3.1.5 GPS Receiver	44
4.3.2 Thrusters	45
4.3.2.1 Cold Gas Thruster System	45
4.3.2.2 Micro-Electrospray Thruster System	45
4.3.3 Reaction Wheels	45
4.3.3.1 Three Wheel Control	46
4.3.3.2 Four Wheel Control	46
4.3.4 Magnetic Torque Rods	46
4.3.5 Solar Sails	47
4.3.6 Control Moment Gyroscope	47
4.3.7 Control Sensors: Advantages and Disadvantages	47
4.3.8 Actuators: Advantages and Disadvantages	48
4.4 Communication and Data Handling	49
4.4.1 Radar Bands	50
4.4.2 Transmission Architecture	51
4.4.3 Transmitter	52
4.4.4 Antenna Configurations	53
4.4.4.1 Wire Antenna	53
4.4.4.2 Patch Antenna	54
4.4.4.3 Arrays	54
4.4.4.4 Helical	55
4.4.4.5 Antenna Configuration Comparison	55
4.4.5 Data Handling	56
4.4.5.1 On Board Computer Architectures	57
4.4.5.1.1 Microcontrollers	57
4.4.5.1.2 Rad-Hard	58
4.4.5.1.3 FPGAs	59
4.4.5.1.4 Distributed	59
4.4.5.1.5 On-Board Computer Comparison	60
4.4.5.2 Operating Systems	60
4.4.5.2.1 Linux	61
4.4.5.2.2 VxWorks	61
4.4.5.2.3 FreeRTOS	61
4.4.5.2.4 Operating System Comparison	61

4.6 Thermal	62
4.6.1 Sprayable Thermal Control Coatings/Paints and Tapes	63
4.6.2 MLI Blanket	63
4.6.3 Sunshields	63
4.6.4 Thermal Louvers	64
4.6.5 Deployable Radiators	64
4.6.6 Electric Heaters	65
4.6.7 Cryocoolers	65
4.6.8 Thermoelectric Coolers	65
4.6.9 Fluid Pumps	65
4.6.10 Summary of Advantages and Disadvantages	66
4.7 Ground Station	68
4.7.1 General Design	68
4.7.2 Ground Station Designs: Advantages and Disadvantages	69
4.7.2 Ground Network	70
4.7.2.1 Ground Network Models	70
4.7.2.2 Ground Networks: Advantages and Disadvantages	71
4.7.3 Commercially Available Ground Station Networks (CAGSN)	71
4.7.3.1 AWS Ground Station	72
4.7.3.2 KSAT Lite	72
4.7.3.3 Leaf Line	73
4.7.3.4 NASA's Near Earth Network	74
4.7.3.5 CAGSN: Advantages and Disadvantages	75
4.8 Propulsion	76
4.8.1 Thrusters	76
4.8.1.1 Kinetic Thrusters:	77
4.8.1.1.1 Cold Gas Propulsion:	77
4.8.1.1.2 Warm Gas Propulsion:	78
4.8.1.1.3 Resistojets:	78
4.8.1.2 Chemical Propulsion:	79
4.8.1.2.1 Solid Rocket:	79
4.8.1.2.2 Liquid Rocket:	80
4.8.1.2.2.1 Monopropellant:	80
4.8.1.2.2.2 Bipropellant:	80
4.8.1.3 Electric Propulsion:	81
4.8.1.3.1 Electrostatic:	81
4.8.1.3.1.1 Hall Effect:	81
4.8.1.3.1.2 Electrospray	82
4.8.1.3.1.3 Gridded Ion Thruster	82
4.8.1.3.2 Electromagnetic Propulsion:	83
4.8.1.3.2.1 Pulse Plasma Thruster/Vacuum Arc Thruster:	83

4.8.2 Miscellaneous Methods	84
4.8.2.1 Solar Sail	84
4.10 Structures Alternatives	85
4.10.1 Materials	85
4.10.1.1. Aluminum 6082	85
4.10.1.2 Aluminum 7075	86
4.10.1.3 Aluminum 6061	86
4.10.1.4 Aluminum 5052	86
4.10.2 Summary of Advantages and Disadvantages	86
5. Trade Study Process and Results	87
5.1 Electrical Power Subsystem	87
5.1.1 Battery Trade Study	87
5.1.2 EPS Architecture Trade Study	88
5.1.2.3 Weights	89
5.1.2.4 Pugh Analysis EPS Architecture	89
5.1.3 Solar Panel Trade Study	90
5.2 Payload Trade Study	91
5.2.1 Antenna Shape/Style	91
5.2.1.1 Motivation	91
5.2.1.2 Metrics	91
5.2.1.3 Weights	92
5.2.1.4 Trade Study Process	92
5.2.2 Frequency Bands	93
5.2.2.1 Motivation	93
5.2.2.2 Metrics	93
5.2.2.3 Weights	94
5.2.2.4 Trade Study Process	94
5.3 GNC Trade Study	95
5.3.1 Control Sensors	95
5.3.1.1 Motivation	95
5.3.1.2 Metrics	95
5.3.1.3 Weights	96
5.3.1.4 Trade Study Process	96
5.3.2 Control Actuators	97
5.3.2.1 Motivation	97
5.3.2.2 Metrics	97
5.3.2.3 Weights	98
5.3.2.4 Trade Study Process	98
5.4 Communications and Data Handling	99
5.4.1 Band	99
5.4.1.1 Metrics	99

5.4.1.2 Weights	99
5.4.1.3 Pugh Analysis Radar Band	99
5.4.2 Communication Architecture	100
5.4.2.1 Motivation	100
5.4.2.2 Metrics	100
5.4.2.3 Weight	100
5.4.2.4 Pugh Analysis Communication Architecture	100
5.4.3 Transmitters	101
5.4.3.1 Motivation	101
5.4.3.2 Metrics	101
5.4.3.3 Weight	101
5.4.3.4 Pugh Analysis Transmitters	101
5.4.4 Antenna	102
5.4.4.1 Motivation	102
5.4.4.2 Metrics	102
5.4.4.3 Weights	102
5.4.4.4 Pugh Analysis Antenna	102
5.4.5 On-Board Computer	103
5.4.5.1 Motivation	103
5.4.1.2 Metrics	103
5.4.5.3 Weights	103
5.4.5.4 Pugh Analysis	103
5.4.6 Operating System	104
5.4.6.1 Motivation	104
5.4.6.2 Metrics	104
5.4.6.3 Weights	104
5.4.6.4 Pugh Analysis	104
5.6 Thermal Trade Study	105
5.6.1 External System Metrics	105
5.6.2 External System Weights	105
5.6.3 Internal System Metrics	106
5.6.4 Internal System Weights	106
5.6.5 External System Pugh Matrix	107
5.6.6 Internal System Pugh Matrix	107
5.7 Ground Station Trade Study	108
5.7.1 Frequency Band	108
5.7.1.1 Motivation	108
5.7.1.2 Metrics	108
5.7.1.3 Weights	109
5.7.1.4 Trade Study Process	109
5.7.2 Ground Network	110

5.7.2.1 Motivation	110
5.7.2.2 Metrics	110
5.7.2.3 Weights	110
5.7.2.4 Trade Study Process	111
5.7.3 Commercially Available Ground Station Networks	112
5.7.3.1 Motivation	112
5.7.3.2 Metrics	112
5.7.3.3 Weights	112
5.7.3.4 Trade Study Process	113
5.8 Propulsion Trade Study	114
5.8.1 Thruster Trade Study	114
5.8.1.1 Motivation	114
5.8.1.2 Metrics	114
5.8.1.3 Weights	114
5.8.1.4 Trade Study Process	115
5.11 Structures Trade Analysis	116
5.11.1 Motivation	116
5.11.2 Metrics	116
5.11.3 Weights	117
5.11.4 Trade Study Process	117
6. Design Solution	118
6.1 Power Design Solution	118
6.1.1 Prototype Summary	119
6.1.2 Power Prototype Solution	119
6.1.3 Power Exploded View	132
6.2 Payload Design Solution	132
6.2.1 Prototype Summary	132
6.2.2 Prototype Solution	133
6.2.2.1 Function Generator & Quad Op-Amp	134
6.2.2.2 Voltage Controlled Oscillator and Attenuator	136
6.2.2.3 Power Amplifier and Low Noise Amplifier	137
6.2.2.4 Power Splitter and RF Mixer	139
6.2.2.5 Receiver and Transmitter Antennas and Associated Components	140
6.3 GNC Design Solution	143
6.3.1 Prototype Summary	143
6.3.2 Prototype Solution	143
6.3.2.1 Photo Sensor	144
6.3.2.2 Accelerometer	145
6.3.2.3 Reaction Wheel Motors	146
6.3.2.4 3D Printed Disks	148
6.4 Communications and Data Handling	151

6.4.1 Prototype Summary	151
6.4.2 Prototype Solution	151
6.5 Ground Station Design Solution	159
6.5.1 Prototype Summary	159
6.5.2 Prototype Solution	159
6.5.2.1 Transceiver & Adapter with Antenna	160
6.5.2.2 Microcontroller	162
6.5.2.3 Micro SD Card & Adapter	164
6.5.2.4 Adafruit Perma-Proto Board	165
6.5.2.5 Breadboard Power Supply Module	167
6.6 Thermal Design Solution	168
6.6.1 Prototype Summary	168
6.6.2 Prototype Summary	168
6.6.2.1 All-Polyimide Thermofoil Heaters	169
6.6.2.2 Thermostat and K-Type Temperature Sensor	170
6.6.2.3 Thermal Louver	170
6.6.2.4 Cooling Fan	171
6.6.3 CAD Models	172
6.8 Propulsion Design Solution	172
6.8.1 Prototype Summary	172
6.8.2 Prototype Solution	173
6.8.2.1 CO2 Cartridge	174
6.8.2.2 Keg CO2 Regulator	175
6.8.2.3 Half Union 7/16"-20UNF Flare Male to 1/4" FNPT	176
6.8.2.4 1/2" MNPT to 1/4" MNPT	177
6.8.2.5 NPT Solenoid Valve	178
6.8.2.6 Male to Male NPT Converter	179
6.8.2.7 3D Printed Nozzle	180
6.8.2.8 Plumbers Tape	181
6.8.2.9 Metal Repair Glue	182
6.8.3 CAD Models	183
6.9 Structures Design Solution	183
6.9.1 Prototype Summary	184
6.9.2 Prototype Solution	184
6.9.2.1 Aluminum Alloy 6061	184
6.9.2.2 Screws and hex nut	185
7. Vehicle Layout	188
7.1 Propulsion	188
7.1.1 System Features	188
7.1.2 Properties	188
7.1.3 Critical Elements	188

7.1.4 Image	189
7.2 Payload	189
7.2.1 System Features	189
7.2.2 Properties	190
7.2.3 Critical Elements	190
7.2.4 Images	190
7.3 Ground Station	191
7.3.1 System Features	191
7.3.2 Properties	192
7.3.3 Critical Elements	192
7.3.4 Images	193
7.4 GNC	193
7.4.1 System Features	193
7.4.2 Properties	194
7.4.3 Critical Elements	194
7.4.4 Images	195
7.6 Thermal	196
7.6.1 System Features	196
7.6.2 Properties	196
7.6.3 Critical Elements	196
7.6.4 Images	197
7.7 Structures	197
7.7.1 System features	197
7.7.2 Properties	197
7.7.3 Critical elements	198
7.7.4 Images	198
7.8 Power	199
7.8.1 System Features	199
7.8.2 Properties	199
7.8.3 Critical Elements	199
7.8.4 Images	200
7.9 Communications and Data Handling	202
7.9.1 System Features	202
7.9.2 Properties	202
7.9.3 Critical Elements	203
7.9.4 Images	203
8. Manufacturing	205
8.1 Ground Station Manufacturing	205
8.1.1 Objective	205
8.1.2 Fabrication Procedure	205
8.1.2.1 Integrating the Micro SD Card Adapter	205

8.1.2.2 Integrating the nRF24L01 Transceiver	206
8.1.2.3 Integrating Buttons	207
8.1.2.4 Housing the Completed Circuit	208
8.1.3 Manufacturing Process	208
8.1.4 Facility/Location	208
8.1.5 Equipment, Tools, and Machines Required	208
8.2 Payload Manufacturing	208
8.2.1 Objective	208
8.2.2 Fabrication Procedure	208
8.2.2.1 Modulator Circuit Assembly	209
8.2.2.2 Video Amplifier Circuit Assembly	209
8.2.2.3 RF Circuit Assembly	210
8.2.2.4 Cantenna Assembly	211
8.2.3 Manufacturing Process	212
8.2.4 Facility/Location	212
8.2.5 Equipment, Tools, and Machines Required	212
8.2 Propulsion Manufacturing	213
8.2.1 Objective	213
8.2.2 Fabrication Procedure	213
8.2.3 Manufacturing Process	213
8.2.4 Facility/Location	214
8.3 GNC Manufacturing	214
8.3.1 Objective	214
8.3.2 Fabrication Procedure	215
8.3.2.1 Integrating Photoresistors	215
8.3.2.2 Integrating Accelerometer	218
8.3.2.3 Integrating motors	218
8.3.2.4 Housing the Completed System	219
8.3.3 Manufacturing Process	221
8.3.4 Facility/Location	221
8.3.5 Equipment, Tools, and Machines Required	221
8.6 Thermal Manufacturing	221
8.6.1 Objective	221
8.6.2 Fabrication Procedure	221
8.6.3 Manufacturing Process	221
8.6.4 Facility/Location	222
8.7 Structures Manufacturing	222
8.7.1 Objective	222
8.7.2 Fabrication Procedure	222
8.7.2.1 Cutting of Parts	222
8.7.2.2 Interior skeleton attachment	223

8.7.2.3 Connection of Parts	224
8.7.3 Manufacturing Process	224
8.7.4 Facility/Location	224
8.7.5 Equipment, Tools, & Machines Required	224
8.8 Power Manufacturing	224
8.8.1 Objective	224
8.8.2 Fabrication Procedure	224
8.8.2.1 Constructing Solar Panels	224
8.8.2.2 Making the 12V rail solution	227
8.8.2.3 Constructing Thermal System	229
8.8.3 Manufacturing Process	230
8.8.4 Facility/Location	230
8.8.5 Equipment, Tools, and Machines Required	230
8.9 Communications and Data Handling Manufacturing	231
8.9.1 Objective	231
8.9.2 Fabrication Procedure	231
8.9.2.1 Integrating the Micro SD Card Adapter	232
8.9.2.2 Integrating the nRF24L01 Transceiver	233
8.9.2.3 Failing to add a third SPI device	233
8.9.2.4 Final Step	234
8.9.3 Manufacturing Process	234
8.9.4 Facility/Location	235
• Senior Design Lab	235
8.9.5 Equipment, Tools, and Machines Required	235
8.10 Systems Integration	235
8.11 Summary Table	240
9. Validation, Verification, and Testing	241
9.1 Power VV&T	241
9.1.1 Test 1: Solar Panel Charging Test	242
9.1.2 Test 2: High Powered Relay Test	244
9.1.3 Test 3: Propulsion Actuation Test	245
9.2 Payload VV&T	247
9.2.1 Test 1: Cantenna Calibration	248
9.2.2 Test 2: Power Conservation Via Deactivation	249
9.2.3 Test 3: Transmitter/Receiver Pairing	249
9.2.4 Test 4: SAR Object Detection and Maximum Range	250
9.3 GNC VV&T	251
9.3.1 Test 1: Reaction Wheel Test	251
9.3.2 Test 2: Sun Tracking Test	252
9.3.3 Test 3: Earth Pointing Test	253
9.4 Communications VV&T	254

9.4.1 Test 1: Save Data to SD Card	255
9.4.2 Test 2: Telemetry Data Test	256
9.5 Ground Station VV&T	257
9.5.1 Test 1: Soil Moisture Map Development	258
9.5.2 Test 2: Communication & Data Storage Test	259
9.5.3 Test 3: CubeSat Initiation & Telemetry Return Test	260
9.6 Thermal and Survivability	261
9.6.1 Test 1: Heat Addition Test	261
9.6.2 Test 2: Thermocouple Circuit Test	262
9.9 Propulsion VV&T	262
9.9.1 Test 1: Propulsion Feasibility Test	263
9.9.1.1 Test Design	263
9.9.1.2 Procedure	263
9.9.1.3 Required Facilities/Equipment/Tools/Software	264
9.9.1.4 Expected Results	264
9.9.1.5 Success Criteria	264
9.9.1.6 Results	264
9.9.2 Test 2: Systems Interaction Test	264
9.9.2.1 Test Design	265
9.9.2.2 Procedure	265
9.9.2.3 Required Facilities/Equipment/Tools/Software	265
9.9.2.4 Expected Results	265
9.9.2.5 Success Criteria	265
9.9.2.6 Results	265
9.9.3 Test 3: Structural Integrity Test	265
9.9.3.1 Test Design	265
9.9.3.2 Procedure	265
9.9.3.3 Required Facilities/Equipment/Tools/Software	266
9.9.3.4 Expected Results	266
9.9.3 Success Criteria	266
9.9.4 Results	266
9.10 Structures VV&T	266
9.10.1 Test 1: 20 cm strut bending moment test	267
9.10.1.1 Test Design	267
9.10.1.2 Procedure	267
9.10.1.3 Required Facilities/Equipment/Tools/Software	267
9.10.1.4 Expected Results	267
9.10.1.5 Success Criteria	267
9.10.1.6 Results	268
9.10.1.7 Conclusion	268
9.10.2 Test 2: 30 cm strut bending moment test	268

9.10.2.1 Test Design	268
9.10.1.2 Procedure	269
9.10.1.3 Required Facilities/Equipment/Tools/Software	269
9.10.1.4 Expected Results	269
9.10.1.5 Success Criteria	269
9.10.1.6 Results	269
9.11 VV&T Summary Table	270
10. Risk Management	271
10.1 Risk Description	272
10.2 Likelihood/Frequency & Severity/Consequence	272
10.3 Mitigation Plan	273
10.4 Risk Matrix	274
10.5 Risk Realization	276
11. Project Management	276
11.1 Gantt Chart	277
11.2 Timeline	278
11.3 Tasks/Activities	279
11.3.1 Manufacturing and Test Review	279
11.3.2 System Readiness Review Report	279
11.3.3 VV&T Report	279
11.3.4 Vehicle Inspection Checklist, SOP	279
11.3.5 Final Report	279
11.4 Project Phases	279
11.4.1 Phase 1 (Fall 2024)	279
11.4.2 Phase 2: (Fall 2024)	280
11.4.3 Phase 3: (Spring 2025)	280
11.5 Work Breakdown Structure	281
11.6 Test Plan	282
12. Budget Breakdown	283
12.1 Real World Cost	283
12.2 Budget Overview	284
12.3 Prototype Budget	285
12.4 Prototype Budget Overview	286
12.5 Benefit to Customer	287
13. Lessons Learned	288
14. Individual Contributions	289
15. Conclusion	291
16. References	292
14.1 Literature Review References	292
14.1.1 Power References	292
14.1.2 Payload References	292

14.1.3 GNC References	293
14.1.5 Ground Station	293
14.1.8 Propulsion References	294
14.1.10 Structure References	295
14.2 Design Requirements References	296
14.3 Key Design Alternatives References	296
14.3.1 Power References	296
14.3.2 Payload References	297
14.3.3 GNC References	298
14.3.4 Communication and Data Handling	299
14.3.5 Ground Station	300
14.3.8 Propulsion References	301
14.3.10 Structures References	301
14.4 Evidence of Feasibility References	302
14.4.2 Payload References	302
14.4.3 GNC References	302
14.4.8 Propulsion References	302
14.4.9 Structures References	303
14.5 Design Solution References	303
14.5.1 Power References	303
14.5.2 Payload References	304
14.5.3 GNC References	305
14.5.4 Power References	305
14.5.6 Thermal References	306
14.5.8 Propulsion References	307
14.5.10 Structures References	308
17. Appendix	309
15.9 Propulsion Appendix	309

1. Information

1.1 Project Customers

Table 1.1.1: Customers Information

Customers

Dr. Felix Ewere Course Instructor feewere@ncsu.edu	Shridhar Thakar Course TA ssthakar@ncsu.edu	Dr. Bigham Course Instructor sbigham@ncsu.edu
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1.2 S.M.A.R.T. Squadron CubeSat Personnel

Table 1.2.1: Team Members Information

Team Members	
Dylan Perdue Thermal Lead Financial Lead zamoore2@ncsu.edu (919) 273 9667	Jamison Fuller Structures & Mechanism Lead Test/Safety Lead jrfulle3@ncsu.edu (919) 770-8777
Samuel Martinez Payload Lead CAD/Simulation Lead semarti8@ncsu.edu (984) 789 2538	Zachariah Moore Propulsion Lead Manufacturing Lead zamoore2@ncsu.edu (919) 273 9667
Zumi Cuji Project Manager Ground Station Lead zhcuji@ncsu.edu (910) 508-2478	Jack Lamont Guidance Navigation and Control Lead Systems Engineering Lead jvlamont@ncsu.edu (704) 910 7528

Waylon Ponder

Power/Electronics lead

Data and Communication lead

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1.3 S.M.A.R.T. Squadron Project Organizational Structure

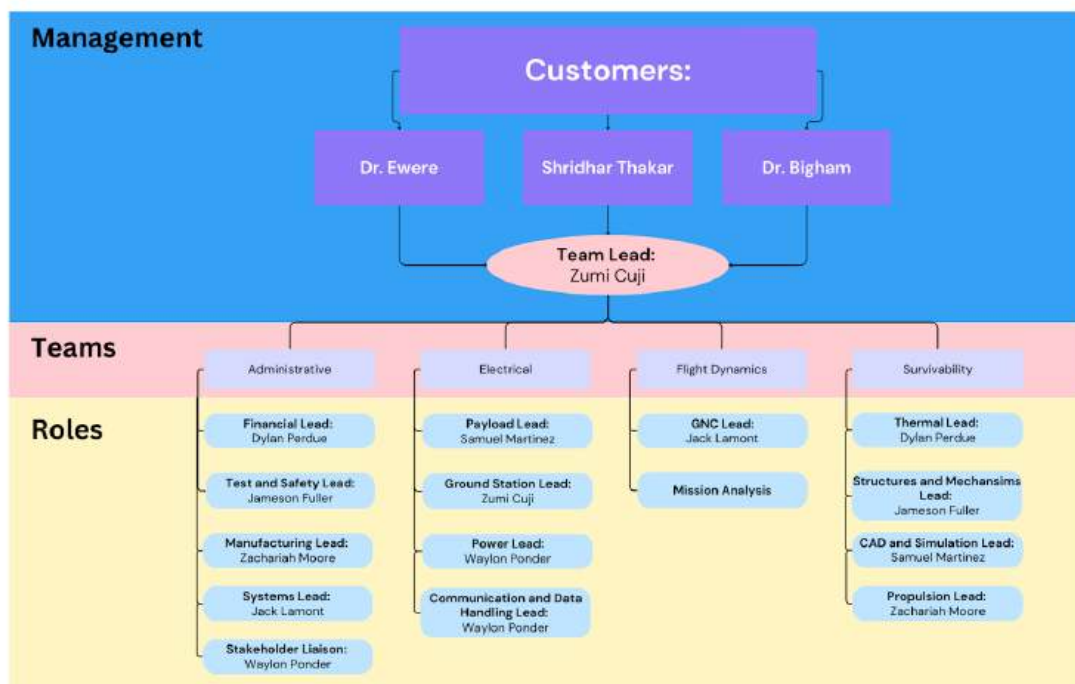


Figure 1.3.1: S.M.A.R.T. Squadron Project Organizational Structure

2. Project Description

2.1 Project Overview

The S.M.A.R.T. CubeSat aims to address search and rescue operations by providing flood mapping data during natural disasters such as hurricanes. Leveraging synthetic aperture radar (SAR) technology, the CubeSat will also measure soil moisture levels during non-disaster periods for applications in agricultural development and DoD terrain scouting. This project seeks to offer a cost-effective and efficient solution to improve data collection frequency and coverage, targeting areas of interest that are underserved by current satellite solutions like Sentinel-1A and SMAP.

2.2 Purpose & Need

The S.M.A.R.T. CubeSat project addresses critical challenges in disaster response and resource management through innovative use of synthetic aperture radar (SAR) technology. By measuring soil moisture levels, the CubeSat enhances search and rescue operations during natural disasters such as hurricanes by enabling predictive flood mapping. Beyond disaster scenarios, it supports applications like agricultural development and DoD terrain scouting, providing valuable data for decision-making. Current solutions like the Soil Moisture Active Passive (SMAP) satellite and Sentinel-1A suffer from limitations in coverage, frequency, and cost, with SMAP requiring 2–3 days for global mapping and both satellites being prohibitively expensive. This project was inspired by a team member's observation that Sentinel-1A failed to adequately cover parts of western North Carolina during Hurricane Helene, highlighting the need for a cheaper and more targeted alternative. By addressing these gaps, S.M.A.R.T. aims to offer a practical and cost-effective solution for soil moisture monitoring.

2.3 Objectives

The primary objective of the S.M.A.R.T. CubeSat is to provide accurate and timely soil moisture data to support predictive flood mapping and other applications. The CubeSat will transmit this data to a ground station and relay it to customers more frequently and efficiently than existing systems. To achieve its goals, the CubeSat must survive the launch process and maintain its orbit throughout the mission. It must orient its radar toward Earth to collect precise data and map soil moisture in designated areas of interest. It must use solar panels for power generation in orbit. Finally, to ensure responsible operations in space, the CubeSat will deorbit at the end of its mission timeline, preventing the accumulation of space debris. These objectives ensure that the CubeSat delivers high-value data while minimizing operational and environmental risks.

2.4 CONOPS

2.4.1 Concept of Operation Overview

The figure below presents the 6-phase concept of operations for the S.M.A.R.T. soil moisture detection satellite. Phase 1 requires the nanosat to ride a typical launch vehicle into low earth orbit. The satellite will be small enough to fit into any extra space in the payload of a typical ISS mission. Phase 2 has the satellite ejected from the launch vehicle's payload to achieve a low Earth orbit not exceeding 2000 km in altitude. Phase 3 will have the satellite deploy the radar dish payload once an optimal orbit is achieved. Phase 4 has the satellite establishing a connection between the ground station using the onboard radar. In phase 5, the satellite will cycle between sun pointing, scanning, and data processing to provide the functionality of the mission. Phase 6 will close out the satellite's life as it falls out of low Earth orbit and burns up on re-entry. This will occur in roughly 4-5 years, as is typical of nanosat missions.

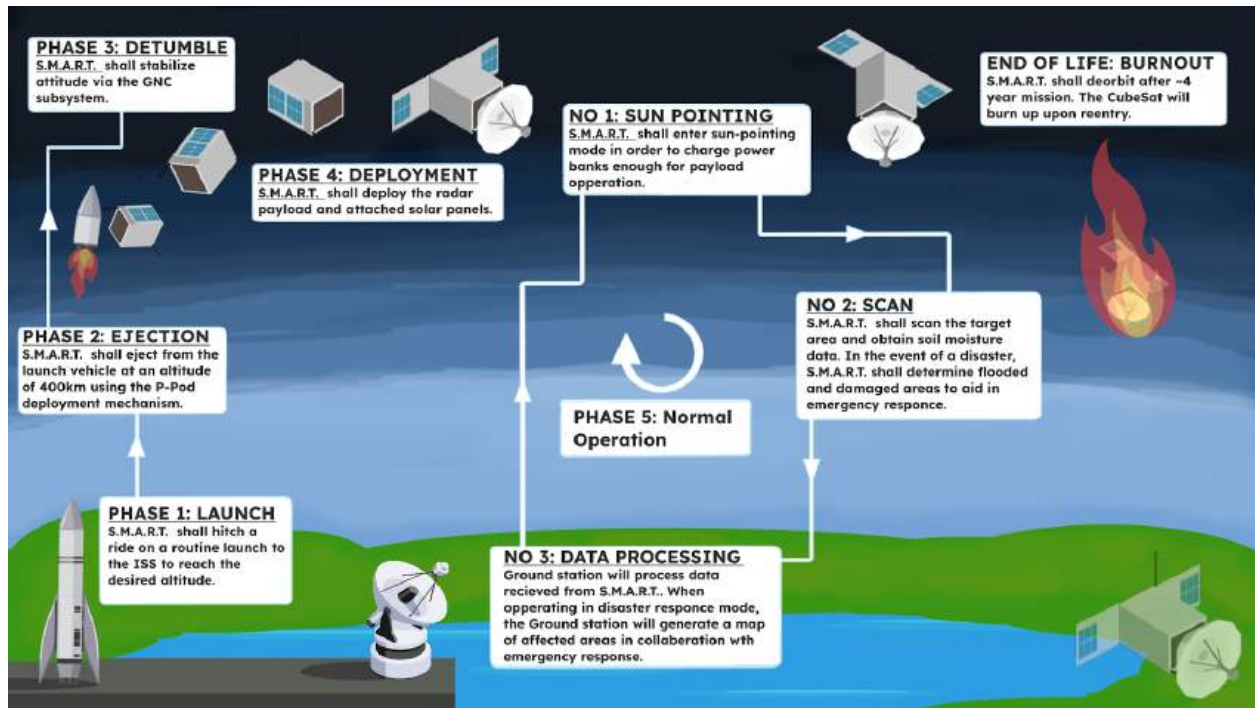


Figure 2.4.1.1: Concept of Operations

2.5 Design Solution

The following figures illustrate the design solution for the S.M.A.R.T. CubeSat, which serves as a scaled-down proof of concept for addressing the project's objectives and validating its feasibility within the constraints of cost, time, and testing.

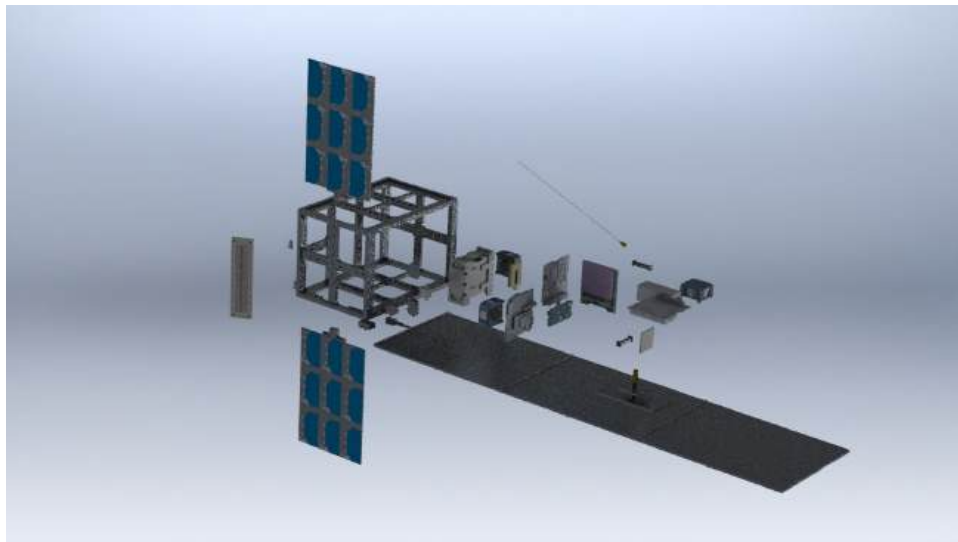


Figure 2.5.1: Internal View of Prototype Satellite

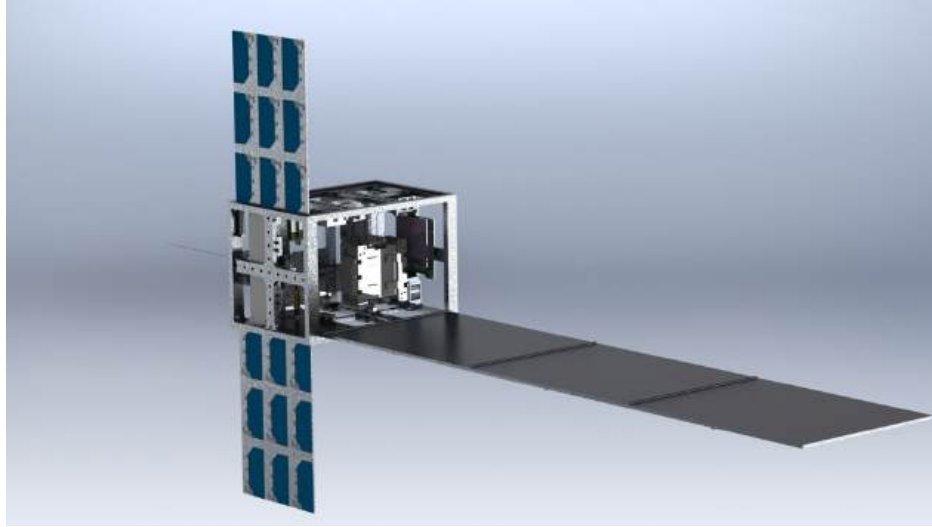


Figure 2.5.2: External View of Prototype Satellite

The CAD renders above showcase the S.M.A.R.T. CubeSat in its deployed configuration, modeling the full-scale system on a smaller scale to demonstrate feasibility and practicality. The deployed configuration illustrates the satellite's operational state.

This design approach offers a cost-effective platform for testing critical functionalities such as synthetic aperture radar (SAR) operation, data transmission, and power generation, while validating structural integrity and subsystem interactions. By scaling down the project, the design provides an efficient, manageable means to prototype and refine technologies that will be used in full-scale implementations. It ensures that the satellite meets mission objectives, such as collecting soil moisture data for flood mapping and agricultural applications, while adhering to industry standards for compact, reliable satellite systems.

The render demonstrates the design's compliance with 12U CubeSat dimensional constraints, reflecting the project's scalability and readiness for practical application. The CubeSat's modularity and compact design ensure proof-of-concept success while minimizing risks associated with cost, time, and resource allocation.

2.6 Functional Block Diagram

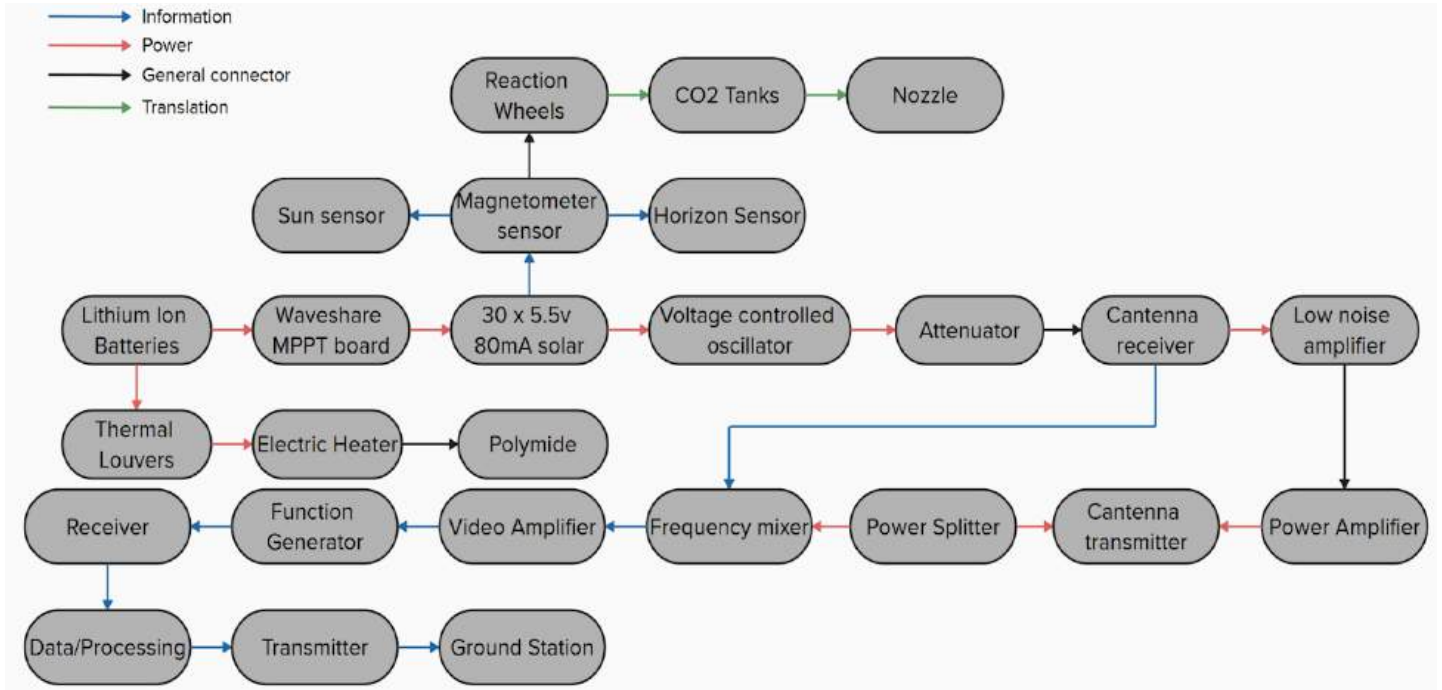


Figure 2.6.1: Functional Block Diagram

3. Functional Requirements

FR1: *The CubeSat shall conform to the U size standard [FRC1].*

DR1.1: The total combination of units shall be 12U, 20x20x30 cm.

DR1.2: The cubesat mass shall be under 24 kg.

- **Motivation:** The mass must not exceed payload restrictions for launch. The contest requires conformity to the U size standard.
- **Verification:** Dimensions and weight will be minimized throughout design. During constructions, the dimensions and weight will be measured continuously.

FR2: *The CubeSat shall maintain operational status for 4 years.*

DR2.1: CubeSat shall have dedicated deorbit propulsion.

- **Motivation:** Four years will allow for ample data collection. Too much longer and the propellant needed will be excessive. A dedicated system will prevent the system from running out of propellant and becoming stranded.

- Verification: Systems Tool Kit will be used to verify the propulsion method.

FR3: *The CubeSat shall operate in low Earth orbit.*

DR3.1: Orbital period shall be 1.6 hours.

DR3.1.1: Satellite shall scan desired location once a day.

DR3.1.2: Satellite should scan desired location three times a day.

- Motivation: Low Earth orbit will not allow for constant surveillance of a region. These are estimates based on orbital period and charging rates.
- Verification: STK will be used to simulate orbit.

DR3.2: The CubeSat shall operate at an altitude of 567 km.

DR3.3: The CubeSat should operate within 200 km to 800 km.

FR4: *The CubeSat shall use synthetic aperture radar to scan for landscape topography [FRC2].*

DR4.1: The radar system shall scan a target area of 10 km².

DR4.1.1: The radar system shall detect top soil moisture levels.

- Motivation: This will allow for soil moisture levels that are relevant to flooding possibilities and agricultural viability.
- Verification: Radar penetration rating will be measured in centimeters.

DR4.2: The radar antenna shall deploy upon reaching operational orbit.

- Motivation: Extended radar antenna would not fit in contest size requirements.
- Verification: Collapsible design will fit in a smaller area.

DR4.3: The radar system shall be able to activate and deactivate to conserve power.

- Motivation: Power to the radar is not required when not taking in data.
- Verification: Radar is used for data collection. A simple switch should suffice for activation and deactivation.

DR4.4: The radar system shall have a maximum range of twice the altitude of the CubeSat.

- Motivation: The radar must be able to receive signals at a power level high enough to be measured.
- Verification: Calculations will be done based on the dimensions of the radar antenna and power output to ensure theoretical maximum range and experiments will be done once constructed to confirm these values.

FR5: *The CubeSat shall survive the loads invoked by the launch.*

DR5.1: The CubeSat shall survive a load of 5G's.

DR5.1.1: The CubeSat shall survive the load for 15 minutes.

DR5.2: The CubeSat should survive a load of 7G's.

DR5.2.1: The CubeSat should survive the load for 20 minutes.

FR6: *The CubeSat shall communicate topographical data of the landscape to the ground station [FRC3].*

DR6.1: A detailed topographical map of soil moisture levels shall be communicated to the ground station.

DR6.1.1: The details of soil moisture variation shall be communicated to the ground station using radio communication.

DR6.1.1.1: Details should be communicated to the ground station using S band radio frequencies.

FR7: *The CubeSat shall be powered using solar energy.*

DR7.1: The power demand shall be met using solar panels.

DR7.1.1: The power bank shall supply enough power to the radar antennae to communicate with the ground station.

DR7.1.2: The power bank shall supply enough power to SAR to complete one pass over the desired target area.

DR7.1.3: The power bank shall supply enough power to GNC to perform attitude maneuvers.

DR7.1.4: The power bank shall supply enough power to the propulsion system to perform maneuvers in orbit.

DR7.1.5: The power bank shall supply enough power to on board data handling computers to deliver information.

DR7.1.6: The power bank shall supply enough power to the thermostat to the thermal system to trigger an active thermal management system.

DR7.2: The GNC system shall be able to maintain sun pointing for charging orbits.

FR8: *The CubeSat shall survive thermal stresses from low earth orbit.*

DR8.1: CubeSat shall survive in temperatures ranging from -65°C to 125°C.

- Motivation: Delicate parts must be within a certain temperature range, outside of this temperature range would cause failure
- Verification: Run thermal simulation software to verify parts stay within the specified temperature range for electronics

DR8.1.1: CubeSat shall be able to dissipate heat generated during SAR scans.

DR8.2: CubeSat shall survive and maintain full functionality with space radiation.

DR8.2.1: CubeSat should be able to dissipate heat generated by the SAR system in between scans.

DR8.3: CubeSat should survive temperatures ranging from -150°C to 150°C.

DR8.3.1: CubeSat should be able to dissipate heat generated by all subsystems.

DR8.3.2: CubeSat shall be able to generate heat during cold cases

DR8.4: CubeSat shall be able to read temperatures internally to determine what actions to take to cool off or heat up the CubeSat.

DR8.5: CubeSat shall maintain a temperature range of 0°C to 50°C to maintain operating temperature range

- Motivation: Electronics need to stay within this temperature range in order to function properly or failure will occur
- Verification: Run simulation to make sure the satellite can hold this internal temperature while considering L.E.O. temperatures

DR8.6: CubeSat structure shall not fracture or break under thermal stresses

FR9: *CubeSat shall adjust attitude using dedicated guidance, navigation, and control system.*

DR9.1: CubeSat shall maneuver via 3 axis reaction wheels.

DR9.1.1: Reaction wheels shall spin at a needed rate to create torque on system

- Motivation: CubeSat shall have the SAR pointed towards Earth at all times in order to create readings of ocean waves.

- Verification: Before launch, torque generated by reaction wheels shall be experimentally tested to ensure correct force being applied to the satellite.

DR9.2: CubeSat shall be in contact with ground station for position coordinate pointing.

DR9.3: CubeSat shall point at desired area for SAR scan.

FR10: *CubeSat shall communicate telemetry data to the ground station.*

DR10.1: The radio communication system shall transmit data to the ground station.

DR10.1.1: The radio communication system shall transmit GNC data to the ground station.

DR10.1.1.1: The communication system should transmit GNC system anomalies to the ground station.

- Motivation: Transfer of information is essential to the mission. The CubeSat would be useless if unable to communicate.

- Verification: Transfer packets of information across the system while on Earth. Information packets will be similar in size and complexity of what is to be expected from data collection.

DR10.1.2: The radio communication system shall transmit SAR data to the ground station.

DR10.1.2.1: The radio communication system should transmit SAR system anomalies to the ground station.

DR10.1.3: The radio communication system shall transmit positional data to the ground station.

DR10.1.3.1: The radio communication system should transmit altitude data to the ground station.

DR10.1.4: The radio communication system shall transmit thermal data to the ground station.

DR10.1.4.1: The radio communication system should transmit thermal system anomalies to the ground station.

DR10.2: The radar communication system shall transmit telemetry data using UHF communications at 2.4 GHz.

FR11: *CubeSat will collect data using an On-Board Data Handling system (OBDH).*

DR11.1: The OBDH system shall collect radio communication data.

DR11.1.1: The OBDH system shall store radio communication.

DR11.2: The OBDH system shall collect SAR topographical data.

DR11.2.1: The OBDH system shall store SAR topographical data.

DR11.3: The OBDH system shall collect GNC attitude data.

DR11.3.1: The OBDH system shall store GNC attitude data.

DR11.4: The OBDH system shall collect thermal data.

DR11.4.1: The OBDH system shall store thermal data.

4. Key Design Alternatives

In the development of the CubeSat prototype, several design alternatives were considered to ensure the most effective and efficient solution. These alternatives address various aspects of the system, including communication methods, power management, propulsion, and structural design. Each alternative was evaluated based on criteria such as cost, performance, complexity, and compatibility with the overall mission objectives. The following sections outline the key design alternatives that were explored, including their advantages, disadvantages, and the reasoning behind the final design choices.

4.1 Electrical Power System

The electrical power system (EPS) of a CubeSat is responsible for power generation, storage, and delivery to all subsystems. Power generation is achieved using photovoltaic cells arranged on a solar panel as is typical for satellites. The power can then be distributed to meet the requirements of each subsystem, with any remaining energy being stored in the on-board power bank. With these functions in mind, the EPS can be broken down into three main elements: the battery, the EPS architecture, and the solar panels.

4.1.1 Battery

Important factors to consider when choosing a battery are sizing, maximum power capacity, voltage, current, and life cycle. The battery is a crucial element in the mission design, as the power drawn from the SAR payload combined with attitude adjustment and data transmission will require a large power bank to maintain operation. The size of this power bank will be estimated based on NASA's RainCube weather CubeSat design. The RainCube microsat had a power output of 30W at its peak [Power 1]. Due to the similarity in SMARTSat's design and application, RainCube will be a reference point for the power requirements to be expected. Due to SMARTSat's more ambitious goal of high-resolution topography capture, a power requirement of >50W power output should be anticipated. The battery used on the RainCube was a 70Wh rechargeable battery pack, capable of powering one single pass with three sun-pointing orbits [Power 2]. For the SMARTSat, similar power storage capacities should be sought after. The voltage and current output of the battery will be determined by the components of the

system and their respective requirements. Voltage and current are only a major concern for designing battery packs in house, as most commercial batteries are designed with typical subsystems in mind.

With a relative capacity of the battery chosen, the chemistry should be decided next. Three common batteries used in CubeSats are lithium-ion (Li-ion), nickel-cadmium (Ni-Cd), and lithium-polymer (Li-Po). The main difference between these battery chemistries is power density, cycle life, and efficiency.

4.1.1.1 Lithium-Ion

Lithium Ion batteries are the most popular choice for energy storage in CubeSat applications. According to a review published by MDPI in 2020, Li-ion batteries made up two thirds of batteries used in nanosatellites in 2010 [Power 3]. As battery technology has advanced, the percentage of CubeSats powered by Li-ion batteries has only increased. The high energy density of Li-ion batteries make them an appealing choice for CubeSat missions. Other appealing characteristics of Li-ion batteries are their high cycle life, and low self discharge rate [Power 4]. These determine the lifetime and efficiency of the battery, overall lengthening the mission duration.



Figure 4.1.1.1.1: Lithium Ion battery produced by Pumpkin, Inc. [Power 5]

The above picture shows a lithium ion battery produced by Pumpkin, Inc. a company that produces satellite components. This battery will act as the main comparison for lithium ion batteries produced commercially.

4.1.1.2 Lithium-Polymer

Lithium polymer batteries are another common energy storage device for CubeSats. They are a subcategory of Li-ion batteries. Lithium polymer batteries typically boast faster charging capabilities when compared to Li-ion batteries as their main advantage. Lithium polymer batteries have comparable energy densities, cycle lives, and cell voltages to Li-ion as well. Despite these similarities, Li-Po batteries are not nearly as common for commercial CubeSats. The main reason for Li-Po batteries being less popular among CubeSat designers likely comes down to the design of the battery itself.

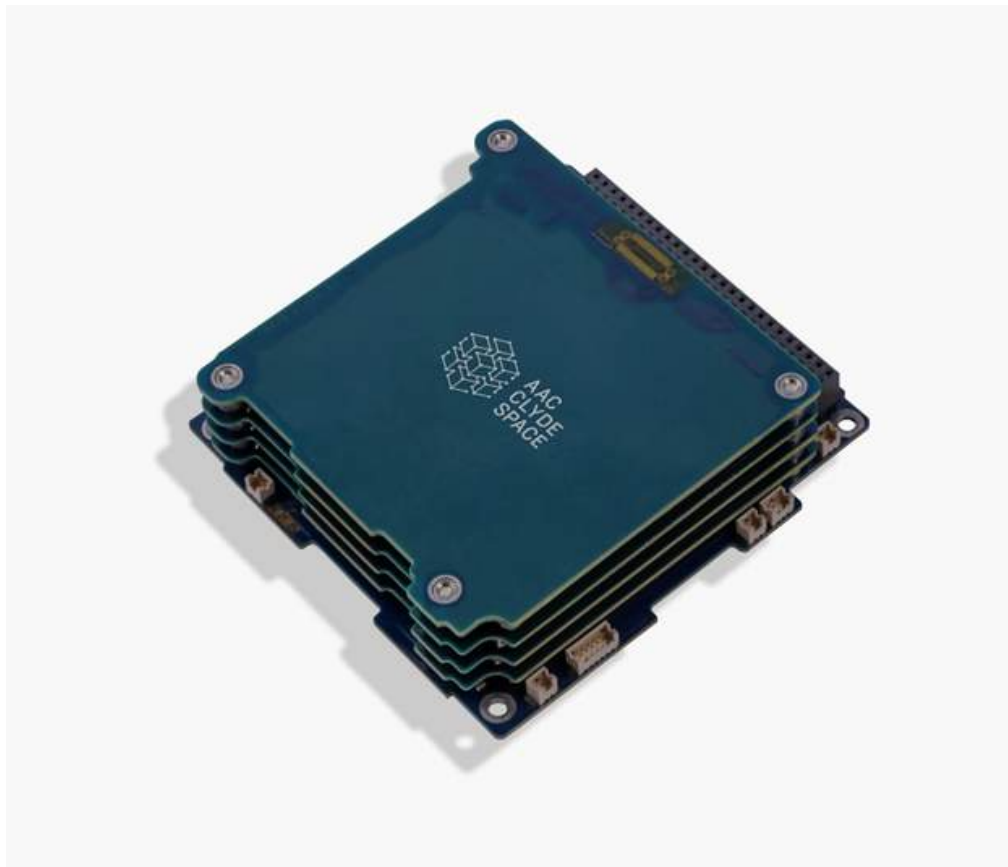


Figure 4.1.1.2.1: Lithium Polymer battery produced by AAC Clyde Space [Power 6]

Pictured above is an 80Wh Li-Po battery produced by AAC Clyde Space. This battery pack will act as the main comparison for lithium polymer batteries.

Lithium Polymer batteries use a pouch design rather than the typical cell configuration. The main concern with this approach is that the pouch can swell under vacuum conditions, possibly leading to a failure of the system [Power 7]. This is a major drawback of the design, as any power failure in-flight will result in a mission failure.

4.1.1.3 Nickel Cadmium

Nickel cadmium batteries are another popular choice for satellite power storage. These batteries have been used in space applications since the 1960s [Power 8]. Ni-Cd batteries also tend to be far cheaper than that of lithium-ion battery configurations. However, the Ni-Cd chemistry has serious drawbacks when used in CubeSat designs, leading to a nearly nonexistent market share in commercial CubeSat batteries. Nickel cadmium power cells have much lower energy densities than Li-ion, leading to larger power banks needed to match the capacity of a Li-ion pack. Figure 51.1.1.3.1 does not showcase a Ni-Cd battery, but rather a good metric for the size of a Ni-Cd battery that the SMARTSat will require.



Figure 4.1.1.3.1: EPS system with integrated battery pack produced by Skylabs [Power 9]

For a Ni-Cd battery pack to be used in the SMARTSat power system, a battery weighing 1-2 kg. The mass of these batteries can be calculated with the following equation:

$$\text{Mass of Battery (kg)} = \text{Capacity Desired (Wh)} / \text{Specific Energy (Wh/kg)}$$

(4.1.1.3.1)

Using the typical specific energy of a Ni-Cd battery of between 45-80 Wh/kg, the mass of the battery is found to be between 1-1.78 kg [Power 10]. With the added weight of the eps system, this only further increases the mass profile of the power system. The battery in Figure 4.1.1.3.1 depicts an eps system with a 158 Wh battery weighing a total of 2 kg [Power 9]. With the previous calculation, it can be assumed that the battery in Figure 4.1.1.3.1 gives a rough approximation of the size of a Ni-Cd battery if used in the SMARTSat system. With that said, a Ni-Cd battery will need to be made in house if it is used due to the lack of commercially available parts.

4.1.1.4 Battery Comparison

Below is a table with the overall advantages and disadvantages of each battery choice. These factors will be used to determine the final considerations for the battery chemistry in the trade analysis.

Table 4.1.1.4.1: Battery Comparison

Battery	Pros	Cons
Li-ion	• Long lasting • High energy density • COTS Available	• Cost • Temperature concerns
Li-Po	• Faster charge rate High energy density COTS Available	• Cost • Less energy dense than Li-ion • Temperature concerns
Ni-Cd	• Second most popular Long heritage • Reliable • Low Cost	• Low energy density • Largely outdated • Likely need to make in house • Much shorter lifespan

4.1.2 Electronic Power System Architecture

The EPS architecture acts as the main power delivery system for the satellite. Power is transferred from the solar panels to the subsystems and battery via a regulated DC bus in most EPS designs. The interface between the solar panels and the EPS is typically a maximum power

point tracker (MPPT) protocol. MPPT is common for CubeSats due to limited power generation in LEO [Power 11].

CubeSat EPS architecture generally comes in three varieties for most designs. Commercial off the shelf (COTS) systems with integrated power storage, COTS systems without integrated power storage, and custom EPS. These will be the main considerations for the design of SMARTSat.

4.1.2.1 Standalone Commercial EPS Architectures

Many COTS EPS designs do not have an integrated power storage unit. This allows for more flexibility in the power management system of the satellite, allowing for a battery to be specially chosen for the application. However standalone EPS does have the added drawback of a higher cost for the final design. Another downside is the final form factor of the power management subsystem. If the battery is not integrated with the EPS system, the final volume of the EPS architecture and battery may take up more space than an integrated power storage unit. To counteract this, the systems could be mounted together with a custom part to ensure rigidity and reliability of the design.

Commercial EPS systems also have the added benefit of being radiation tolerant by design. This is also true for systems with an integrated battery. As electrical components are placed in orbit, they will gradually degrade as ions in solar winds narrow the band gap inside semiconductors and disrupt their crystal structure [Power 12]. EPS and any electronic component commercially manufactured for use in CubeSats is made partially radiation resistant to prevent any damage from the sun's solar winds.

Radiation in low Earth orbit is not a high enough concern to make incredibly rad-hard systems used in deep space, but one major concern is the effects of cosmic rays. Cosmic rays can introduce something known as a single event upset or SEU which can result in soft or hard damage to a circuit. SEUs can introduce voltage rises, current rises, and bit-flips to a circuit. With the EPS, the biggest risk to the integrity of the power delivery system is a single event latch up, where a massive current is introduced to the system from cosmic rays [Power 12]. One way to prevent this is with a current limiter, that will limit the current supplied to the circuit in the event of a latch up. Pictured below is a typical EPS sold by Ibeos.



Figure 4.1.2.1.1: EPS system designed by Ibeos [Power 13]

This power delivery system comes equipped with a radiation resistant design and a watchdog timer to protect against radiation degradation. The Ibeos EPS will act as the main comparison for commercially produced electrical power systems.

4.1.2.2 Commercial EPS architectures with Integrated Power Storage

Integrated power storage in commercially produced EPSs provides the same qualities as commercial EPSs with the added benefit of power storage included in the design. This is not only convenient but also has the added benefit of reducing the overall volume of the EPS system. Another added benefit is that the EPS with an integrated battery is more cost effective than purchasing the battery separately. However, this comes at the cost of versatility and modularity. The figure below is an EPS developed by DHV Technology and will act as the main comparison for commercial EPSs with integrated power storage.

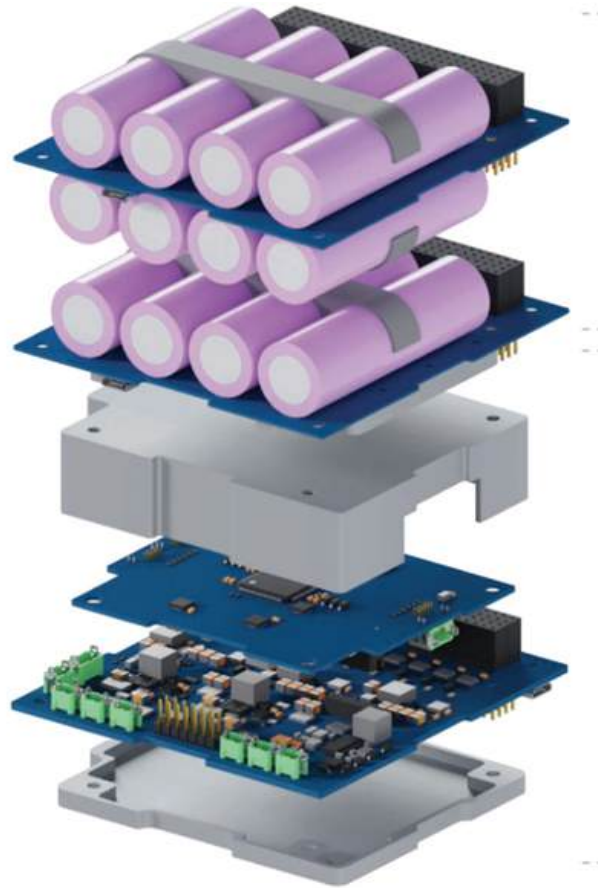


Figure 4.1.2.2: EPS with integrated power bank produced by DHV Technology [Power 14]

Due to the high energy consumption of the SAR payload, the integrated power supply unit may limit the efficiency of SAR operation in tandem with system diagnostics and telemetry management data. If the integrated battery does not provide the power needed for all subsystems to function in all operation modes, then an additional battery may be required, thus negating the benefits of integrated power storage.

4.1.2.3 Custom EPS

The most versatile option for an electrical power system is one manufactured in house. This would ensure that all design conditions are met, and that the system will have no power delivery limitations during operation. Issues with this approach are development time, R&D costs, and a lack of heritage for the component. Rather than having satellites utilize this component during missions as a reference and confirmation of performance, SMARTSat will be the first satellite to test the prototype EPS. This method should only be used if the institution manufacturing the component has a heritage of its own with fabricating highly complex electronic components.

Another major concern of manufacturing an electrical system as complex as the power delivery system is solar radiation and cosmic rays. For a component to be space-grade, it must undergo rigorous testing before such qualifications can be obtained. Manufacturing a radiation resistant circuit for power distribution would be incredibly challenging but would also guarantee that power is transferred efficiently to all subsystems without issue. Cosmic rays can be mitigated partially by using circuit limiters and watchdog circuits to ensure that the design will not fail under the intense conditions present in low Earth orbit.

4.1.2.4 EPS Architecture Comparison

Below is a table depicting the main advantages and disadvantages of each EPS design. This will help determine the criteria used for the trade analysis between each component.

Table 4.1.2.4.1: EPS Comparison

EPS Architecture	Pros	Cons
Stand Alone	• Simple implementation • Widely compatible • Flexible Design	• Higher total cost than integrated battery system
Integrated Battery	• Small form factor • Simplest installation	• Less versatile • Fixed power supply
Custom Built	• Purpose built for mission requirements • Ensures all systems are supplied efficiently • Incredibly versatile	• Requires research and development • Space grade needed • Lack of heritage

4.1.3 Solar Panels

Solar panels act as the main source of power generation for any satellite. The solar panel design will determine the maximum total power that can be generated, the time needed to fully charge, and the efficiency of the charging procedure. Total power generation is determined by the solar panels surface area. The time needed to fully charge is determined by both the power generation, and orientation of the panels. Efficiency of a solar panel is determined by both photovoltaic cell efficiency and the orientation of the panels relative to the sun.

Panel orientation will be the main factor that separates different solar panel designs. Orientation will determine the maximum size of the total solar panel and in turn the total power generated by the system. For orientation, three options will be considered: fixed, deployable, and in house deployable.

4.1.3.1 Fixed Solar Panels

Fixed solar panels are attached and situated to the outside of a CubeSat. These are used for simple, low power demand designs such as 1U-3U CubeSats. The solar panel does not move during operation of the satellite. Despite this being both the simplest and most cost-effective design, the power generation for this method is very limited. Below is an image that depicts a fixed solar panel orientation.

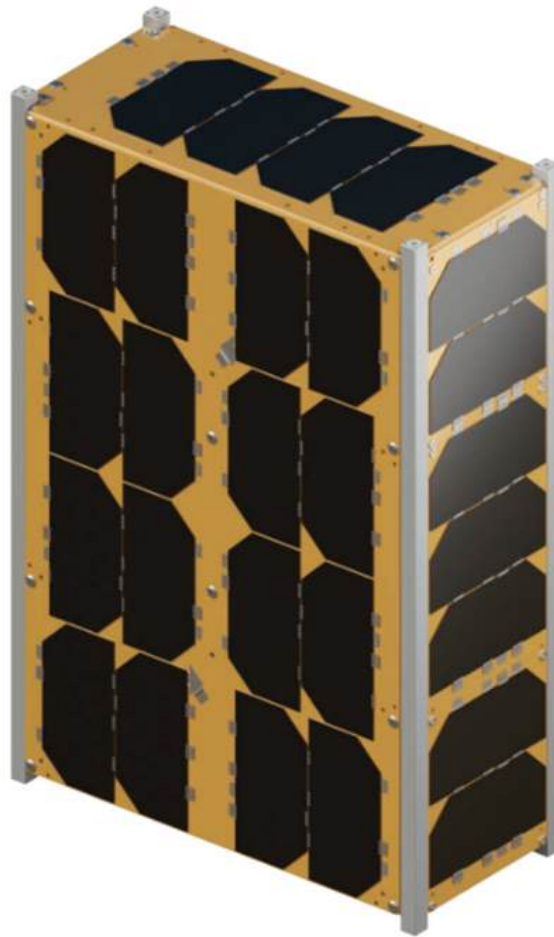


Figure 4.1.3.1.1: Fixed Solar panel orientation [Power 15]

The power generation of a solar panel is dependent on the photo voltaic cells' exposure to sunlight as well as the inclination angle of sunlight relative to the cell for a given sunlit period. With a fixed solar panel design, two approaches can be used. The entire outside of the satellite

can have solar panels attached to maximize passive charging, or only one side can have solar panels, and the system can utilize active charging. Active charging refers to a sun pointing mode where power generation is prioritized, and the GNC system will ensure that the solar panels are facing a direction that maximizes power generation.

Regardless of what orientation is used, SMARTSat will require dedicated charging times in between active orbits due to insufficient charging. This can be expected due to the heavy power costs of the mission and the limited charging capabilities of solar panels. Looking at RainCube, for every active orbit, three sun pointing orbits were needed to power the Ka band radar [Power 1]. This ratio of passive to active orbits is expected for the

4.1.3.2 Deployable Solar Panels

Much like RainCube, deployable solar panels are a much more appealing option for the SMARTSat mission requirements. To increase the surface area exposed to the sun, the solar panels can be deployed from the side, with the photovoltaic cells facing towards the sun during sunlit observation periods while the SAR can be pointed at Earth at the same time. This ensures charging capabilities and full system implementation are possible to achieve in tandem. Deployable solar panels also offer maximum charging potential during sun pointing orbits. With proper coordination between the GNC system, the solar panels can achieve maximum power generation for nearly half of a single orbit. At an orbital period of an hour and a half and an altitude of 400km, this would give the solar panels around 45 minutes of charging time per sun-pointing orbit.

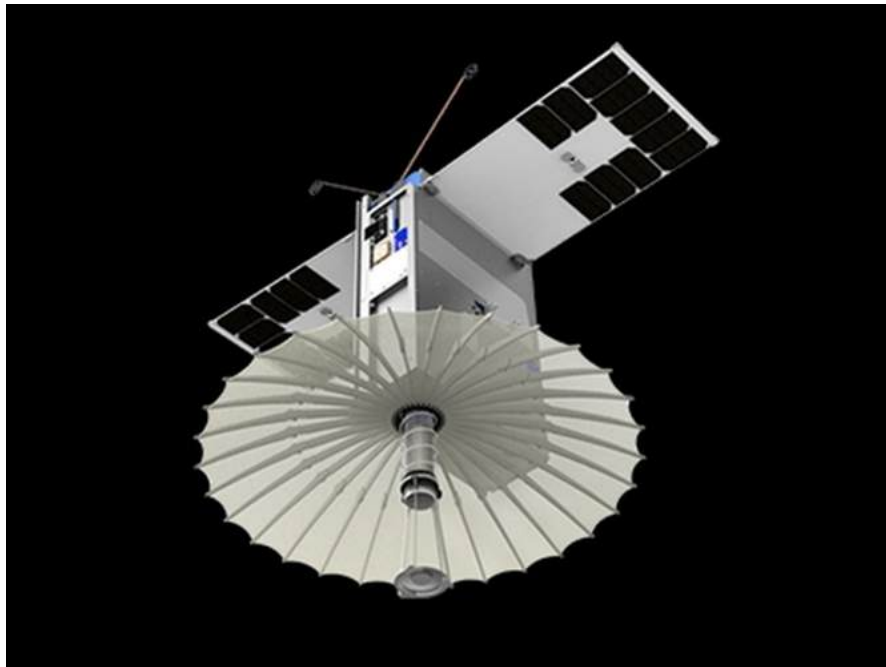


Figure 4.1.3.2: Deployable Solar panel orientation utilized in RainCube [Power 16]

One major downside to this method is an increase in the complexity of the design. Sun pointing will require a power draw from the GNC subsystem for nearly the entire charging orbit, reducing the effectiveness of the charging potential. However, this method will still maximize the power generation of the system, enabling longer data collection and transmission periods. Another issue with complexity is the requirement that the panels need to be deployed before optimal charging operations. This will lengthen the initial orbit phase of detumbling and ground station link establishment.

4.1.3.3 Custom Deployable Solar Panels

While similar to the overall design of the RainCube, the SMARTSat will likely produce a higher power output, resulting in a deployable solar panel design. This brings up the possibility of manufacturing in-house solar panels with existing solar cells on the market to design a deployable solar panel, capable of generating enough power to the subsystems. Making deployable solar panels in house has the added benefits of lower cost implementation and higher versatility in the design. With more cost-effective production, more solar panels can be implemented in the in-house design at the same price of smaller commercial panels, allowing for a higher power generation.

Another benefit to in-house solar panels is the ability to experiment with different deployment mechanisms. One such design could have a multistage deployment mechanism for the panels, allowing for even greater power generation. Initially, two layers of solar panels would be folded over one another. First, one layer would fold out, then the second layer would fold out from the first layer, resulting in a much higher surface area and in turn a higher power generation than most commercial solar panels can achieve. Pictured below is an example of a multistage, fold out design.



Figure 4.1.3.3: Multi Stage Deployable Solar panel orientation [Power 17]

Solar panels are far more feasible to manufacture than an in-house EPS or on-board computer (OBC) due to the simplicity of solar panel circuits when compared to other more complex printed circuit boards (PCBs). While this option will require further research and development in constructing space grade panels, the potential to increase power generation and improve efficiency of the operation of the satellite makes this approach appealing. The capability of in-house solar panel designs from universities has already been proven in the past. The University of Alberta constructed in-house solar panels for the northern SPIRIT constellation consisting of three separate CubeSats [Power 18]. These solar panels performed well in orbit, successfully supplying power to the satellite subsystems.

The main motivation for custom panels is the cost of solar panels for CubeSats. Typical 6U panels cost approximately \$20,000 [Power 19]. According to Mike Piszczor, chief of the Photovoltaic and Electrochemical Systems Branch at Glenn Research Center, space-grade photovoltaic cells cost approximately \$400-\$500 per cell [Power 20]. At 16 cells per 6U face, this places the photovoltaic cells alone at \$8000, still far cheaper than a commercial 6U panel. This account was given in 2016, so the market cost of a single cell may be even low now. Though, without direct contact with a photovoltaic manufacturer, it's difficult to get an accurate estimate.

Cutting the cost of solar panels while providing adequate power generation will not only allow for more of the budget to be allocated to other subsystems, but it would also give NC State

the opportunity to manufacture space grade solar panels in house. This would give students in the future CubeSat club at NC State the opportunity to learn more about what goes into developing and manufacturing space-grade components, deepen their electrical knowledge, and give hands-on experience.

4.1.3.4 Solar Panel Comparison

Below is a table that compares the advantages and disadvantages of each solar panel design. This comparison will help determine the main factors to consider with the trade analysis of the solar panels.

Table 4.1.3.4.1: Solar Panel Comparison

Panel Design	Pros	Cons
Fixed	• Low cost • Ease of implementation • Lack of deployment phase	• Low power generation • Longer charging phase • Lower efficiency
Deployable	• High power generation • High charge efficiency • Shorter charge time	• Sun pointing subroutine required • Heavier than fixed
In-House Deployable	• Lower cost than COTS High charge efficiency	• Sun pointing subroutine required • Heaviest configuration

4.2 Payload Alternatives

Although there are many parts that make up the whole of a SAR, most have only one option that provides the functionality required by our mission parameters. Of what's left, most provide a choice between only two options, like SDR or HDR for signal generation where HDR must be chosen immediately due to the power constraints of our satellite. There are only a couple key alternatives to consider which do not have immediately obvious options. Namely, the antenna shape and style and the exact frequency band.

4.2.1 Antennas

Key to the functionality of the SAR, the receiving antenna shape and style must be chosen based on which type best suits our mission requirements, power restrictions, and budget.

4.2.1.1 RainCube Adjacent Antenna/Dish Antenna

The RainCube's antenna is an example of a CubeSat radar antenna. Naturally, building a similar antenna would be an option to consider. The RainCube antenna's cost is not public information. If built in house, we estimate a material cost of around \$1000. Even if the structure proves to be quite large in comparison to RainCube's, the cost shouldn't be excessive compared to the project's budget. The RainCube antenna functionally opens like an umbrella and forms a dish. This could prove hard to manufacture, but fits the mission size requirements well. Use as a SAR antenna could have issues and result in small swath width.

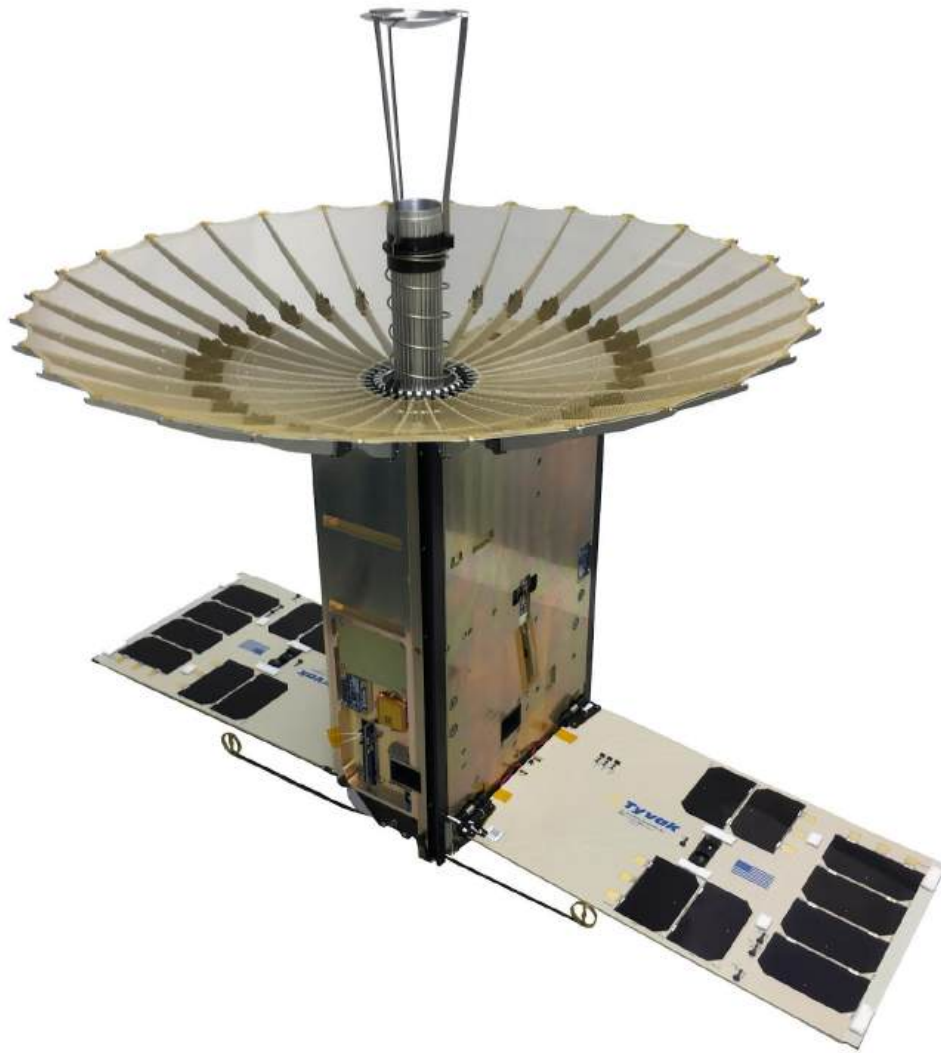


Figure 4.2.1.1: *RainCube with Deployed Radar Antenna* [PAY11]

Unfortunately, RainCube was not a SAR equipped CubeSat. Instead it measured precipitation using Ka-band radar. However, a dish shape can still be considered for the purposes of our mission.

4.2.1.2 RADARSAT-1 Adjacent Antenna/Panel Antenna

RADARSAT-1, while not a CubeSat, used a SAR like what is planned for SMARTSat. Its antenna resembles a set of panels in a line. It is a NADIR S-Band Antenna [PAY12]. This style of antenna is most common among already operational SAR spacecraft.

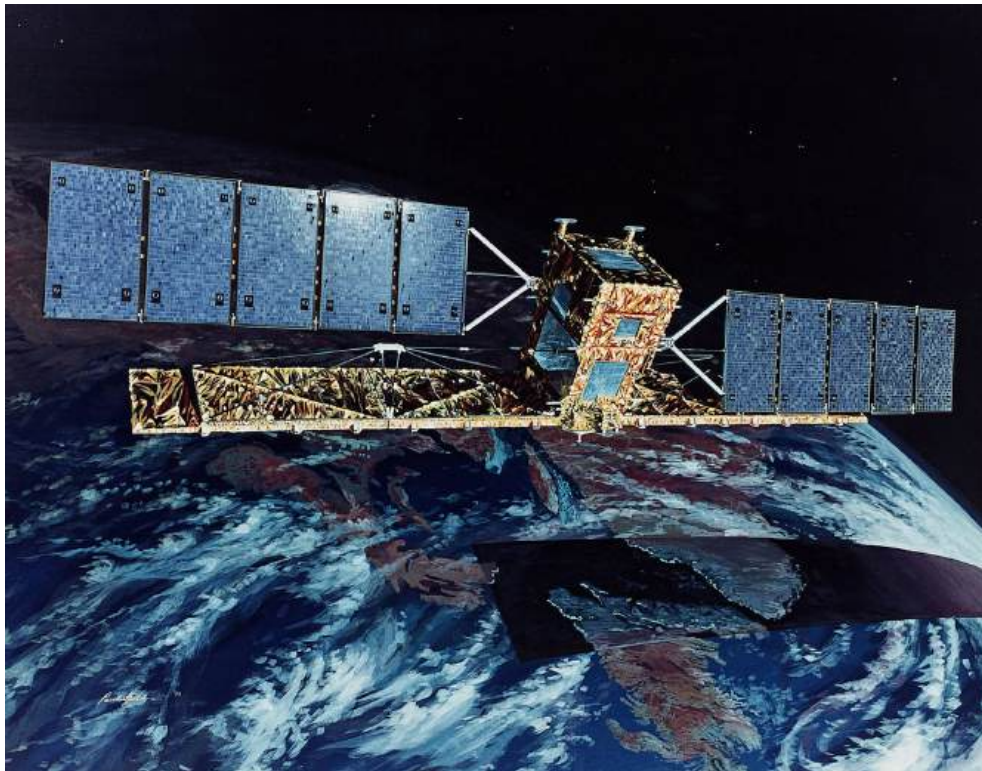


Figure 4.2.1.2: RADARSAT-1 with Deployed Panel SAR Facing its Nadir [PAY13]

4.2.1.3 Sentinel-1 Adjacent Antenna/Array Antenna

Sentinel-1 uses a SAR for monitoring wind and waves. Its targets are very similar to our mission objectives, but are larger. It used an experimental SAR method called TOPS, and a used a C-Band array antenna [PAY14].



Figure 4.2.1.3: Sentinel-1 with Phased Array Antenna Deployed [PAY15].

4.2.1.4 Capella Space Synthetic Aperture Radar Adjacent Antenna/Mesh Antenna

Capella Space SAR uses a large deployable antenna. This would be useful for minimizing undeployed volume on SMARTSat. Capella uses circular mesh reflector antennas that can be stored in a small volume. The antenna is X-Band [PAY16]. A similar design for SMARTSat could prove hard to manufacture, expensive, and difficult to deploy, but would be cutting edge for a commercial SAR and allow for sub-meter resolution at a 20 km swath width if implemented like in Capella's product [PAY17].

4.2.1.5 Horn Antenna

Horn antennas are good for receiving clean signals. They are also very well suited to long distances. The signals received are not focused, and thus the target area is not limited, but noise and resolution would be less than ideal [PAY18].

4.2.2 Radar Bands

Similar to antenna shape and style, the frequency of the radar signal determines some of the key parameters of a SAR. Certain frequencies are better suited for penetration through clouds, weather, vegetation, and even the top layer of soil. Others are better suited for higher resolutions, resistance against solar radiation, and require more power. Our choice of frequency band will determine when and where SMARTSat can collect data.

4.2.2.1 P Band: 250 to 500 MHz

P-band is well suited to penetrating clouds, fog, and even solid earth for a few meters. Their wavelengths are around 30 to 100 cm, which makes them the largest of the SAR bands [PAY19]. This option offers the best penetration, though it is comparable to L-band and will have great difficulty operating during periods of high radiation. In addition, lower bands will require larger antennas to operate effectively.

4.2.2.2 L Band: 0.5 to 1.5 GHz

L-band is also subject to solar interference, and should not be used during the day, or during major solar activity. L-band has good penetration and low power requirements. L-band wavelengths range from 15 to 30 cm [PAY19].

4.2.2.3 S Band: 2 to 4 GHz

S-band is less subject to solar interference, but should not be used during major solar activity. They are well suited to penetrating clouds and fog. S-band wavelength ranges from 7.5 to 15 cm, and offers better penetration than C, X, and K Bands, but worse penetration than the lower bands [PAY19].

4.2.2.4 C Band: 4 to 8 GHz

C-band are well suited to penetrating clouds, fog, and precipitation, but cannot penetrate thick canopies or much solid earth. Rain cores and hydrometeors can reduce the signal strength of C-band. C-band wavelength is 3.75 to 7.5 cm, and offers better resistance against radiation, but will endure scintillation at high levels [PAY19].

4.2.2.5 X Band: 8 to 12 GHz

X-band may allow precipitation to reduce signal strength. Rain cores and hydrometeors can reduce the signal strength as well. X-band wavelength is around 2.4 to 3.75 cm. It offers no penetration against precipitation, but can better withstand radiation interference [PAY19].

4.2.2.6 K Band: 12 to 26 GHz

K-band can be blocked by nearly any obstruction. Rain cores, hydrometeors, precipitation, and clouds can reduce the signal strength. K-band wavelengths are from 0.75 to 2.4 cm. They are great for detecting smaller objects and resisting radiation, but require higher power to break through interference [PAY19].

4.3 Guidance, Navigation and Control

4.3.1 Attitude Control Sensors

ACS requires attitude sensors and on board computers to determine the current positioning and the needed torques to get to desired angle. This is done by a combination of ground station computing and on board sensing. There must be two separate sensors on board the spacecraft. Each sensor will provide a unit vector pointing towards a reference position. In combination, these vectors will provide the needed information to calculate the current attitude.

4.3.1.1 Magnetometer

Instrument that measures Earth's magnetic field strength and direction. This gives information on the orientation of Earth based on the current position of the satellite. The reading from the onboard magnetometer is sent to the ground station to be compared to an accurate model of Earth's magnetic field in order to find the orientation of the CubeSat. This process requires complex algorithms and there's always a risk of drift factors. The on board sensor will be one of the smallest, as there is not much for it to compute. Magnetometers are very common and can be store bought, but for this instance it must be highly accurate and space grade. The accuracy of this system is 0.5 - 3 degrees. This device also needs to be isolated from electromagnets. [GNC5]

4.3.1.2 Sun Sensor

This device just finds the location of the sun. It finds the position by detecting the intensity difference between radiation arriving from the Sun. Two closely spaced photosensitive elements get illuminated and when the difference between them reaches zero is the location of the Sun. The variation that the sun's rays hit each panel can be used to determine relative position. This sensor is only useful when the sun is in observation, limiting its capabilities greatly. An example of this configuration is shown below. [GNC6]

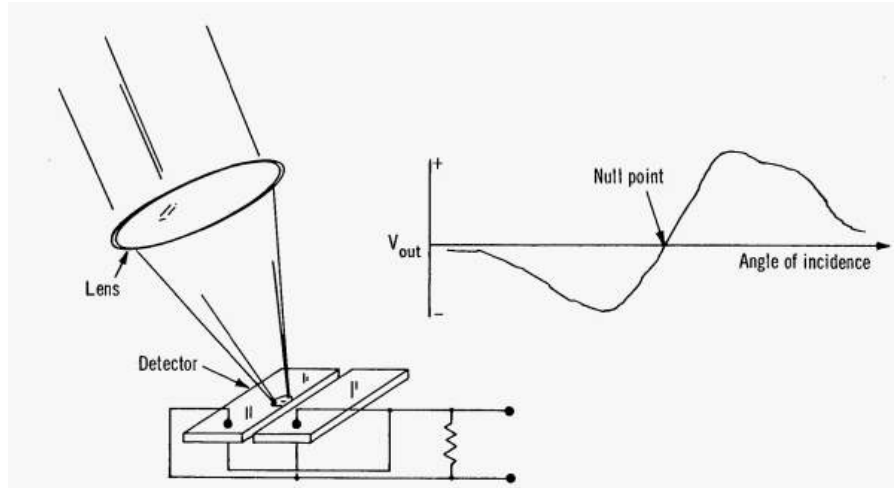


Figure 4.3.1.2.1: Sun Sensor Example. [GNC6]

4.3.1.3 Earth Sensor/Horizon Sensor

This sensor is critical for Earth orbit satellites. The Earth sensors/ horizon sensors use infrared cameras to sense the discontinuity between the horizon and the cosmic background by determining direction of electromagnetic radiation. This type of sensor is relatively cheap and accurate, so it is usually used as a check system. The Earth's radiation is in the infrared spectrum of 3-30 μm . The results from this sensor are very accurate, but it is harder to maintain for low Earth orbit satellites. [GNC7]

4.3.1.4 Star Tracker

Considered the most accurate way of attitude determination. This is because it uses fixed stars to determine position. The star tracker uses pattern recognition of star constellations in its field of view to determine a known star that can be used as an inertial reference frame. There is very long computational time for star trackers. This means that another sensor will have to be used for real time attitude adjustments. This system is best suited for missions that require extremely accurate results.

4.3.1.5 GPS Receiver

Two GPS antennas need to be mounted on opposite sides of the satellite body. The frequency and phase measurements from each antenna can be compared to determine current attitude. This method is limited to the availability of the GPS signals and requires sophisticated algorithms to run accurately.

4.3.2 Thrusters

Impulse thrusters are somewhat common on small satellites/CubeSats. Thrust vectors are misaligned with the spacecraft's center of mass, allowing for torque applied to the system. A reverse thrust is needed to stop the momentum of a spinning satellite which results in a less controlled system. The two main types of thrusters used for attitude control are cold gas and micro-electrospray designs. Thruster design is analyzed in depth in the propulsion system section, 5.9. [GNC8]

4.3.2.1 Cold Gas Thruster System

Cold gas systems have high control authority making it increasingly precise for specific movements while in orbit. The cold gas system works through expansion of pressurized gas. This system has no combustion and a relatively simple design. An example of this plan is shown as below. Despite the simplicity, its monolithic size and substantial volume (from the fuel tank) make it hard to implement on a smaller satellite such as a CubeSat.

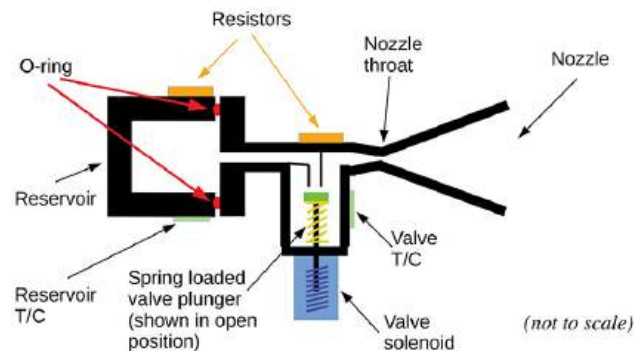


Figure 4.3.2.1.1: Cold Gas Propulsion System. [GNC9]

4.3.2.2 Micro-Electrospray Thruster System

The MEP system provides less control but a more discrete design allows for strategic placings of the propulsion system. As a result, it takes more time to stabilize the satellite, but it is more efficient when using propellant mass. Due to the optimized size, there are much less design restrictions imposed on other subsystems, which is especially important in small satellite designs such as CubeSats.

4.3.3 Reaction Wheels

Reaction wheels, sometimes referred to as momentum wheels, allow for three axis control without the need for external sources of force such as propellants. Flywheel devices put on each of the axes spin in order to create torque on the system. Through conservation of momentum, when one of these wheels spin at a different rate there will be a resultant imposed moment. A drawback to this design is possible disastrous vibrations if momentum wheels are not

perfectly balanced. These systems are much more precise than thrusters and are commonly used in Earth observation satellites that require strict dynamics. These wheels are also offered by manufacturers in different sizes to fit the needs of each individual space system differently. This system is the most used version of attitude control for satellites. The technology has been flight proven and has a significant amount of heritage. [GNC10]



Figure 4.3.3.0.1: Reaction Wheel Example. [GNC11]

4.3.3.1 Three Wheel Control

For a 3 wheel control system the total minimum torque required is found to be in a configuration where one wheel is aligned x-axis, and the other two wheels are tilted. The most accurate attitude correction comes from an arrangement of one wheel on each axis, as the error was less than 0.001 deg. This arrangement is not possible however, due to complete mobility loss if one of the motors goes down. [GNC12]

4.3.3.2 Four Wheel Control

The four wheel design actually turns out to be more efficient as it consumes less total current to induce the same correction torque on the system. The most ideal configuration for four reaction wheels is one wheel on the x- and z-axes, while the two other wheels are tilted. This results in a total attitude pointing accuracy of around 0.001 deg. This design is also resistant to hardware malfunctions, as three wheels can still control the system.

4.3.4 Magnetic Torque Rods

Magnetorquers use an arrangement of electromagnets and a coil of electric current to create a magnetic field. The magnets are attached to the satellite itself, so the force exerted on the magnets from the field will pivot the spacecraft. This allows for electric current to be converted directly to torque. The absence of moving parts make this system very reliable to use for long

term missions. This system has high power draws, but is relatively reliable. Their small size makes them a great supplemental torque source. [GNC13]

4.3.5 Solar Sails

Solar sails use radiation pressure from sunlight to push against a giant reflective sail. Photons from the traveling light transfer the momentum to the sail, and when the center of gravity is not in the center of the sail, a moment is created on the sail. A controller is needed to position the sail to create the needed torque to correct the craft's orientation. These sails need to be very large in order to gather extremely small momentum forces from the light photons. This application is only used on deep space traveling missions. [GNC14]

4.3.6 Control Moment Gyroscope

This device reads the rotation of the satellite without dependence on outside reference frames. A flywheel or rotor spins on an axis at a constant speed and is on a motorized gimbal. The gimbal changes the orientation of the rotors axis, altering the direction of the rotors angular momentum. This creates a resultant torque on the system. A minimum of three CMGs are required for three axis control. This system requires very little power to operate. [GNC15]



Figure 4.3.6.1: Control Moment Gyroscope Example. [GNC15]

4.3.7 Control Sensors: Advantages and Disadvantages

Table 4.3.7.1: Control Sensor Analysis

General Design	Advantage	Disadvantage
Magnetometer	- Low cost - Less mounting constraints - Works best in LEO	- High power draw in comparison - Less accurate results

Sun Sensor	- Common in Satellites - Accurate results - Low power consumption - Makes sun pointing orbit easy	- No use at night - Need 2 on board
Earth Sensor	- Less complex system - Makes Earth pointing easier	- Less accuracy in LEO
Star Tracker	- Most accurate	- Complex algorithms required - Very long response time
GPS Receiver	- Reliable - Cheap	- Dependent on outside satellites - Deployable boom needed - GPS antenna design constraints

4.3.8 Actuators: Advantages and Disadvantages

Table 4.3.8.1: Actuator Analysis

General Design	Advantage	Disadvantage
Thrusters	- Work with propulsion system already on board	- Size for fuel tank - Cost - Requires exact counter torque
Reaction Wheels	- Accuracy - Size - Low power draw - Simple design	- Unreliable at times, - Momentum dumping
Magnetorquers	- Size - Very inexpensive - No mechanical parts	- Less powerful - Less accurate
Gyroscope	- Gives attitude information simultaneously	- Less accuracy - Complex design - Less powerful

4.4 Communication and Data Handling

The communication subsystem is responsible for relaying all data and information collected from the mission to the ground station, as well as responding to the ground station for system inquiries, diagnostics, or potential firmware upgrades for the on-board computer (OBC). Two-way communication is very important for maintaining the spacecraft over the duration of the mission. To achieve this, a radio transceiver capable of transmitting and receiving data is connected to the satellite OBC. The transceiver operates at a range of radio frequencies outlined by the Institute of Electrical and Electronics Engineers (IEEE), ranging from 300 MHz to 40 GHz [CDH 1].

CubeSats produce a wide range of data sets during their lifetime. Payload data sets will typically take up the most bandwidth due to the satellite's purpose-built nature being centered on payload data acquisition. The other subsystems generate telemetry management data, which consists of subsystem diagnostics, subsystem health, and positional data (attitude/altitude configuration). This data is interpreted and processed by the OBC and transferred via a data-bus to the transceiver for signal modulation, amplification, and transmission. Due to the high complexity of data configuration and transmission, the payload and communication subsystems sometimes have a dedicated processor to more efficiently transfer data to the ground station.

It is important for the SMARTSat to design a communication system with a high enough data transfer rate to ensure that no information is lost between orbit passes. One way to increase data transfer rates is to increase the transmission radio frequency to a higher band. As the frequency of radio bands increases, the rate of data transfer increases as well. However, this comes at the cost of higher power requirements and atmospheric losses on transmission. Increasing the power requirement of the communication system should only be done if absolutely necessary, as the power management of the CubeSat platform is already quite limited. An increase in atmospheric losses will also require an increase in the gain of the antennae. Gain is essentially a measurement in the directionality of an antenna, with low gain associated with an omnidirectional antenna, and higher gain associated with an antenna that must be pointed towards the target.

Another motivation to move communication to a higher band frequency is signal congestion. Lower bands are the easiest radio communication devices to implement, leading to heavy signal crowding at lower frequencies. This has motivated many small satellite designers to utilize higher radio frequencies for communication, such as X and K band communications [CDH 1]. These are best for faster inputs, larger data packets, and quicker response times for the system, at the cost of power and more sophisticated hardware.

4.4.1 Radar Bands

Radio frequencies can be divided into a series of bands associated with their energies at each band frequency. The lowest allowed frequency for communication as designated by the IEEE is UHF bands, and the highest is Ka [CDH 1]. With this allowable range in mind, it is important to choose a frequency with a higher data transfer rate while also balancing the power consumption of higher bands with the atmospheric losses associated with higher frequencies.

The RainCube utilized two radio communication devices for data transmission. The first is a UHF transceiver for general two-way communication between the satellite and the ground station, and the second is an S-band transmitter used to send important mission data to the ground station over a wider bandwidth [CDH 2]. This design uses multiple communication devices to achieve low power draw for general satellite communication, and high power draw for mission data relay. Such a configuration may be an attractive option for optimal satellite control with a rapid response time.

Below is a table comparing the performance of multiple radio bands. This will help determine what communication band should be utilized for the communication subsystem.

Table 4.4.1.1: Radar Band Comparison

Radar Band	Pros	Cons
VHF	• Common • Low power cost	• Lower data transfer • Signal Crowding
UHF	• Very common • Low power cost • Widely commercially available	• Low data transfer • Signal Crowding
S	• Growing popularity • High data transfer • Less signal crowding	• High power cost • Atmospheric losses
C	• High data transfer • Less signal crowding	• Higher power cost • Not typically commercially available • Atmospheric losses
X	• High data transfer • Less signal crowding	• Higher power cost • Atmospheric losses

K	• Highest data transfer • Less signal crowding	• Highest power cost • Atmospheric losses • High overall cost
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Due to the wide selection of different frequencies, the preferred radio frequencies will require an initial selection phase that removes the most undesirable frequencies from the trade analysis.

4.4.2 Transmission Architecture

The method of transmission for the communication system considers the devices that will be used to modulate and transmit the signal and receive and demodulate incoming signals on board the CubeSat. Methods include a transmitter and receiver setup, or a transceiver setup. There also is a combined approach which would utilize a transceiver for general communications and a high band transmitter for payload data downlink. Using a single transceiver is the most power efficient method of communicating with the ground station but will either require a high band transceiver for efficient payload data delivery, or lower band deliveries resulting in slower data delivery speeds. Transmitter and receiver setup has the benefit of having different receiving signals and transmission signals, creating a more versatile design. The major drawback of this method is a higher power consumption, more volume lost, and a higher cost of implementation.

Combining these approaches will result in a transmitter and transceiver setup, allowing for simultaneous communication with the ground station while transmitting SAR readings in parallel. This design has higher transmission efficiency, while suffering from higher costs, higher communication profile within the bus, higher power draw, and a more complex design.

This table outlines the general advantages and disadvantages of each setup. These factors will influence the selection of the later trade analysis.

Table 4.4.2: Transmission Architecture Comparison

Transmission Architecture	Pros	Cons
Transmitter/Receiver	• Flexible design • Send and receive different frequencies	• Less spatially efficient • High power cost • Higher cost of implementation
Transceiver	• Smallest form factor	• Less bandwidth flexibility

	• Efficient two-way communication	
Combined Approach	• Best bandwidth flexibility • Two-way communication and fast data downlink	• Higher cost than transceiver alone • Requires two separate antenna

4.4.3 Transmitter

Radio Transmitters take two basic forms with CubeSat communication systems, hardware radios and software defined radios. Hardware radios use the standard method of generating a radio frequency from an oscillating circuit, while software defined radios, or SDRs, generate the radio frequency digitally and allow for much more versatility in tuning the final frequency. SDRs have high frequency ranges, allowing for many different transmission wavelengths and in turn a variety of bandwidths to choose from. SDRs also typically come equipped with on-board field programmable gate arrays which help process incoming signals quickly and efficiently. Radios can also be made in house, but would require radiation considerations and cosmic ray protection in place to function well in the satellite, making an in-house communication system undesirable.

This table compares the general advantages and disadvantages to using hardware, custom hardware, and SDR devices for radio communication. The factors considered will help generate the selection criteria for the transmitter analysis.

Table 4.4.3.1: Transmitter Comparison

Transmitter Design	Pros	Cons
Hardware	• Simple implementation • Proven to work • Highly commercially available	• Less flexible with bandwidth • Processor required
SDR	• On board FPGA • Bandwidth flexibility • Great for payload data downlink	• Higher cost

Custom	• Most versatile design • Purpose built for mission	• High complexity and research requirements • No heritage • Space grade required
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4.4.4 Antenna Configurations

Antenna choice is important for each individual mission design. The main antennas considered for communication with the ground station will be monopole/dipole, patch, patch arrays, and helical. One major consideration for CubeSat communication will be the altitude of the final orbit, which will help determine the range required from the communication system. Antenna performance is largely dependent on the bandwidth chosen, so once the trade analysis for the radar bands is completed, the trade analysis for the antennas can be done.

4.4.4.1 Wire Antenna

Wire antennas, or monopole and dipole antennas, are incredibly common for CubeSats in LEO. These antennas are best for lower frequencies such as UHF and VHF. These appear as long wires that are capable of transmitting and receiving data from the ground station. One unfortunate drawback of wire antennas is that they must be deployed during the mission, which will add to the overall complexity of the design. Pictured below is a typical wire antenna design.

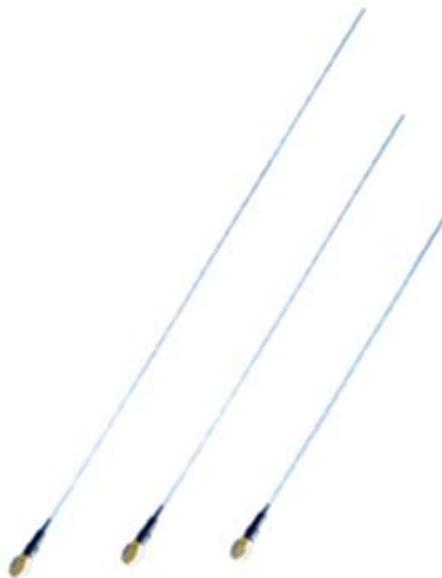


Figure 4.4.4.1.1: Wire Antenna Manufactured by AAC Clyde Space [CDH 4]

4.4.4.2 Patch Antenna

Patch antennas look like a small piece of reflective metal. These antennas are flat and require no deployment. They also work best at higher frequencies such as X band and S band. Patch antennas are also relatively low cost when compared to other alternatives [CDH 3]. The simplicity of their design and their ability to communicate at high frequencies make them appealing for high bandwidth communication. Some bandwidth is lost when using the patch design, but this will still provide wideband transmission [CDH 3]. Below is an example of a patch antenna.



Figure 4.4.4.2: Patch Antenna Manufactured by the Ecuadorian Space Agency [CDH 5]

4.4.4.3 Arrays

Arrays, or using multiple spaced-out antennas, are useful for applications that require higher gain. For patch antennas, this is common due to atmospheric losses that can occur with higher radio bands. These arrangements are of course more expensive than a single patch antenna but provide more reliable communication.

4.4.4.4 Helical

Helical configurations are a way of storing powerful deployable antennas in a 2D form factor. These antennas are placed on the outside of the satellite in a spring coil that can then be deployed once in orbit, forming a helical 3D shape. They are useful for both low and high frequencies, but work well with a high gain, high frequency configuration [CDH 3]. One drawback of helical antennas is that they tend to be more expensive than patch antennas. The operation of this antennae is also fully dependent on successful deployment. The image below presents a helical antenna arrangement.



Figure 4.4.4.4.1: Helical Antenna Manufactured by Oxford Space Systems [CDH 6]

This figure illustrates the collapsable deployment of the antenna, and how this system can store a high gain instrument in a 2D form factor on launch.

4.4.4.5 Antenna Configuration Comparison

Below is a table that compares the previous antenna configurations. These factors will help determine the selection criteria for the later trade analysis.

Table 4.4.4.5: Antenna Comparison

Antenna	Pro	Cons
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Wire	• Work well with lower frequencies • Low cost	• Deployed • Limited at higher frequencies
Patch	• Work well with higher frequencies • No deployment • Low cost • High gain	• Less functionality with lower frequencies
Helical	• Work well with multiple frequencies • High gain	• Deployed • Higher cost • Less heritage
Array	• Highest gain • Reliable data transfer • Redundant	• More expensive

The patch antenna in figure 10 gives a better idea of the kind of advantages a non-deployable antenna has for radio communication in space.

4.4.5 Data Handling

The command and data handling (C&DH) subsystem is responsible for monitoring the subsystem's telemetry data, processing communications with the ground station, and processing the payload data. Monitoring the health of the satellite during the mission is crucial for ensuring that all systems are functioning properly and to verify the reliability of the data collected. Damage to the satellite from solar radiation and cosmic rays can also be reported to allow the ground station to assess the damage as it happens and make any necessary adjustments to the system. The C&DH system is the most vulnerable to SEUs and SELs when compared to the other subsystems, so it is important that the on-board computer comes equipped with circuit protection in place should the device encounter high energy cosmic rays. Payload data is also important to process as the collection and transmission of said data is the very reason a satellite is constructed.

This subsystem also determines the modes that the satellite will enter under specific conditions, including the many deployment protocols needed for a successful mission launch. These modes include detumbling, deployment, sun pointing, safe mode or low power mode, communication mode, fault detection, recovery mode, and SAR operation. Some modes should

be able to operate in parallel such as communication and sun pointing. SMARTSat should be capable of switching between the desired modes autonomously in orbit to maximize efficiency. The ground station should only need to send system updates if any adjustments to the data configuration need to be made, or if a failure or a fault is detected, and the system needs to be reset or altered in some way.

The C&DH subsystem can be divided into two main sections: the system architecture, and the operating system. These will act as the main two alternative design choices for the command and data handling subsystem.

4.4.5.1 On Board Computer Architectures

Computer architecture is important for collecting and storing different kinds of mission data. There are three common types of processors used in space: microcontrollers, Rad-Hard systems, and field programmable gate arrays or FPGAs. The main differences between these systems are the computational power each is capable of on their own, and the complexity of their design. These separate architectures can also be combined in a distributed approach, which is common for microcontroller and FPGA systems. The central computer also heavily influences the lifespan of the mission based on the circuit's built-in radiation resistance.

4.4.5.1.1 Microcontrollers

Microcontrollers are simple, commercially available processors meant to handle relatively low complexity tasks actions for in-flight commands. Utilizing microcontrollers is especially popular for LEO missions due to their simplicity in design and ease of implementation. However, microcontrollers struggle with higher doses of radiation, and thus can greatly limit the lifetime of the mission [CDH 7]. Microcontrollers are also not the most effective choice of processor for data collection. The speed at which the payload data can be processed by the microcontroller has the potential to limit the mission effectiveness due to longer times between data transfer with the ground station [CDH 7]. When it comes to basic telemetry data, however, the microcontroller is a good option for most CubeSat systems. Pictured below is a typical microcontroller commercially available for CubeSats.



Figure 4.4.5.1.1.1: Kryten-M3 On-Board Computer [CDH 8]

The Kryten-M3 microcontroller will be the main point of reference for microcontrollers in this comparison. This on-board computer offers a wide range of data transmission devices, as well as enough radiation tolerance to allow SMART to last five years in orbit. This computer is designed to handle single event upsets and latch ups using built-in safety features and fault detections [CDH 8]. For microcontrollers, this design will act as a great reference for the performance our system will need.

4.4.5.1.2 Rad-Hard

Radiation hardened systems are meant to withstand higher doses of radiation than typical space-grade components to allow for longer mission durations as well as deep space applications. For our Earth orbit, rad-hard systems typically aren't used so long as radiation exposure remains low. For observations in highly irradiated regions such as the South Atlantic anomaly, radiation exposure may be high enough to justify using a rad-hard system. However, for any other low orbit, radiation hardening will only serve to extend the life of the mission. Rad hardened systems typically have a total ionizing dose or TID of 100 to 1000 krad [CDH 7]. CubeSats typically don't operate at these ranges, but many rad processors still exist for CubeSat applications. The ESA has developed a line of processors known as the LEON series for use in deep space [CDH 7].

4.4.5.1.3 FPGAs

FPGAs, or field programmable gate arrays, are a type of configurable processor that can be programmed to perform repeated tasks reliably and efficiently. FPGAs are great for all kinds of data acquisition from telemetry, instrument measurements, and radio communications. Many SDRs come equipped with an FPGA processor for rapid communications processing. These systems are typically less power hungry than other computing systems in CubeSats [CDH 7]. One difficulty that FPGA based computers face is a higher difficulty reconfiguring the system post launch. This typically can be done using a stored update loaded onto a flash memory drive [CDH 9]. Such operations may require another on-board computer to write and send an update to the flash memory, should an update mid-flight be needed.

The figure below depicts an onboard computer equipped with an FPGA processor.



Figure 4.4.5.1.3.1: OBC-1 On-Board Computer [CDH 10]

This OBC is equipped with 96 Gb of flash storage, a watchdog timer, and radiation protection rated for 3-5 years of operation. The OBC-1 will be the main comparison for FPGAs in the trade analysis of the on-board computer.

4.4.5.1.4 Distributed

Distributed data handling solutions utilize multiple computers with different capabilities to minimize power output and maximize data acquisition and transmission. The way this is typically done is to have a core processor handle all the main subsystem processes, while another processor focuses on the payload data handling. Especially when utilizing an FPGA for faster payload data processing, this method proves to be more energy efficient than running the entire system on a single processing unit. The major downside to this approach is a higher cost to utilize two separate processors.

4.4.5.1.5 On-Board Computer Comparison

Below is a table that highlights the advantages and disadvantages of the discussed on-board computers. This will determine the later selection criteria in the trade analysis.

Table 4.4.5.1.5.1: On-Board Computer Comparison

OBC Configuration	Pros	Cons
Microcontroller	• Functions well with simple operations • Easier to update in orbit	• Less capable with payload data processing • Limits lifespan
Rad-Hard	• Radiation tolerant • Longest lifespan	• High cost • Limited applications in LEO
FPGA	• Most capable with payload data processing • Capable with handling simple processes • Performs repeated actions in parallel • Highly flexible design	• More difficult to update in orbit • Limits lifespan
Distributed	• Optimal power distribution • Highest data processing for telemetry and payload • Redundant capable	• Higher cost and higher design complexity • Limits lifespan

4.4.5.2 Operating Systems

While the chosen operating system for a given OBC is not always up to the consumer, it's important to still consider what the most preferred option is when choosing an operating system for the mission. The design considerations for the SMARTSat mission will be open-source development, real-time OS, and preferably free. These considerations will allow for a highly versatile system capable of performing all mission subroutines autonomously, while not contributing to the overall cost of the final design.

4.4.5.2.1 Linux

Linux is the most widely compatible operating system for use in CubeSats, making it a very desirable choice for SMARTSat [CDH 7]. Many commercial hardware devices use Linux as a baseline operating system for their processes due to Linux's versatility and open source development. KubOS is a free framework for developing CubeSat operating systems [CDH 7]. This framework supports Linux development allowing developers to alter the operating system to their needs of the mission. While not initially real time, the Linux kernel patch PREEMPT_RT gives real time OS capabilities to the system, making processors running off Linux a perfect fit for the mission.

4.4.5.2.2 VxWorks

VxWorks is another highly compatible operating system capable of real time computation. This OS is developed by the Wind River System operating in the US. VxWorks is known for having low latency and minimal jitter, making it an efficient and reliable choice for an OS [CDH 7]. One major drawback of using VxWorks is that the license costs nearly \$20,000 [CDH 11]. Using VxWorks would take a substantial portion of the budget just towards developing the OS when other free alternatives with proven heritages already exist.

4.4.5.2.3 FreeRTOS

FreeRTOS is another popular choice for an open source, real-time operating system. This OS is best suited for applications where computations remain relatively simple, and resources such as power and memory are limited [CDH 7]. These features make FreeRTOS a good choice for the main computer in the space bus, but a poor choice for the payload computer. Should a distributed architecture be used, FreeRTOS is a strong candidate for a reliable and power efficient primary OBC.

4.4.5.2.4 Operating System Comparison

The table below is a comparison between the different operating systems discussed with the advantages and disadvantages of each highlighted. The trade analysis of the preferred operating systems will use these features as a basis for the selection criteria.

Table 4.4.5.2.4.1: Operating System Comparison

OS	Pros	Cons
----	------	------

FreeRTOS	• Free • Open source • Real time OS • Widely adopted	• Simple
Linux	• Free • Open source • Real time capable • Moderate complexity • Most supported OS	• Requires kernel patch to enable real time capabilities
VxWorks	• Highest complexity and versatility in orbit • Real time OS	• \$20,000 License

4.6 Thermal

The thermal control system of the CubeSat is essential for the entire system of electronic components and radar to function properly despite the heat differences that can occur while in low earth orbit. Furthermore, we must consider the fact that the sun will not always be shining on the CubeSat which will add another dimension of issues to consider for thermal properties. There must be a balance between efficiency, weight, volume, cost, and sustainability when considering these different options for thermal systems.

Different thermal control systems exist which are used in combination with each other which is something important we must consider for later when deciding what system(s) we would like to use. There are two main types of systems which are passive and active. Passive systems maintain component temperatures without using powered equipment. These are beneficial to spacecraft with limited mass, volume, and power. Active systems rely on input power for operation and are more effective in maintaining tighter temperature control [Therm 2]. It is worth mentioning that the systems are not implemented separately as many satellites and CubeSats utilize both types of systems. The following sections will go into detail about different types of systems for passive and active systems with some noted features that are not full on systems, but are good to consider as potentially being part of the main systems.

4.6.1 Sprayable Thermal Control Coatings/Paints and Tapes

Thermal coatings and tapes are a common type of passive thermal control. The heat transfer for these is driven by thermal radiation to or from the surrounding environment. Surface properties can be modified by these specialized paints, coatings, surface finishes, or adhesive tapes by altering solar absorptivity and IR emissivity [Therm 2]. Paints and tapes are diverse in what their thermal properties are like with matte black paints having high absorptivity of solar heating and having a high IR emissivity (how much heat is released by the spacecraft) or matte white having low solar absorptivity and high IR emissivity. Fluorinated Ethylene Propylene (FEP) tapes offer similar properties to matte white paints, but are easy to apply and remove, more expensive, and longer lifetime than paints.

4.6.2 MLI Blanket

MLI blankets are another passive thermal control that consist of multiple inner layers of a thin material with low IR emissivity and durable outer layer that limits radiative heat transfer through layers of reflectors [Therm 2]. MLI blankets also protect against solar fluxes and are common for maintaining temperature ranges for components in orbit with large satellites, but smaller satellites have issues to consider. MLI blankets performance drops if compressed, pose snagging hazards in tight fitting, pusher-spring style deployers, and decrease in efficiency with size decrease.



Figure 4.9.2.1: MLI Blanket Example

4.6.3 Sunshields

Sunshields are another type of passive control that are often deployable devices made up of a material with low solar absorptivity that reduces the amount of solar flux impinging on a spacecraft by blocking the view of the sun. Sunshields can be made of different materials that will provide the sunshield with different lifetimes. Since the sunshield is blocking the sun, sunshields are typically used to keep scientific instruments very cold like on the Webb Telescope

[Therm 11]. In the Webb example, the temperature difference between the hot and cold side of the telescope is 570°F.

4.6.4 Thermal Louvers

Thermal louvers are thermally activated shutters that regulate how much heat the louvers surface can dissipate. As louvers open, the average IR emissivity of the surface changes which changes heat dissipation [Therm 8]. This is challenging for small satellites, but NASA seems to have a patent for them. The internal spacecraft components are thermally coupled to the side of the spacecraft. Bimetallic springs serve as a passive control mechanism for opening and closing flaps. As spacecraft heats up, the springs expand due to differences in thermal expansion rates of their two fused metals [Therm 8]. These systems can be passive or active.



Figure 4.9.4.1: Thermal Louver Example [Therm 8]

4.6.5 Deployable Radiators

This is a dedicated surface for dissipating excess heat through radiative heat transfer and has high IR emissivity and low solar absorptivity. These are stored during transport or when not needed and deployed when excess heat dissipation is required. This is challenging due to volumetric constraints. This is good over paints if limited external surface area is a problem like through solar cells [Therm 2]. Deployable radiators increase available radiative surface area.

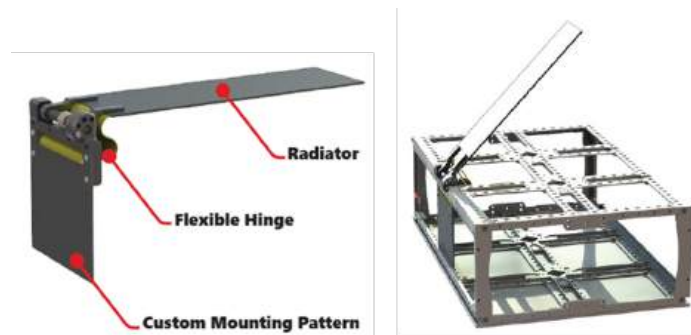


Figure 4.9.5.1: Deployable Radiators Example [Therm 2]

4.6.6 Electric Heaters

This is the first example of active thermal systems that are most commonly kapton heaters which consist of a polyamide film with etched foil circuits that produce heat when a current is applied. These are typically controlled by a thermostat or temperature sensor and used in cold environments to maintain battery temperature. These are helpful since low mass requires little heat addition power to maintain temperature limits. These may require additional passive control for spacecraft systems.

4.6.7 Cryocoolers

These are another form of active thermal control systems that are refrigeration devices that are designed to cool around 100K and below. These are used on instruments or subsystems requiring cryogenic coolings such as high precision IR sensors. These are associated with longer instrument lifetimes, low vibrations, high thermodynamic efficiency, low mass, and supply cooling temperatures less than 50K [Therm 2].

4.6.8 Thermoelectric Coolers

Similar to cryocoolers, these are also active thermal control systems that cool components, but these are miniature solid-state heat pumps which provide localized cooling via Peltier effect which is cooling resulting from passing electric current through junction formed by two dissimilar metals. These have no moving parts, reliable, noiseless, lightweight, and compact. However, these have low efficiency below temperatures of 130K and low performance with large temperature differences. These are fragile to mount, highly sensitive to thermal expansion stresses, and external stresses can be mitigated by adding a conductive strap on the cold side [Therm 2].

4.6.9 Fluid Pumps

This is the final active thermal control system that involves a circulating pump that moves liquid through tubing connected to a heat exchanger and heat sink. Heat source is mounted to a heat exchanger and pumped fluid carries the heat from source to a heat sink like a radiator then the cooled liquid is returned to the heat source. This is capable of cooling multiple locations, but these are associated with high power consumption and high mass [Therm 2]. ATA is advanced, active thermal control technology for small satellites that are capable of high power thermal rejection and zonal temperature control along with integrated heaters and a PID-based control algorithm.

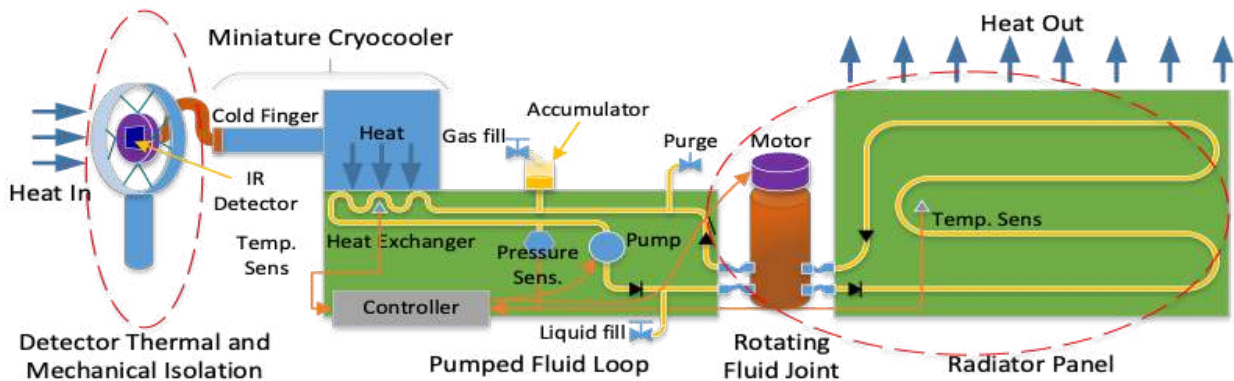


Figure 4.9.9.1: Fluid Pump example [Therm 2]

4.6.10 Summary of Advantages and Disadvantages

Table 4.9.10.1: Thermal System Analysis

Thermal System	Pros	Cons
Paints/Tapes	-Easy to apply and remove -Inexpensive -Tapes can be added later -Tapes have longer lifetime than paints -Coatings cover non-flat surfaces easier -Coatings properties vary	-Tapes must be handled carefully to maintain properties -Difficult to bond to curved surfaces -Coating can darken due to atomic oxygen bombment -Coating/paints degradation -surface area limitation
MLI Blankets	-Prevents heat dissipation -Protects against solar flux	-Performance drops if compressed -Pose snagging hazard for deployable -decrease in efficiency with size
Sunshields	-Covers wide area of thermal protection -Stablizes internal temperature -Mass	-Already have one deployable -Not good for changing environment like LEO -Risk of deployment failure -Volume constraints
Thermal Louvers	-Allows changing IR emissivity -create more stable thermal environment	-Complexity -volume constraint -efficiency in LEO

	-One spring fails, still can function -Can do either passive or active -Lightweight and uses additive manufacturing -Easily reproducible/adaptable	-Potential higher costs
Deployable Radiators	-can be stored and deployed when needed -good for surface area constraints over paints -greatly enhances thermal performance by increasing area -good if working with solar cells	-volumetric constraints -mechanical complexity -could interfere with deployed systems -more expensive than paints and fixed radiators -rapid thermal cycling like in LEO
Electric Heaters	-High thermal precision -Good for cold environments for battery temperature and electronics -simple design -compatibility	-Power input (little) -can't cool down parts -mass and volume with other parts to work it -requires active monitoring -complexity
Cryocoolers	-Used for instruments requiring cryogenic temperatures -Longer lifetime, low vibrations -high thermodynamic efficiency -Low mass -Temperatures to less than 50K -No orbit limitation	-Power input -Volume -Mechanical complexity -other parts to get rid of heat -cost -limited use cases
Thermoelectric Coolers	-no moving parts -reliable -noiseless -lightweight and compact -tight temperature regulation	-High Power input -Low efficiency below 130K -Low performance with large temperature differences -Fragile to mount -Highly sensitive to thermal expansion stresses -other parts to get rid of heat
Fluid Pumps	-cooling multiple locations -high heat transfer efficiency -varying conditions	-High power consumption -High mass -mechanical complexity -volumetric constraints

4.7 Ground Station

4.7.1 General Design

Ground stations are an essential component of any CubeSat mission, facilitating communication between the spacecraft and mission operators on Earth. The choice of ground station design is highly dependent on the CubeSat's mission parameters, such as orbit altitude, data transfer needs, and power availability. Several different approaches to the ground station communication design are available, ranging from simple low-cost amateur setups to sophisticated multi-antenna arrays capable of high data rate transmissions.

UHF/VHF Based Ground Stations

UHF/VHF systems are common for CubeSat missions due to their simplicity and reliability. These stations operate in the Ultra High Frequency (UHF) and Very High Frequency (VHF) bands, commonly using omnidirectional antennas. These setups are popular among amateur radio operators and small satellite missions due to their low cost and ease of setup. However, UHF/VHF systems offer lower data rates and are often subject to atmospheric interference. Key advantages of UHF/VHF stations include low power requirements and wide availability of equipment. A typical setup might cost around \$5,000-\$20,000, depending on the quality of the antenna and receiver system used. A major drawback is that UHF/VHF stations typically have lower bandwidth, limiting the amount of data that can be transmitted per pass. [GS6]

S-Band Based Ground Stations

S-Band ground stations operate in the microwave band (2-4 GHz) and offer a good balance between data rate and power requirements. These systems are more complex than UHF/VHF setups, requiring more precise pointing of directional antennas and higher power, but they also support much higher data rates, which is ideal for missions requiring the transmission of large volumes of scientific data or high-resolution imagery. The cost of S-Band stations is significantly higher than UHF/VHF systems, typically ranging from \$50,000 to \$200,000, depending on antenna size, ground equipment, and receiver sensitivity. The increased data rate, however, makes these stations a popular choice for larger CubeSat missions. [GS6]

X-Band Based Ground Stations

For CubeSat missions that require even higher data throughput, X-Band systems (7.25-8.4 GHz) provide the ability to transmit data at very high rates. These systems are ideal for deep space missions or CubeSats in higher orbits, where lower frequency bands may be limited by interference or bandwidth constraints. X-Band stations typically involve larger parabolic dishes and sophisticated tracking systems to maintain a stable connection. The high cost of X-Band ground stations, often exceeding \$500,000, limits their use to larger CubeSat projects

with extensive scientific payloads or high-definition imaging needs. These systems also require more power from the CubeSat, impacting its overall design. [GS6]

Ka-Band Based Ground Stations

Ka-Band ground stations, operating in the 26.5-40 GHz range, represent the cutting edge of satellite communications. These systems offer extremely high data rates, making them an ideal choice for advanced CubeSat missions with a focus on real-time data streaming or large-scale data collection. However, they come with significant challenges, including the need for very precise antenna pointing, high power requirements, and susceptibility to atmospheric attenuation, especially due to rain. Ka-Band stations are among the most expensive, with systems costing upwards of \$1,000,000. They are typically used for high-budget missions or government-funded projects. Their ability to transmit large amounts of data in a short time frame, however, makes them an attractive option for ambitious CubeSat missions. [GS6]

Optical Based Ground Stations

Optical communication ground stations, which use lasers to transmit data between the satellite and ground station, are a relatively new technology in CubeSat communications. These systems offer very high data rates and are not limited by the radio frequency spectrum, making them an attractive option for future CubeSat missions. However, they are highly dependent on clear weather conditions and require extremely precise pointing. Optical ground stations are currently in the experimental phase and are very expensive to implement. The costs can reach millions of dollars, and they require significant research and development investment. However, as the technology matures, it is expected to become more accessible for CubeSat missions with critical data transfer needs. [GS7]

4.7.2 Ground Station Designs: Advantages and Disadvantages

Table 4.7.2.1: Ground Station Design Analysis

General Design	Advantage	Disadvantage
UHF/VHF	- Low-cost equipment - Widely available for small satellites - Reliable for basic communication	- Limited data rate - Crowded frequencies - Shorter range and low bandwidth
S-Band	- Higher data rates compared to UHF/VHF - Suitable for medium-sized missions - Good balance of range and power requirements	- More expensive than UHF/VHF - Increased regulatory requirements

X-Band	- High data rates - Ideal for transmitting large data volumes - Long-distance communication capability	- Expensive equipment - Requires higher power - More complex regulatory requirements
Ka-Band	- Extremely high data rates - Suitable for deep-space communication - Supports high-bandwidth applications	- Very expensive equipment - High susceptibility to weather interference - More complex to operate
Optical	- Very high data rates - Secure communication (hard to intercept) - Not affected by frequency congestion	- Highly susceptible to weather - Complex and expensive to build and maintain - Requires precise alignment between satellite and ground station - Still in development

4.7.2 Ground Network

4.7.2.1 Ground Network Models

The design of ground station networks for CubeSat missions can vary significantly based on mission objectives, budget constraints, and operational requirements. Below are several design alternatives that can be considered to enhance communication and data handling capabilities.

Single Ground Station Model

This is the simplest model where one ground station serves as the sole communication interface between CubeSats and mission operators. This setup is cost-effective and straightforward to manage, but it limits communication opportunities as it relies solely on the satellite's pass over a single location. This model may suffice for missions with low data transmission needs or limited operational budgets. [GS8]

Distributed Ground Station Network

A network of multiple ground stations located in various geographic locations allows for extended communication windows and more frequent data downlinks, especially for Low Earth Orbit (LEO) satellites. This model reduces risks of downtime by creating redundancy across the stations, although the complexity and cost of operations increase with the addition of each station. [GS9]

Hybrid Ground Station Network Model

In this approach, a combination of traditional ground stations and cloud-based services

are used for communication and data handling. Ground stations relay data to cloud-based systems for processing and storage. This model leverages both physical infrastructure and cloud technology, making it flexible and scalable for various mission needs, while also mitigating the high costs associated with deploying multiple stations globally. [GS8] [GS9]

Cloud-Based Ground Station Network Model

This model involves using cloud computing resources for data processing and storage, often in conjunction with remote ground station facilities. Cloud-based solutions enable satellite operators to access data globally in real time without the need for large physical infrastructures. Although highly scalable and flexible, concerns such as data security, latency, and ongoing cloud service fees must be addressed. [GS8]

4.7.2.2 Ground Networks: Advantages and Disadvantages

Table 4.7.2.2.1: Ground Network Analysis

Ground Network Model	Advantages	Disadvantages
Single Ground Station Model	- Lower initial cost and simpler to operate. - Easier to manage and maintain.	- Limited communication windows, especially for low Earth orbit (LEO) satellites. - Risk of single point of failure.
Distributed Ground Station Network Model	- Increased coverage area, resulting in more frequent communication windows. - A redundant system reduces the risk of downtime.	- Higher setup and operational costs due to multiple stations. - More complex to manage and coordinate.
Hybrid Ground Station Model	- Combines traditional and cloud benefits. - Flexible and scalable.	- Initial setup can be costly. - Requires integration of physical and cloud components.
Cloud-Based Ground Station Network Model	- Scalable and flexible, allowing for real-time processing and storage from anywhere. - Reduces the need for extensive physical infrastructure.	- Data security and latency issues due to reliance on cloud services. - Recurring costs for cloud storage and processing

4.7.3 Commercially Available Ground Station Networks (CAGSN)

Companies around the world offer ground station communication as a service. These companies have a network of antennas around the world to increase coverage and reliability.

Depending on the specific antenna station and company, these networks are able to operate on the frequencies mentioned in section 4.7.1.. Depending on the company, these networks are able to schedule and communicate with satellites very rapidly. The following provides general descriptions of the researched commercially available ground stations in addition to NASA's NEN.

4.7.3.1 AWS Ground Station

Amazon Web Services (AWS) Ground Station provides a scalable and cost-effective solution for satellite communications. It offers the ability to schedule satellite communications, downlink data, and process it using AWS services. AWS Ground Station operates with a global network of antennas, which can reduce latency and increase data coverage. Integration with other AWS cloud services provides added benefits for data storage, processing, and analytics, making it an attractive option for organizations looking for a seamless infrastructure for satellite-ground communications. AWS offers coverage based on 12 worldwide antennas which are interconnected through the previously mentioned cloud service. [GS5]



Figure 4.7.3.1.1: AWS Ground Station Antenna Locations [GS5]

4.7.3.2 KSAT Lite

KSAT Lite is a lightweight and flexible ground station service provided by Kongsberg Satellite Services (KSAT). It is tailored for small satellites and offers near real-time communication with a low-cost pricing model. KSAT Lite operates a network of ground stations strategically located around the world, ensuring global coverage and fast satellite pass scheduling. With their experience in both polar and equatorial orbits, KSAT Lite supports a wide range of satellite missions. They offer communication on their lite service through S-band

uplink/downlink, X-band downlink, and Ka-band downlink. The network has more than 300 antennas in 28 worldwide locations dedicated to its coverage/service. [GS9]



Figure 4.7.3.2.1: KSAT Lite Antenna Locations [GS9]

4.7.3.3 Leaf Line

Leaf Line is a relatively new player in the ground station market, focusing on cost-efficient solutions for small and medium-sized satellite operators. Their ground station network is designed to provide reliable communications at competitive rates, and they focus heavily on automating the scheduling and data transfer process to minimize operational complexity. Leaf Line is particularly attractive for startups and smaller operations looking to manage costs. The network has 26 antennas in 16 worldwide locations dedicated to the ground stations coverage/service. [GS10]



Figure 4.7.3.3.1: Leaf Line Antenna Locations [GS10]

4.7.3.4 NASA's Near Earth Network

NASA's Near Earth Network (NEN) provides reliable ground communication services for low-Earth orbit (LEO) and near-Earth spacecraft. The network consists of ground stations located worldwide and is managed by NASA's Space Communications and Navigation (SCaN) program. NEN supports a wide range of missions, offering real-time communication, tracking, telemetry, and data transfer services. Its integration with NASA's broader infrastructure provides a high degree of reliability and access to government-operated systems. [GS5.9]



Figure 4.7.3.4.1: NASA's NEN Antenna Locations [GS5.9]

4.7.3.5 CAGSN: Advantages and Disadvantages

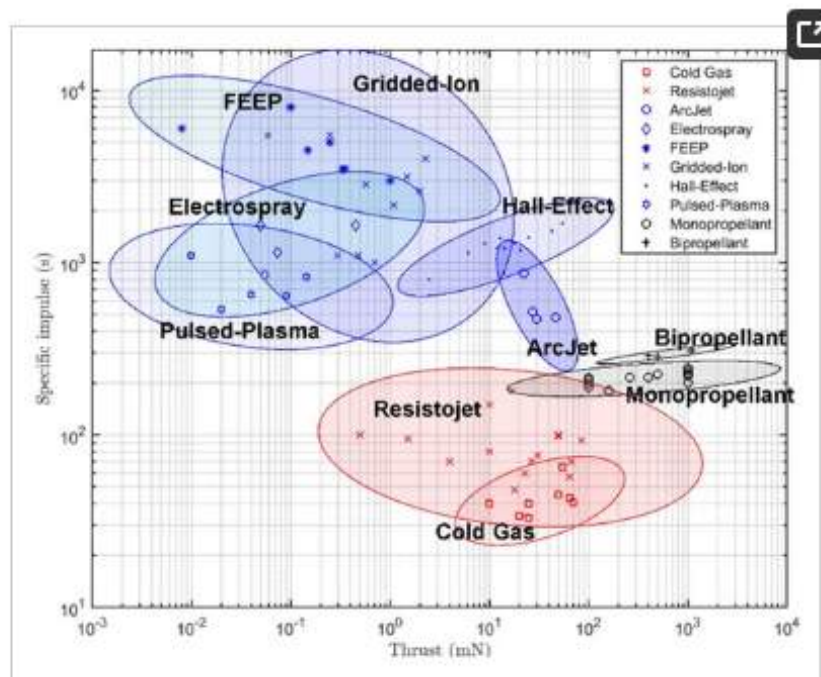
Table 4.7.3.5.1: CAGSN Analysis

CAGSN	Advantages	Disadvantages
AWS Ground Station	- Integration with AWS cloud services for data processing and storage - Global network with low latency - Pay-per-use model - Supports multiple frequencies and satellite types	- Additional costs from AWS services - Dependency on internet/cloud infrastructure - Limited customization for specific needs
KSAT Lite	- Global coverage with large network - Optimized for small satellites - Near real-time communication - Experience with polar and equatorial orbits - Flexible pricing	- Limited scalability for larger missions - Smaller network than AWS - Potentially higher costs for frequent use or large data
Leaf Line	- Competitive pricing for startups - Automated scheduling and data transfer - Easy-to-use interface - Growing global network	- Less coverage compared to AWS/KSAT - Limited capacity for large missions - Newer service, less proven over time
NASA's Near Earth Network	- High reliability and trusted service backed by NASA - Extensive global coverage - Real-time communication and tracking - Supports diverse mission types - Access to NASA's infrastructure and expertise	- Primarily focused on NASA/government missions - Higher costs than commercial services - Scheduling priority may delay commercial missions

4.8 Propulsion

4.8.1 Thrusters

Thrusters are a key component of space propulsion systems. The propulsion system itself will need to be built around the needs of a type of thruster. For example, a resistojet thruster will require some heating element, and thus a connection to power supply. The performance of these systems can be largely divided into four subtypes: kinetic, chemical, electrical, and non propulsion methods of locomotion. Kinetic thrusters tend to be simple, cheap, and have lower overall performance. Chemical thrusters are more complicated and have high thrust and higher ISP than kinetic options. Electrical options have high ISP, but tend to have low thrust. The non propulsion options can have very variable performance, since the category is a catch all for non propulsion options. The performance of different types of propulsion are included in the figure below.



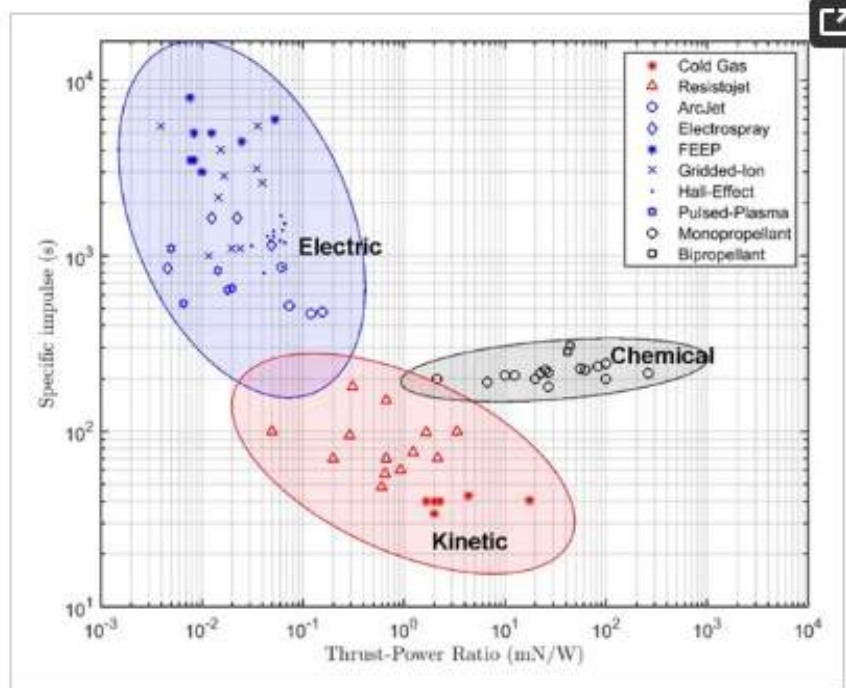


Figure 4.8.1.1: Thruster Performance Chart. Source: [PROP5]

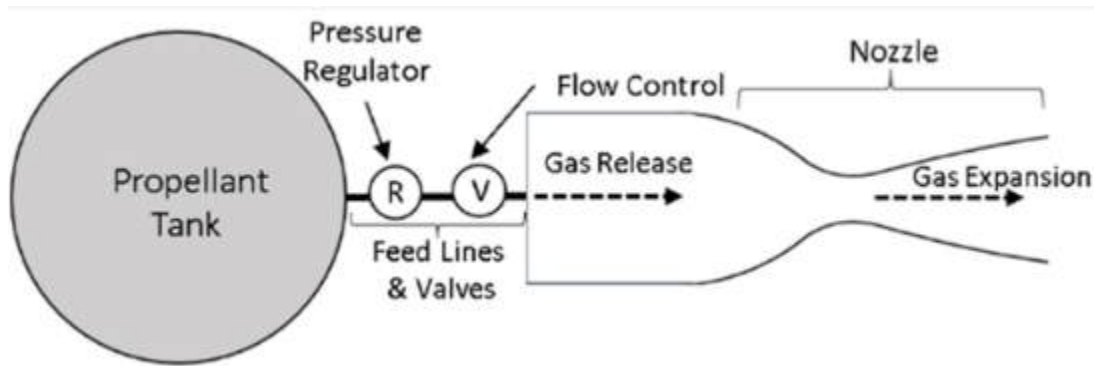
4.8.1.1 Kinetic Thrusters:

Kinetic thrusters convert a pressurized propellant into kinetic energy, using a converging diverging nozzle. Generally, these are the least efficient propulsion systems, since nothing is done to maximize thrust. They do have particular use cases due to their simplicity and low cost. A rough cost estimate for these systems would be around \$45,000 [PROP9]. A source for monopropellants found that the cost of one could be \$20,000 and most sources seem to agree that kinetic is generally cheaper, so the exact cost range is hard to determine [PROP10]. The power consumption of the kinetic thrusters is examined for each subcategory of thruster.

4.8.1.1.1 Cold Gas Propulsion:

Cold gas propulsion is limited by tank pressure, where the propellant is stored as either a liquid or a gas. The key design factor in these systems are the propellant used; density and molecular weight decide thrust, and storage density decides how large the tank needs to be for a mission. Some systems store a propellant as a saturated liquid. This makes for smaller volume storage tanks without the need to pressurize. The most common use for cold gas propulsion is in attitude adjustment systems for small satellites. To summarize, the benefits of cold gas propulsion systems are that they are simple, compact, and cheap, but in exchange they provide low thrust and ISP. Cold gas provides around 100 to 1000 mN of thrust and 10 to 100 s ISP [PROP5].

Figure 4.8.1.1.1: Cold Gas Thruster [PROP5]



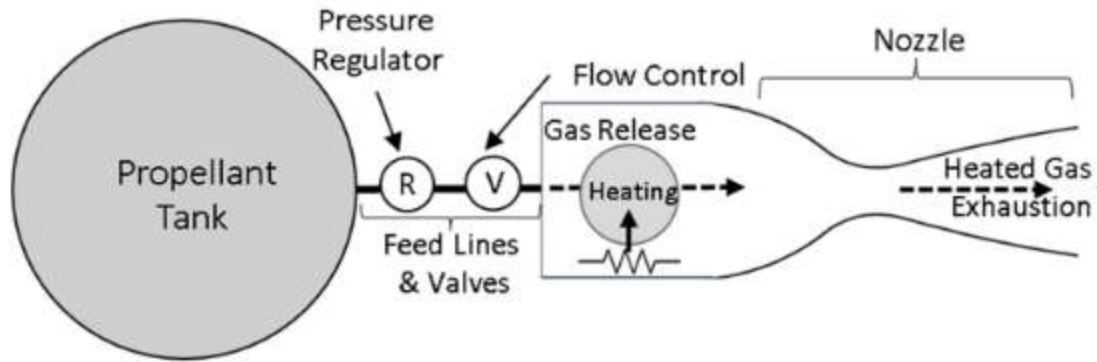
4.8.1.1.2 Warm Gas Propulsion:

Warm gas propulsion is essentially almost identical to cold gas propulsion. It is slightly more complicated, but in exchange, it offers slightly improved performance. A warm gas propulsion system used a propellant storage to supply thrust using a converging diverging nozzle. Density, molecular weight, and heat addition decide the thrust, and storage density decides the propellant tank volume. Propellant can be stored as a liquid or a gas, or a saturated liquid to allow for high storage density without a pressurization system. Warm gas propulsion is commonly used in attitude control of small satellites. To summarize, a warm gas propulsion system is slightly more efficient than a cold gas propulsion system. It is simple, compact, and cheap, but less so than a cold gas system. In exchange, it provides slightly more thrust and ISP than a cold gas system. A cold gas system produces around 100 to 1000 mN of thrust and 10 to 100 s ISP [PROP5]. A rough power consumption estimate for this system is 15 W [PROP11].

4.8.1.1.3 Resistojets:

Resistojets are similar to cold and warm gas propulsion systems, but include an electrical heating element in the gas release chamber before the nozzle. A storage tank, filled with either a liquid, gas, or saturated liquid (for high storage density without a pressurization system) is used to provide a mass flow through a nozzle, producing thrust. Heat, propellant density, and molecular weight decide the thrust, and the storage density decides the size of the tank. Resistojet propulsion provides better ISP than gas and warm gas systems, but at the cost of a larger power consumption, and some complexity. These systems are good for smaller satellite missions. A resistojet system produces around 1 to 10,000 mN of thrust, and around 1000 s ISP [PROP5]. For a resistojet, the power consumption can be around 100 to 450 W [PROP12].

Figure 4.8.1.1.3.1: Resistojet Thruster [PROP5]



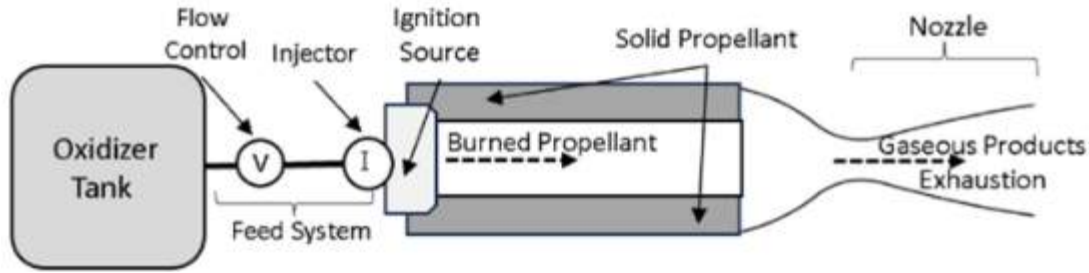
4.8.1.2 Chemical Propulsion:

Chemical Propulsion converts energy from exothermic chemical reactions into thrust by heating propellant. These systems tend to perform with high thrust, and moderate ISP. Chemical propulsion generally performs better than the kinetic thrusters, but there are cases where kinetic propulsion may be better suited to a mission. Chemical propulsion is also generally more expensive than kinetic propulsion. Monopropellant options can cost around \$20,000, but the costs can range a large ways in either direction for other types of chemical propulsion. [PROP10]. Monopropellant should be a good estimate of the lower end cost of chemical options. Another source found that monopropellants could cost around \$230,000-\$240,000, so there is likely a large range of prices [PROP9]. Solid options can cost around \$710,000 [PROP9]. From various listings, the power consumption seems low for these options, somewhere under 10 W [PROP13].

4.8.1.2.1 Solid Rocket:

Solid rocket propulsion uses a solid propellant in the thrust chamber that reacts to an oxidizer when ignited to produce large thrust. These systems are often used for deorbit, orbital insertion, and boosters. The most important design criteria are the surface area of the propellant and the propellant burn rate. One major drawback to a solid rocket system is actually the large thrust produced. A powerful attitude adjustment system is needed to counteract the action of the propulsion system. The use of these systems is also regulated for CubeSat missions because of the pyrotechnic nature of this propulsion system type. The team will most likely drop this option due to its regulations, but it should produce around 100,000 mN of thrust, the high end of the possible monopropellant and bipropellant thrust. Its ISP is likely similar to the other chemical options, around 200 to 700 s ISP. [PROP5]

Figure 4.8.1.2.1.1.: Solid Propellant Thruster [PROP5]

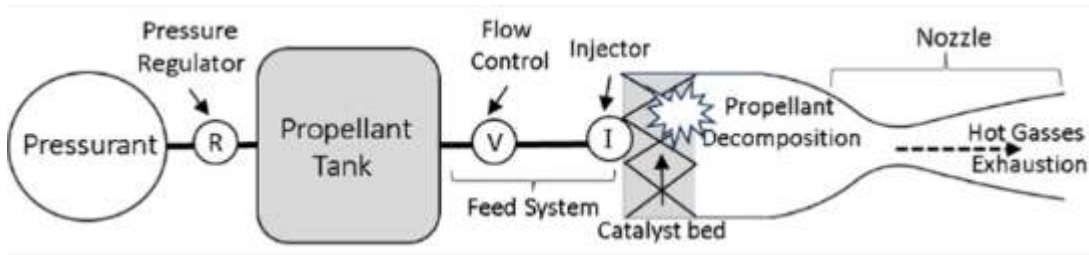


4.8.1.2.2 Liquid Rocket:

4.8.1.2.2.1 Monopropellant:

Monopropellant propulsion systems use the decomposition of liquid propellant with a catalyst to produce energy that gets converted into thrust. Includes a pressurization system, propellant storage, a feed system, a flow control valve, an injector feed, a catalyst bed, and a nozzle. The system offers high thrust and storage density, which means a smaller tank. It is good for small satellites that need large impulsive maneuvers. Monopropellants produce 100 to 100,000 mN of thrust, and around 200 to 700 s ISP [PROP5].

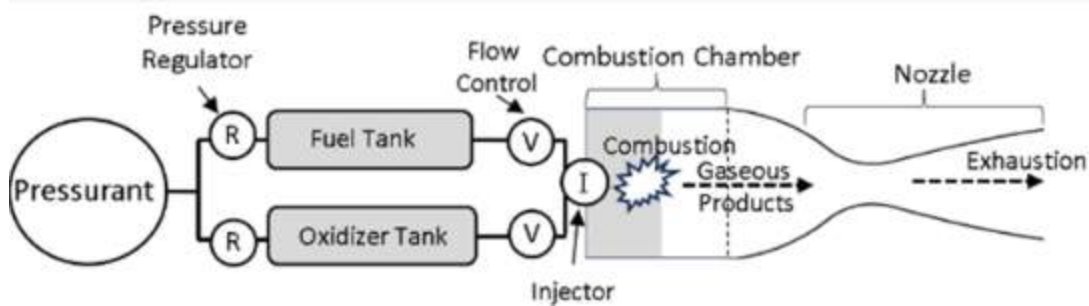
Figure 4.8.1.2.2.1.1: Monopropellant Thruster [PROP5]



4.8.1.2.2.2 Bipropellant:

Bipropellant uses a liquid oxidizer fuel system that exhausts hot gasses to produce thrust. Provides increased ISP from monopropellant, but the propellant storage and feed systems are more complex. Bipropellant systems provide large thrust and are good for larger spacecraft, but are not extensively used in small satellites. The systems could prove to supply an added efficiency compared to monopropellant. The Bipropellant system includes a pressurization system, a fuel tank, an oxidizer tank, a feed system, a flow control valve, an injector feed, a catalyst bed, and a nozzle. Bipropellant produces around 1000 to 100,000 mN of thrust, and roughly 600 to 800 s ISP [PROP5].

Figure 4.8.1.2.2.2.1: Bipropellant Thruster [PROP5]



4.8.1.3 Electric Propulsion:

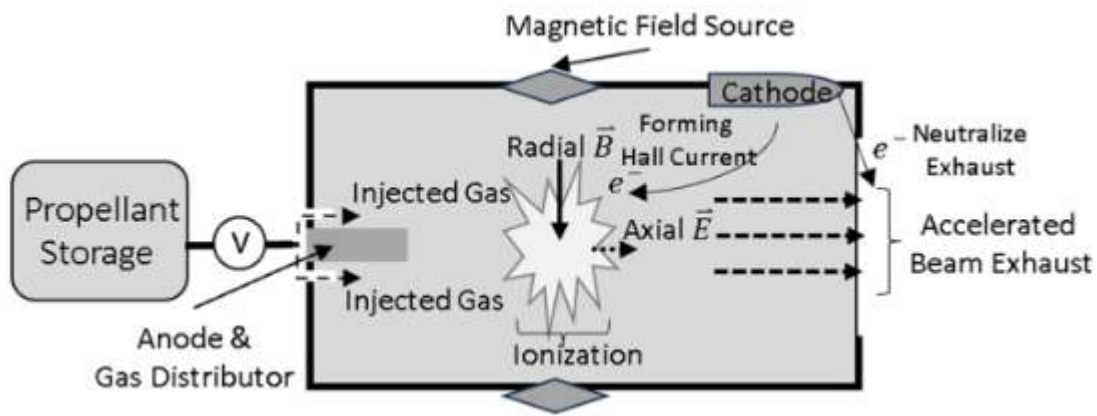
Electric propulsion uses electrical power to accelerate a current. There are a large variety of ways that these propulsion systems do this, which will be covered in more depth in their sections. The primary defining factor of these systems is that electrical power is their primary source of energy. They tend to function at higher ISP with a tradeoff in thrust, though some options can perform decently well. These tend to be the most expensive propulsion option, and are good for endurance missions. Prices can vary for these, but one source claims around \$100,000 to \$300,000 for an electric propulsion system, depending on the type [PROP9]. These options are notable for their power supply, which should be higher than other options, given that is the main source of their thrust. The power consumption for these options is high, with a range of around 8 to 400 W [PROP13].

4.8.1.3.1 Electrostatic:

4.8.1.3.1.1 Hall Effect:

Hall effect thrusters work by using magnetic current to collide electrons together. These collisions cause the flow to ionize, and then electrostatically accelerate the current. This option generally has a very high ISP. Hall effect produces around 1 to 1000 mN of thrust and around 10,000 to 30,000 s ISP [PROP5].

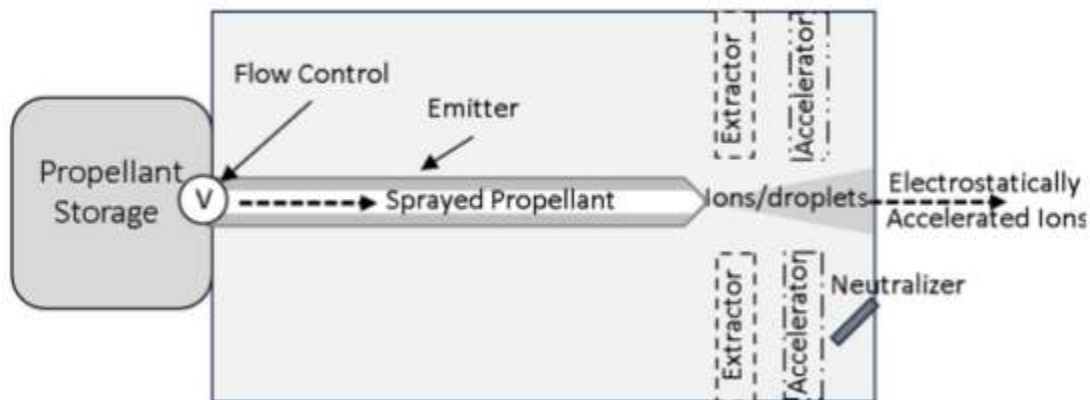
Figure 4.8.1.3.1.1.1: Hall Effect Thruster [PROP5]



4.8.1.3.1.2 Electrospray

Electrospray propulsion is conceptually complicated. Propellant is ejected from an emitter and an electric field forms a Taylor cone. This interaction with the cone involves surface tension of the propellant. The static field formed accelerates the ions out of the thruster. Electrospray performance is dependent on voltage difference and the charge of the electrons. Electrospray is good for high ISP precision maneuvering missions. It doesn't require pressurized storage, and thus takes up a small space. Electrospray produces around .01 to 1 mN of thrust and around 7000 to 50,000 s ISP [PROP5].

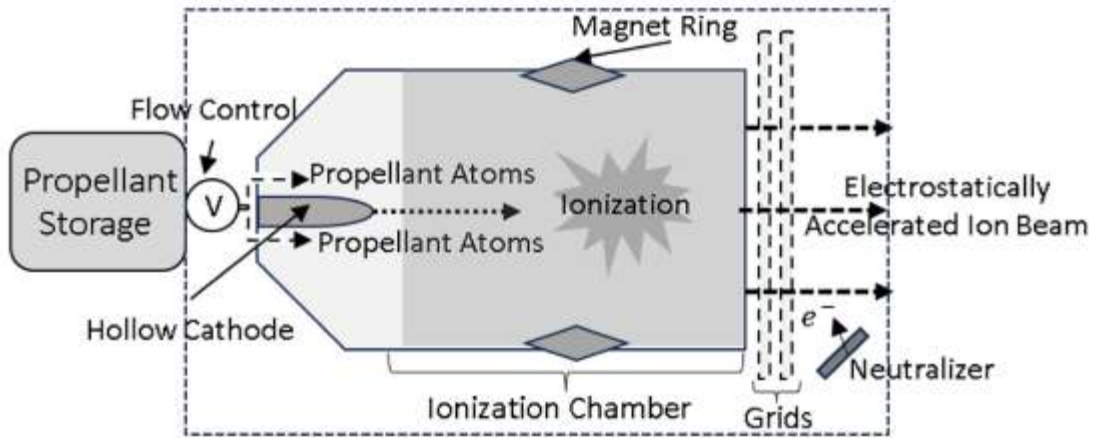
Figure 4.8.1.3.1.2.1: Electrospray Thruster [PROP5]



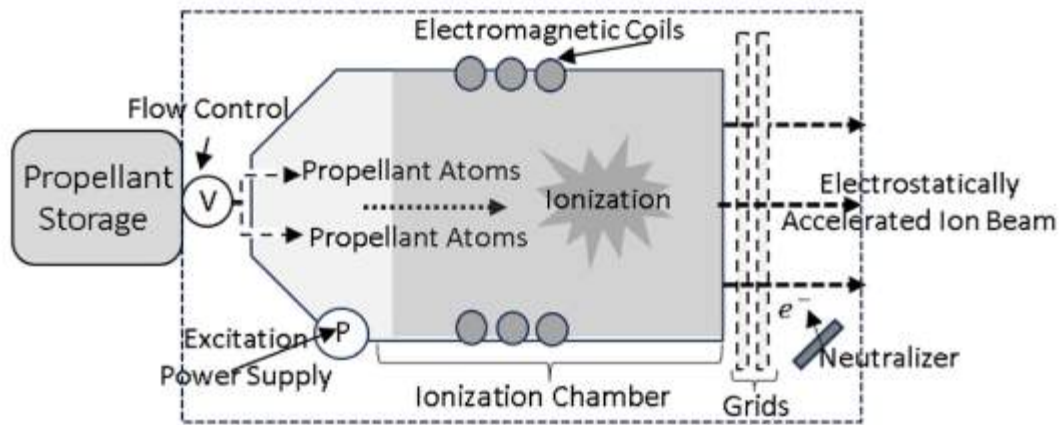
4.8.1.3.1.3 Gridded Ion Thruster

Gridded Ion thrusters use plasma discharge to ionize their propellant. They are classified as electrostatic because they utilize electrostatic grids to accelerate flow. A neutralizer is required to remove the charge of the expelled propellant, and the device itself has a large power demand. Gridded ion thrusters produce around .05 to 1 mN of thrust and 5000 to 100,000 s ISP [PROP5].

Figure 4.8.1.3.1.3.1: Gridded Ion Thruster (two different configurations) [PROP5]



(a)



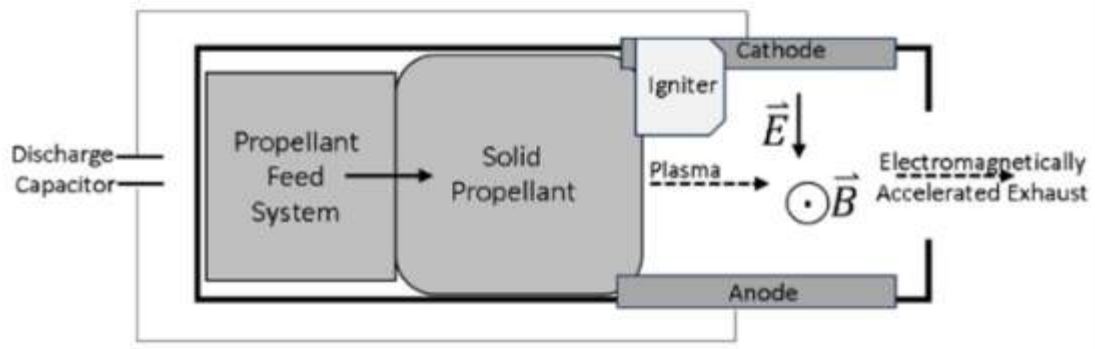
(b)

4.8.1.3.2 Electromagnetic Propulsion:

4.8.1.3.2.1 Pulse Plasma Thruster/Vacuum Arc Thruster:

Pulse plasma and vacuum arc thrusters use discharge to ionize propellant. Magnetic forces propel the propellant out. In a vacuum arc thruster, the propellant itself acts as a cathode, but pulsed plasma thrusters have a separate cathode. These options are similar enough that they are placed under the same option. They perform as a very general electrical option, they provide good ISP and lower thrust. Pulsed plasma produces around .001 to 1 mN of thrust and 5000 to 30,000 s ISP [PROP5].

Figure 4.8.1.3.2.1.1: Pulse Plasma/Vacuum Arc Thruster [PROP5]



4.8.2 Miscellaneous Methods

4.8.2.1 Solar Sail

Solar sails are very unique as an option from this list. They don't use propellant or require power supply, instead taking up a large surface area outside of the satellite. Oftentimes these systems are used for deorbit, and could be an effective option for that. It has an "infinite" ISP, and the thrust is dependent on surface area [PROP5]. It produces around 0.000009 N/m^2 [PROP14].

Figure 4.8.2.1.1: Solar Sail [PROP5]

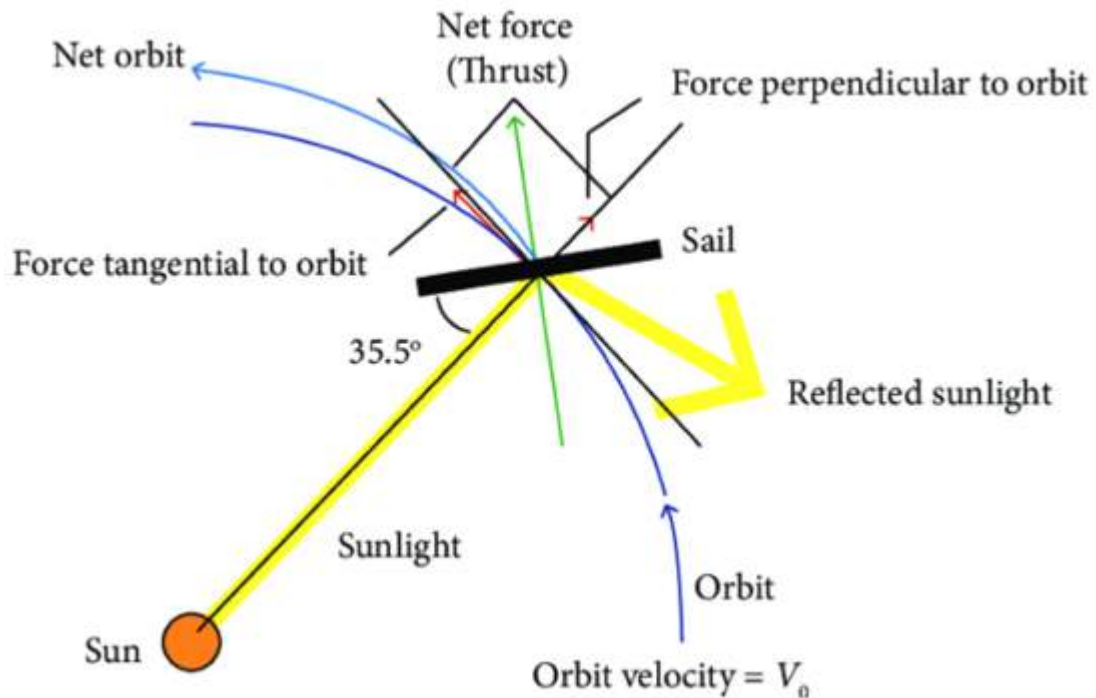


Table 4.8.3: Propulsion System Analysis

	Kinetic	Chemical	Electrical
Pros	- Cheap and simple - Small - Decent Thrust - Low power consumption	- Impressive thrust - Low power consumption	- Impressive ISP - Most efficient option
Cons	- Poor ISP - Generally poor performance	- Potentially regulated - Complicated	- Very complicated - Very low thrust - High power consumption

4.10 Structures Alternatives

Ensuring structural integrity during various stages of the mission is essential to the continual operation of the S.M.A.R.T cubesat. The frame of the cubesat needs to be able to withstand the loads invoked by the launch, outlined in *FR5*. Upon completion of launch into leo orbit, the cubesat structure also has to withstand quick maneuvers from the propulsion system to adjust position whenever deemed necessary, as well as the orbital speed it will be traveling. As a result, to maintain structural integrity/durability throughout the duration of the mission, the S.M.A.R.T cubesat frame and side panels will be manufactured from materials commonly found in other cubesats and rovers.

4.10.1 Materials

Narrowing down which material to use for the cubesat chassis plays a pivotal role because it will determine the attributes that the satellite will have. Distinguishing which material yields the best thermal, physical, and mechanical properties that will be best suited for the mission will ensure the survivability of the satellite during the different mission stages. Furthermore, making sure that the S.M.A.R.T satellite is fully operational after the high stresses it will face during launch is vital in extracting the best performance out of the cubesat during its orbital lifespan.

4.10.1.1. Aluminum 6082

This alloy is largely composed of magnesium and silicon balanced with aluminum, which gives it a 10-18% higher tensile strength compared to aluminum 6061 [Struct 9]. Additionally, contrary to aluminum 6061, this alloy yields a higher corrosion resistance due to its lower content of copper. This alloy's usage is the main downside associated with it because it is not widely used compared to other alloys such as 7075 and 6061. However, on the flipside, this alloy

yields a high melting point which could indicate it could perform to the degree required for launching.

4.10.1.2 Aluminum 7075

Compared to its counterparts, this alloy boasts the strongest tensile strength because of its high utilization of zinc as the primary alloying element. This specific alloy also exhibits good resistance to fatigue, but there is a major tradeoff in the form of corrosive resistance. Copper makes up 1.2-1.6% of the composition of this alloy, resulting in a lower corrosive resistance as compared to its other competitors. Furthermore, it has a melting temperature that is slightly lower than aluminum 6082 [Struct 10]. Nevertheless, this material has a wide range of applications outside of the aerospace industry because of its low density and usability.

4.10.1.3 Aluminum 6061

In the pool of contenders, aluminum 6061 yields the second highest tensile strength, coming up short from aluminum 7075. The main alloying elements found in this material are magnesium and silicon, making it less dense than aluminum 7075. This alloy is suitable for medium-high strength requirements and is one of the easier alloys to work with. As a result, this alloy's versatility allows it to take the shape of various product forms such as sheet, strip plate, rod, etc. [Struct 11]. More importantly, the melting point for aluminum 6061 is similar to 6082, which is around 1100 Fahrenheit. This alloy exhibits performance metrics and physical properties that indicate it could perform well during the different stages of the mission.

4.10.1.4 Aluminum 5052

This alloy is mainly composed of chromium and magnesium, chromium in particular allows this material to be more resistant to stress corrosion cracking. Additionally, the presence of chromium strengthens the base metal and weld seam, as well as reduces the tendency for welding cracks [Struct 12]. Aluminum 5052 is the lightest alloy compared to its counterparts, which could prove to be an essential asset that could be utilized when trying to save weight. Furthermore, this alloy's melting point is 1125-1200 Fahrenheit, making it the highest range out of the other alloys. Another important note is that this material has a high fatigue strength and good plasticity, allowing it to be hardened during cold work.

4.10.2 Summary of Advantages and Disadvantages

Table 4.8.2: Material Analysis

Material	Pros	Cons
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Aluminum 6082	-high melting point -good corrosion resistance -good formability	-low presence in ASTM -not commonly found in cubesats -low tensile strength
Aluminum 7075	-highest tensile, shear, and compressive strengths -hardest material -widely used -low thermal conductivity	-low melting point -high density -most expensive
Aluminum 6061	-good corrosion resistance -similar strengths to 7075 -high melting point	-highest thermal conductivity -high malleability -low fatigue strength
Aluminum 5052	-lightest alloy -highest melting point -high fatigue strength	-not as common on cubesats -lowest elongation % before break

5. Trade Study Process and Results

Using the information collected from the alternative designs of each subsystem, a trade analysis can be conducted on each subsystem's components. The trade analysis will include choosing specific metrics to judge each system component based on the mission requirements and assigning a weighted score to each metric based on its importance to the function of the component. Using these weighted metrics, a Pugh Matrix will be constructed to compare each component and determine the optimal design for the satellite application. The Pugh Matrices will also be repeated multiple times with varied weights to ensure that the final choice for the subsystem component is the correct choice. All system leads will participate in the Pugh Matrix review to mitigate any bias towards specific components.

5.1 Electrical Power Subsystem

5.1.1 Battery Trade Study

5.1.1.1 Motivation

The battery selected for the SMARTSat will determine the number of charging orbits, maximum SAR scan duration, total lifetime, and the weight constraints placed on the system. Metrics chosen to determine the desired battery are cost, charge rate, energy density, commercial availability, and cycle life.

5.1.1.2 Metrics

- **Cost:** Cost is not a major concern of power storage. It is important to supply power to all systems for continuous operation. If the power supplied is not sufficient, the mission loses efficiency too due to high charge times.
- **Charge Rate:** Charge rate is an important metric for battery performance. In low Earth orbit, the opportunity to scan the target area will be short, so faster charge periods will increase the rate at which the SAR can collect data. Energy Density (0.3) - Energy density of a battery is the maximum capacity over mass. High energy density allows for more efficient charge storage without compromising the mass constraints of 2 kg per unit.
- **Commercial Availability:** The ability to obtain commercially produced batteries is critical to the fabrication process as high quality batteries are difficult to produce in house.
- **Cycle Life:** The cycle life of a battery is the number of charge cycles it can undergo before the cell degrades and is no longer usable. Ideally, a high cycle life is used for CubeSats to maintain longer mission lifetimes.

5.1.1.3 Weights

- **Cost:** 0.5
- **Charge Rate:** 0.3
- **Commercially Available:** 0.2
- **Cycle Life:** 0.15

5.1.1.4 Pugh Analysis Battery

Table 5.1.1.4.1: Battery Pugh Matrix

Selection Criteria	Weight	Li-ion	Li-Po	Ni-Cd
Cost	0.5	3	3	6
Charge Rate	0.3	8	9	3
Commercially Available	0.2	8	6	2
Cycle Life	0.15	8	8	4
Weighted Sum		6.9	6.6	4.9

5.1.2 EPS Architecture Trade Study

5.1.2.1 Motivation

The power delivery board will supply power generated by the solar panels and stored in the Li-ion battery to each subsystem. It's important that this system is simple and reliable to ensure each system receives the required voltage input and power supply for operation. Design considerations for this system are versatility, cost, and ease of implementation.

5.1.2.2 Metrics

- **Versatile Design:** The versatility of the design is rated highly for this trade analysis. The EPS must be capable of supplying power to multiple components and allow for changes in power consumption as different modes are enabled.
- **Cost:** The cost of the EPS system is the lowest rated criterion, but still has considerable impact on the trade analysis. Because the battery and solar panels will already take a large portion of the budget, it would be more desirable to obtain a lower cost board without compromising the overall design and implementation of the system.
- **Ease of Implementation:** The final EPS design should allow for easy integration with the other subsystems. Higher complexity may cause issues with compatibility between different subsystems and other connection points. This is the second highest rated criterion.

5.1.2.3 Weights

- **Versatile Design:** 0.45
- **Cost:** 0.2
- **Ease of Implementation:** 0.35

5.1.2.4 Pugh Analysis EPS Architecture

Table 5.1.2.4.1: EPS Architecture Pugh Matrix

Selection Criteria	Weight	Commercial	Commercial + Integrated Battery	Custom
Versatile Design	0.2	7	4	9
Cost	0.2	6	5	3
Ease of Implementation	0.35	8	8	2
Weighted Sum		5	3.9	3.1

5.1.3 Solar Panel Trade Study

5.1.3.1 Motivation

The orientation of solar panels directly affects the power generated in each charging orbit. However, the cost of adding more panels alone will jeopardize the budget of the design. Therefore, a balance must be struck so that each system is powered, and enough of the budget remains so that no system loses functionality. The design metrics are cost, weight, power generation, and commercial availability.

5.1.3.2 Metrics

- **Cost:** Cost is an important consideration for solar panel design. As solar panels get larger, they will get much more expensive, especially with deployable designs.
- **Mass:** Mass variations between options are sufficiently large enough to take into consideration. There is a non-flexible ceiling for max weight per CubeSat unit.
- **Power Generation:** This is the main concern of the solar panel design. Less power generated will result in more time needed to charge and shorter SAR operation periods.
- **Commercial Availability:** The commercial availability of solar panels is weighted lower than other criteria as manufacturing custom solar panels is far more feasible than other components.

5.1.3.3 Weights

- **Cost:** 0.15
- **Weight:** 0.3
- **Power Generation:** 0.4
- **Commercial Availability:** 0.15

5.1.3.4 Pugh Analysis Solar Panels

Table 5.1.2.4.1: Solar Panel Pugh Matrix

Selection Criteria	Weight	Fixed	Deployable	In-House
Mass	0.15	8	6	5
Cost	0.3	5	3	8
Power Generation	0.4	2	6	7
Commercially Available	0.15	6	4	0
Weighted Sum		4.4	4.8	5.95

5.2 Payload Trade Study

The payload is composed of electrical equipment required to produce the EM signal which is focused and fired towards the target using a transmitter at a certain frequency band. The signal scatters back towards the satellite after making contact with the target and the backscatter is detected by the antenna where it is then converted from an electrical signal into usable raw data before being sent to SMARTSat's central processing unit. Of these parts, the most important decisions to be made are the antenna's shape and style and the frequency band to use for the signal. As mentioned previously, the choice between using software defined radio or hardware defined radio was made without a trade study due to the overwhelming power requirements and complexity associated with SDRs. This trade study will compare our options for these design features through a number of pugh matrices defined by the motivation for the study, an explanation of the metrics and weights used, and then a summary of the results. After our selection is made, a sensitivity analysis will be performed to ensure integrity of the process.

5.2.1 Antenna Shape/Style

5.2.1.1 Motivation

When calculating an antenna's ability to receive various signals, the most important aspect other than size is the shape or style of the antenna. Different antennas vary in their ability to focus, detect, and discriminate backscatter signals. This trade study will allow us to determine which shape or style is best suited for our mission requirements.

5.2.1.2 Metrics

For the purposes of this trade study, we will use the following metrics to select the best option.

- **Ease of Deployment:** The complexity of the deployment mechanism.
- **Durability:** The overall durability of the antenna, especially during launch and deployment.
- **Backscatter Reception:** How well the antenna detects backscatter.
- **Backscatter Focus:** How well the antenna focuses backscatter into the detecting portion of the antenna.
- **Antenna Size:** The overall size of the antenna based on minimum usable size.
- **Pedigree:** Amount of previous use in similar satellites and missions.
- **Cost:** Cost of the construction and implementation of the antenna.

These metrics provide an encompassing view of the aspects of each alternative antenna style and represent a fair swath of each antenna's strengths and weaknesses. As such, the consideration of each of these metrics should provide a suitable choice from the group.

5.2.1.3 Weights

- **Ease of Deployment:** 0.05
- **Durability:** 0.15
- **Backscatter Reception:** 0.2
- **Backscatter Focus:** 0.15
- **Antenna Size:** 0.25
- **Pedigree:** 0.1
- **Cost:** 0.1

Since the size of our CubeSat is severely limited, the radar's size, before and after deployment, is the most important metric to consider. Closely behind is the effectiveness of the antenna at detecting backscatter and focusing said backscatter on the detecting portion of the antenna. Least important is the ease of deployment of the antenna, followed by the cost and pedigree. All of the antenna styles we are considering have been deployed in space before successfully, however some consideration must be made for the deployment mechanism since it will have to be made custom for our particular volume and satellite internals position.

5.2.1.4 Trade Study Process

Table 5.8.1.4.1: Antenna Style/Shape Pugh Matrix

	Weights	Dish	Panel	Phased Array	Mesh	Horn
Ease of Deployment	0.05	5	8	3	5	10
Durability	0.15	5	7	5	3	8
Backscatter Reception	0.2	5	6	7	5	4
Backscatter Focus	0.15	5	4	5	4	1
Antenna Size	0.25	5	9	8	9	3
Pedigree	0.1	5	7	6	2	1
Cost	0.1	5	7	1	4	8
Sum	1	5	6.9	5.75	5.15	4.3
Rank	-	4	1	2	3	5

Sensitivity Analysis

Above is the final pugh matrix that best represents our mission requirements. The horn style antenna consistently came last due to its near inability to be used as a radar receiver. By far the best option available is the panel antenna. Even with cost weighted near the least among the metrics, the phased array antenna, which would technically be the most capable antenna style, was too expensive. Similarly the mesh antenna is a newer technology and does not currently represent a superior option over a panel antenna even considering its exceptional stowed volume. As such, our choice is a panel shaped antenna.

5.2.2 Frequency Bands

5.2.2.1 Motivation

Choosing the frequency band of our SAR will determine, among other parameters such as exact models of necessary electrical equipment, the capability of the SAR to perform its mission during various scenarios. Different frequency bands provide varying levels of obstacle penetration, maximum scanning resolution, necessary antenna size, and power consumption. As such, a trade study is necessary to determine the best choice of frequency band for our mission requirements

5.2.2.2 Metrics

For the purposes of this trade study, we will use the following metrics to select the best option.

- **Input Power:** The power required to send a signal at that frequency.
- **Interference/Obstacle Penetration:** The ability of a frequency to penetrate clouds and weather and resist radiation interference.
- **Base Resolution:** The minimum resolution possible for the frequency regardless of SAR use.
- **Antenna Size:** The general antenna size required to receive a signal at this frequency.
- **Pedigree:** Amount of previous use in similar satellites and missions.

These metrics provide an encompassing view of the aspects of each frequency band and represent a fair swath of their strengths and weaknesses. In particular, power will be a defining metric as it determines the number of scans we can do over a given period of time. Overall, the consideration of each of these metrics should provide a suitable choice.

5.2.2.3 Weights

- **Input Power:** 0.20
- **Interference/Obstacle Penetration:** 0.20
- **Base Resolution:** 0.10
- **Antenna Size:** 0.40
- **Pedigree:** 0.10

Although we are very power limited due to being a CubeSat, power is not the chief concern when deciding on a frequency band. Most important is the necessary size of the antenna due to our small mass and volume constraints. Secondary to antenna size is power and interference resistance or obstacle penetration. Without considering each of these the reliability of our SAR drops considerably. Least of concern is the image resolution and the previous use of a band by a similar satellite or to do a similar mission. Image resolution isn't a primary concern since SAR imaging generally provides resolution in the tens of meters regardless of signal frequency, but higher resolutions would be easier to achieve with higher frequencies and could provide future functionality and better granularity for our images. Pedigree is even less necessary since direct comparisons are unlikely due to the lack of SAR platforms on existing CubeSats, though there are multiple studies and current research regarding the idea.

5.2.2.4 Trade Study Process

Table 5.8.1.4.1: Frequency Band Pugh Matrix

	Weights	S Band	P Band	L Band	C Band	X Band	K Band
Input Power	0.2	5	7	6	3	2	1
Interference / Obstacle Penetration	0.2	5	8	7	4	2	1
Base Resolution	0.1	5	2	3	4	7	8
Antenna Size	0.4	5	2	3	7	8	9
Pedigree	0.1	5	3	7	10	6	1
Sum	1	5	4.3	4.8	5.6	5.3	4.9
Rank	-	3	6	5	1	2	4

Sensitivity Analysis

Above is the final pugh matrix that best represents the mission requirements we have. Although the P band has great input power, and interference resistance, it ended up being the worst choice. Some of the Pugh matrices had it edging closer to the top rank, but ultimately antenna size was deemed too important of a metric for it to win in the final study. The winner of the study is the C band. This frequency band is the most often used band for SAR platforms and provides a good balance of interference resistance, maximum resolution, and input power.

5.3 GNC Trade Study

The guidance, navigation, and control system is tasked with the specific objective for orienting the CubeSat in whichever way is needed for that time. The two main modes this system will have is Earth pointing and Sun pointing. The Sun pointing mode shall be used when directioning the solar panels towards The Sun for recharge. Earth pointing will be used for when the SAR is performing readings.

5.3.1 Control Sensors

5.3.1.1 Motivation

The chosen sensor for the CubeSat will work as the eyes. This sensor will interact with the actuator in order to correctly orient the on board instruments at Earth. This is why accuracy is the highest ranked out of all the categories. Each option is rated from 1-10 based on this criteria. Some options can be immediately voided, as they are not possible to implement on this CubeSat. For example, most star tracker systems will be extremely expensive, this is not feasible for the tiny budget given. For any given satellite, a minimum of two sensors are needed in order to calculate the attitude of the craft.

5.3.1.2 Metrics

Evaluating different sensor options for CubeSat missions involves considering several key factors:

- **Cost:** The GNC sensors are extremely important for orientation of the spacecraft, but the cost is not the most important factor in this determination. Most of the options are similar in price, and therefore it is not used as a determining factor in deciding the ideal baseline design for sensors in the GNC system. It is still important to consider partially however, since the budget for this project is small.
- **Accuracy:** The accuracy of the proposed solutions is the most important determining factor in design selection. Obviously this is an important characteristic, but any accuracy to within 0.1 degree. The difference comes from the way in which these sensors measure

their intended reference. It is important to note that for a specific sensor, there are more expensive versions of the sensor that will be increasingly accurate.

- **Power:** The power on any CubeSat is a valuable resource. It is important to minimize power for subsystems that are not the payload. In this case, the lower weight is because most of these systems don't require much power to begin with. The estimated power for a simple sensor is about 0.1 Watts.
- **Feasibility/Heritage:** Past designs are important to consider in determining a sensor system. Past CubeSat missions in low Earth orbit seem to always prefer an Earth sensor and a Sun sensor on board.

5.3.1.3 Weights

The trade study assigns weights to each of these metrics to determine the optimal control sensor design:

- **Cost:** 0.15
- **Accuracy:** 0.40
- **Power:** 0.15
- **Feasibility/Heritage:** 0.30

Cost and power considerations are the lowest on this list because of the similarities between all of these sensors. None of them will take a massive power draw, and none will cost too much. Accuracy is the key determining factor, because it is for the application of an Earth observing spacecraft.

5.3.1.4 Trade Study Process

Table 5.2.1.4.1: Control Sensor Pugh Matrix

	Weight	Earth Sensor	Sun Sensor	Magnetometer	Star Tracker
Cost	0.15	5	8	7	1
Accuracy	0.4	7	4	6	9
Power	0.15	5	7	6	7
Feasibility	0.3	6	9	8	1
Total Score	1	6.1	6.55	6.75	5.1

Rank	-	3	2	1	4
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Sensitivity Analysis

From the pugh matrix, the best suited option for this CubeSat is a Sun sensor with a magnetometer. This is a common setup for CubeSats with lower power consumption and high accuracy. The Earth sensor is another good option, however it has limited flights in past CubeSat projects. The sensitivity of this matrix is tested through varying weights of data and the same two options are still the ideal choice. This is because both of the other options have detrimental characteristics, The Earth sensor is considered an emerging technology for CubeSat designs, and the star tracker is too expensive.

5.3.2 Control Actuators

5.3.2.1 Motivation

The actuators will be the system that applies the torque onto the satellite. The on board computer will take the data from the sensors and calculate a needed torque for the system. It is important that this system can orient the spacecraft for sun pointing, and Earth pointing when in low Earth orbit.

5.3.2.2 Metrics

Evaluating different attitude control actuator designs for CubeSat missions involves considering several key factors:

- **Cost:** The price for the actuators are all similar. This is still an important design consideration as the budget is on tight constraints.
- **Size:** The size is the most important parameter for the GNC actuators. This is because for a CubeSat, the size of our entire system is significantly smaller than a normal satellite. Most of these systems have alternative smaller options, which makes the power important.
- **Power:** The power consumption for each of these systems is important, as on a CubeSat the power is extremely limited. Any system that's not the payload should minimize the power draw in order to maximize the payload's ability to fulfill the intended mission path.
- **Feasibility/Heritage:** Past CubeSats most common mode of torque production comes from reaction wheels. This is because of the small size and high accuracy. The momentum dumping is important because it helps extend the life of the wheels without increasing the budget a significant amount.

5.3.2.3 Weights

The trade study assigns weights to each of these metrics to determine the optimal control actuator design:

- **Cost:** 0.15
- **Size:** 0.40
- **Power:** 0.15
- **Feasibility/Heritage:** 0.30

Cost and power are the least weighted metric. This is because of the similarity between all of these options. This satellite being a CubeSat means it will be relatively small, therefore the needed moment created by the system will not have to be large. On the same note, size is heavily restrictive, therefore the size of this actuator must be considered as the most important criteria. Feasibility is also an important consideration to take into account what designs were previously successful in CubeSat missions.

5.3.2.4 Trade Study Process

Table 5.2.2.4.1: Control Actuator Pugh Matrix

	Weight	Thrusters	Reaction Wheels	Magnetic Torque Rods	Control Moment Gyroscope	Reaction Wheel with Momentum Dumping
Cost	0.15	3	5	7	2	4
Size	0.4	1	5	6	5	5
Power	0.15	9	4	8	2	5
Feasibility	0.3	3	8	4	3	9
Total Score	1	3.1	5.75	5.85	3.5	6.05
Rank	-	5	3	2	4	1

Sensitivity Analysis

From this Pugh matrix, reaction wheels with momentum dumping into magnetorquers are the best design choice. When differing the weights, the reaction wheels will win when feasibility is important, but magnetic torque rods win when power and size are more important. The reaction wheels with momentum dumping are the best of both worlds in this situation. With only

a small increase in price for the magnetic torque rods, the inconsistent reaction wheels can be countered with momentum dumping into the torque rods.

5.4 Communications and Data Handling

5.4.1 Band

5.4.1.1 Metrics

- **Data Rate:** The rate of data transfer will determine how long it takes for the ground station to receive data from the SAR and process it into usable information. Because this function is tied directly to the purpose of the mission, data rate is weighted highly.
- **Signal Crowding:** Signal crowding arises from having so many satellites communicating across similar frequencies. Proper planning will require securing communication channels that allow for reliable communication with the satellite's on-board computer. While not the primary concern of this trade study, signal crowding still must be taken into consideration.
- **Atmospheric Losses:** Radio signals can lose some data transfer to atmospheric interference. This generally happens at higher frequencies due to less penetration power. However, many methods of counteracting this effect exist and allow for high band communication. Because of this, atmospheric loss is not a primary concern.
- **Power Consumption:** At higher frequencies, power consumption becomes a concern. Because the system will attempt to transmit data at any point it is over the ground station, it's important to manage the power draw at these instances. High power consumption will limit the capability of the communication system.

5.4.1.2 Weights

- **Data Rate:** 0.35
- **Signal Crowding:** 0.15
- **Atmospheric Losses:** 0.1
- **Power Consumption:** 0.4

5.4.1.3 Pugh Analysis Radar Band

Table 5.4.1.3.1: Radar Band Pugh Matrix

Selection Criteria	Weight	UHF	S	X	K
Data Rate	0.35	4	6.5	7	8
Signal Crowding	0.15	1	6	8	9.5

Atmospheric Losses	0.1	8	6	4	3
Power Consumption	0.4	8	5	2	1
Weighted Sum	1.0	5.55	5.775	4.85	4.925
Ranks	-	2	1	4	3

5.4.2 Communication Architecture

5.4.2.1 Motivation

Choosing the preferred architecture for the communication system will determine the transmission capacity of the comms system, the complexity of the design, and the power profile of the system. For a successful implementation, certain design criteria will determine the best communication architecture for the SMARTSat mission.

5.4.2.2 Metrics

- **Data Rate:** Maximizing the data rate of the communication system is important for generating a quick response during a disaster. This makes data rate the highest weighted criteria for the communication architecture.
- **Power Consumption:** Using higher bandwidths and more transmitters will require a much higher power supply. With limited power generated from the solar panels, it's important to minimize the power drawn for each system. Especially with radio transmission, high power draw is a major concern.
- **Cost:** Higher frequency transmitters skyrocket in price as frequency and power draw increase. This makes cost a high valued criterion for the communication architecture.

5.4.2.3 Weight

- **Data Rate:** 0.4
- **Power Consumption:** 0.25
- **Cost:** 0.35

5.4.2.4 Pugh Analysis Communication Architecture

Table 5.4.2.4.1: Communication Architecture Pugh Matrix

Selection Criteria	Weight	Transmitter/Receiver	Transceiver	Combined Approach
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Data Rate	0.4	5	3	8
Power Consumption	0.25	4	7	3
Cost	0.35	3	6	4
Weighted Sum		4.05	5.05	5.35

5.4.3 Transmitters

5.4.3.1 Motivation

With the transmission architecture chosen, the optimal transmitter can be selected based on the selection criteria. The criteria for a successful transmitter in the SMARTSat mission are cost, flexibility, and an integrated processing unit.

5.4.3.2 Metrics

- **Cost:** The cost of transmitters can quickly reach out of the necessary price range when higher bandwidths and SDRs are considered, so it's important to keep the budget in mind when selecting a transmitter for payload.
- **Flexibility:** Having an adjustable system is always beneficial to the capabilities of the system. Particularly with radio transmission, it's useful to be able to change frequencies should there be an issue communicating at a particular bandwidth.
- **Integrated Processor:** Having a payload-linked transmitter with an integrated processor will help streamline the data transmission protocol and give the ground station access to the SAR data quicker and more efficiently. While not required, it will increase the strength of the design.

5.4.3.3 Weight

- **Cost:** 0.5
- **Flexibility:** 0.35
- **Integrated Processor:** 0.15

5.4.3.4 Pugh Analysis Transmitters

Table 5.4.3.4.1: Transmitter Pugh Matrix

Selection Criteria	Weight	Hardware	SDR
Cost	0.5	7	2

Flexibility	0.35	4	6
Integrated Processor	0.15	0	8
Weighted Sum		4.9	4.3

5.4.4 Antenna

5.4.4.1 Motivation

Choosing the appropriate antenna for the satellite design ensures that data can be efficiently and reliably sent from the transmitter so that the ground station has a better chance at receiving each dataset with minimal packet loss. Important considerations for antenna choice are gain, band compatibility, lack of deployment.

5.4.4.2 Metrics

- **Gain:** The gain or directionality of the antenna is important for increasing the range of communication. With limited ground station visibility, gain can help maximize each opportunity the satellite has to transmit data.
- **Band Compatibility:** The chosen band will determine the antennae that best suits that communication method. This compatibility is highly rated to ensure efficient satellite communications.
- **No Deployment:** Deployable systems place a heavy reliance on the system's ability to successfully arrange itself. Should the antennae not deploy when a high gain requirement is needed, the system cannot effectively communicate with the ground station. No deployment systems are preferred here.
- **Cost:** Most antennae do not cost more than a few thousand dollars commercially. That said, it's still important to weigh the price of each antenna configuration.

5.4.4.3 Weights

- **Gain:** 0.35
- **Band Compatibility:** 0.45
- **No Deployment:** 0.2
- **Cost:** 0.1

5.4.4.4 Pugh Analysis Antenna

Table 5.4.3.4.1: Antenna Pugh Matrix

Selection Criteria	Weight	Wire	Patch	Helical
Gain	0.25	4	6	7

Band Compatibility	0.45	5	8	7
No Deployment	0.2	0	10	0
Cost	0.1	6	8	4
Weighted Sum		3.85	7.9	5.3

5.4.5 On-Board Computer

5.4.5.1 Motivation

The on-board computer will manage the data collected from the payload, the telemetry from other subsystems, and manage any incoming data from the ground station. The determining factors of the OBC design are cost, processing power, volume, and power consumption.

5.4.1.2 Metrics

- **Cost:** The cost of the on-board computer is important to consider, especially with distributed architectures. Balancing the cost of the final system with performance is crucial to maintaining the budget. An incredibly powerful setup may have the strongest computational power but may also limit the budget distribution to other systems.
- **Processing Power:** The very crux of the operation of the on-board computer is the computational power of the system. Based on the application of SAR imaging, a highly capable system will be required, placing this criterion as the highest rated.
- **Volume:** The size of the overall data handling system should be considered. Because this is only a major concern with distributed architectures, this criterion is rated rather low.
- **Power Cost:** The on-board computer will be on most of the time if not the entire time the satellite is operational. Because of this, it's important to consider the impact this will have on the overall power draw of the satellite.

5.4.5.3 Weights

- **Cost:** 0.2
- **Processing Power:** 0.4
- **Volume:** 0.15
- **Power Cost:** 0.25

5.4.5.4 Pugh Analysis

Table 5.4.5.4.1: On-Board Computer Pugh Matrix

Selection Criteria	Weight	Microcontroller	FPGA	Distributed
Cost	0.2	6	8	4

Processing Power	0.4	5	7	9
Volume	0.15	3	3	6
Power Cost	0.25			
Weighted Sum		3.65	5.75	6.25

5.4.6 Operating System

5.4.6.1 Motivation

The operating system will determine the satellites methods of data transfer/ storage, and set the complexity ceiling of the OBCs capabilities. The design metrics for the OS are free to use, open source, and real time processing.

5.4.6.2 Metrics

- **Free:** Having a free operating system for our CubeSat design is important for not adding unnecessary pricing to our final cost. This is the highest rated criterion
- **Data Handling Capability:** With a SAR, it is important that the satellite possesses the necessary computational power to process the data from not only the payload, but the telemetry of the other systems as well.
- **Real Time:** Having a real-time operating system for CubeSats is preferred due to faster execution and lower latency. This is rated lowly here as all systems compared are real-time capable.

5.4.6.3 Weights

- **Free:** 0.5
- **Data Handling Capability:** 0.35
- **Real Time:** 0.15

5.4.6.4 Pugh Analysis

Table 5.4.6.4.1: Operating System Computer Pugh Matrix

Selection Criteria	Weight	FreeRTOS	VxWorks	Linux
Free	0.5	10	0	10
Data Handling Capability	0.35	2	8	6

Real Time	0.15	10	10	8
Weighted Sum		7.2	4.3	8.3

5.6 Thermal Trade Study

5.6.1 External System Metrics

- **Cost:** Cost is always an important factor. For thermal, this would mean increasing thermal precision while staying under the defined budget. Systems that cost more will be ranked lower
- **Volume:** CubeSats have a volume constraint which means we have to allocate certain amount of space for each of our subsystems, so systems that take up lots of volume are ranked lower
- **Adaptability:** This is the most important thermal metric. The goal of thermal is to maintain a temperature range within the CubeSat, so all components will function normally throughout the lifetime of the mission. If the system isn't dynamically changing due to the environment, they will be ranked lower. For example, paints are permanently on and can't change, so when you go to the dark side of the planet, it will be releasing heat that it shouldn't when heat conservation is necessary. This will also be graded based on how much we can change the design to fit our desired requirements
- **Mass:** CubeSats have certain mass requirements that cannot go over. This means collaborating with other subsystems to make sure the final design weight comes under the mission required weight. Systems that weigh more will be ranked lower

While cost is an important metric, the other factors will matter more when it comes to survivability of the CubeSat and mission success. Power required is very essential to mission success due to our payload requiring a radar that will take up most of the power budget, so the power budget must be allocated with only a certain amount being given to thermal systems, so this will gain some more weight compared to mass. CubeSats have a required mass requirement that will require the thermal systems to not take up too much mass from other subsystems, so we want the system to be as lightweight as possible. Adaptability is given the highest because these parameters matter the most because this wagers on whether the mission will fail or succeed. The entire mission problem for thermal is based on LEO where there is a warm side and cold side of the planet. If the adaptability is too relaxed, the electrical components of the CubeSat could fail.

5.6.2 External System Weights

- **Cost:** 0.1

- **Volume:** 0.2
- **Mass:** 0.2
- **Adaptability:** 0.5

5.6.3 Internal System Metrics

- **Cost:** Cost is always an important factor. For thermal, this would mean increasing thermal precision while staying under the defined budget. Systems that cost more will be ranked lower
- **Power Required:** CubeSats have a defined power budget based on the power systems. This means minimizing power consumption of active thermal systems by using passive systems or using less power consuming active systems when possible if power consumption is too great. Systems that require more power will rank lower
- **Compatibility with External System:** This is the most important thermal metric. The goal of thermal is to maintain a temperature range within the CubeSat, so all components will function normally throughout the lifetime of the mission. The external and internal system must work together to accomplish this. If the external gives mainly expulsion of heat then we would need a system that can heat up.
- **Mass:** CubeSats have certain mass requirements that cannot go over. This means collaborating with other subsystems to make sure the final design weight comes under the mission required weight. Systems that weigh more will be ranked lower

While cost is an important metric, the other factors will matter more when it comes to survivability of the CubeSat and mission success. Power required is very essential to mission success due to our payload requiring a radar that will take up most of the power budget, so the power budget must be allocated with only a certain amount being given to thermal systems, so this will gain some more weight compared to mass. CubeSats have a required mass requirement that will require the thermal systems to not take up too much mass from other subsystems, so we want the system to be as lightweight as possible. Compatibility with the External System is given the highest because these parameters matter the most because this wagers on whether the mission will fail or succeed. The entire mission problem for thermal is based on LEO where there is a warm side and cold side of the planet. If the internal system can't work with the external system, the system will fail because it is lacking one aspect of the heat transfer like if the system can only cool, we have no ability for heat addition.

5.6.4 Internal System Weights

- **Cost:** 0.1
- **Power Required -** 0.3
- **Mass:** 0.2
- **Compatibility with External System:** 0.4

5.6.5 External System Pugh Matrix

Sunshield will not be added as an option since we already have a deployable which brings volume constraint issues as is.

Table 5.6.5.1: External Thermal Pugh Matrix

	Weight	Paint/Tapes	MLI Blanket	Thermal Louvers	Deployable Radiator
Cost	0.2	10	8	8	6
Volume	0.2	10	9	7	8
Mass	0.1	10	9	8	8
Adaptability	0.5	5	5	9	9
Sum	1	7.5	6.8	8.3	8.1

Sensitivity Analysis

As you can see from the chart, thermal louvers and deployable radiator are pretty close together so either of these options are best. The external system will be used to look at compatibility for each of these other parts for internal.

5.6.6 Internal System Pugh Matrix

Table 5.6.6.1: Internal Thermal Pugh Matrix

	Weight	Heaters	Cryocoolers	TEC	Fluid Pump
Cost	0.1	9	4	5	7
Power	0.4	8	6	4	2
Mass	0.2	9	7	7	8
Compatibility	0.3	8	4	4	9
Sum	1	8.3	5.4	5.5	5.8

Sensitivity Analysis

As you can see from the chart, electric heaters win by a pretty reasonable margin. The external and internal will seem to be a good match with the louver or radiator getting rid of heat to make sure the CubeSat doesn't overheat and the electric heater adds heat to make sure the CubeSat doesn't cool down too much.

5.7 Ground Station Trade Study

A ground station consists of three major components, the frequencies of communication, the network model, and the options available. This trade study will discuss the analysis of these components and provide determinations on what would be best for our proposed CubeSat. A motivation, metric breakdown, pugh matrix analysis, and sensitivity analysis will be provided for each component. Due to the somewhat few options discussed, a pugh matrix selection screening will be ignored/skipped.

5.7.1 Frequency Band

5.7.1.1 Motivation

For CubeSat missions, selecting the appropriate frequency band is critical to ensure reliable communication. Different frequency bands offer varying capabilities in terms of data rate, coverage, availability, and regulatory constraints. A trade study helps determine which frequency band best suits a CubeSat's communication needs.

5.7.1.2 Metrics

To evaluate the performance of different frequency bands, the following metrics are considered:

- **Data Transfer Rate:** The capacity for transmitting data over the communication link.
- **Range:** The effective communication range of the frequency band.
- **Regulatory Restrictions:** Licensing and usage constraints imposed by international and local regulatory authorities.
- **Atmospheric Interference:** How weather and atmospheric conditions affect signal quality.
- **Antenna Size/Complexity:** The size and complexity of the antennas required for communication.
- **Cost:** Costs related to licensing, equipment, and operational maintenance.

These metrics offer a comprehensive look into the characteristics of radio communication frequencies. Each frequency band has variable DTR, range, regulatory constraints, interference, complexity, and cost to operate on. In order to fairly compare the frequency bands, all of these categories must be considered.

5.7.1.3 Weights

- **Data Transfer Rate:** 0.40
- **Range:** 0.30
- **Regulatory Restrictions:** 0.10
- **Atmospheric Interference:** 0.10
- **Antenna Size/Complexity:** 0.10

The data transfer rate and range were rated the highest because they determine the operational possibilities and create the bottleneck of performance for the communication. In order to not bottleneck the communication capabilities these categories must be considered the most important. Small range is able to be countered with more antenna ground coverage, which is why it is slightly less important than the data transfer rate. The remaining criteria offer a deeper look into the development process but outline challenges that are relatively easy to overcome.

5.7.1.4 Trade Study Process

Table 5.7.1.4.1: Frequency Band Pugh Matrix

	Weights	VHF/UHF	S-Band	X-Band	Ka-Band
Data Transfer Rate	0.30	5	6	8	10
Range	0.30	5	7	9	9
Regulatory Restrictions	0.10	5	7	5	4
Atmospheric Interference	0.10	5	8	6	4
Antenna Size/Complexity	0.10	5	6	4	2
Cost	0.10	5	6	4	2
Total Score	1.00	5	6.6	7	6.9
Rank	-	4	3	1	2

Sensitivity Analysis

Determining the frequency is highly dependent on the data transfer rate and range due to its imposition of a bottleneck on performance. Changing the weights for these criteria sections and adjusting the weight to the cost and regulatory restrictions criteria sections provides a different result for the best option, VHF/UHF. However, this pugh matrix has been determined to reflect the most accurate selection criteria and weights for our project. Furthermore, in the key design alternatives section, it was shown that almost all of these frequencies are able to be communicated on by all of CAGSN's. Which diminishes the importance of the frequency determination, this solely determines the transceivers that the satellite will be equipped with.

5.7.2 Ground Network

5.7.2.1 Motivation

CubeSats require reliable communication with ground stations to receive commands and transmit collected data. The design of a ground station network is crucial for ensuring that CubeSat missions achieve their objectives. Effective communication between the CubeSat and mission control enhances data retrieval, operational flexibility, and mission success.

5.7.2.2 Metrics

Evaluating different ground station network designs for CubeSat missions involves considering several key factors:

- **Cost:** Budgetary constraints play a significant role in determining which network model is feasible.
- **Coverage:** The network's ability to provide consistent global or regional communication coverage.
- **Data Rate:** The efficiency with which the network can handle data transmission.
- **Reliability:** Redundancy and resilience to ensure uninterrupted communication.
- **Scalability:** The ease of expanding the network to support additional missions or future growth.
- **Operational Complexity:** The level of effort required for management, maintenance, and operation.
- **Flexibility:** The network's adaptability to diverse mission requirements and changing conditions.

5.7.2.3 Weights

The trade study assigns weights to each of these metrics to determine the optimal ground station network design:

- **Cost:** 0.25

- **Coverage:** 0.20
- **Data Rate:** 0.15
- **Reliability:** 0.20
- **Scalability:** 0.10
- **Operational Complexity:** 0.05
- **Flexibility:** 0.05

Cost and coverage are prioritized due to budget constraints and mission reliability. Data transfer rate and reliability also receive substantial weight, as they directly impact data handling and communication stability. Scalability, operational complexity, and flexibility are weighted lower, reflecting their roles as supporting factors. This weighting approach provides a balanced assessment of network performance, focusing on essential criteria for CubeSat mission success.

5.7.2.4 Trade Study Process

Table 5.7.2.4.1: Ground Network Pugh Matrix

	Weight	Single Ground Station	Distributed Ground Station	Hybrid Ground Station	Cloud-Based Ground Station	Hybrid + Distributed + Cloud Based Ground Station
Cost	0.25	5	3	4	5	4
Coverage	0.20	5	10	8	7	10
Data Transfer Rate	0.15	5	9	7	8	10
Reliability	0.20	5	9	7	8	10
Scalability	0.10	5	8	6	9	10
Operational Complexity	0.05	5	4	5	6	4
Flexibility	0.05	5	7	9	10	10
Total Score	1.00	5	8	7.3	7.15	8.2
Rank	-	5	2	3	4	1

Sensitivity Analysis

Varying the weight of data transfer rate provides a larger variance in total score, with the combination of the three options coming out on top. However, this matrix has been determined to be the most accurate and representative of the importance of each criteria to the selection of a ground network model. As shown, the gap between the distributed ground station and the combination is minimal, this is because the combination is essentially a distributed ground station with extra performance capabilities. The same goes for a hybrid ground station, as it depends on multiple ground station antenna locations. Generally, the combination provides the best performance with minimal change in price.

5.7.3 Commercially Available Ground Station Networks

5.7.3.1 Motivation

For CubeSat missions, choosing a Commercially Available Ground Station Networks (CAGSN) is essential for ensuring reliable data transfer, cost-efficiency, and global coverage. Each CAGSN offers unique benefits and challenges in terms of frequency capabilities, network reliability, scheduling flexibility, and cost structures. This trade study helps identify the most suitable CAGSN for CubeSat missions by examining critical operational and budgetary factors.

5.7.3.2 Metrics

The following metrics are used to evaluate the effectiveness of each CAGSN:

- **Coverage:** The geographical reach and the number of ground station antennas available across different locations to support continuous communication.
- **Latency:** The delay in receiving satellite data on Earth, impacting time-sensitive operations.
- **Cost-Effectiveness:** Includes both initial setup and ongoing operational costs, considering pricing structures like pay-per-use and subscription models.
- **Reliability:** Consistency in service availability and overall network uptime.
- **Scalability:** Flexibility to support higher data volumes and mission expansion over time.

These metrics provide a holistic view of each CAGSN's strengths and limitations, ensuring a fair comparison. Each network varies in terms of coverage, latency, cost, reliability, and scalability, all of which are crucial for CubeSat communication.

5.7.3.3 Weights

To prioritize the needs of typical CubeSat missions, each metric is assigned a weight based on its significance:

- **Coverage:** 0.30
- **Latency:** 0.25

- **Cost-Effectiveness:** 0.20
- **Reliability:** 0.15
- **Scalability:** 0.10

Coverage and latency are weighted the highest since they directly impact the frequency of communication opportunities and the speed of data access, respectively. While coverage allows for more consistent data transfer, low latency minimizes delays in receiving data. Cost-effectiveness is also prioritized, as CubeSat missions often operate within limited budgets. Reliability is essential to mission success, especially for long-duration missions, while scalability allows for flexibility in meeting future mission demands.

5.7.3.4 Trade Study Process

Table 5.7.3.4.1: Commercially Available Ground Station Networks Pugh Matrix

	Weight	NASA's NEN	AWS Ground Station	KSAT Lite	Leaf Line
Coverage	0.30	5	3	4	2
Latency	0.25	5	4	5	3
Cost-Effectiveness	0.20	5	6	7	8
Reliability	0.15	5	4	4	3
Scalability	0.10	5	6	5	3
Total Score	1.00	5	4.30	4.95	3.70
Rank	-	1	3	2	4

Sensitivity Analysis

Adjusting the weights for the criteria, across the board, has minimal effect on the analysis. This is due to the baseline design generally and consistently outperforming the other CAGSN's. This pugh matrix represents the final draft of multiple iterations of analysis for the chosen CAGSN's. As shown, KSAT Lite is highly competitive with the baseline. However, due to the extensive system that NASA has implemented, the baseline outperforms all other options. This is relevant as it reinforces the fact that NASA will be the best option for ground station communications. This is beneficial as the UNP proposal is directed towards NASA and the DOD.

5.8 Propulsion Trade Study

5.8.1 Thruster Trade Study

5.8.1.1 Motivation

Firstly, the prices are analyzed to remove potentially nonviable options. Our total budget is \$240,000. Kinetic thrusters can cost around \$45,000. Chemical thrusters can cost \$230,000-700,000. Electrical options can cost \$100,000-300,000. The only financially viable option is the kinetic thruster, or possibly the electrical option. The cost of launch will potentially expend another \$60,000. This could leave the rest of the CubeSat with only \$80,000 to work with. As these costs are estimates, the team finds it best to avoid potential financial pitfalls. The kinetic thrusters will be chosen as our propulsion system. The solar sail is neglected as an option due to its lack of heritage, along with its low thrust. Another consideration was the power input for the electrical system, which further devalued such options for our already power heavy design.

5.8.1.2 Metrics

Evaluating different propulsion designs for CubeSat missions involves considering several key factors:

- **Cost:** The propulsion system is an expensive part. Cost is very important
- **ISP:**
- **Thrust:**
- **Power:** The power consumption for each of these systems is important, as on a CubeSat the power is extremely limited. Any system that's not the payload should minimize the power draw in order to maximize the payload's ability to fulfill the intended mission path.

5.8.1.3 Weights

The trade study assigns weights to each of these metrics to determine the optimal propulsion system.

- **Cost:** 0.1-.35
- **ISP:** 0.35-.2
- **Thrust:** 0.35-.2
- **Power:** 0.1-.35

5.8.1.4 Trade Study Process

A Pugh matrix is made to analyze the kinetic options. Cold gas is chosen to be the baseline option. Multiple iterations of the Pugh matrix are performed to find the best option.

Table 5.8.1.4.1: Thruster Pugh Matrix

	Weight	Cold Gas	Warm Gas	Resistojets
Cost	.1	5	4	1
ISP	.35	5	6	8
Thrust	.35	5	6	8
Power	.2	5	3	1
Sum	1	5	5.2	5.9

	Weight	Cold Gas	Warm Gas	Resistojets
Cost	.2	5	4	1
ISP	.25	5	6	8
Thrust	.25	5	6	8
Power	.3	5	3	1
Sum	1	5	4.7	4.5

	Weight	Cold Gas	Warm Gas	Resistojets
Cost	.35	5	4	1
ISP	.2	5	6	8
Thrust	.2	5	6	8
Power	.35	5	3	1
Sum	1	5	4.85	3.9

Sensitivity Analysis

From these Pugh matrices, we can tell that the Resistojet option becomes worse the higher we rate the cost and power consumption. The Resistojet option should actually not be

selected using the same reasoning we eliminated the electrical options. The cost would eat into the budget too much, with a resistojet estimated to cost \$100,000, the most expensive option out of the kinetic. Also notable is how the ISP and thrust are weighted compared to each other won't not affect our grading since the team values them equally. Our team values the low cost and power more than the high thrust and ISP since our budget is tight and we value our payload more than propulsion. We need the power budget elsewhere. With all of these considerations in mind, the team chose the cold gas propulsion system as our design. It performs better overall than the warm gas option. In the budget analysis section the weight for our cost proved necessary. The same can be said for our power budget.

5.11 Structures Trade Analysis

5.11.1 Motivation

The structure of the CubeSat is important to ensure the survival of the systems inside. By choosing a strong material, the team ensures that the CubeSat can survive the vacuum of space for its entire 4 year mission.

5.11.2 Metrics

- **Cost:** Cost is weighted at 10% because the budget isn't as important as the functionality of the alloy. Ensuring that the alloy can survive the stresses and extreme temperatures plays a bigger role than the overall cost ranking of each individual alloy. Additionally, the overall budget was \$3,000 under the \$240,000 for two years, therefore, money can be allocated to various subsystems if the time calls for it.
- **Density:** Density is weighted at 20% because the weight was an important design requirement listed earlier in the document. Therefore, ensuring that whatever material that was chosen met this requirement was of the utmost importance because that would allow the cubesat to comfortably fall under the maximum allowable weight.. Additionally, this would also allow for most of the weight to go to the payload rather than the frame of the cubesat.
- **Strengths:** Strengths had the highest weight because the cubesat needs to be able to survive the launch and function properly during the orbital period of 4 years. Any malfunction within the frame or any component of the cubesat would undoubtedly result in mission failure and prohibit the cubesat from surveying the targeted area properly.
- **Melting Point:** Melting point was weighted second highest because the cubesat will face harsh temperatures, specifically during the launching phase. Ensuring that the selected alloy would not melt from the temperatures during launch will ensure that the frame will hold up and that the cubesat will be able to carry out its mission in orbit.
- **Thermal Conductivity:** Thermal conductivity was weighted low because even though having a low value would be ideal, our cubesat will have a thermal regulatory system on board so that it will be able to cool or warm itself when needed.

5.11.3 Weights

- **Cost:** 0.1
- **Density:** 0.2
- **Strengths:** 0.35
- **Melting Point:** 0.25
- **Thermal Conductivity :** 0.1

5.11.4 Trade Study Process

Table 5.11.4.1: Structures Pugh Matrix

	Weight	Aluminum 6082	Aluminum 7075	Aluminum 6061	Aluminum 5052
Cost	.1	6	2	4	5
Density	.2	5	3	5	6
Strengths	.35	4	8	7	3
Melting Point	.25	5	2	6	7
Thermal Conductivity	.1	5	6	5	5
Weighted Sum		4.75	4.7	5.85	5

Sensitivity Analysis

As seen in the pugh matrix, the aluminum alloy 6061 scored the highest with alloy 5052 right behind. Alloy 6061 is the best all around material to use for the cubesat frame because it scored high in the areas that were heavily weighted. Increasing the thickness of the frame would allow this alloy to be just as strong and robust as the 7075 alloy. Furthermore, its density is also lower than its counterpart which will allow for the team to save weight and prioritize the weight going into the payload where all the mission critical devices will be. With that being said, the chosen alloy that will be used for the material of the cubesat chassis will be aluminum alloy 6061.

6. Design Solution

The Design Solution section outlines the approach taken to develop the CubeSat system, detailing the components, subsystems, and integration strategies that support the mission objectives. This solution is informed by the requirements of the CubeSat, ensuring that each element is selected and designed to meet the needs of the mission while staying within the constraints of size, weight, power, and cost.

In this section, we will explore the decisions made regarding the hardware and software components, as well as how they were integrated to create a functional prototype. The prototype serves as a smaller-scale model, allowing us to test and verify the effectiveness of these design choices before scaling them to the full CubeSat system. This approach ensures that the final design will be both feasible and optimized for mission success.

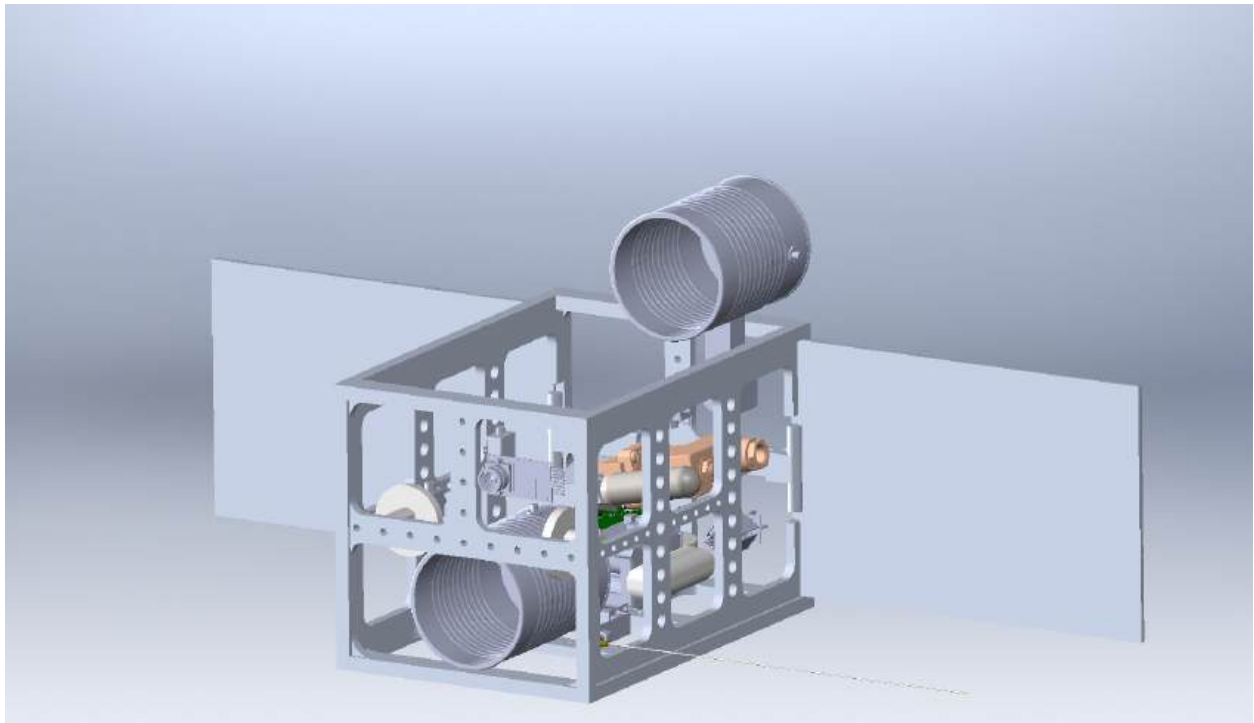


Figure 6.1: Prototype CAD Model

6.1 Power Design Solution

The power system of the SMARTSat will be made nearly entirely out of space grade components. Power itself will be provided using two 6U solar panels generating a combined 40W charging potential. These panels will use space grade cells and inhouse assembled circuitry. A total of 12 40x80 EXA solar cells will be used on each panel. For power storage, a Pumpkin Space lithium ion battery pack rated for 72-100Wh will be used. Power delivery will be accomplished using an Ibeos small sat power board with

3.3V, 5V, and 12V regulated buses and a 24V unregulated bus. This ensures that the correct voltage is delivered to each component.

6.1.1 Prototype Summary

For the prototype design, commercial hobby electronics will provide an adequate simulation of the final design. To simulate the solar panels, 30 small solar panels will be used to generate 13.2 Watts of energy. Power can then be stored in 3-6 18650 lithium ion batteries rated at 3.7V and 5000 mAh. An MPPT board will be used to convert the power generated by the solar panels into something usable. Power delivery will utilize a soldered circuit board connecting the battery to the components. The heater and solenoid are the most power demanding devices, and will use a 5V to 12V transformer in conjunction with two relays to switch the components on. To deploy the solar panels, two 20kg servos will be used.

6.1.2 Power Prototype Solution

Table 6.1.2.1: Parts List

	Actual Part	Price	Weight	Dimensions
3 x SUNYIMA 5.5V 80mA Mini Solar Panels	Solar Panels	3 x \$16.99	14g	2.36"x2.36"
Waveshare Solar Power Manager Module	MPPT Board	\$19.99	45g	45mm x 45mm x 10.5mm
2 x 20KG Servo Motor High Torque RC Servo	Servos	2 x \$13.59	100g	1.59" x 1.77"
6 x 18650 Li-ion Battery 5000mAh	Power Storage	\$29.99	337.36g	18.01mm x 67.02mm
MT3608 DC-DC Step Up Boost Power Converter	3.3V - 5V Step Down	\$5.99	10g	0.55" x 0.67" x 0.55"
HiLetgo AMS1117-3.3 DC-DC	5V - 12V Step Up	\$6.99	1.8g	0.52" x 0.67"

Step-Down Power Module				
5V One Channel Relay Module Relay Switch	Relay	\$7.39	10g	0.52" x 0.67"
18650 Battery Holder Single (10pcs)	Additional battery storage	\$5.99	5g	20mm x 21mm x 76mm
MakerFocus DC to DC Boost Converter	5V - 12V Step Up	\$7.99	30g	0.71"D x 18.9"W x 0.71"H
ElectroCookie Prototype PCB Solderable Breadboard	Protoboard	\$8.49	10g	3.5" x 2.05"

The above table shows the breakdown for the power subsystem. The total mass for the power subsystem comes out to 1252.88g - 2279.96g depending on 3-6 battery configuration, and the total price for all components totals \$148.50. Some masses present are estimated based on the average mass of small electric components. The total variance should be negligible as the batteries take up the majority of the mass budget. The following sections will go into greater detail for each component's specifications.

6.1.2.1 Solar Panels

The chosen solar panels are manufactured by SUNYIMA. These will generate all power for the prototype.

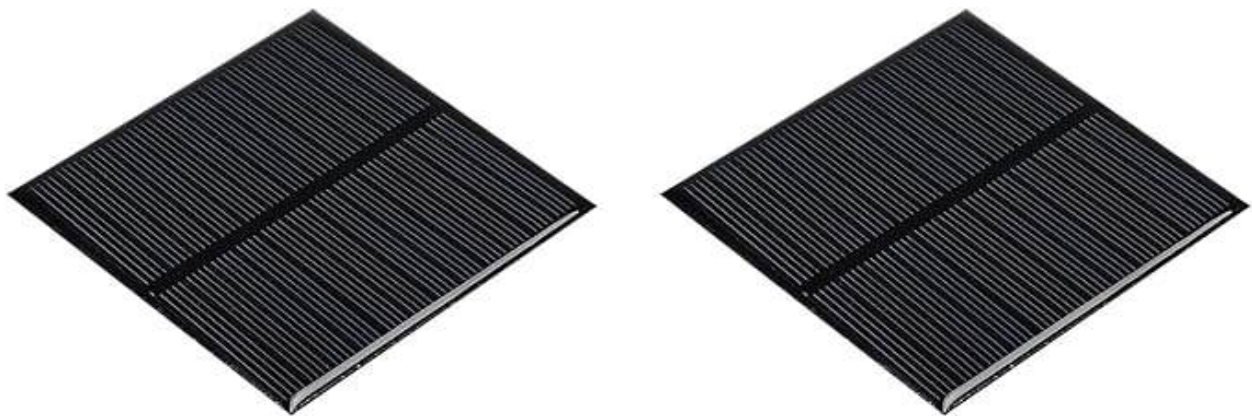


Figure 6.1.2.1.1: Solar Cells [Pow1].

Specifications:

- 5.5V.
- 80mA.
- 0.44W.

Role in Prototype:

- Convert solar energy into electric energy.
- Charge power bank.
- Simulate final design construction.

Justification:

With 24 in hand, the maximum power output sits around 10.56W. This is certainly less efficient than the full scale design capable of producing 40W power generation, but provides an accurate depiction of the manufacturing process that will be required at full scale. These panels will act as the cells for the full design. They will be oriented in a 3x4 grid on each 6U solar panel. This layout perfectly mimics the full scale design. One key difference when using mini solar panels as opposed to solar cells is that the circuitry is all embedded, meaning no additional circuitry needs to be designed. In the full scale design, an inhouse solar panel circuit will need to be created. The following model shows the full solar panel assembly.

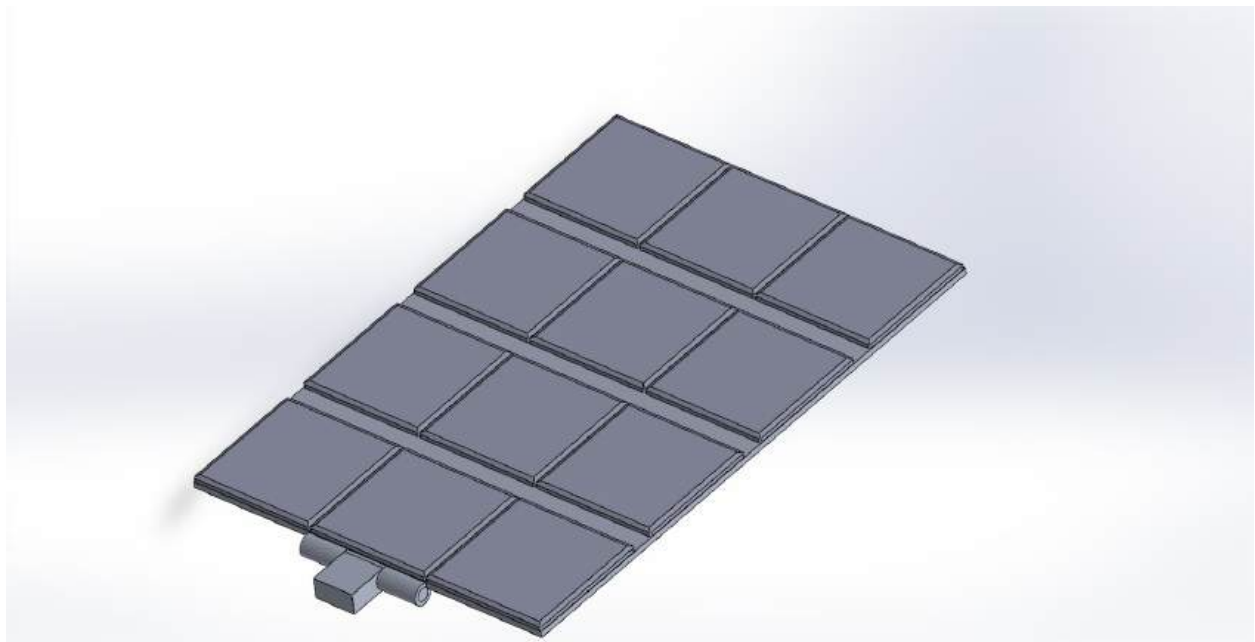


Figure 6.1.2.1.2: Solar Panel CAD Model.

The solar panels must be assembled in a way that provides enough voltage to the MPPT board. To do this, each column of 4 solar panels will be wired in series to bring the voltage to 22V, within the range of the Waveshares 6V - 24V operating range. The higher input voltage will ensure that the MPPT Board achieves maximum efficiency. The following figure outlines this configuration.

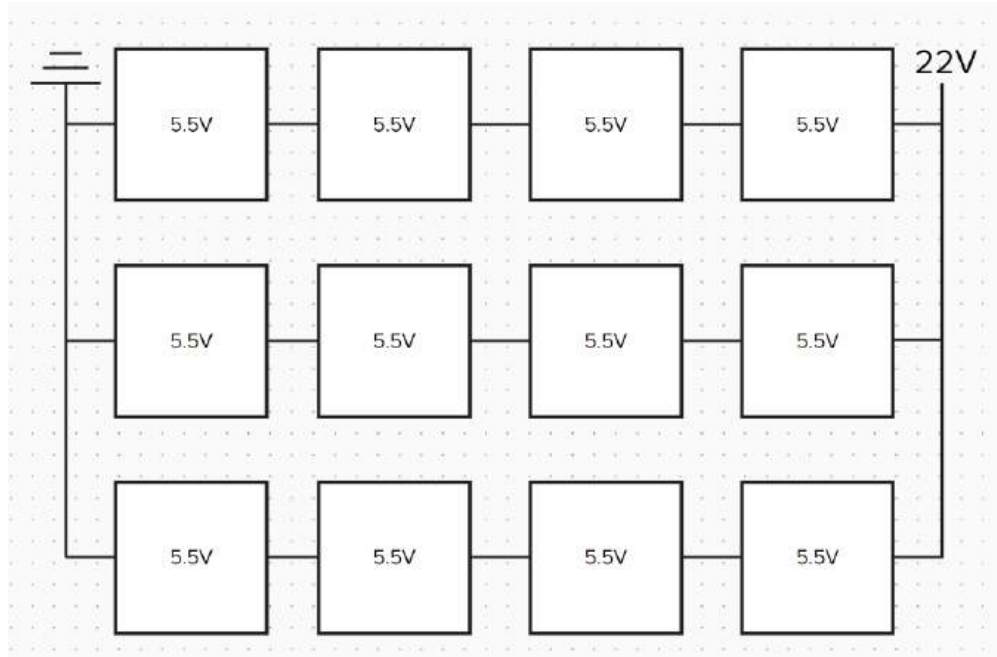


Figure 6.1.2.1.3: Solar Panel Circuit Model.

6.1.2.2 Power Distribution

In the full scale design, the EPS system is a single unit consisting of voltage busses, voltage converters, and MPPT charging. However, this kind of system is not readily available for most robotics applications, and a power delivery system must be designed from the ground up. To start, the MPPT board can be observed. The Waveshare MPPT board is chosen for the prototype.

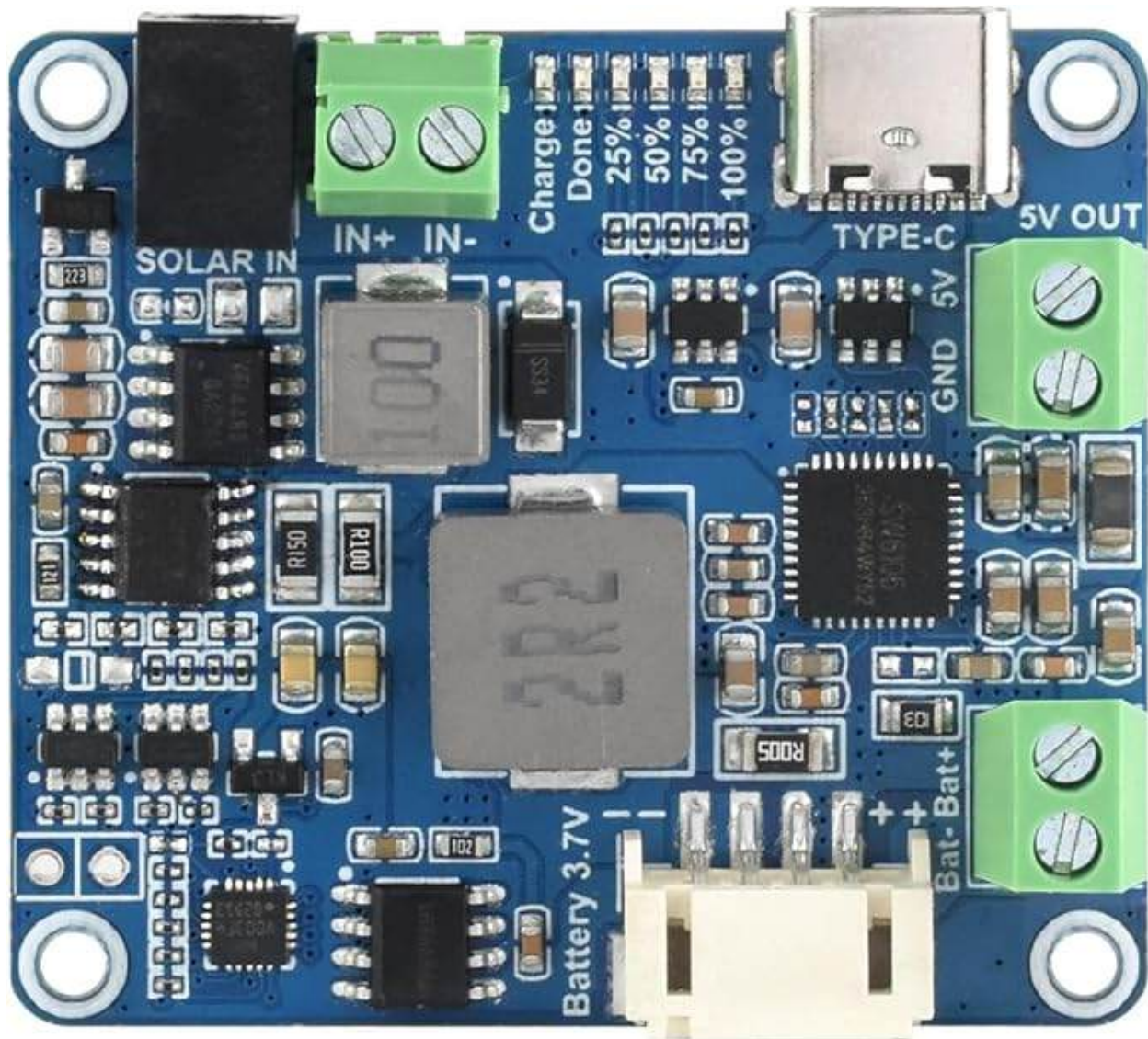


Figure 6.1.2.2.1: MPPT Board [Pow2].

Specifications

- 6-24V in.
- Direct wire in.
- 5V out.
- USB-C out.
- 3.7V Li-ion battery compatible.
- Included 3 x battery charging pack (housing, no included batteries).
- Dual battery output.
- Charging indication lights.

Role in Prototype

- Interface between solar panels and power bank.

- Deliver power directly to the microcontroller.
- Provide MPPT for optimal charging.
- Provide a backup power bank solution.

Justification

For any solar panel charging system, a maximum power point tracking (MPPT) capable circuit is needed. This helps ensure that the energy from the solar panels is collected as efficiently as possible. In the full scale design, this is achieved using the IBEOS EPS system. For the prototype, a Waveshare MPPT board is chosen. This MPPT board takes a 6-24V input from a solar panel and converts it to 5V out. In addition, it has built in battery pack compatibility, designed for lithium ion 18650 batteries. These features make this board the ideal choice for the MPPT design solution. Below is an estimated 3D model of the MPPT board generated from multiple images.

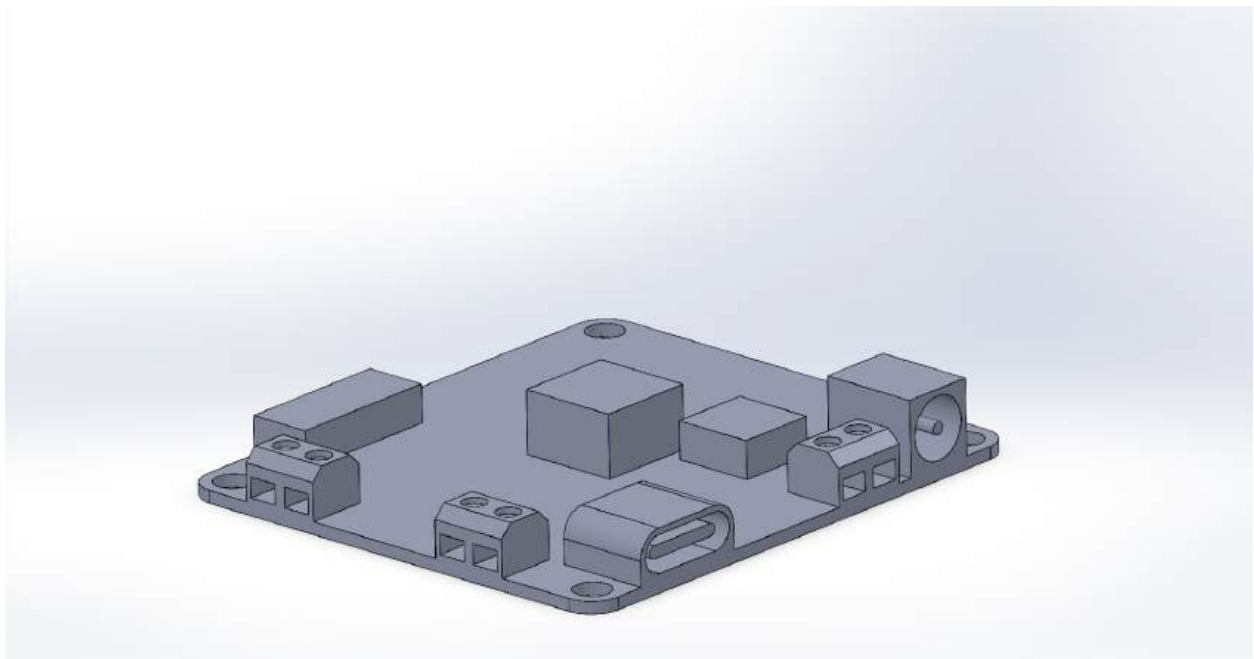


Figure 6.1.2.2.2: MPPT Board CAD Model.

However, this board alone does not replace the EPS in the full design solution. This board only has two outputs; that being a USB-C and a 5V out. Without knowing for sure, it can be assumed that the USB-C supply is also rated at 5V. Some of the electronics in the prototype design are only rated for 3.3V, and others require 12V. In addition, there are far too many electronics for this board alone to power. Thankfully, Arduino microcontrollers provide a method of delivering power to many 5V electronics at a time. In addition, the voltage converter circuits will provide the necessary voltage to each component.

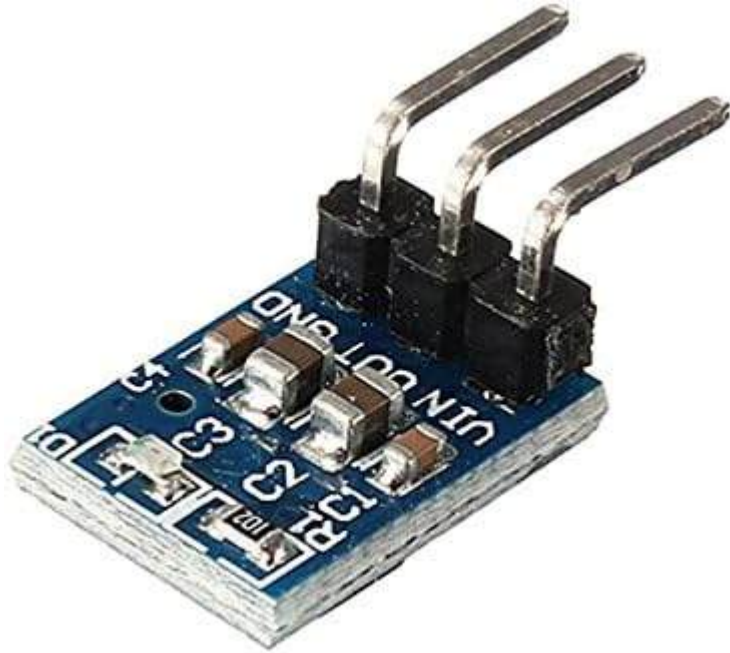


Figure 6.1.2.2.3: Step Down Converter [Pow3].

Specifications

- 4.2-10V in.
- 3.3V out.

Role in Prototype

- Convert 5V to 3.3V for micro SD reader.

Justification

The step down circuit solves the issue for powering 3.3V electronics. This is required by the micro SD Card reader specifically. The micro SD card reader can fail if too high a voltage is applied. In addition, an over-voltage can also damage the micro SD card itself and destroy any data stored on the device. For this reason, the buck converter is absolutely necessary.

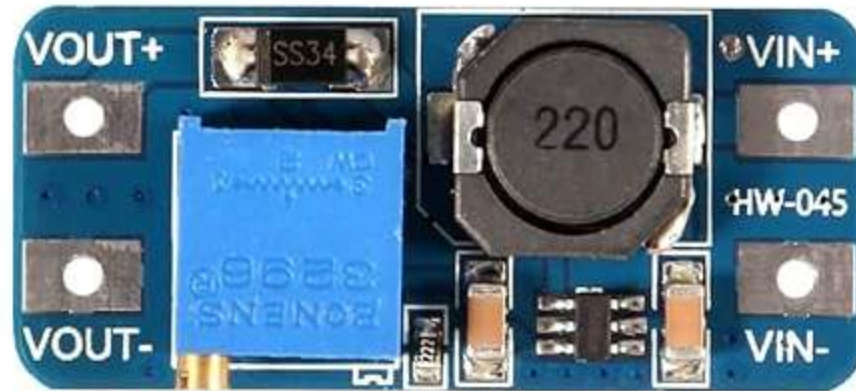


Figure 6.1.2.2.4: Step Up Converter [Pow4].

Specifications

- 2-24V in.
- 5-28V out.
- Tunable output.

Role in Prototype

- Convert 5V to 12V.
- Provide necessary voltage to solenoid.

Justification

For the 12V electronics, a step up conversion circuit can be used to deliver the required voltage. The component that requires 12V is the solenoid valve for the propulsion system. While a simple manufactured transformer can deliver the required voltage, it is more efficient and effective to use a step up board with a tunable output. The following image depicts the chosen step up converter. This converter is rated for 2-24V in and 5-28V out. This output is once again tunable with the potentiometer. While this component is required for the solenoid alone, other components such as the servo are rated for higher voltages that can be met using this **device**. With the voltage converters and the MPPT board, the needs of electrical power delivery are satisfied for the prototype design.

6.1.2.3 Power Storage

The solar panels alone cannot power the system, so a power bank solution is required for full system functionality. This system must be compatible with the MPPT board as well as accurately simulate the final design solution. For the power storage solution, 6 lithium ion batteries were chosen for the mock design.



Figure 6.1.2.3.1: 18650 Lithium Ion Batteries [Pow5].

Specifications

- 3.7V out.
- 4.2V required for charge.
- 5000mAh.
- 18.5W per battery.
- 18650 form factor.

Role in Prototype

- Provide power to the entire system.
- Store energy for later use.

- Accurately mimic final design.

Justification

The standard 18650 Li-ion form factor is used both in the prototype and the full scale design. In addition, the chosen batteries as well as the storage capacity of 5000mAh brings the total storage available to 111Wh, 11Wh above what the full scale capacity is set at. It must be noted that the provided battery pack from the MPPT board only houses 3 batteries, so a 6-battery power bank must be 3D printed. This will bring the power capacity for the prototype as close to the full scale as possible. Below is a basic design for a 6-battery power bank. The provided MPPT board will display charging capacity using LEDs and indicate when the bank is full.

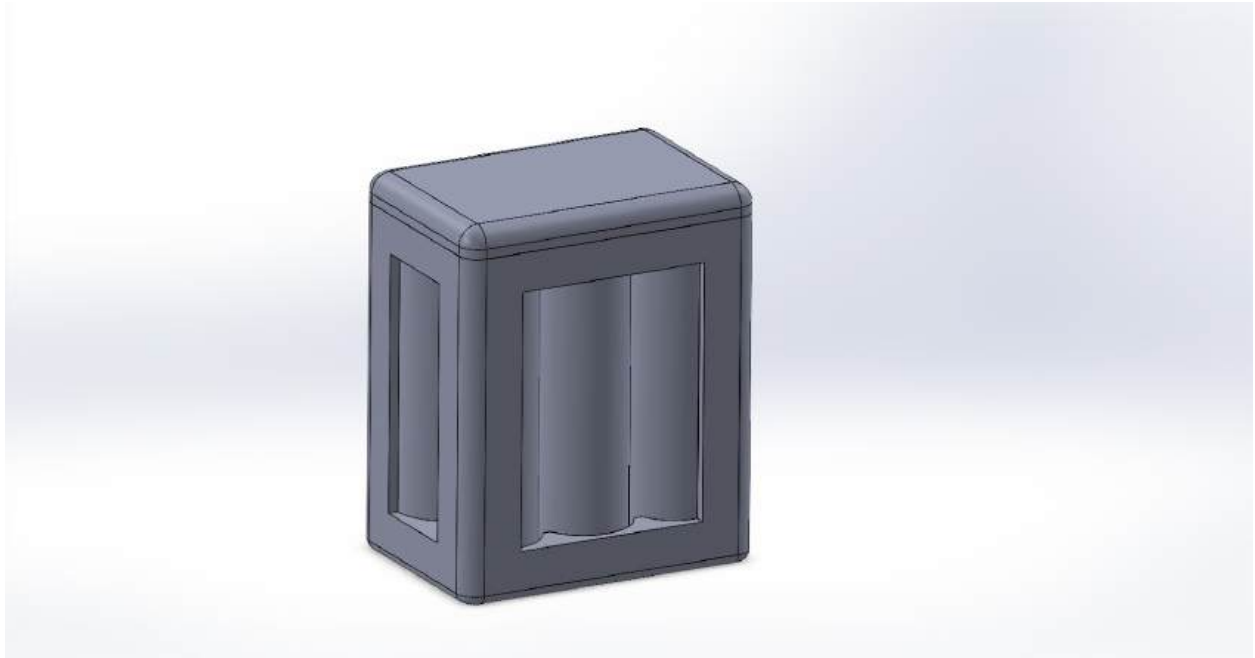


Figure 6.1.2.3.2: Battery Pack CAD Model.

The provided MPPT board will display charging capacity using LEDs and indicate when the bank is full.

With the battery capacity known, it can be calculated how long it will take the batteries to be charged to 100% using the solar panels alone. The following equation is used to calculate this time.

$$t_{charge} = \frac{C}{P_{generated}}$$

Here, C is charge, t is time, and P is power. With a capacity of 111Wh and a maximum of 10.56W, the minimum charge time from zero to 100% is 10.51 hours. This time is far longer than

what is observed in the full scale design. However, the power output of each system is not measured to be incredibly high in each operational mode. This will also provide a much longer period for testing each individual system before the full charge is depleted.

6.1.2.4 Solar Panel Deployment

To deploy the solar panels, a servo motor capable of lifting the entire panel can be used. This will allow a smooth deployment of the system as to not impose any unnecessary stress on the components unlike a spring loaded design. The chosen servo for this solution is a 20kg RC servo.



Figure 6.1.2.4.1: Servo Motor [Pow6].

Specifications

- Amplifier type: Digital controller.
- Command signal: Pulse width modification.
- Bearing: Double ball bearings.
- Control angle: 270°.
- Pulse width range: 500 ~ 2500usec(0 ~ 270 °).
- Dead band width: 3 usec.
- Neutral position: 1500usec.
- Stall current: 2100mA±10% (4.8 V) -2700mA±10% (6.8V).
- Peak stall torque: 20.5kg·cm (284.7oz·in) -22.8kg·cm (316.63oz·in)
- No load speed: 0.18sec/60° (4.8 V); 0.14sec/60° (6.8V).

Role in Prototype

- Deploy solar panels.
- Ensure smooth deployment.

- Will operate at 5V.

Justification

Servos are a good choice for panel deployment as they allow for a gentle transition between the undeployed and deployed phase. To ensure that the system is stable and strong in its deployed state, a latch mechanism that locks it in place can be used. 20kg is chosen as this output should be able to handle the weight of the full solar panel. The panel is made of aluminum 6061 at a density of 2.7g/cm. The dimensions of the panel are 28cm x 18cm x 0.25cm. With the added 12g from the solar panels themselves, the following equation can be used to calculate the required torque in kg·cm.

$$\tau_{req} = (\rho(V) + m_{solarcells}) \frac{l}{2}$$

Here, τ is torque, ρ is density, V is volume, m is the mass of the solar cells, and l is the length of the solar panel. With the known values, the required torque is 4.931 kg·cm. At a rating of 20kg·cm, this servo is more than capable of deploying the panel with a 5V supply voltage.

6.1.2.5 High Power Relay

Some components such as the payload and the heater will require more power than the arduino can supply. For this reason, a relay attached to a regulated voltage bus on the battery pack can be used to power these components with no interaction between the high voltage and the arduino.

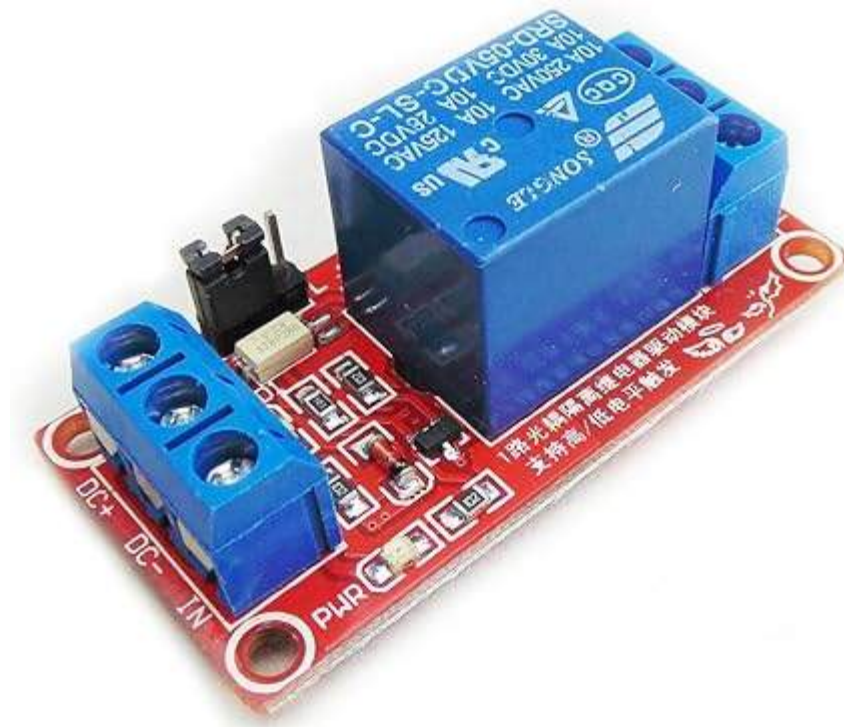


Figure 6.1.2.5.1: Relay Switch [Pow7]

Specifications

- Rated for 10 Amps
- Rated for 30V DC
- Rated for 250V AC
- Normally open
- Operating voltage of 5V
- Maximum 300W DC

Role in prototype

- Act as a switch for high power electronics.
- Minimize damage to the microcontroller.
- Ensure that the heater pad is adequately powered at 10W.

Justification

One component in need of high power is the heater pad. For the full scale model, this is a crucial system to power, as many electronic components won't function properly at below zero degrees centigrade. This component will ensure the safe delivery of high power delivery while keeping other components safe. The arduino will control the voltage delivery to the relay and ensure that it will not deliver power to the heater unless the thermocouple calls for it.

6.1.3 Power Exploded View

The figure below shows the exploded view of the power subsystem for SMARTSat. Voltage converters and the relay are not included in this model. This concludes the power design solution for the prototype of SMART Sat.

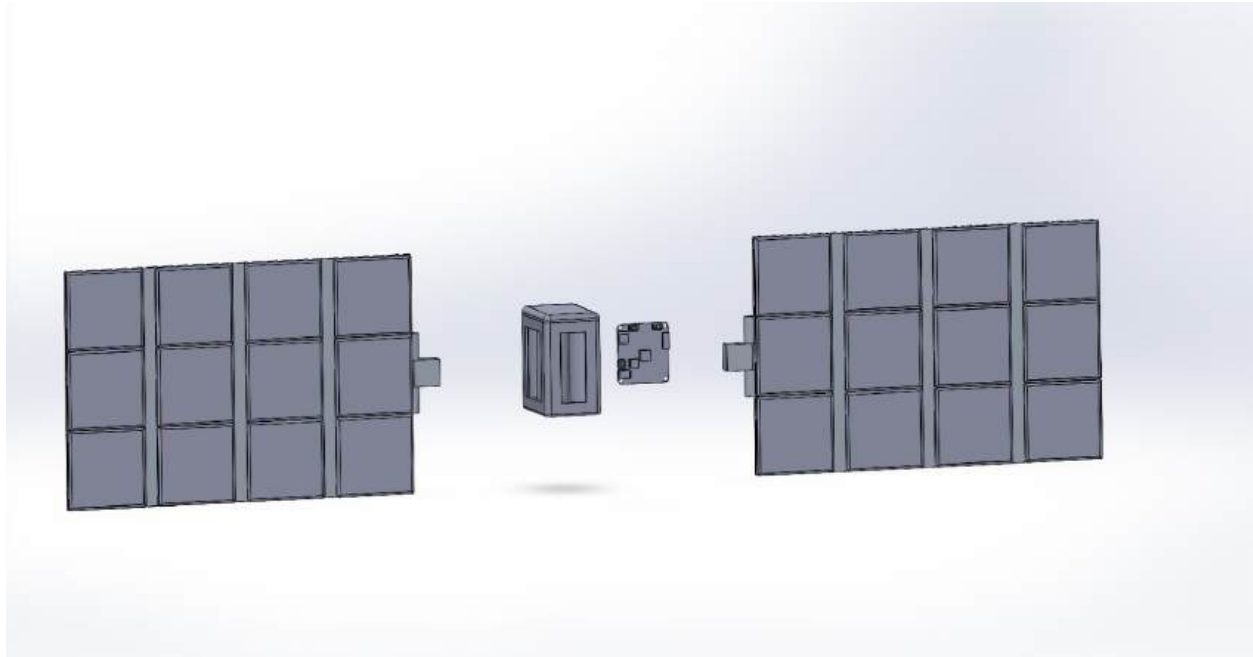


Figure 6.1.3.1: Power Exploded View

6.2 Payload Design Solution

The payload solution for SMARTSat is a SAR with a self-fabricated receiver and transmitter antenna utilizing COTS driving equipment to generate the signal pulse in C-band. The driving equipment alone does not fit inside our prototype budget and fabricating a working receiver antenna is well beyond the scope of our senior design program. To estimate the full-scale payload and show an example of a simple SAR system working in our prototype SMARTSat infrastructure, we will still self-fabricate the antennas, but we will do so under reduced and lax design requirements, such as a 1 km maximum range, two dimensional instead of three dimensional SAR images, and drastically reduced power requirements. This allows us to fabricate our antennas using cheaper materials and to use COTS driving equipment that is, in some cases, one hundred times cheaper than it will be for the full-scale design.

6.2.1 Prototype Summary

The prototype payload will consist of self-fabricated transmitter and receiver antennas made using metal cans and monopole wire connected to COTS driving equipment in the form of circuit components and integrated circuit chips. To enable communication with the faux ground station, we will use a wired USB connection between the output of the SAR circuit and a laptop that will function as the data processor for the payload side of the ground station. The circuit for

the SAR system will include signal amplifiers, attenuators, splitters, mixers, and a video amplifier along with breadboard assembled circuit components connecting to function generator ICs that will generate the signal pulses necessary for range calculation. Credit must be given to the MIT OpenCourseWare instructors who so generously provided free access to information and circuit designs needed to design our payload solution. Those instructors are Gregory L. Charvat, Jonathan H. Williams, Alan J. Fenn, Steve Kogon, and Jeffrey S. Herd [PAY50].

6.2.2 Prototype Solution

The following table represents a comprehensive list of the parts necessary to build the prototype payload circuit and antennas. Individual wires, adapters, circuit components, and fasteners will be aggregated into a single section each for brevity.

Table 6.2.2.1 Parts List

	Actual Part	Price	Weight	Dimensions
TI LM324	Quad Op-Amp	\$ 0.34	0.928 g	0.635 x 1.93 0.457 cm
RQRA-2328-2536-CR	Voltage Controlled Oscillator	\$ 30.00	UNAVAILAB LE	1.27 x 1.27 x 0.589 cm
CATTEN-03R0	Inline SMA Attenuator	\$ 18.55	UNAVAILAB LE	UNAVAILABLE
MAAL-009053-T R3000	Power Amplifier/Low Noise Amplifier	\$ 2.35 x 2	UNAVAILAB LE	UNAVAILABLE
ZX10-2-42-S+	RF Power Splitter	\$ 54.65	UNAVAILAB LE	2.286 x 1.88 x 1.448 cm
ZX05-43MH-S+	RF Mixer	\$ 72.62	UNAVAILAB LE	1.88 x 2.286 x 1.372 cm
WRL-21225	SMA Male to Male Adapter	\$ 3.69 x 4	UNAVAILAB LE	UNAVAILABLE
Can	Antennas	TBD	TBD	10 cm Diameter
901-9889-RFX	Bulkhead Solder Cup Connector	\$ 7.32 x 2	UNAVAILAB LE	UNAVAILABLE
IC XR2206CP Monolithic Function Generator	Function Generator Chip	\$ 7.95	UNAVAILAB LE	UNAVAILABLE

200 to 102K Ohm Resistors	Circuit Part	~\$ 2.00 total	Varies	Varies
0.1 uF to 10 uF capacitors	Circuit Part	~\$ 1.00 total	Varies	Varies
10K to 2M Ohm Potentiometers	Circuit Part	~\$ 2.00 total	Varies	Varies
Audio and Circuit Wires	Circuit Part	~\$ 40.00 total	Varies	Varies
EXP-300E	Circuit Part	\$ 8.35	UNAVAILAB LE	550 Tie Points, 16.5 x 4.45 cm
Screws, Fasteners, and Washers	Structure	\$ 15.00	Varies	Varies

Although many of the dimensions and masses of the chosen parts are unavailable, they are almost entirely negligible. Circuit components are generally less than 10 cubic centimeters in volume on the large end. The only parts large enough to be of note have their dimensions listed. Since this system will never be launched into space, the mass and size matter little compared to the functionality of each component and its ability to integrate with the payload and other subsystems.

6.2.2.1 Function Generator & Quad Op-Amp

These are the only integrated circuit chips in the payload system and represent key components on either end of the radar circuit. The function generator is what generates the radio signal pulse in the form of a voltage ramp that feeds into the VCO. The quad op-amp takes the mixed transmitter and receiver signal and processes it into readable digital signals that can be sent to the ground station to be processed into radar images.



Figure 6.2.2.1.1: XR2206 and Pinout [PAY25].

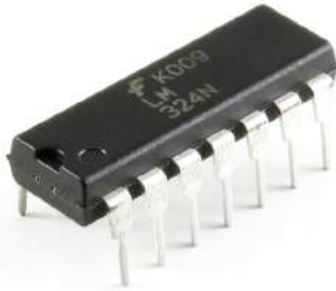


Figure 6.2.2.1.2: LM324N [PAY26].

Specifications:

XR2206 Function Generator:

- Supply Voltage: 10V to 26V (18V Recommended)
- Duty Cycle: 1% to 99%
- Sweep Frequency Range: 2000:1
- Sine Wave Distortion: 0.5%

LM324 Quad Op-Amp:

- Number of Channels: 4
- Supply Voltage: 3V to 32V (Halved for dual supply voltage)
- Supply Current: 175 μ A
- Output Current: 20 mA
- GBP: 1.2 MHz
- Slew Rate: 500 mV/ μ s
- Operating Temperature: 0-70 C
- Voltage Gain: 100 dB

Role in Prototype:

- The function generator will create the voltage ramp necessary to give to the VCO to generate a radar pulse that can be measured to find range to the target instead of attempting to directly measure the signal return time, which would be impossible with current technology.
- The quad op-amp will function as a video amplifier that turns the radio frequency signal generated by the RF mixer into readable digital signals that can be processed by the ground station to get usable data.

Justification for Selection

The XR2206 and LM324 both are rated for our chosen frequency of 2.4 GHz and can integrate with the power subsystem with typical 3V-18V power requirements. We need a way to generate a pulsed voltage signal to correctly operate the VCO and produce the necessary ramping frequency. The XR2206 function generator chip is designed to perform operations such as this in a circuit and will easily integrate into the payload system to provide us with this voltage ramp. Likewise, the payload needs to be able to take the signals received by the antennas and turn them into digital signals that can be read by the faux ground station laptop. The quad op-amp is capable of performing these operations easily due to its four channels and fast slew rate.

6.2.2.2 Voltage Controlled Oscillator and Attenuator

The VCO is arguably the most important part of the radar circuit. It will take the ramping voltage from the function generator and turn it into our desired ramping radio signal. The attenuator will then take this signal and reduce its power to a baseline value that will be amplified to a useful power for the transmitter antenna.



Figure 6.2.2.1: RQRA-2328-2536-CR [PAY27].



Figure 6.2.2.2.2: CATTEN-03R0 [PAY28].

Specifications:

RQRA-2328-2536-CR VCO:

- Tuning Voltage: 0.5V to 4.5V
- Supply Voltage: 5V
- Frequency Range: 2.328 to 2.536 GHz (2.432 GHz Center)
- 2nd Harmonic: -15 dBc
- Icc Max: 35mA
- Pushing: 1.5 MHz/V
- Power: 7 +/-2 dBm
- Operating Temperature: -40 to 85 C

CATTEN-03R0 Attenuator:

- Attenuation Value: 3 dB
- Frequency Range: 0 to 3 GHz
- Power: 1 W
- Impedance: 50 Ohms

Role in Prototype:

- The RQRA VCO will take the voltage ramp from the function generator chip and use it to generate a ramped radio signal pulse that will be sent to the transmitter antenna to be fired at the target for range measurements.
- The CATTEN attenuator will take the roughly 7 dBm signal from the VCO and attenuate it by 3 dB to a base value that can be amplified to a useful power for the transmitter. It will do this with minimal impact on the signal due to a maximum 1 W power requirement and a 50 Ohms impedance.

Justification for Selection

The RQRA VCO's frequency range is perfectly situated around our target frequency and will easily integrate into the payload and power subsystems with a 5V supply power. The output power of the RQRA VCO is higher than required based on the feasibility equations used early and should help future proof our prototype if we wish to increase power to the SAR. The attenuator simply must operate in our chosen frequency range and reduce the signal to a baseline value before amplification. The CATTEN Attenuator fits all of these requirements and is one of the cheaper options available.

6.2.2.3 Power Amplifier and Low Noise Amplifier

The two amplifiers in the payload circuit will perform similar tasks at opposite ends of the radar. The power amplifier will take the output of the VCO and increase it to a useful power level for the transmitter. The low noise amplifier will take the signal captured by the receiver antenna and amplify it to a readable value for the quad op-amp.

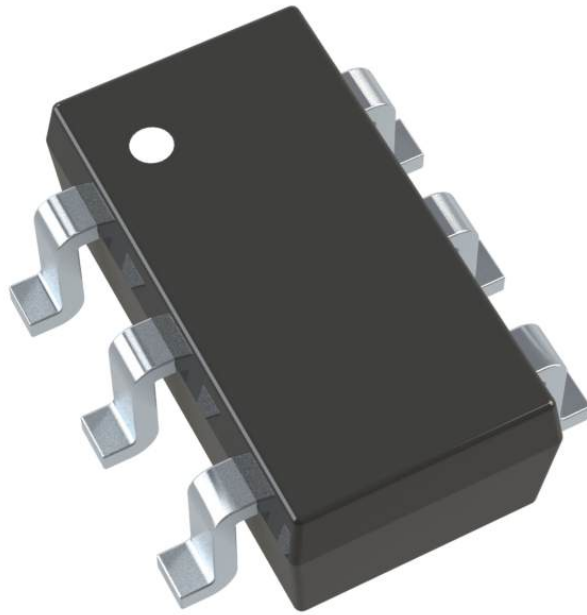


Figure 6.2.2.3.1: MAAL-009053-TR3000 Representation [PAY29].

Specifications:

MAAL-009053-TR3000 RF Amplifier:

- Frequency Range: 800 MHz to 3 GHz
- Supply Voltage: 3V
- Supply Current: 80 mA
- P1: 18.5 dBm
- Gain: 14 dB
- Noise Figure: 1.4 dB

Role in Prototype:

- The MAAL RF amplifier will perform both amplification roles. It will take the output of the VCO and Attenuator combination and increase the power of the signal with minimal noise and it will amplify the input signal from the receiver antenna to a readable level for the quad op-amp.

Justification for Selection

The MAAL RF amplifier can perform both amplifier roles required by the payload circuit without any drawbacks. Its operation frequency range easily fits our chosen frequency and it

requires a standard 3V power input with less than 0.1 Amps of current draw. It will increase the power of the signal by 14 dB, easily meeting the expectations of the feasibility equations. The low noise figure will allow cleaner images to be taken by the radar. Since we only need to source two of the same part, using this RF amplifier will also ease procurement.

6.2.2.4 Power Splitter and RF Mixer

Two similar, but inverted requirements of the circuit are the splitting of the transmitter signal and the mixing of the transmitter signal and the received signal. In order to take advantage of the feasibility equations governing the generation of a readable frequency, we must take the transmitted signal and evenly distribute it between the transmitter antenna and a mixer that can allow it to be compared with the received signal and be sent to the quad op-amp for RF to digital conversion.



Figure 6.2.2.4.1: ZX10-2-42-S+, ZX05-43MH-S+ Representation [PAY30][PAY31].

Specifications:

ZX10-2-42-S+ Power Splitter:

- Insertion Loss: 0.2 dB
- Frequency Range: 1.9 to 4.2 GHz
- Supply Current: 800 mA
- Power Input: 1 W max
- Operational Temperature: -40 to 85 C
- Minimum Isolation: 17 dB
- Maximum Imbalance: 5 degrees

ZX05-43MH-S+ RF Mixer:

- Conversion Loss: 9.8 dB max
- Frequency Range: 0.824 to 4.2 GHz

- LO Frequency: 0.824 to 4.2 GHz
- IF Frequency: 0 to 1.5GHz
- Third Order Intercept: 22 dBm
- Operational Temperature: -40 to 85 C
- Maximum RF Power: 50 mW

Role in Prototype:

- The ZX10 Power splitter will take the amplified signal from the VCO and split it evenly between the transmitter antenna for firing at the target and the signal mixer where it can be compared against the signal from the receiver antenna.
- The ZX05 RF Mixer will perform the function of the frequency feasibility equation and combine the transmitted and received signals so they can be processed by the quad op-amp and the ground station to determine range to the target.

Justification for Selection

These components are the most expensive purchases for the payload subsection, they represent around a third of the budget allotted to the payload for the prototype. This is because they perform some of the most technically challenging, if simple, operations in the circuit. The ZX10 power splitter must perfectly evenly distribute the signal from the VCO between the transmitter and the RF mixer or noise will overpower the comparison and range cannot be measured accurately. The high cost of the splitter reflects the quality and capability of the part as seen by the 0.2 dB insertion loss, low maximum imbalance, and large frequency range which fits our chosen frequency. The ZX05 must similarly perform the function of accurately mixing the two signals without loss or interference. Its frequency range also includes our chosen frequency and expected readable frequency of roughly 1 kHz.

6.2.2.5 Receiver and Transmitter Antennas and Associated Components

These are the self-fabricated components and their necessary COTS components that will integrate them into the rest of the radar circuit and the payload subsystem as a whole. An exact size for the cans cannot be determined until procurement, as they will be consumer products purchased or collected independently for very low cost. The other components can be defined within reason. Wherever an exact representative image is impossible, a CAD model will supplement for easier visualization.



Figure 6.2.2.5.1: Transmitter Cantenna With Bulkhead Connector and Monopole Wire [Internal].

Specifications:

Transmitter/Receiver Cantenna:

- Estimated Diameter: ~10 cm
- Estimated Length: ~15 cm
- Material: Aluminum
- Estimated Weight: ~100 g
- Maximum Gain: 8 dBi
- HPBW: 73 degrees
- Gain at 1 m: 8 dBi
- Guide Wavelength: 18.38 cm
- Monopole Length: 3.1 cm
- Monopole Spacing: 4.6 cm

901-9889-RFX Bulkhead Solder Cup Connector:

- Connection Type: SMA
- Contact Termination: Solder Cup
- Max Frequency: 18 GHz
- Mounting Type: Bulkhead - Rear Side Nut
- Material: Brass
- Operational Temperature: -65 to 165 C
- Mating Cycles: 500
- Impedance: 50 Ohms

WRL-21225 SMA-SMA Male to Male Adapter:

- Connection Type: SMA, Inline, Straight
- Center Gender: Male to Male
- Mounting Type: Bulkhead - Rear Side Nut

CSE-SGAM-305-SGAM Male to Male Coaxial Cable:

- Connection Type: SMA
- Connectors: Male to Male
- Max Frequency: 8 GHz
- Cable Impedance: 50 Ohms
- Material: Copper
- Operational Temperature: up to 105 C
- Length: 30.48 cm

Role in Prototype:

- The transmitter and receiver cantennas will be fabricated from identical cans with a diameter of near 10 cm. They will act as a circular waveguide and aperture for the radio signal sent and received. This allows the SAR to focus its radio signal in a general direction and target something to be imaged.
- The RFX bulkhead connectors will allow the cantennas to connect to the rest of the radar circuit and allow us to install a monopole wire inside the cans at the right spacing. The solder cup will allow the installation and removal of monopoles should one be damaged and need replacing [PAY32].
- The WRL adapters allow short connections between circuit components such as the VCO and the Attenuator which can be placed next to each other [PAY33].
- The CSE coaxial cables will carry the radio signal from the ramped pulse generation circuit to the transmitter and then the received signal from the receiver antenna to the RF mixer [PAY34].

Justification for Selection

These components are mostly chosen for compatibility with the rest of the payload subsystem. The wires and the adapters are SMA type, the same as the circuit components chosen, and the bulkheads are designed to connect through the cans and carry the RF signals into the cables to their destination. The cantennas themselves will be machined into acceptable tolerances if they are procured with unexpected dimensions, but will generally serve their purpose regardless of exact dimensions. Having cans take the place of a purpose built circular waveguide reduces cost tremendously as cans can be procured for free if necessary. There will be almost no loss in effectiveness due to the chosen frequency and low range of the prototype. The extension mechanism for the receiver antenna is included for completeness sake. It will be constructed of mostly left over materials from the construction of the structure and other subsystems if necessary. It is impossible to accurately simulate the full-scale extension mechanism with the budget available so this representation of the mechanism was included simply to represent that the receiver antenna will be the one that needs deployment.

6.3 GNC Design Solution

The final design solution for attitude and spacecraft dynamics will utilize a combination of two sun sensors working in tandem with a magnetometer for sensing capabilities, and 3 reaction wheels paired with 3 magnetorquers for imposing physical torques on to the system. The Sun sensor will be vital for charging orbits, while the magnetometer will help with target pointing, and orbital position analysis.

6.3.1 Prototype Summary

The prototype will be much simpler than the overall design solution, and will be a demonstration of the CubeSat capabilities. The prototype includes an LDR photo sensor that will track our Sun, working in conjunction with a mini infrared sensor that will act as the magnetometer for Earth pointing. For physical movements, the reaction wheels will be simulated by using toy motors with 3D printed fly wheels attached to the ends of them. The torques induced by these wheels will be too small to move the structure on Earth, which is why weights will be added to the end of the flywheels via tungsten cubes.

6.3.2 Prototype Solution

The prototype for SMARTSat will be a proof of concept model that emphasizes the strict budget constraints and ease of assembly required in a one semester project. There will be microcontrollers for the interfacing between motors and sensors, and an in house code will need to be created to be run on these controllers. The following table lists the components used in the prototype design:

Table 6.3.2.1 Parts List

	Actual Part	Price	Weight	Dimensions
Teyleten LDR PhotoSensitive Sensor Module	Sun Sensor	\$ 8.99	132.11 g	3.2004 x 1.397 x 0.508 cm
HiLetgo GY-521 MPU6050 3 Axis Accelerometer Gyroscope	Magnetometer	\$ 25.99	17.86 g	7.62 x 6.35 x 1.27 cm
Uxcell Micro Motor	Reaction Wheel Motor (X3)	\$ 9.99	290.02 g	5.26 x 2.4 cm
3-D Printed Flywheel	Reaction Wheel Disk (X3)	\$ 0.00	N/A	23.25 mm long, 27 mm diameter

2mm Aluminium Alloy Flange Coupling Connector	Universal Mounting Hub (X3)	\$ 14.98	2.8 g	0.715 x 0.05 cm
Mini RC 20A Brushed ESC	Speed Controller	\$ 14.99	49.895 g	3.81 x 2.29 x 0.76 cm
Tungsten Cubes	Weights	\$ 9.95	56.7 g	0.635 x 0.635 x 0.635 cm

6.3.2.1 Photo Sensor

The Teyleten Robot photo sensor is a tiny, compact digital light detection module that will have arduino compatibility capabilities. This light sensor will be able to adjust the potentiometer to change the detection brightness for our controller capabilities. When the brightness of ambient light reaches a certain threshold, the DO end outputs a high level. When ambient light brightness exceeds the set threshold, the DO end outputs a low level. The DO output will be directly connected to a microcontroller which will detect the change of the light brightness of the environment. The sensor is shown below. [GNC17]



Figure 6.3.2.1.1: LDR PhotoSensitive Light Sensor. [GNC17]

Specifications:

Teyleten LDR PhotoSensitive Sensor Module:

- Connector Type: Screw
- Number of Contacts: 1
- Mounting Type: bolt mounted
- Material: Copper
- Brand: Teyleten Robot
- Voltage: 5 Volts (DC)
- Dimensions: 1.26" x 0.55"
- Temperature Rating: 55 Degrees C

Role in Prototype:

- Sun Tracking: This device will act as a Sun sensor for Sun pointing mode. In turn, this will allow for the testing of light tracking capabilities in the prototype. This is directly correlated with the Sun tracking that the full design is required to do.
- Aid In Attitude Control: The microcontrollers that are supplied via the power and communications section will be directly connected to this device. This will act as a control system for spacecraft orientation.
- Detumbling: Having this device will aid in the detumbling verification testing that the prototype will have done. By allowing the sensor to be sensitive to only certain strength of lights, it will aid in the detumbling of the satellite whereas the other sensor will be most likely overrun with unimportant data.

Justification for Selection

The LDR sensor will act as the best substitute sun sensor for the prototype model. The ability to adjust the brightness sensor is perfect for the sensing needs in indicating the difference between ambient light and a prominent source. This device is also easily compatible with arduino, which the electrical lead for the prototype prefers. All of these functions and this device comes in a compact design at the low price of \$9. This small device will fulfill all of the attitude determination needs and assist in other aspects as well.

6.3.2.2 Accelerometer

The HiLetgo MPU6050 accelerometer is a sensor that measures motion, orientation, and acceleration in six degrees of freedom. It is easily used with the arduino and allows for simple attitude control. The accelerometer will measure the direction of gravity in three axes to send information on the current tilt of the sensor. The analog outputs allow for very precise measurements of slight tilts, allowing for control during the Earth pointing mode. [GNC18]

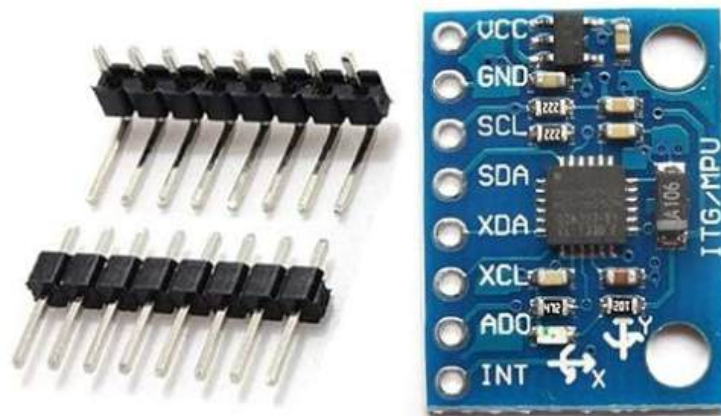


Figure 4.3.2.2.1: Accelerometer. [GNC18]

Specifications:

Arducam Mini Camera:

- Brand: HiLetgo
- Model Name: 680727635251
- Connectivity Technology: MPU-6050
- Processor: SDA, SCL
- Item Weight: 17.86 grams
- Power Source: battery
- Dimensions: 7.62 x 6.35 x 1.27 cm
- Power Inputs: 5V/3.3V

Role in Prototype:

- Targeted Pointing: To prototype the idea of targeted pointing while in orbit, this sensor will be used with an arduino to ensure pointing capabilities.
- Attitude Determination: This accelerometer will aid in the orientation of the model satellite. This sensor will allow for in depth precision pointing of the CubeSat.
- Orbital Positioning: The sensor allows for not only attitude sensing, but also orbital positioning. This is similar to the role of the magnetometer in how it serves two purposes.

Justification for Selection

The accelerometer will simulate the functions of the magnetometer in the prototype. The tiny design allows for precision placement along the body of the prototype, which ensures the optimal angle for sensing. The analog outputs also allow for the use of this product in multiple ways. Being able to potentially use this device as an altitude measurement device allows for multiple uses in attitude determination. This will allow for the controller to find the real Earth when verification testing which is the exact function that the magnetometer fulfills. This device is also only \$ 25 which is extremely cheap for all the allowable functions.

6.3.2.3 Reaction Wheel Motors

The Uxcell micro motors are lightweight and inexpensive ways to convert electrical energy into rotational energy. A small shaft of diameter of 2 millimeters spins out the front end and will need a supplementary part for connection to another object. This is the function of the universal mounting hub. There are two terminals on the back which current will run to spin it. There must be ESC to route the high amperage away from the arduino and directly to the motors. The signal for the ESC's will be coming from an arduino controller which will moderate the current to input the correct RPM for said positioning. The motors and controllers are shown below. [GNC19]

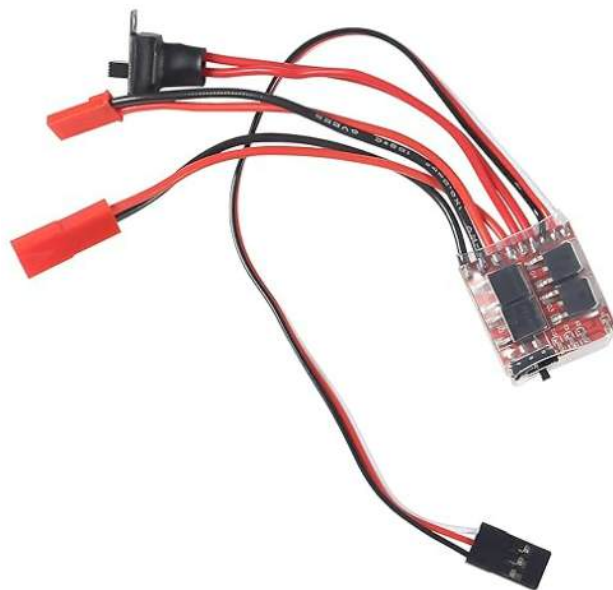


Figure 4.3.2.3.1: Micro Motor & ESC's. [GNC19]

Specifications:

Uxcell Micro Motor:

- Model Name: gear
- Speed: 9000 RPM
- Volts: 9 V
- Material: Steel
- Product Weight: 10.23 ounces
- Product Dimensions: 5.26 x 2.4 cm

Role in Prototype:

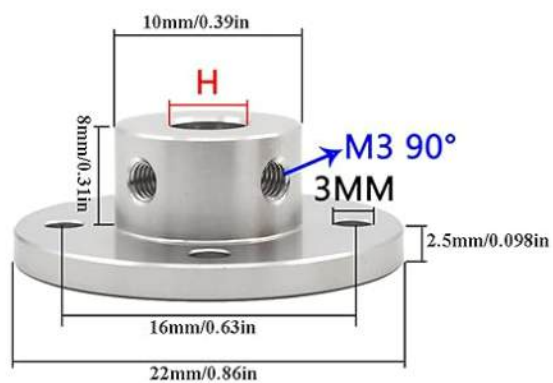
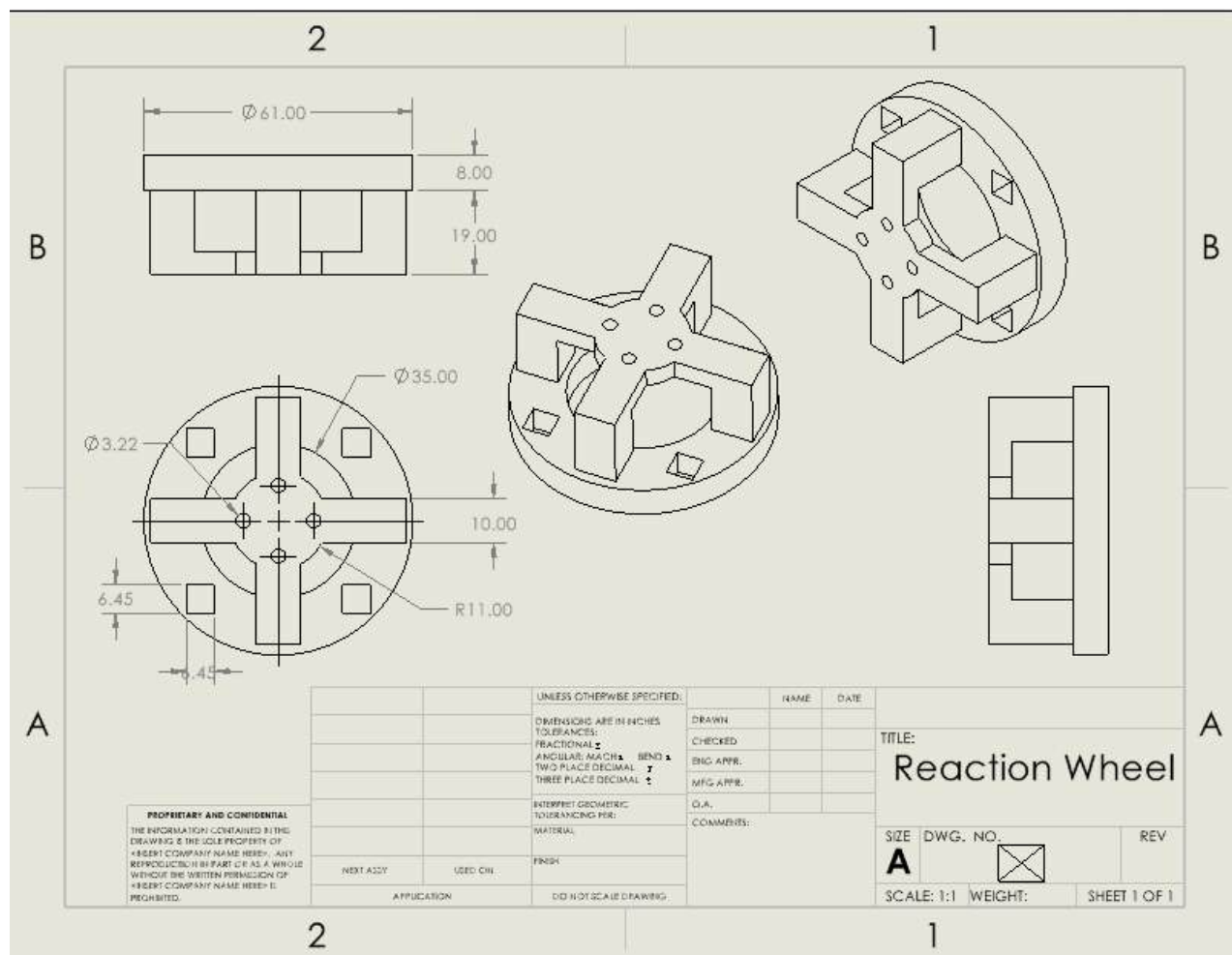
- Actuator for Attitude Control: The spinning motors will be placed on the three separate axes of this prototype. When each one spins, there will be a mass attached to the end which induces a torque onto the CubeSat.
- Detumbling: The detumbling process will be completed solely by this actuator. The spinning masses will allow for the satellite to regain control when spinning in unwanted directions.

Justification for Selection

There is not much difference in the market of toy DC motors compared to the other products on this list. This super cheap model allows for the purchase of multiple motors at one time. The small and compact size allows for easy installation into small spaces such as on the CubeSat. The simple 2 terminal design will be easily integrated into the electrical system. This model also operated well at a range of voltages, which will aid in the design process. This product also stands out because of the high strength metal in which it is made from, proving suitable for satellite applications.

6.3.2.4 3D Printed Disks

The reaction wheel disk will be 3D printed to fit on the end of the toy motors. The printed disk will be attached to the motors to create a homemade reaction wheel. This will be done via a 2mm aluminum alloy flange coupling connector. This hub connects to the spinning rotor and allows for screws to be inserted in it. This will then be easy for the 3D printed wheels to attach to. An example of the mounting hub along with complete reaction wheel mock design are shown below.



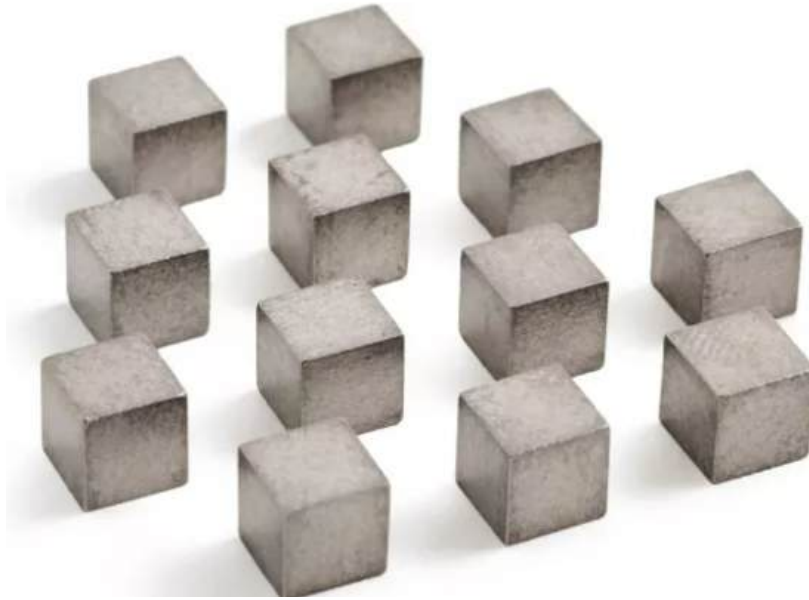


Figure 6.3.2.4.1: Overall reaction wheel prototype. [GNC20]

Specifications:

3D Printed Disk:

- Size: 38 mm long, 27 mm diameter
- Material: Filament

Role in Prototype:

- Torque Generation: The spinning mass will induce a torque onto the system as a result of conservation of angular momentum. This allows for attitude control over the prototype satellite. The input current is first run through a microcontroller which will change the RPMs that the disks will spin at, creating an attitude controlled satellite.
- Size Considerations: The 3D printed part is sized to where it takes up a very similar amount of space to the real design. The diameter and length make it to where whatever space the prototype fits in, the real design should as well. This is the reason for the

Justification for Selection

3D printing is a better option than buying an off the shelf part because it allows for full customization of the disk. In order for the mock reaction wheels to meet the same size restrictions as the real reaction wheels, the design of the disk will need to be held to strict values. This also allows for ease of integration for using the mounting hub, as the spots for bolts will be customized to fit perfectly. This design option also allows for a cheaper method, as 3D printing at NC State is relatively inexpensive.

6.4 Communications and Data Handling

For the full scale design solution, only space grade components will be utilized. General telemetry data will be handled by the Kryten M3 microcontroller with a data storage of 4gb. Payload data will be calculated using the OBC-1 field programmable gate array computer with 64gb of storage. Data will be replayed to the ground station across two channels. UHF band communication will be used for basic telemetry, while S band will be used for payload data delivery. Each communication channel will require a separate antenna to perform efficiently, so a wire antenna will be used for UHF, and a patch will be used for S band communication.

6.4.1 Prototype Summary

The prototype will be much simpler than the final design. Due to the lack of a space-like environment, most space-grade considerations will not be necessary. rad hardening, solar radiation, and cosmic radiation will not be issues necessary to combat. The aluminum housing will act as a strong RF shield to ensure that electronic signals flow steadily.

For the prototype, commercial microcontrollers and hobbyist electronics can be used. This will simulate the communication and data handling between each subsystem and will allow the SMARTSat team to prepare for any challenges with building the full scale design.

6.4.2 Prototype Solution

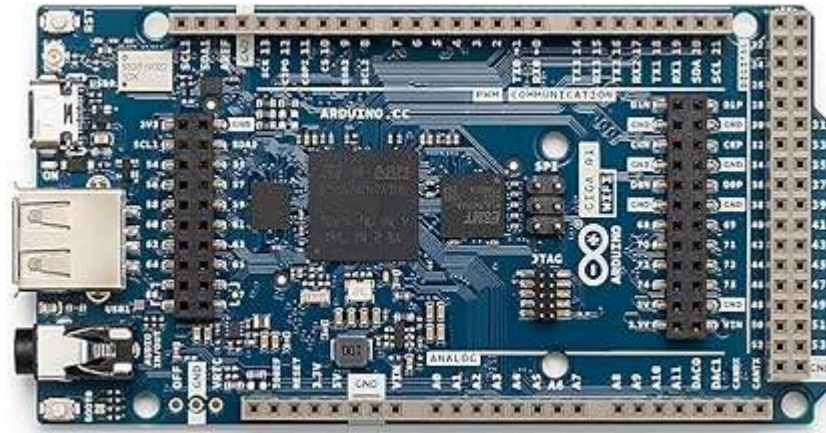
Table 6.4.2.1 Parts List

	Actual Part	Price	Weight	Dimensions
Arduino Giga R1 WiFi	Microcontroller	\$58.26	50g	53mm 101mm
NRF24L01+ Breakout Adapter with 3.3V Regulator	S Band Comms	\$13.99	13g	1.9cm x 5.08cm
Micro SD TF Card Adapter Reader	Data Storage Interface	\$6.99	49.9g	4.5cm x 2.5cm
PNY 32GB Elite Class 10 U1 microSD	Data Storage	\$6.99	0.5g	N/A

The chosen components amount to \$86.23, and weight approximately 113.4g. Some components are estimated based on similar PCB weights. These masses can be confirmed once components are ordered.

6.4.2.1 Microcontroller

For the onboard computer, a microcontroller will be used for both payload and telemetry data. No computation will be performed by the satellite, only data acquisition and delivery. Basic commands will be sent to and received from other electronic components to perform thermal system commands, read and write data, and receive commands from the ground station. The chosen microcontroller is the Arduino Giga R1.



6.4.2.1.1 Arduino Giga R1 [CC&D1].

Specifications

- STM32H747XIH6 Microcontroller
 - Dual-core
 - 32-bit Arm® Cortex®-M7 core with double-precision FPU and L1 cache up to 480 MHz
 - 32-bit Arm® 32-bit Cortex®-M4 core with FPU up to 240 MHz
 - Full set of DSP instructions
 - Memory Protection Unit (MPU)
- Murata® 1DX Wi-Fi®/Bluetooth® Module
 - Wi-Fi® 802.11b/g/n 65 Mbps
 - Bluetooth® Low Energy (version 5.X via Cordio stack, version 4.2 via Arduino Stack)
 - Micro UFL connector for external antenna
- Memory
 - STM32H747XI
 - 2 MB Flash
 - 1 MB RAM
 - AT25SF128A-MHB-T
 - 16 MB NOR Flash
 - QSPI Interface
 - AS4C4M16SA
 - 8 MB SDRAM
- I/O
 - Digital I/O Pins: 76
 - Analog input pins: 12

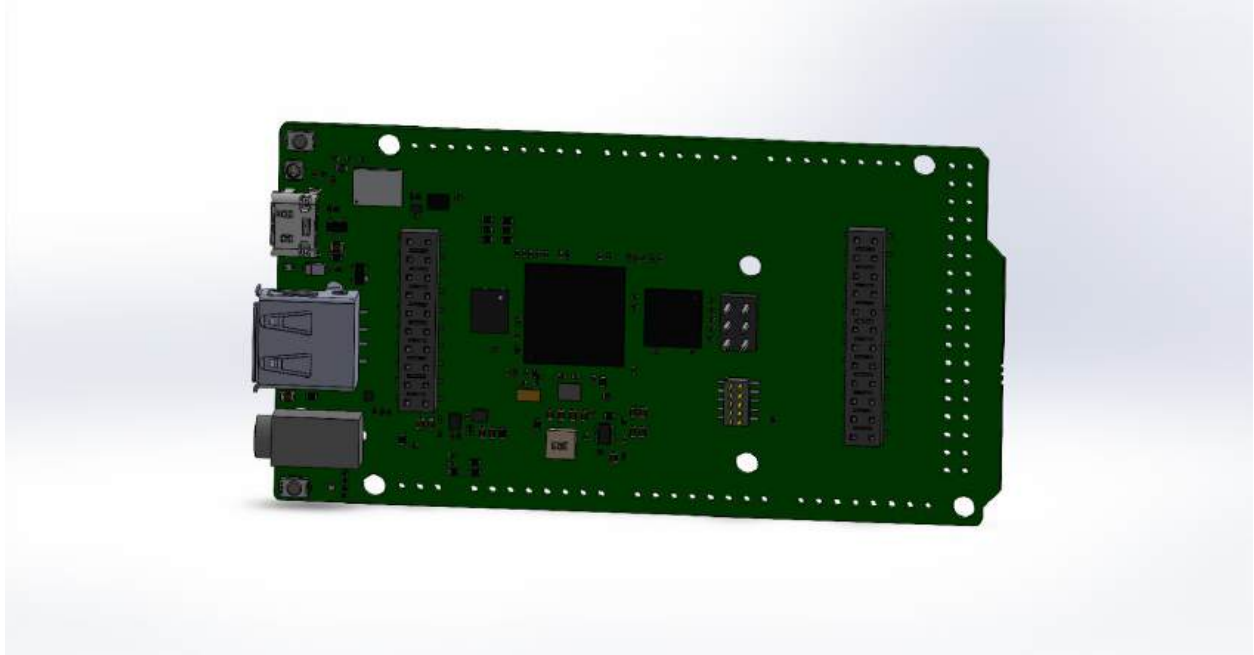
- PWM pins: 13
- Analog output pins (DAC0/DAC1): 2
- USB Host: USB 2.0 A
- USB Peripheral: USB-C®
- Logic level: 3.3V
- VRTC: To power the RTC while the board is off
- OFF pin: To turn off the board
- Communication
 - 4x UART
 - 3x I2C
 - 2x SPI
 - 1x CAN (an external transceiver is required)
- Secure Element ATECC608A-MAHDA-T Module
- USB
 - USB Host USB 2.0 Type A
 - Host
 - USB Peripheral USB-C®
 - Programming Port
 - HID
- Connectors
 - Camera: 20 pin Arducam camera connector
 - Display: D1N, D0N, D1P, D0P, CKN, CKP, D68-D75
 - Audio jack: DAC0, DAC1, A7
 - JTAG connector
- Power
 - Circuit operating voltage: 3.3V
 - Input voltage (VIN): 6-24V
 - DC Current per I/O Pin: 8 mA

Role in Prototype

- Deliver commands to electronic components.
- Interpret signals from components
- Communicate with ground station via radio
- Deliver commands from ground station
- Collects and store SAR data
- Read data from micro SD and send to ground station

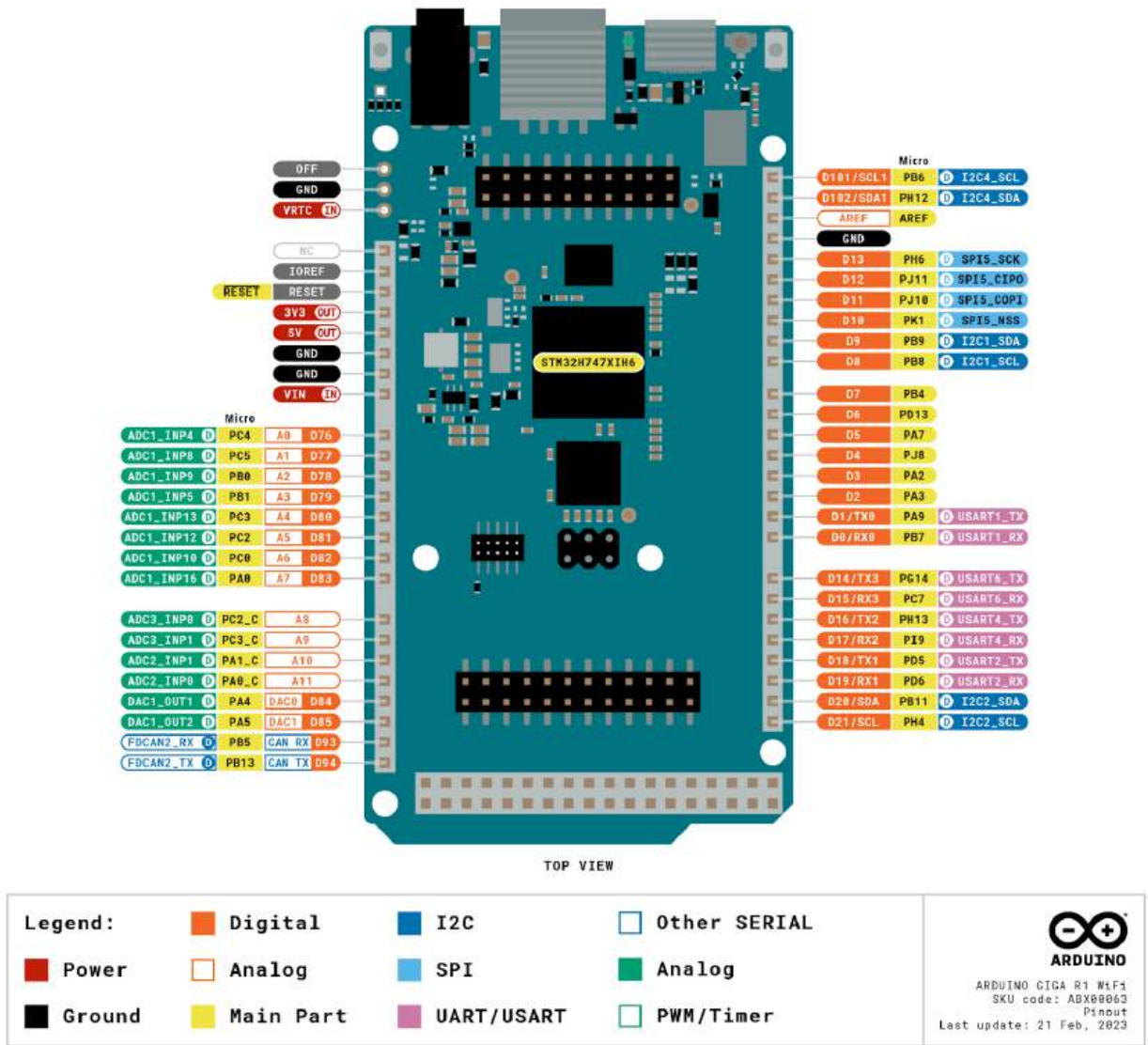
Justification

The Arduino Giga R1 is the preferred microcontroller for this project mainly due to its high number of I/O pins. The final circuit will call for at least 15 5V connections, 1 I2C, 13 digital signals, and 3 SPI connections. It wouldn't be possible to use only one microcontroller if the classing Arduino Uno was used. In addition, the Giga R1 allows for far more flexibility when wiring the completed design. There are plenty of pins to support connections to the 3 servos, camera, 4 photosensors, radio, thermal interface, micro SD interface, relay, and 3 DC motors. The main motivator for this microcontroller selection is the I/O versatility.



6.4.2.1.2 Arduino Giga CAD model[CC&D1].

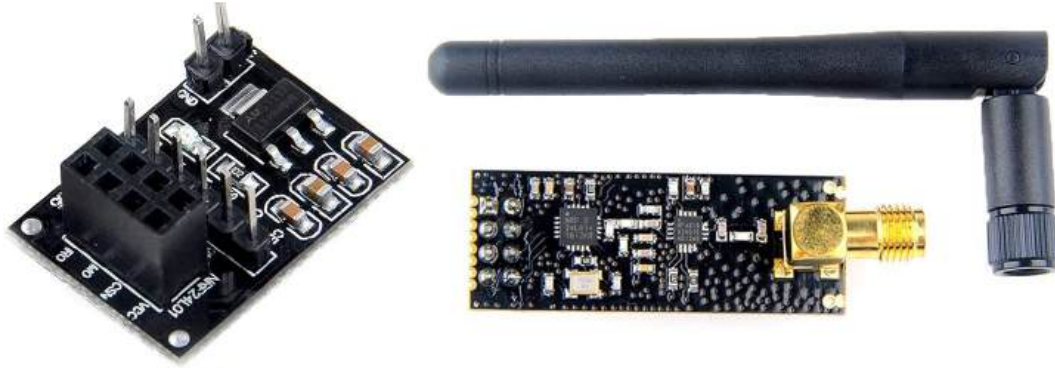
While this model is capable of WIFI communication, it is not an accurate model of the data transmission protocol that the SMARTSat will follow. Instead, a radio communication device will be installed on the I2C, SPI, and power pins to send and receive data from the ground station. The following image shows the extensive pinout diagram for the Giga R1, not including the 5V bus located at the bottom of the board. \



6.4.2.1.3 Arduino Giga pin out [CC&D1].

6.4.2.2 S Band Communications

S band communication will be used in place of a two channel communication. This requires that SAR data be obtained from the satellite at the same time as telemetry data. While this does limit the system to a degree, it also ensures that data can be passed at a fast enough rate during orbit. The chosen S band communication device is the NRF24L01.



6.4.2.2.1 NRF24L01 + Voltage Conversion [CC&D2]

Specifications

NRF24L01+PA+LNA Wireless Transceiver RF Transceiver Module:

- PA Gain: 20dB
- LNA Gain: 10dB
- Frequency: 2.4GHz~2.5GHz
- Range: 800+ meters line of sight
- Multi-frequency: 125 frequency
- Transmit Operating Current(Max): 115mA
- Receiving Operating Current(Max): 45mA
- Board dimensions: 40.7mm x 15.5mm x 12.2mm
- Operating Voltage: 3 ~ 3.6V (Recommended 3.3V)
- Maximum power consumption of 0.3795W

NRF24L01+ Breakout Adapter:

- This is a simple socket board which is for NRF24L01 Wireless module
- Small power on SMD LED indicator
- Voltage Regulator Input Voltage (connected to VCC): 4.8V to 12V
- Voltage Regulator Input Voltage Max: 15V (we recommend staying below 12V)
- Pin Quantity: 8Pin
- On-board: AMS1117-3.3 chip
- Compatible: Arduino and Raspberry Pi

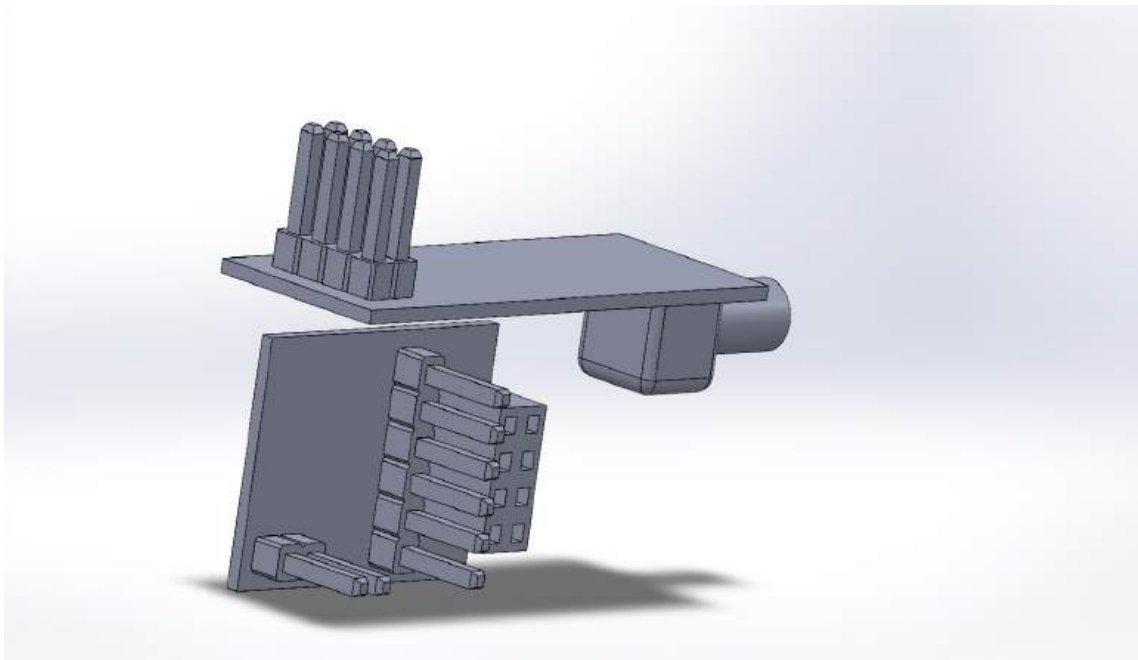
Role in Prototype

- Transmit data to the ground station at 2.4 GHz
- Transmit payload data as well as telemetry data
- Convert 5V to 3.3V

Justification

The NRF module provides an S Band connection that allows for high speed data transmission from 800+ meters. This will help simulate the working model and the interaction between the communication protocol and data acquisition. One important note about this module

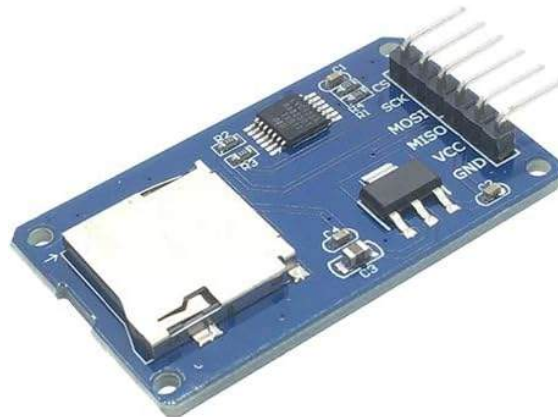
is that it requires a 3.3V supply. This is provided using the included breakout board which will convert the 5V signal to the operating voltage of 3.3V. Below is a Cad model of the communications system for the prototype design.



6.4.2.2.2 NRF24L01 + Voltage Conversion CAD model

6.4.2.3 Data Storage

The data storage capabilities of the full scale design is not difficult to replicate with the prototype design. For this solution, a standard 32 Gb microSD card is chosen in place of the on board flash memory used with CubeSat OBCs. Compared to the 68Gb total flash memory on the full scale design, this model falls on the same order of magnitude.



6.4.2.3.1 Micro SD Interface [CC&D3][CC&D4]

Specifications

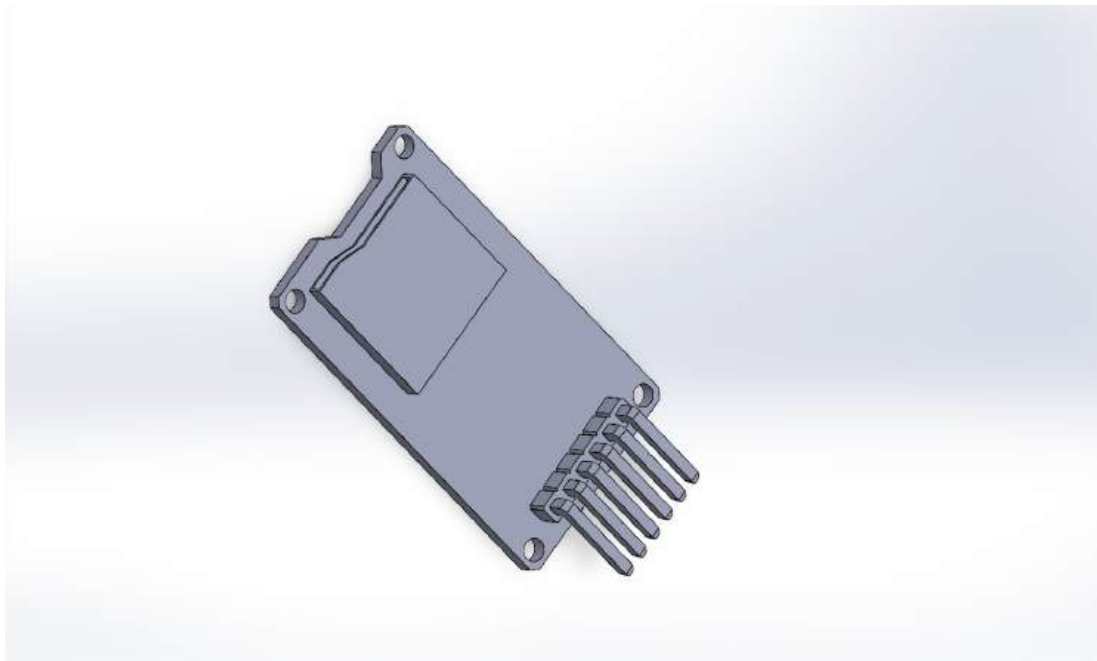
- 3.3V
- SPI interface
- 32 Gb storage

Role in Prototype

- Interface between SD and Arduino
- Store data when transmission is not possible
- Clear data when information has been sent to ground station
- Partition data packets to optimize transmission

Justification

The Micro SD reader interfaces well with an Arduino microcontroller using the SPI pins located on the motherboard. This should not interfere with other chosen SPI devices such as the camera and thermal sensor. The micro SD reader also allows for high versatility in the storage limit for the model design, as a larger micro SD can be used in place of the current 32Gb storage solution. Exact numbers for the transmitter are not known, so it is unclear at this moment what the data transmission speed is with this interface included. However, since the Arduino Giga R1 only holds 4Mb of on board data, this storage solution is crucial for the design. SAR data acquisition cannot be limited by the onboard storage. Below is a CAD model of the MicroSD interface.

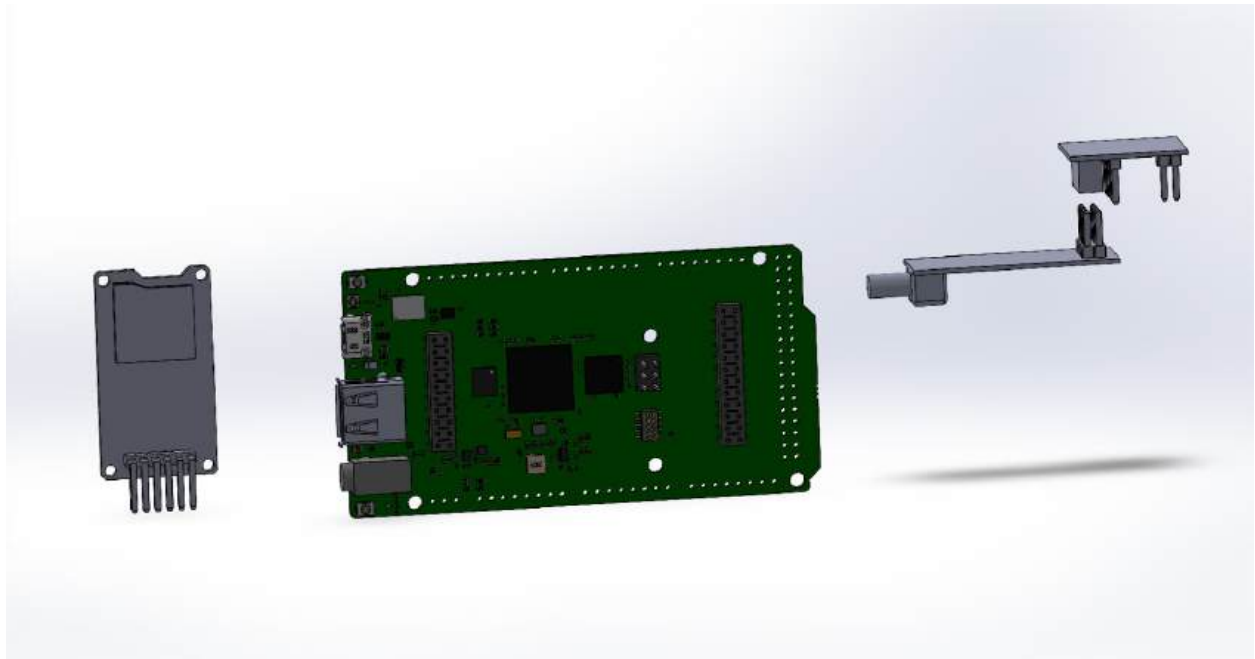


6.4.2.3.2 Micro SD Interface CAD model

6.4.3 Communication and Data Handling Exploded View

The image below is the exploded view of the full command and data handling system. The wires here are not modeled, as such an orientation is not useful in a CAD model such as this.

Instead, these connections will be represented in the circuit design. This concludes the command and data handling design solution for the prototype.



6.4.3.1 Command and Data Handling Exploded View.

6.5 Ground Station Design Solution

The final communication solution for the satellite will leverage NASA's Near Earth Network (NEN) due to its proven capabilities, robust infrastructure, and compatibility with CubeSat missions. However, the NEN implementation is beyond the scope of this two-semester project, given the complexity of integration, scheduling, and hardware access requirements. To model the NEN on a small scale, the following prototype will be constructed and tested.

6.5.1 Prototype Summary

A simplified transceiver prototype has been developed to simulate and validate the CubeSat's communication capabilities with a ground station. It is intended to model the behavior of a ground station that is capable of uplinks and downlinks. This prototype employs a UHF-band transceiver operating at 2.4 GHz, supporting line-of-sight (LOS) communication for a range of up to 800 meters. While limited in range, it offers a practical solution for testing critical subsystems and data handling pipelines. The prototype ground station will be tested and provide a proof of concept for the implementation of a ground station network such as the NEN.

6.5.2 Prototype Solution

The prototype consists of a modular design that emphasizes functionality, ease of assembly, and system compatibility. It interfaces with the CubeSat's onboard data handling

system, transmitting mock telemetry and raw SAR data as a proof of concept. The following table lists the components used in the prototype design:

Table 6.5.2.1 Parts List

	Actual Part	Price	Weight	Dimensions
NRF24L01+PA+LNA Wireless RF Transceiver Module	Transmitter/Receiver/Transceiver	\$ 13.99	43.94176 g	12.0904 x 11.811 x 2.3114 cm
HiLetgo ESP-WROOM-32 ESP32 ESP-32D	Central Processor	\$ 9.99	28.3495 g	9.6012 x 5.9944 x 1.8034 cm
PNY 32GB Elite Class 10 U1 microSDHC Flash Memory Card	Data Storage System	\$ 6.99	9.071847 g	1.4986 x 1.0922 cm
Micro SD TF Card Adapter Reader Module	Data Storage System	\$ 6.99	49.89516 g	UNAVAILABLE
Breadboard Power Module	Portable Power Supply	\$ 10.99	10.886217 g	UNAVAILABLE
Adafruit Perma-Proto Breadboard	Circuit Board	\$ 4.50	9.07185 g	8.13 x 5.08 x 0.160 cm
Laptop/Desktop	Data Handling & Conversion	\$ 0.00	N/A	N/A

It is important to mention, this subsystem will not be launched or on the satellite itself. Dimensions are generally irrelevant as the ground station is on Earth. However, the following specifications of the chosen hardware is incredibly relevant to the communication and processing capabilities of the prototype ground station and satellite.

6.5.2.1 Transceiver & Adapter with Antenna

The MakerFocus NRF24L01+ Wireless Transceiver with Adapter and Antenna is a compact, low-cost radio-frequency (RF) module that provides the wireless communication backbone for the ground station prototype. Its long-range, low-power characteristics make it an excellent choice for simulating uplink and downlink communication with the CubeSat. With high gain amplifiers and a breakout adapter, it ensures reliable communication for the ground station

prototype while supporting flexible integration with microcontrollers like the ESP32. The following provides a visual for the selected combination. [GS12]

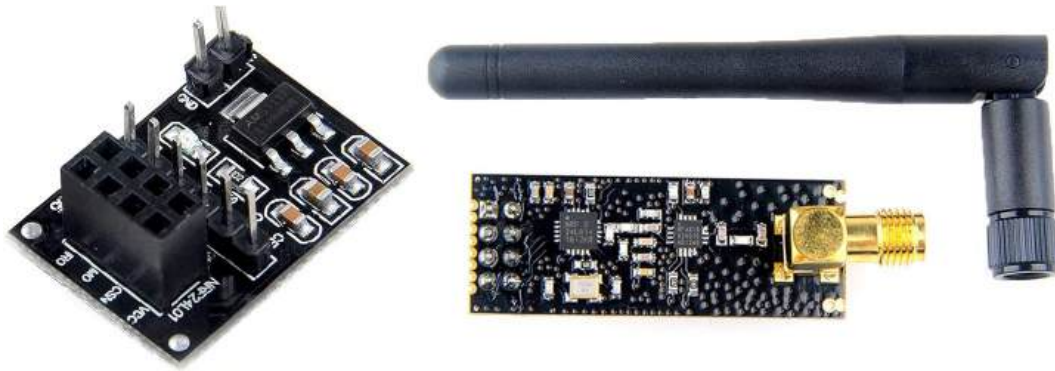


Figure 6.5.2.1.1: Transceiver & Adapter with Antenna Image

Specifications:

NRF24L01+PA+LNA Module:

- Frequency Range: 2.4 GHz to 2.5 GHz ISM band
- PA Gain (Power Amplifier): 20 dB
- LNA Gain (Low Noise Amplifier): 10 dB
- Multi-Channel Support: 125 selectable frequencies
- Operating Voltage: 3.0V to 3.6V (recommended 3.3V)
- Power Consumption:
 - Transmit Current: Max 115 mA
 - Receive Current: Max 45 mA
- Line-of-Sight Range: 800+ meters
- Dimensions: 40.7 mm x 15.5 mm x 12.2 mm

NRF24L01+ Breakout Adapter:

- Voltage Regulator: Onboard AMS1117-3.3 chip to support a wide range of input voltages
- Input Voltage Range: 4.8V to 12V (recommended <12V)
- Max Input Voltage: 15V
- Compatibility: Works seamlessly with Arduino, Raspberry Pi, and other 3.3V/5V microcontrollers
- Additional Features:
 - Simple socket board for NRF24L01 module
 - SMD LED indicator for power status
 - Pin Count: 8 pins

Role in Prototype:

- Long-Range Communication: Supports uplink and downlink for mock telemetry and SAR data up to 800 meters, making it ideal for validating ground station communication with CubeSat models.
- Power Efficiency: With efficient power usage, the module is well-suited for prolonged testing while simulating real-world communication scenarios.
- Multi-Frequency Capability: Supports 125 frequency channels, allowing flexibility in testing interference handling and multi-channel communication.

Justification for Selection

The PA+LNA configuration of the transceiver module ensures proven reliability by enhancing transmission range and maintaining signal clarity, even in noisy environments. Its flexible design supports integration with various microcontrollers and voltage levels, making it a versatile choice for prototyping. At just \$13.99 for a comprehensive kit that includes three modules, antennas, and adapters, it offers a cost-effective solution for development and testing. Additionally, its 800-meter line-of-sight range ensures practical performance, enabling the prototype to handle real-world communication scenarios even within constrained test environments.

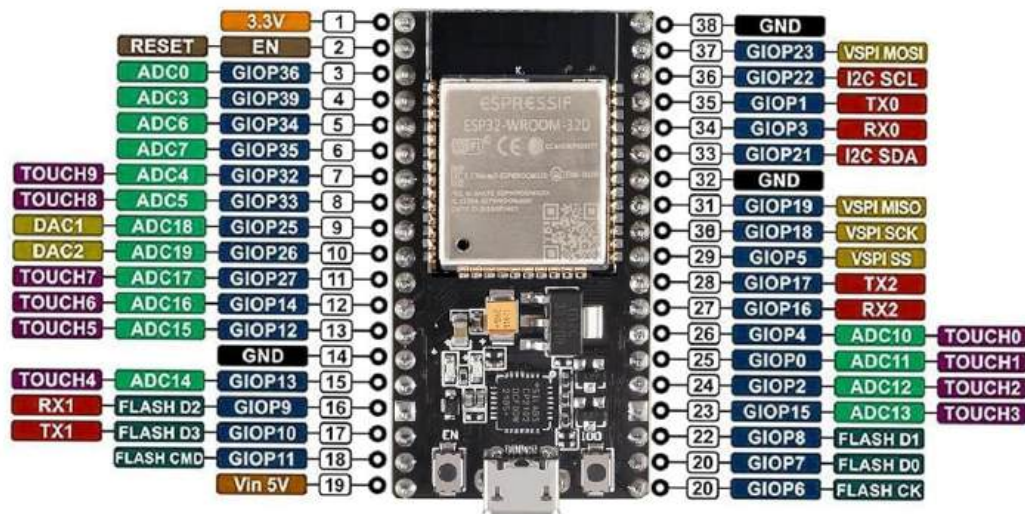
6.5.2.2 Microcontroller

The ESP32 Development Board is selected for its exceptional performance and versatility. This microcontroller integrates Wi-Fi and Bluetooth capabilities, which, although not utilized in this prototype, future-proof the design for extended testing scenarios. The board is compact, affordable, and supports the processing and interfacing needs of the ground station prototype. The following provides a visualization of the microcontroller. [GS13]



Figure 6.5.2.2.1: Microcontroller Image

The following provides the pin blueprint to allow for correct and efficient implementation/connection with other hardware components.



Justification for Selection

The ESP32 microcontroller serves as the core of the ground station prototype, offering a reliable balance between functionality and simplicity. Its processing power enables real-time handling of telemetry and mock SAR data, making it integral to the communication system. The ESP32 is a cost-effective solution that balances affordability with functionality. Its modular design ensures compatibility with the current prototype while allowing scalability for future iterations, including NEN simulation. This combination of features makes the ESP32 an ideal choice for modeling critical aspects of satellite communication in a controlled and budget-friendly manner.

6.5.2.3 Micro SD Card & Adapter

The PNY Elite 32GB microSD Card and HiLetgo microSD Card Adapter Module form the data storage subsystem of the ground station prototype. Together, they provide efficient and reliable data logging and retrieval capabilities, enabling seamless handling of telemetry and mock SAR data from the CubeSat. The following provides a visualization of the combination. [GS14][GS15]



Figure 6.5.2.3.1: Micro SD Card & Adapter Image

Specifications

PNY Elite 32GB microSD Card:

- Storage Capacity: 32GB (ample for extended testing scenarios).
- Class Rating: Class 10 UHS-I for read speeds up to 100 MB/s and fast data transfers.
- File System: Compatible with FAT32 or exFAT for microcontroller compatibility.
- Durability: Resistant to magnet, shock, temperature, and water, making it ideal for robust use cases.

HiLetgo microSD Card Adapter Module:

- Interface: SPI (Serial Peripheral Interface), compatible with most microcontrollers, including the ESP32.

- **Voltage Compatibility:** Onboard voltage regulator supports both 5V and 3.3V systems.

Role in Prototype

- **Data Logging:** The microSD card is used to store raw telemetry data and mock SAR packets received from the satellite. This ensures that all communication exchanges are logged for subsequent analysis.
- **Command Uplink Preparation:** Preloaded uplink commands are stored on the microSD card and read by the ESP32 during the uplink process, enabling repeatable and testable communication workflows.
- **Interface with Microcontroller:** The HiLetgo adapter bridges the microSD card with the ESP32 microcontroller via SPI, enabling fast and reliable data transfers. The onboard voltage regulator ensures compatibility between 5V microcontroller boards and the 3.3V microSD card.
- **Data Retrieval:** The adapter allows easy connection to external systems, such as laptops or desktops, for transferring test data and logs for analysis.

Justification for Selection

The storage subsystem combines high-speed data handling, modular design, robustness, and cost-effectiveness to meet the project's needs. The Class 10 UHS-I PNY Elite microSD card ensures fast write speeds, making it ideal for logging large volumes of telemetry and SAR data in real time. Paired with the HiLetgo adapter, the setup offers seamless integration with the ESP32 through plug-and-play functionality, while accommodating different voltage levels for versatile compatibility. The PNY Elite card's durability ensures reliable performance under various environmental conditions, making it suitable for extended testing. Additionally, the subsystem delivers exceptional value at a low price, providing high performance and reliability without exceeding budget constraints.

6.5.2.4 Adafruit Perma-Proto Board

The Adafruit Perma-Proto Board serves as a robust and semi-permanent solution for circuit prototyping, offering a stable foundation for integrating various electronic components. Its design mimics a traditional breadboard, allowing for an easy transition from prototyping to a more durable setup.

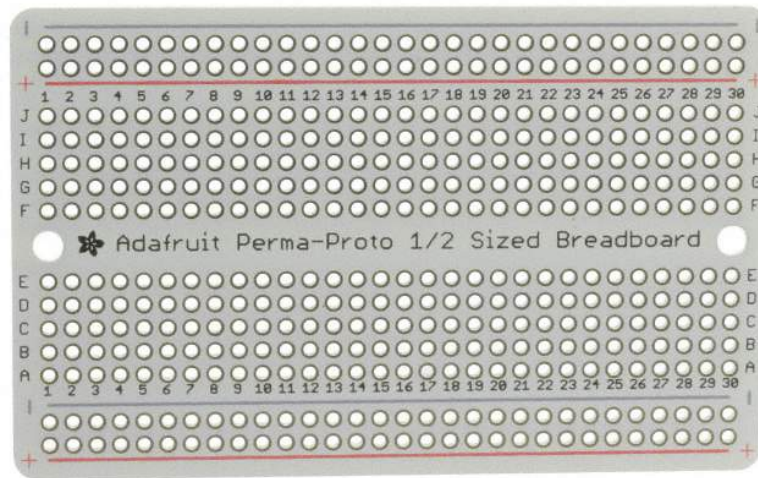


Figure 6.5.2.4.1: Adafruit Perma-Proto Breadboard

Specifications

- Board Type: Half-sized Perma-Proto Breadboard
- Dimensions: 7.5 cm x 5 cm (2.9" x 1.9")
- Material: High-quality FR4 PCB with gold-plated finish
- Connections: 30 rows of 5-hole strips with power rails on both sides
- Hole Plating: Through-hole for robust and secure soldering
- Silkscreen Labels: Grid labels for easy component placement and circuit organization
- Mounting: Four mounting holes for easy integration into enclosures or larger projects

Role in Prototype

- Circuit Integration: The Adafruit Perma-Proto Board allows for a more durable and semi-permanent attachment of components, enhancing the portability and structural integrity of the circuit.
- Housing Design Reference: The rigid, structured PCB provides a stable form factor that facilitates precise enclosure design, ensuring long-term protection and portability of the prototype.
- Reliable Connections: The plated-through holes and gold finish ensure reliable electrical connections, reducing signal interference and improving circuit durability over extended use.
- Power Distribution: Integrated power rails simplify power routing across the circuit, enabling efficient energy management for different subsystems.

Justification for Selection

The Adafruit Perma-Proto Board offers an ideal balance between modular prototyping and long-term circuit stability. Its layout mirrors a traditional solderless breadboard, allowing seamless transition from initial prototyping to a more permanent setup without redesigning the circuit layout. The use of high-quality FR4 material and gold-plated contacts ensures durability

and reliability, making it well-suited for repeated testing and extended field use. Additionally, the pre-labeled grid enhances organization, reducing errors and simplifying troubleshooting during integration with other subsystems. The board's compact size and mounting holes further enhance its suitability for embedded applications, making it a cost-effective and practical solution for the ground station prototype.

6.5.2.5 Breadboard Power Supply Module

The Breadboard Power Supply Module provides a convenient and efficient way to power prototyping circuits by delivering stable 3.3V and 5V outputs. Its integration into the breadboard simplifies circuit development while ensuring reliable power distribution to critical components.

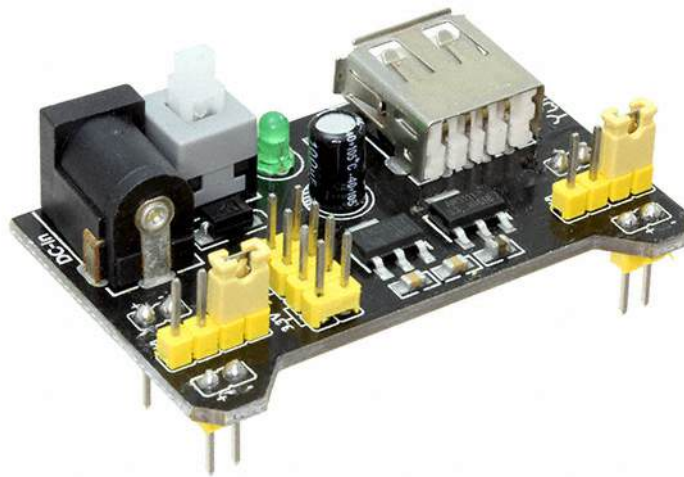


Figure 6.5.2.4.1: Breadboard Power Supply Module

Specifications

- Input Voltage: 6.5V - 12V DC (via barrel jack) or 5V via USB
- Output Voltage: 3.3V and 5V selectable output rails
- Current Output: Maximum 700mA per rail
- Compatibility: Standard breadboard-compatible pin spacing
- Onboard Switches: Independent 3.3V/5V selection for each power rail
- LED Indicator: Power status indicator for easy troubleshooting

Role in Prototype

- Power Distribution: Supplies regulated 3.3V and 5V power to components such as sensors, microcontrollers, and communication modules.
- Modular Integration: Plugs directly into the breadboard, eliminating complex wiring and improving reliability.
- Dual Voltage Support: Ensures compatibility with both low-power 3.3V devices and higher-power 5V components.

- **Overvoltage Protection:** Safeguards sensitive electronic components from damage due to voltage fluctuations.

Justification for Selection

The Breadboard Power Supply Module provides a stable and efficient way to power the ground station prototype's electronic components. Its ability to deliver both 3.3V and 5V ensures compatibility with a wide range of subsystems, eliminating the need for multiple external power adapters. The onboard voltage selection switches enhance flexibility, allowing developers to adjust power requirements dynamically. The module's integration into the breadboard simplifies circuit design, reduces clutter, and improves system reliability. The inclusion of a power status LED aids in quick diagnostics, making it a practical and cost-effective power solution for the prototype.

6.6 Thermal Design Solution

The actual design of the thermal system could be a couple of different solutions. A recommendation for if we get chosen for the contest design, the final design should involve thermal coating that prevents heat transfer by having a low emissivity and low solar absorptivity along with thermal louvers that will passively control heat through bimetallic springs and a survival heater if the CubeSat gets too cold. We could use this for our testing model while we have a prototype that will be much simpler. According to a paper from Utah State University, a prototype was made that cost over \$2000 that doesn't include testing, but only getting the material to build it. Therefore, a simpler way of thermal control is presented.

Moving forward due to the difficulty and prototyping the thermal louvers, the design solution will be a heater, thermostat, and thermal coating where the emissivity and be changed to fit the CubeSats need in further simulation and testing

6.6.1 Prototype Summary

A simplified thermal control system has been developed to simulate and validate the CubeSat's thermal capabilities and survivability. It is intended to model the CubeSat in the shadow of the Earth to see if it will be able to add enough heat to survive since the cooling of the CubeSat from sunlight to shadow is the challenge. This prototype employs a polyimide thermofoil heater and a thermocouple. A thermal coating will not be present on the prototype due to cost restraints.

6.6.2 Prototype Summary

The thermal system emphasizes functionality, ease of implementation, and system compatibility. The thermal system will interface with the main processing unit in order to determine what actions to take regarding heat transfer which is mostly telling the heater to turn on or off. The following table list the components need to complete this prototype:

Table 6.6.2. Parts List

	Actual Part	Price	Weight	Dimensions
All-Polyimide Thermofoil Heaters	Space Grade Heater	\$104	0.080 Oz	0.50 in x 2 in x 0.010 in
Thermocouple and K Type Temperature Sensor	Temperature Sensor	\$18	0.01 Oz	15 mm x 28 mm
Thermal Louver	Aluminum flaps with bimetallic springs that change with temperature	\$20	1 Oz	30 cm x 20 cm
Cooling Fan	This wouldn't be part of actual design	\$7.99	1.13 Oz	1.18 x 0.3 x 1.18 inches

6.6.2.1 All-Polyimide Thermofoil Heaters

These heaters are thin, lightweight that allows you to apply heat where its needed to reduce overall cost. These heaters are low mass and factory laminated to save space and cycle time. These heaters are suitable for vacuum environments, so if this testing is available, we can simulate the heater in space.

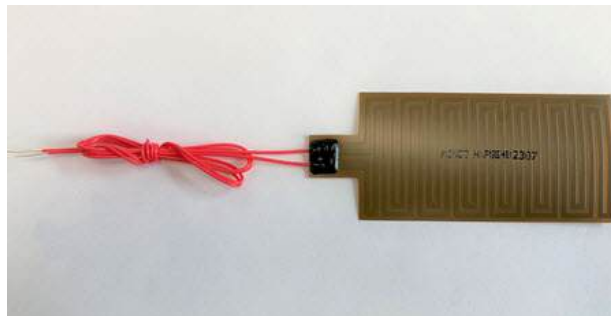


Figure 6.6.2.1: Thermofoil Heater

Specifications

- Mounting: Pressure Sensitive Adhesive
- Temperature Range: -32°C to 100°C
- Resistance: 1011.6 Ω
- Voltage: 0.1 to 160 V
- Wattage: 25.6 Max
- Watt Density: 33 W/in²

Justification for Selection

The heater provides a decent heat contribution and can have a range of values depending on how much wattage we want to send to the heater. The heater is also the cheapest option I could find that we could use to prototype the actual heater we would use. The heater also takes up very little mass and space along with being easily mounted through adhesion.

6.6.2.2 Thermostat and K-Type Temperature Sensor

The thermostat allows for temperature reading that should connect to our arduino that tells it the current temperature reading within the CubesSat then the arduino determines whether to turn the heater on or off depending on the desired operating temperature range.



Figure 6.6.2.2: Thermocouple and K-Type Temperature Sensor

Specifications

- Operating Range: 3.0 - 5.5 V
- Operating Current: 50mA
- Operating Temperature: -20°C to 85°C
- Accuracy: Error within 1.5°C
- Set Screw: M6

Justification for Selection

The thermostat is relatively cheap with the power specifications being within acceptable ranges to not take up all power from other subsystems. The operating temperature range is also expectable for testing in cold cases in our CubeSat

6.6.2.3 Thermal Louver

The thermal louver is part of the final design solution that would help passively release heat by changing the emissivity of the two radiator surfaces on the CubeSat. However, this is a prototype, so it will be more of a model for how it would look and function, but not be able to do its task since it would be very expensive and complicated mechanically.



Figure 6.6.2.3: Thermocouple and K-Type Temperature Sensor

Justification for Selection

The thermal louver may not be functional, but we still not to atleast model it to show how it would work. There isn't any specification because this is a 3D printed part. It will be mounted by using strong adhesive or creating drill holes.

6.6.2.4 Cooling Fan

The cooling fan has been added as part of the prototype due to concerns from the power lead of components overheating because of the difference between space and Earth heat transfer characteristics. The heat transfer characteristic is convection.



Figure 6.6.2.4: Cooling Fan

Specifications

- Operating Voltage: 5V
- Power Connector: 2-pin
- RPM: 5000 RPM

Justification for Selection

The cooling fans are relatively cheap and will resolve concerns with convection heating that needs to be added to the design solution due to this additional heating.

6.6.3 CAD Models

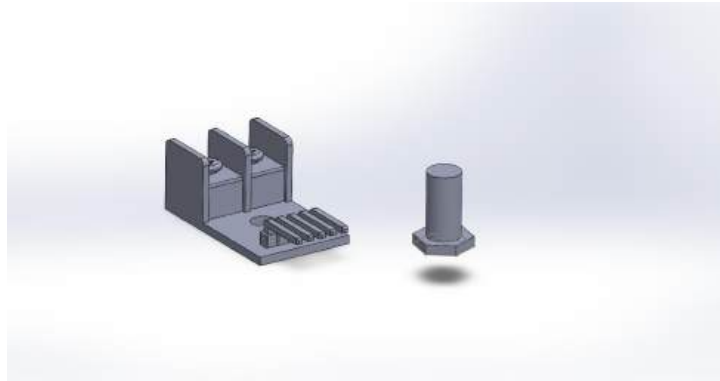


Figure 6.6.3.1 Thermocouple and K-Type Temperature Sensor

The CAD model for the heater is a thin flat sheet, so the CAD drawing will be provided

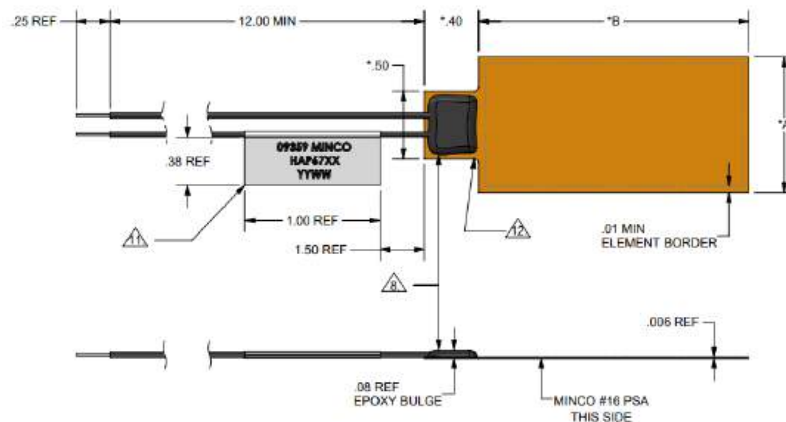


Figure 6.6.3.2: CAD Drawing of Heater Element

6.8 Propulsion Design Solution

The actual design for the propulsion system is far beyond the budget of the senior design class. As such, a smaller and less efficient version of the thruster will be settled on for testing. A real CubeSat cold gas thruster will be optimized for space, which is something the prototype will also fall short in. The prototype solution will aim to match an ISP of 100 and a thrust of 1 N as closely as possible.

6.8.1 Prototype Summary

In order to make a small cold gas thruster for testing, small parts are required, which immediately removes most potential options for fuel tanks such as airsoft air tanks. Instead, CO₂ cartridges will be used. This is a different gas than expected for the actual CubeSat, but should function similarly to the real gas. The CO₂ cartridge threading cannot be directly connected to

many store bought parts. A bike regulator is chosen to down regulate the pressure. This also allows the CO2 cartridge to screw into something. The bike regulator will connect directly into a Schrader valve. This valve will be secured firmly using an adhesive after being sealed with plumbers tape. The Schrader valve will convert to a ½” MPNT threading via a set of bushings. The ½” MNPT thread pipes will screw into a solenoid valve which will prevent flow when not needed. The other end of the solenoid valve will connect directly into a 3D printed nozzle, which will accelerate flow to a max mach number possible. This nozzle will be designed for atmospheric use, and will most likely change after testing back pressure to be a more accurate value. All parts used will be rated for at least 160 psi, which is the current estimated back pressure. An emphasis is placed on choosing higher rated parts to be safe. To ensure strong connection between strange part fittings, heavy metal repair glue will be used.

6.8.2 Prototype Solution

The prototype propulsion system has some design concessions made for functionality. The parts used are hard to convert to a usable thread size, and as such permanent sealing solutions must be used. The parts themselves will also likely only work on a small timescale, due to the regulator needed. The nozzle does not fit into the chart because its dimensions and weight are flexible depending on needs.

Table 6.8.2.1 Parts List

	Actual Part	Price	Weight	Dimensions
CO2 Cartridge	Fuel tank	\$15.99	16 g	111.76 mm x 88.9 mm x 88.9 mm
Keg Regulator (0-60 psi)	Space grade pressure regulator	\$21.49	50 g	79.502 mm x 60.452 mm x 63.5 mm
Half Union 7/16"-20UNF Flare Male to ¼” FNPT	Part of piping system	\$11.99	48.76118 g	105.918 mm x 100.076 mm x 22.098 mm
½” MNPT to ¼” MNPT	Part of piping system	\$12.49	181.437 g	58.42 mm x 35.56 mm x 12.7 mm
NPT Solenoid Valve	Space grade solenoid valve	\$29.95	680.389 g	89.916 mm x 59.944 mm x 115.062 mm
Male to Male NPT Converter	Part of piping system	\$11.21	75 g	12.7 mm x 12.7 mm x 50.8 mm
3D Printed Nozzle	Space Grade Nozzle	\$0.00	N/A	N/A

Plumbers Tape	Sealant	\$3.25	N/A	N/A
Metal Repair Glue	Sealant	\$15.88	N/A	N/A

Nozzle design is tentative, and subject to changes when needed. Weight is negligible. Current estimates for size are similar to solenoid.

6.8.2.1 CO2 Cartridge

The brand of CO2 cartridge is an unimportant decision. Instead, the canister must be inexpensive and about 900 psi. “16g Threaded CO2 Cartridges - for All CO2 Bike Tire Inflators with Threaded Connection - Quick Air Refill for Bicycle Tires - Cartridge for CO2 Pump - Road or MTB Bikes.” are chosen from Amazon.



Figure 6.8.2.1.1 CO2 Cartridge [PROP23]

Specifications

16g Threaded CO2 Cartridges - for All CO2 Bike Tire Inflators with Threaded Connection - Quick Air Refill for Bicycle Tires - Cartridge for CO2 Pump - Road or MTB Bikes.:

- 900 Psi (roughly)

- 16 g
- 5 per container
- Threaded 3/8"-24 TPI

Role in Prototype

- Functions as pressurized fuel tank
- Required for thrust and ISP

Justification for selection

This set of CO2 cartridges comes in the correct threading to convert to NPT via the schrader valve. This is extremely important. It is cheap compared to other options, and it comes with 5 canisters, which will be good for testing where multiple cartridges will be needed since it isn't possible to refill.

6.8.2.2 Keg CO2 Regulator

The keg regulator is used to bring the higher 900 psi back pressure down to a lower 0-60 psi. This is needed in order to ensure parts won't break. The pressure must be low enough that material failure won't occur during testing.



Figure 6.8.2.2.1 Keg CO2 Regulator [PROP24]

Specifications

**Keg Charger Dispenser, 0-60 PSI Keep Stable Pressure Cornelius Corny Beer Growler
Outdoor Picnic Party Carbonation Portable Mini CO2 Regulator 7/16-20 UNF Brass
Adapter Gas Ball Lock Disconnect :**

- Made for 16 g CO2 cartridges
- Connects to male 7/16-20 UNF fitting
- Regulates to 0-60 psi

Role In Prototype

- Functions as regulator
- Unideal, but essentially only working cheap solution that fits
- Low pressure
- Guarantees initial pressure will be low

Justification for Selection

While this option has major flaws, competing options like airsoft canisters and scuba tanks are too large. It will work for systems testing, which is the main need of the system. It will also accelerate flow to at least mach 1.

6.8.2.3 Half Union 7/16"-20UNF Flare Male to 1/4" FNPT

This fitting is needed to convert the less common UNF fitting type to a NPT type which fits the solenoid. Ideally a direct Threaded 7/16"-20UNF to MNPT 1/2" fitting would be used, but such items are not on the market.

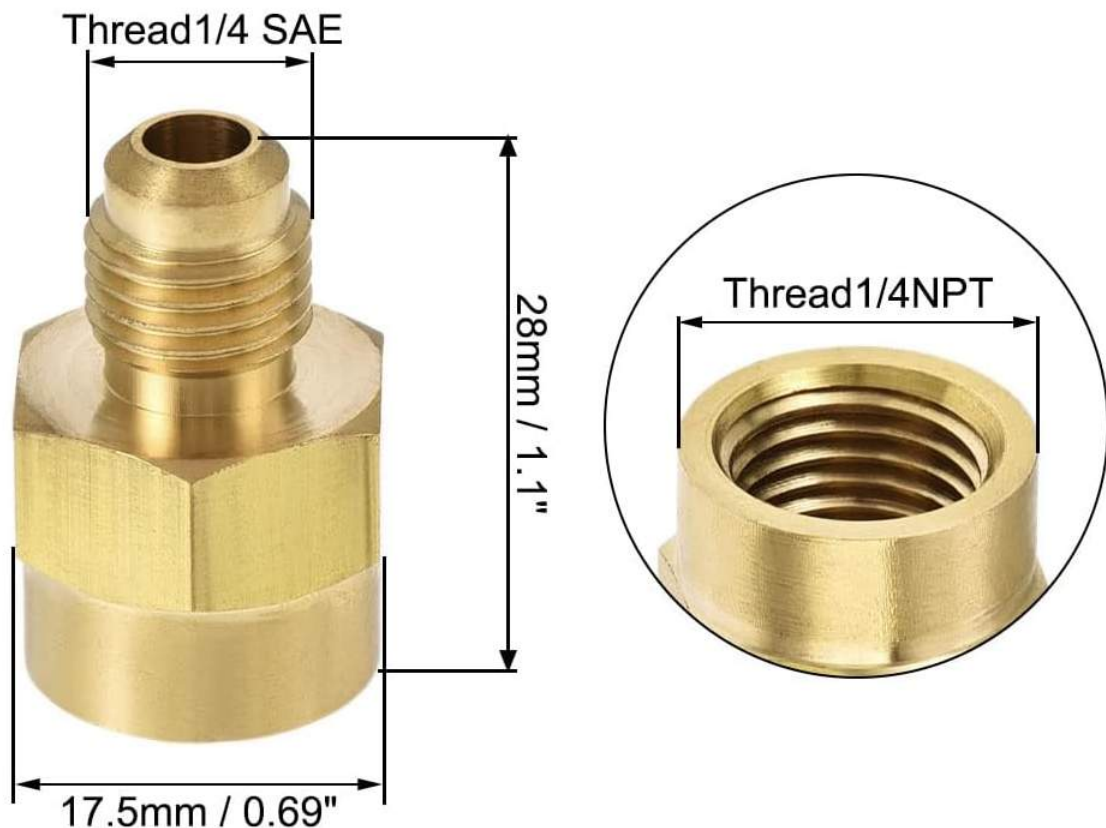


Figure 6.8.2.3.1 Half Union 7/16"-20UNF Flare Male to 1/4" FNPT [PROP25]

Specifications

uxcell Brass Pipe fitting, 7/16"-20UNF Flare Male to 1/4NPT Female Thread, Tubing Adapter Hose Connector, for Air Conditioner Refrigeration, 2Pcs :

- 7/16"-20UNF Flare Male to 1/4" NPT Female Thread

Role In Prototype

- Serves to change pipe to NPT thread.

Justification for Selection

This fitting is needed to convert from the keg regulator to a standard threading that a solenoid could be found for. Very little options exist for this transition.

6.8.2.4 1/2" MNPT to 1/4" MNPT

This fitting is needed to convert the smaller 1/4" MNPT thread size to 1/2" MNPT thread size needed for the solenoid.

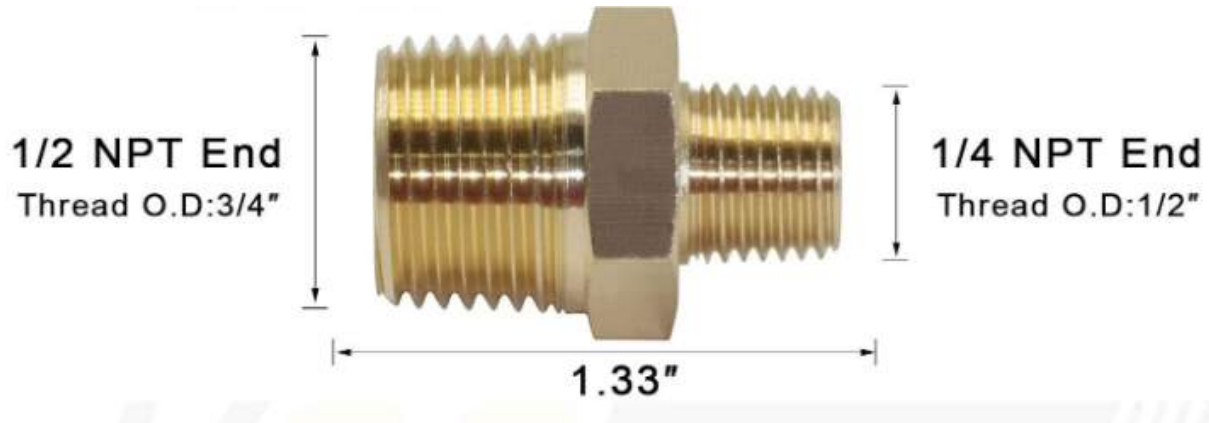


Figure 6.8.2.4.1 1/2" MNPT to 1/4" MNPT [PROP 26]

Specifications

KOOTANS 1/2 NPT Male x 1/4 NPT Male Solid Brass Nipples, Heavy Brass Pipe Adapter Fittings Hex Reducing Nipples Connectors 4Piecess :

- 1/2" MNPT x 1/4" MNPT

Role In Prototype

- Serves to change thread size to fit with solenoid chosen.

Justification for Selection

This fitting is needed to convert from the keg regulator to a standard threading that a solenoid could be found for. Very little options exist for this transition.

6.8.2.5 NPT Solenoid Valve

The solenoid valve ensures that flow will not exit the nozzle until thrust is needed. For the prototype, this will mainly be for demonstration, due to the pressure buildup that should occur due to the pressure regulator.



Figure 6.8.2.5.1 NPT Solenoid Valve [PROP27]

Specifications

1/2" NPT Brass Electric Solenoid Valve 12V DC Normally Closed VITON from U.S. SOLID :

- 12V DC
- Brass
- 1/2" FNPT to 1/2" FNPT
- 0-7 bar

Role In Prototype

- Controls flow exhaust

Justification for Selection

Other solenoids were looked at, but the easiest threading to convert to was NPT. The solenoid is rated for pressure ranges needed for operation, and DC current is needed (which ruled out a competing solenoid on McMaster).

6.8.2.6 Male to Male NPT Converter

A pipe is needed to bring flow into a nozzle. A MNPT to MNPT fitting is chosen for its ability to connect to the solenoid.



Figure 6.8.2.6.1 Male to Male NPT Converter [PROP28]

Specifications

- ½" MNPT to ½" MNPT
- Rated for 7200 psi

Role In Prototype

- Piping for propellant

Justification for Selection

This part is cheap and effective for its purpose. It has a much higher rated pressure than needed, so it should perform well. NPT threading is needed for connections.

6.8.2.7 3D Printed Nozzle

The nozzle is used to accelerate the cold gas using pressure. The higher the velocity at the exit, the higher the ISP, which is good for space thrusters.

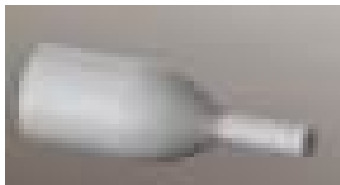


Figure 6.8.2.7.1 3D Printed Nozzle

Specifications

- $P_c = 60$ psi
- $P_{atm} = 0$ (in space) or 1 bar (at SSL)
- PLA plastic
- Shape can change depending on the actual pressure measurements taken

Role In Prototype

- Accelerates flow to generate thrust and improve ISP
- Gives estimate for actual thruster performance

Justification for Selection

The 3D printed nozzle was chosen for ease of construction and flexibility. Changes can be made very quickly if needed. It is also free at Hunt Library, which makes it great for this prototype.

6.8.2.8 Plumbers Tape

Plumber tape is used to seal gaps in between threading. It otherwise serves no purpose but is very important to actual performance. If gaps were present, pressure in the feed lines would drop, reducing ISP.



Figure 6.8.2.8.1 Plumbers Tape [PROP29]

Specifications

- 1/2" wide, 520" long

Role In Prototype

- Fills gaps between threading to maintain pressure

Justification for Selecting

This particular tape option was cheap online. It comes with a large amount of tape, and should be capable of filling all gaps in the threading. It is also necessary to ensure performance.

6.8.2.9 Metal Repair Glue

Metal repair glue serves to completely seal gaps in parts that need reinforcement. The mix is made for metal repair (as the name suggests), and has metal paste in it for extra strength.



Figure 6.8.2.9.1 Metal Repair Glue [PROP30]

Specifications

- Temperature range of at least 32 deg F to 212 deg F
- Impact resistant

Role In Prototype

- Serves to reinforce and seal gaps in connected parts
- Last barrier to prevent failure of system in case of leaks

Justification of Selection

Other options are available for metal glue, but impact resistance was chosen specifically due to quick pressure changes present in the system. This option also proves to have a large temperature range which could prove important to its function. Some failsafe sealant is needed to prevent failure in case of a time crunch

6.8.3 CAD Models

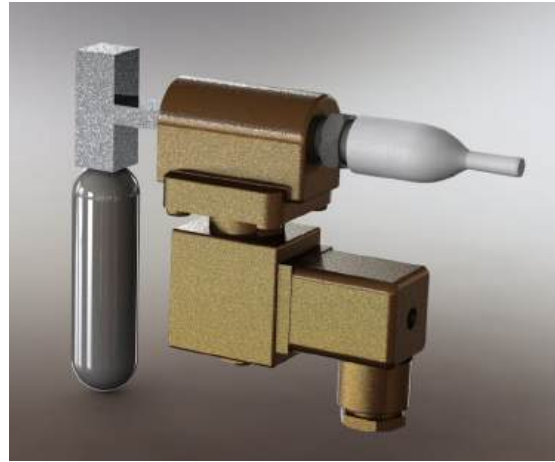


Figure 6.8.3.1 Condensed and Exploded View Propulsion CAD Models

In the exploded view, 1 pictures the CO2 cartridge, 2 pictures the regulator, 3 pictures the solenoid, 4 pictures the thread conversion fittings to turn the regulator threading to NPT, 5 pictures the MNPT to MNPT fitting, and 6 pictures the 3D printed nozzle.

6.9 Structures Design Solution

The real design for the frame of the cubesat will cost around \$13,800 to manufacture [Struct 4], it will be made from aluminum alloy 6061. The real design will be coated in a hard anodized finish to increase the durability of the framework and ensure that it can handle the extreme temperatures. It will also feature the rail deployment method that was discussed earlier in the document. Additionally, the framework will be held together by screws that are also coated

in a hard anodized finish. Unfortunately, the manufacturer, *Endurosat*, does not provide the specifications on what type of screws were used to hold the frame together.

6.9.1 Prototype Summary

With the allocated \$2,000 budget for the prototype, the manufacturing of the frame will cost \$274.81, which consists of buying a 15x15” sheet of aluminum alloy 6061 that has a thickness of ¼”. Furthermore, 6 10x17” sheets of aluminum alloy 6061 with a thickness of ⅛” will be purchased and function as the side panels of the cubesat. Additionally, 4 packs of 316 stainless steel hex drive flat head screws and 1 pack of 18-8 stainless steel hex nuts will be the remaining \$45.38 of the total purchase. As stated above, the prototype design will not have a hard anodized finish on the frame because adding that coating will cost a lot more money than the allocated prototype budget allows. The total dimensions of the frame will be 20x20x30 cm with a thickness of ¼”.

6.9.2 Prototype Solution

The dimensions of the frame are fixed, however, the water jet machine in the senior design shop can be used to cut/trim the aluminum sheet to whatever dimensions are required for the cubesat frame.

Table 6.9.2.1 Parts List

	Actual Part	Price	Weight	Dimensions
Frame Material [Struct 15]	Aluminum alloy 6061	\$39.99	5.39 lbs	(15x15”) & ¼” thickness
Side Panel Material [Struct 16]	Aluminum alloy 6061	\$21.99 (per sheet)	2.03 lbs	(10x17”) & ⅛” thickness
Nuts [Struct 17]	18-8 stainless steel hex nut	\$6.14 (per pack of 100)	Not listed	M2x0.4 mm thread size, 4mm in width,
Screws [Struct 8]	316 stainless steel hex drive flat head screw	\$9.81 (per pack of 25)	Not listed	M2 thread size, 0.4 thread pitch 5mm length

6.9.2.1 Aluminum Alloy 6061

This alloy will serve as the selected material for the skeletal frame of the S.M.A.R.T sat and will have a mass around 2.43 kg, which is subject to variation, depending on the chosen thickness of the frame.

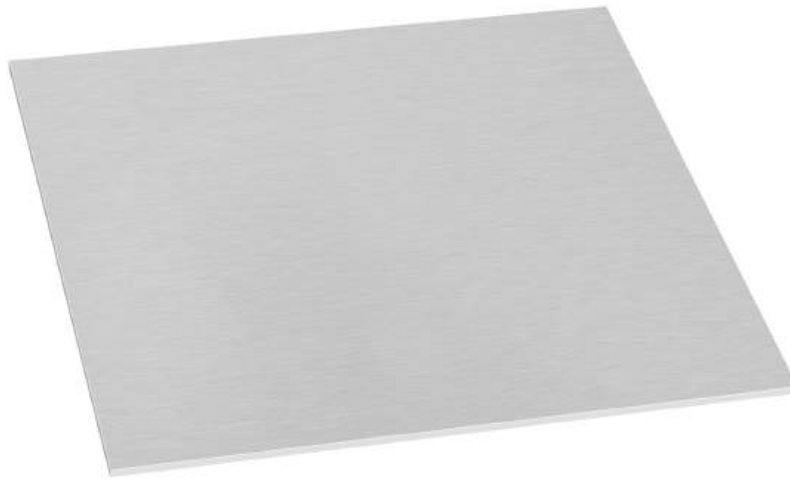


Figure 6.9.2.1.1: Selected aluminum alloy 6061 sheet with (15x15 in) & ¼” thick [Struct 15]

Specifications

- 15x15” & ¼” thick
- 5.39 lb

Justification for selection

This selected alloy provides a robust and durable design and is significantly lighter than aluminum 7075. Furthermore, adjusting this alloy’s thickness can increase its strength and is very easy to work with when making design alterations, specifically when it’s in the sheet form.

6.9.2.2 Screws and hex nut

The chosen screw and hex nut combination will serve as the method for securing components from the different subsystems onto the inside of the cubesat side exoskeleton and side panels. The frame in its entirety will be welded together to ensure maximum robustness and rigidity.



Figure 6.9.2.2.1: 316 stainless steel hex drive flat head screw (5mm length) [Struct 8].

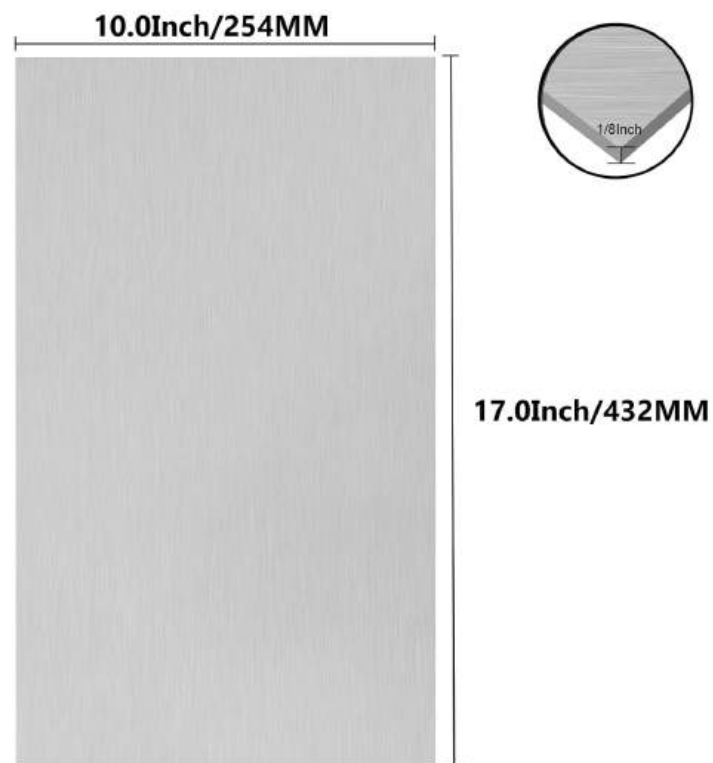


Figure 6.9.2.2.2: Aluminum alloy 6061 sheet [Struct 16]



Figure 6.9.2.2.3: 18-8 Stainless steel hex nut [Struct 17]

Specifications

- M2 thread size, 0.4mm thread pitch, 5mm length (316 stainless steel hex drive flat head screw)
- M2x0.4 mm thread size, 4mm in width (18-8 stainless steel hex nut)

Justification for selection

The selected screws and hex nut fit the design parameters for the S.M.A.R.T sat frame, which is why they were the materials selected for utilization in the construction process. Additionally, the prices for the screws and nuts are fairly cheap compared to the other options, which is another reason why these materials were chosen.

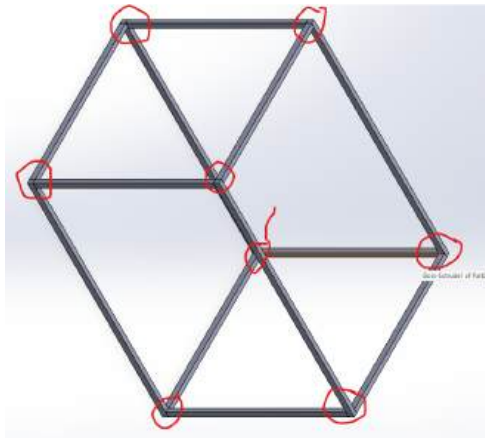


Figure 6.9.2.2.4: CAD model of cubosat exoskeleton with location of welds

7. Vehicle Layout

7.1 Propulsion

7.1.1 System Features

The system is composed of a pressure tank, regulator, solenoid, nozzle, and tubes to connect these. A set of wires attached to the solenoid are used to supply voltage and current, and actuate the structure. Pressure builds behind the solenoid, which is released in bursts and accelerates to mach 1. This allows the ISP to be higher, which is more important than thrust in orbit. 2 pipe elbow fittings allow the system to fit into the CubeSat.

7.1.2 Properties

- Nozzle: Accelerates 60 psi flow to mach 1.
- Solenoid: 12 V, 1.4 A, 15 W.
- Keg Regulator: Takes 900 psi air to 0-60 psi.
- CO2 Canister: Single use 900 psi tank.

7.1.3 Critical Elements

The critical elements are listed below.

- Solenoid: Needed to control when thrust is supplied.
- Nozzle: Needed to improve ISP
- Keg Regulator: Needed to down regulate pressure to be usable with other parts.

7.1.4 Image



Figure 7.1.4: Propulsion System

7.2 Payload

7.2.1 System Features

The payload consists of four major components. Two are breadboard/PCB circuits, the modulator and the video amplifier circuits, one is the RF circuit, and the last are the antennas themselves. The circuits will receive power and reference voltages from the onboard power supply, the system as a whole is expected to draw less than one Watt during operation. Planned full system testing will have the payload continuously feeding data to the onboard controller when activated. The data collected will be sent to the ground station wirelessly through the communications subsystem where it will then be processed into a .wav file that can be viewed and assessed. The theoretical maximum range of the system is roughly 1 km. Some manufacturing and procurement issues may reduce this during initial testing. Currently, all non-RF circuits are wired through a breadboard. After successful testing of the circuits, they will be soldered into a perma-proto board. Integration into the CubeSat will be through bolts connected to the various RF components, antennas, and hot glue to affix the circuit board.

7.2.2 Properties

- RF Circuit: Output power ~50 mW to the antennas, antenna efficiency ~90% in the designed frequency range.
- Antennas: Design frequency 2.4-2.5 GHz ISM band, 10 cm in diameter cans, bulkhead connector using SMA standard connectors, monopole wire made of aluminum, soon to be upgraded to copper.
- Weight: ~300 grams not including mounting hardware.

7.2.3 Critical Elements

- Modulator Circuit: Takes a small amount of input power and a reference voltage to generate the 20 microsecond voltage up-ramp and 20 microsecond voltage down-ramp supplied to the VCO.
- Video Amplifier Circuit: Takes input readings from the RF circuit and processes them into readable audio data for the ground station.
- RF Circuit: Generates the RF signal chirp needed for radar function and mixes the received signal with the sent signal to provide parsable data to the ground station.
- Antennas: Identical antennas that function as the transmitter and receiver of the SAR.

7.2.4 Images

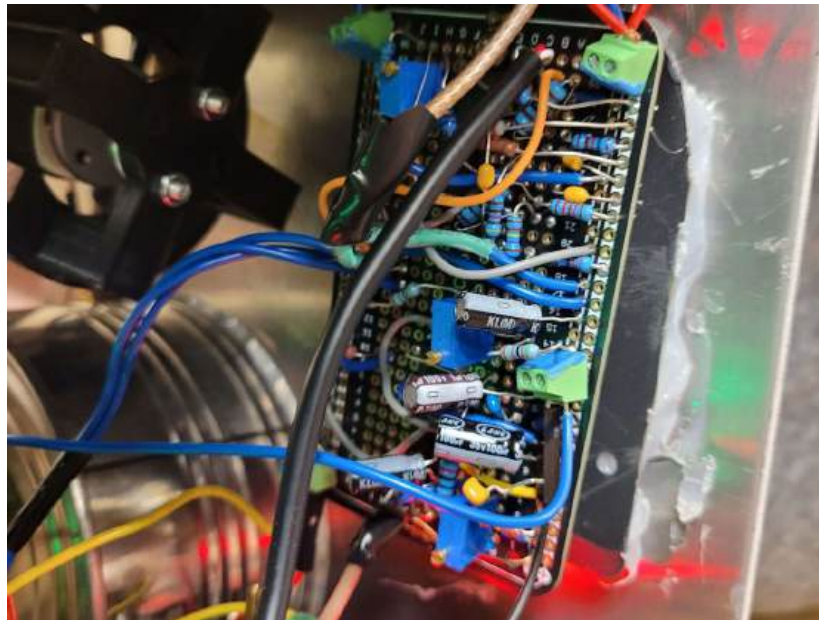


Figure 7.2.4.1: Completed Modulator Circuit (left) and Completed Video Amplifier Circuit (right), Both in a Through-Hole Solder Board.



Figure 7.2.4.2: Fabricated Cantenna with Monopole Wire Installed.

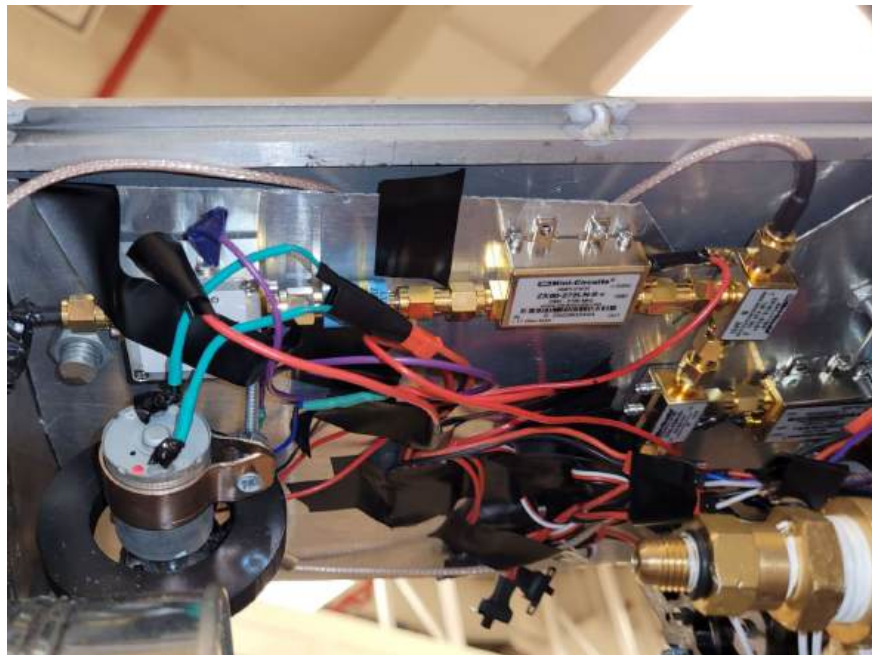


Figure 7.2.4.3: Assembled RF Circuit.

7.3 Ground Station

7.3.1 System Features

The ground station is a communication setup designed for seamless interaction with remote devices through a microcontroller. Initially, the design assumed the microcontroller could power both the SPI devices. However, this assumption proved incorrect. To address this

limitation, a breadboard power supply module was integrated into the system, providing the necessary power to both devices. Additionally, the system was transferred to a perma-proto board to enhance durability and ensure permanence, which was not fully considered in the original design solution. Further integration details are discussed in the manufacturing section. The total weight of the raw system elements is approximately 151.22 grams. The housing has yet to be printed and has not been included in this weight. Issues are assumed to arise from the housing design and manufacturing process.

7.3.2 Properties

- **Power Supply:** A dedicated breadboard power supply module was added to the design to ensure stable power delivery to both devices.
- **Durability:** Transitioned to a perma-proto board for permanent assembly, ensuring greater resilience and reducing the likelihood of failure due to temporary connections.
- **Weight:** The entire system setup weighs 151.22 grams, contributing to the portability and practicality of the ground station for its intended applications.

7.3.3 Critical Elements

- **Microcontroller:** Powers the communication interface and coordinates between the ground station and remote systems.
- **SPI Devices:** Interact with the microcontroller to facilitate data transfer and communication.
- **Power Module:** Ensures sufficient and stable power supply to all components, addressing the initial design oversight.
- **Perma-Proto Board:** Provides a reliable and permanent solution for housing the system components, ensuring long-term stability.

7.3.4 Images

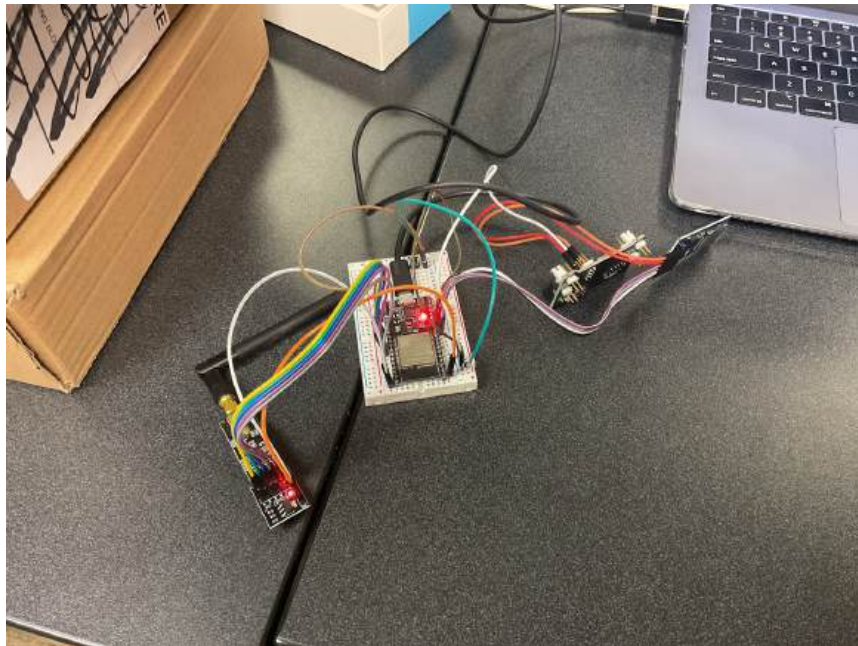


Figure 7.3.4.1: Fully Integrated Raw Ground Station

7.4 GNC

7.4.1 System Features

The guidance, navigation, and controls subsystem is responsible for the attitude control in the CubeSat. This system is composed of the arduino, the motors, some speed controllers, light sensors, and the accelerometer. The light sensors will provide digital outputs to the arduino alerting if there is a light source sensed. The accelerometer will be passively measuring the gravity vector being applied to the satellite, being sent as an analog signal to the arduino. The arduino will then need to interface with the electronic speed controller to provide voltage to the motors. This is an important step as the motor current pulls should not be running through the arduino itself.

The changes from the original baseline design include adding ESC's, ordering new mounting hubs, and switching the Earth pointing sensor to an accelerometer. The baseline design solution had an Earth pointing sensor prototyped as an arduino mini camera. During production, the camera was manufactured to output live images to a file. The image files would then need to undergo an advanced image processing to be useful in the SMART Sat design. This processing would require a lot of computing which is not capable on the arduino giga. The only other option for this system to be feasible would be to do the computing in the ground station and send results to the GNC system. Our design team decided to make the control system completely autonomous to minimize the possible failure modes. This is when the switch to using an accelerometer was

made, this turned out to be very useful as it works well with the arduino. The other main change to this section is adding 3 electronic speed controllers for the motors. This is for protection against over-voltage of the arduino.

7.4.2 Properties

- **Power Supply:** A dedicated breadboard connected directly to the battery power supply module was added to the design to ensure stable power delivery to all devices.
- **Autonomy:** With the power connected, the GNC system will read and react to light sources/no light sources without any reliance on other systems. There will be an option to switch the mode it is currently set to via ground station control for proof of concept in later designs.
- **Weight/Size:** The entire system setup weighs about 1000 grams as a rough estimate. The motors and flywheels will be the parts that take up the most space, estimating about 7.3 cm length with a diameter of 2.7 cm for the final reaction wheel combination. There will be three of these and will be positioned off the three axes.

7.4.3 Critical Elements

- **Microcontroller:** Powers the communication interface and coordinates between the sensors and speed controllers.
- **Motor Circuits:** Interact with the microcontroller to facilitate correct power magnitude to each motor via directional inputs taken from sensors.
- **Photosensitive Resistors:** Multiple resistors acting as high/low outputs to light sensitivity. The sensors can be calibrated to react to higher and lower light via the screw on the back.
- **Accelerometer:** Single sensor that measures Earth's gravitational vector in three axes. Analog outputs allow for linearly mapping motor speed to how far away the satellite is to pointing directly at Earth.

7.4.4 Images

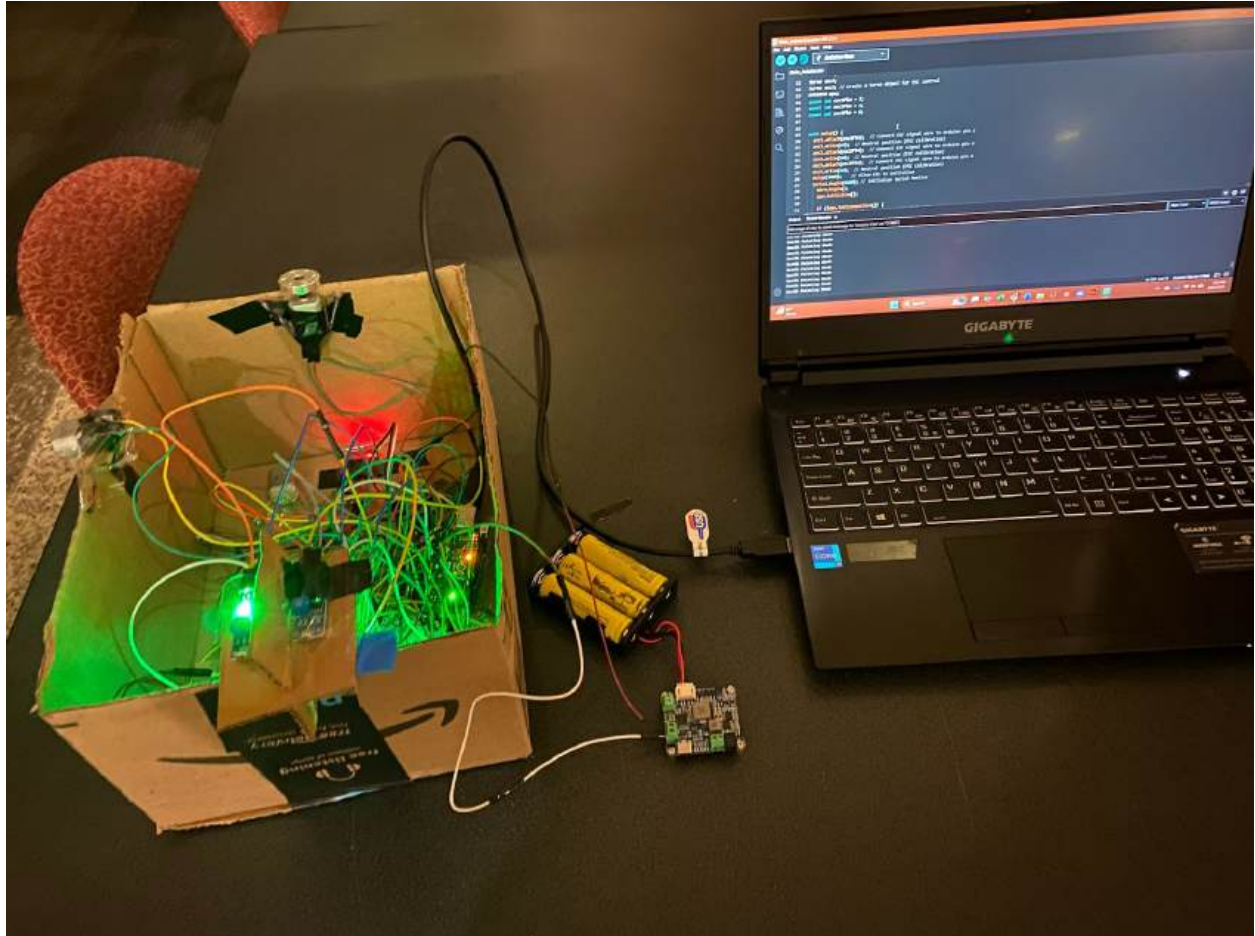


Figure 7.4.4.1: Reaction Wheel with both Sensors Setup, outside of frame

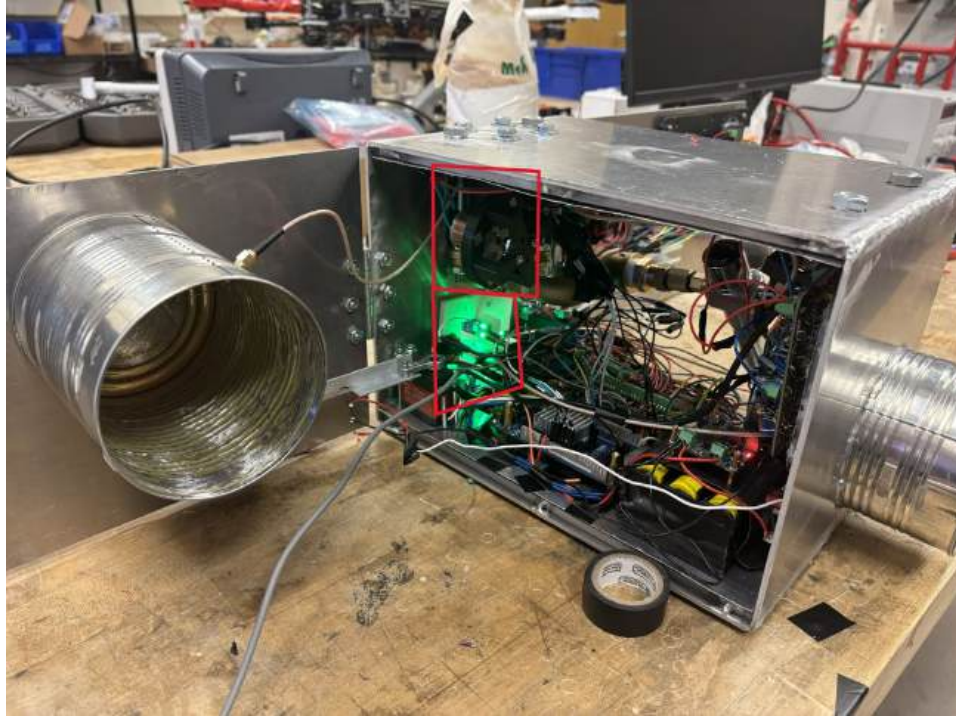


Figure 7.4.4.2: Reaction Wheel with both Sensors Setup, inside of frame

7.6 Thermal

7.6.1 System Features

The main system is composed of a heater and thermocouple. Beside these two main components, the system is connected to a relay, arduino, and wires. There is also a new addition of cooling fans as suggested by the power lead that would help with convection cooling since we're on Earth and that is a big part of the problem. Compared to the baseline design, there are now cooling fans to help with convection heating. The system works by having a thermocouple connected to a relay that detects temperature until it reaches a certain threshold then tells the heater to turn on.

7.6.2 Properties

- Power supply: A 12V battery will be connected to the subsystem
- Weight: The subsystem together will weigh about 3 Oz.
- Heat Flux: 10W
- Heater: Resistance 1011 Ohms, V: 0.1 to 160, Wattage: 25.6 Max

7.6.3 Critical Elements

- Heater: Provides heating element to the system to prevent freezing
- Thermocouple: Reads to temperature of the component to determine the action of turning the heater on or off

- Cooling Fan: Provides cooling effect to prevent overheating of components
- Relay: Connects the thermocouple to the arduino to determine temperature then determine heat on or off.

7.6.4 Images



Figure 7.6.4.1: Thermocouple Relay Setup

7.7 Structures

7.7.1 System features

The exoskeletal frame of the cubesat is composed of a total of twelve aluminum alloy 6061 struts that all have a thickness of $\frac{1}{4}$ " and width of $\frac{1}{5}$ ". The only varying dimension these struts have is the length. Four of these struts are 30 cm long (11.81") and the rest are 20 cm long (7.87"). Additionally, side panels made from aluminum alloy 6061 with a thickness of $\frac{1}{8}$ " will be placed on the exterior of the frame. The four side panels will be 20x30 cm and the front and back panels will be 20x20 cm. Compared to the baseline design there will now be an interior skeleton inside the frame that will be attached to the insides of the panels via standoff screws. The interior skeleton will provide a surface for the components of other subsystems to attach to without compromising the structural integrity of the frame. The attachments will be made using nuts/bolts.

7.7.2 Properties

- Material: Aluminum alloy 6061
- Frame Thickness: $\frac{1}{4}$ "
- Side panel thickness: $\frac{1}{8}$ "

- Interior skeleton thickness: <<< side panel thickness
- Weight: 2.4 kg (estimation)

7.7.3 Critical elements

- Welds provide connection to strut members maintaining shape and rigidity of frame without deformation
- Interior skeleton provides mounting surface for components of other subsystems
- Side panels shield interior components from outside environment

7.7.4 Images

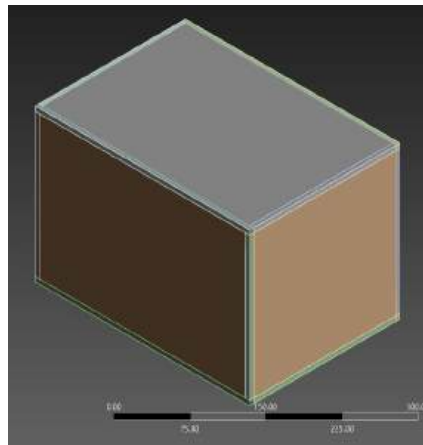


Figure 7.7.4.1: Ansys model of cubesat frame with side panels



Figure 7.7.4.2: Shape of frame with struts (*frame is currently being welded so no images of final product are available yet*)



Figure 7.7.4.3: 30 cm struts (left) and 20 cm struts (right) for exoskeleton of frame

7.8 Power

7.8.1 System Features

The power delivery system consists of two solar panels, 6 lithium ion batteries, an MPPT board, power and signal delivery boards, 5V - 12V transformer, and two high power relays. The panels are made from 24 total 5v 80mA mini solar panels. The MPPT board, power and signal delivery boards serve to deliver 5V power delivery from the battery to each electronic component as well as a 5V signal from the arduino to the components that require it. The relay and solenoid valve require the most power, and thus use a high power relay to deliver a 12V signal from the batteries to the components. The 12V signal is obtained from the 5 - 12V step up transformer.

7.8.2 Properties

- Power Supply: The entire system will be charged by the sun.
- High power delivery: The 12V rail connected to relays will meet the power and voltage requirements for the solenoid and heater pad.
- Durability: To ensure the components receive a steady input as well as remain secured during operation, most components will be directly soldered to protoboards.
- Weight: The entire system setup weighs up to 2.3kg grams with the 6 x lithium ion battery configuration.

7.8.3 Critical Elements

- Power Supply: 6 x 18650 lithium ion batteries. 3.7V and 5000mAh.
- Panels: 24 5v 80mA mini solar panels. 10-14V shop light voltage, 22-24V sunlit voltage
- MPPT Board: Circuit protection, battery protection, maximum power point tracking, power delivery, 5V regulation, USBc out, wire screw interface, input voltage regulation.
- High power delivery: The 12V rail connected to relays will meet the power and voltage requirements for the solenoid and heater pad.
- Deployment: Two 20kg servos will mount to the interior and move the panels outward.
- Protoboards: 3+ protoboards will have components soldered to them to ensure stable and reliable connections. These will take the place of printed circuit boards for power delivery.

7.8.4 Images

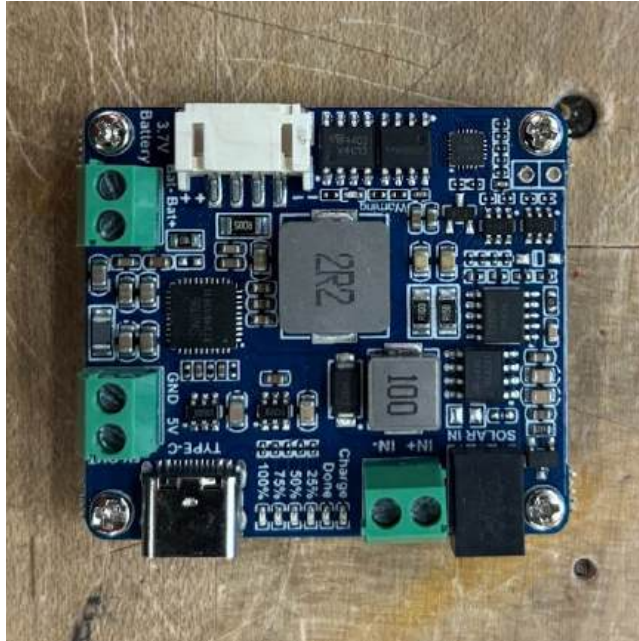


Figure 7.8.4.1: MPPT Board



Figure 7.8.4.2: 5-12V Transformer



Figure 7.8.4.3: Power system components (missing new transformer and protoboards)

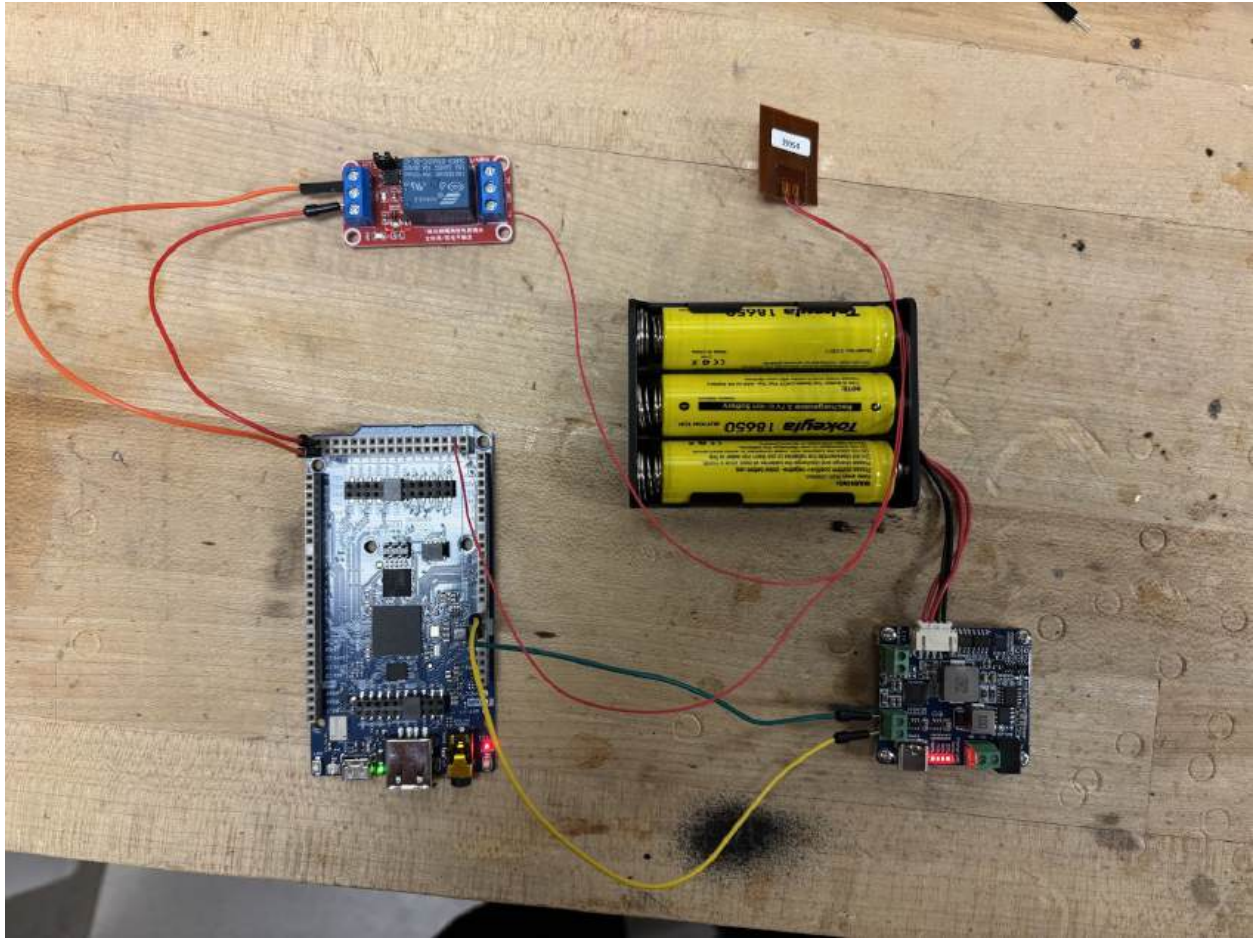


Figure 7.8.4.4: First iteration of relay setup (Did not supply adequate power)

7.9 Communications and Data Handling

7.9.1 System Features

The communications and data handling system is responsible for storing data about the satellite and communicating this information to the ground station. The system is comprised of an Arduino Giga R1, NRF 2.4Ghz radio, and a micro SD card reader.

7.9.2 Properties

- **Power Supply:** The system will be supplied with 5V of power from the power delivery subsystem.
- **Durability:** The system will be soldered to protoboards to keep connections stable during operation. The radio will be mounted to the top of the satellite, and thus will have wires soldered to its connections rather than to a breadboard.
- **Weight:** The entire system setup weighs approximately 150g. Grams. The weight is tentative as the proto boards are assembled.

7.9.3 Critical Elements

- Arduino: Manages the data handling protocols of each subsystem and sends the data to .txt files to the SD card.
- Micro SD Card: Saves the data for each subsystem as telemetry. Stores the raw SAR data to be processed at the ground station.
- Micro SD Card Reader: Interfaces between the SD card and the arduino.
- Radio: Sends data stored on the SD card to the ground station. Extends to 800m in line of sight communication. Fast data transfer.

7.9.4 Images

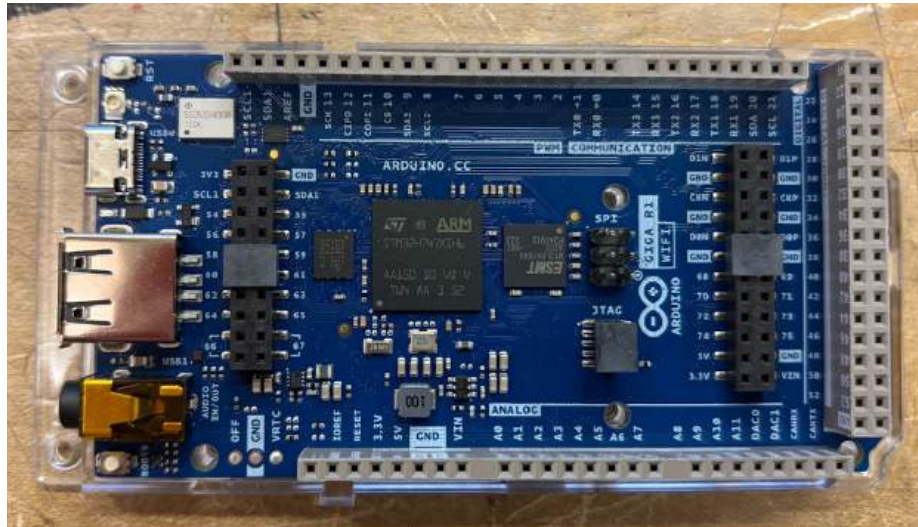


Figure 7.2.4.1: Arduino Giga



Figure 7.2.4.2: SD Card reader

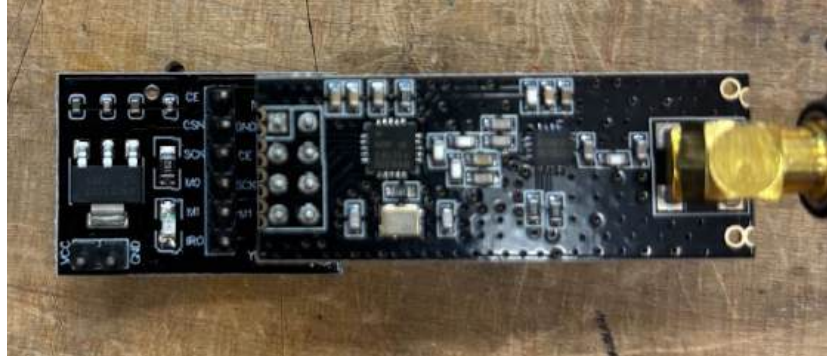


Figure 7.2.4.3: Radio

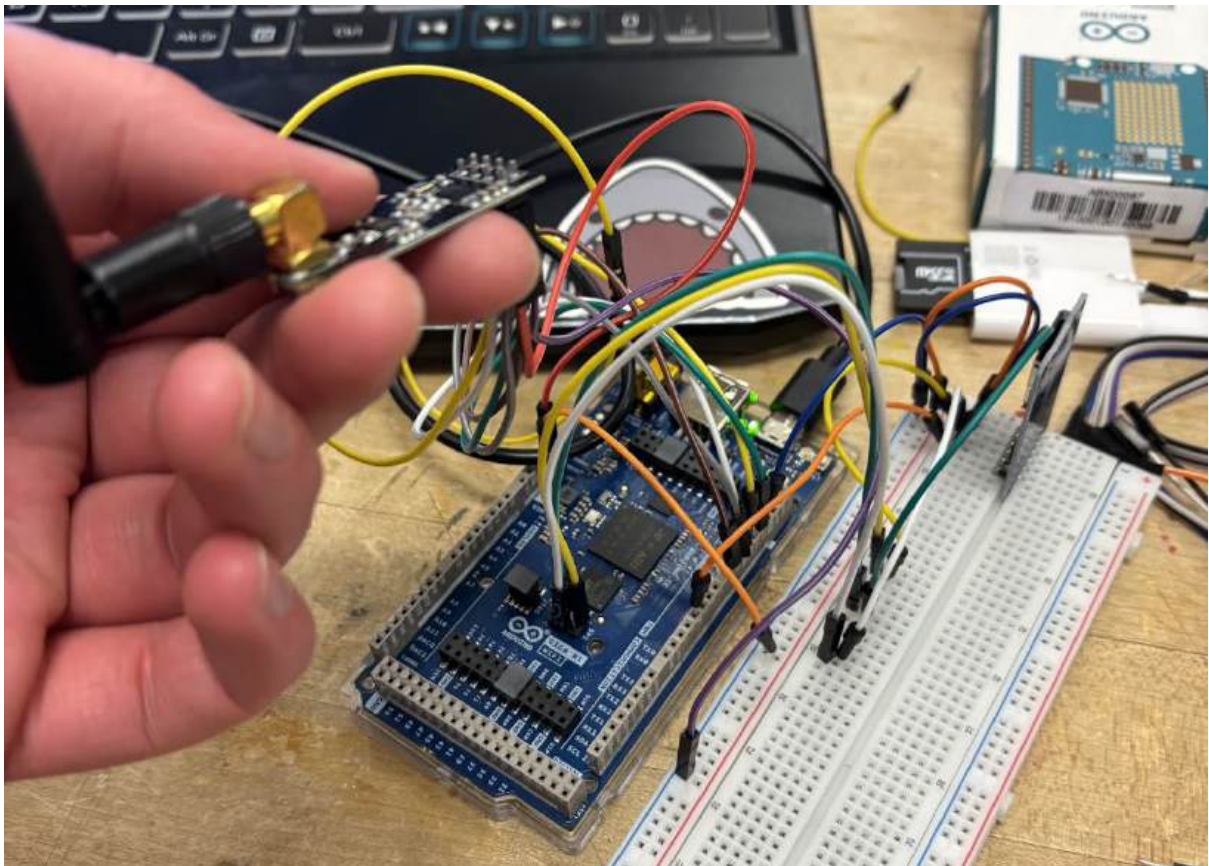


Figure 7.2.4.4: Comms system wired to breadboard

8. Manufacturing

8.1 Ground Station Manufacturing

8.1.1 Objective

The objective of the ground station manufacturing process is to design and assemble a communication system capable of establishing reliable, continuous communication with the cubesat. This ground station will facilitate the transmission and reception of data throughout the mission, ensuring proper functioning and data flow between the satellite and the control center.

8.1.2 Fabrication Procedure

The prototype was built on a breadboard. This allowed for an easy to debug and build setup that will provide a reference for the final prototype. Once the breadboard prototype is complete, the system will be transferred to an Adafruit proto-board. The similar design of the Adafruit proto-board compared to the breadboard will allow for direct transfer of the completed circuit. Once the protoboard is constructed, the housing will be designed using measurements of all necessary components. This design will then be 3D printed and assembled to ensure the ground station prototype is mobile, durable, and aesthetically pleasing. Below is a reference pin out diagram for the following sections.

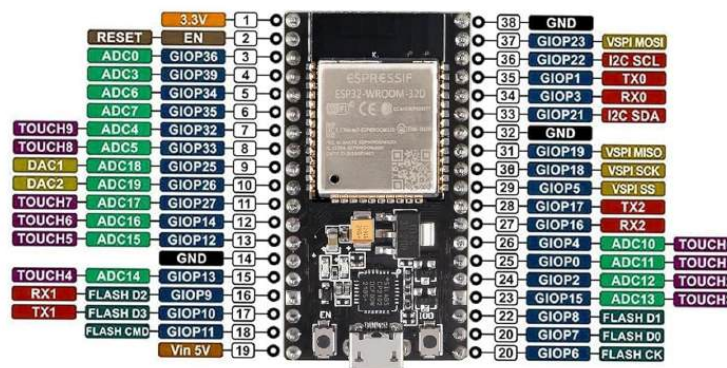


Figure 8.1.2.1: ESP32-WROOM-32D Pinout Diagram

8.1.2.1 Integrating the Micro SD Card Adapter

Since the micro SD card adapter runs on the Serial Peripheral Interface, SPI, communication protocol it requires master in slave out, MISO, master out slave in, MOSI, and serial clock, SCK, pins. Like any other device it also requires a pin for its chip select, CS, ground, GND, and voltage at the common collector, VCC, pins. For implementation with the

ESP32, pins 11 through 15 were used. Although these are not dedicated SPI pins according to the pin out diagram, this connection setup allows for the adapter to function on the microcontroller's VSPI bus. This had to be done as the micro SD card adapter refused to share the same SPI bus as the transceiver. This is handled in the code used to control the device. The device also requires 5V to function. However, the ESP32 does not have a 5V pin. This was resolved by purchasing a power module and integrating it within the micro SD card adapter circuit. A more permanent solution is to be drafted. This will be completed before construction of the proto-board and housing. Below are images of the integrated adapter:

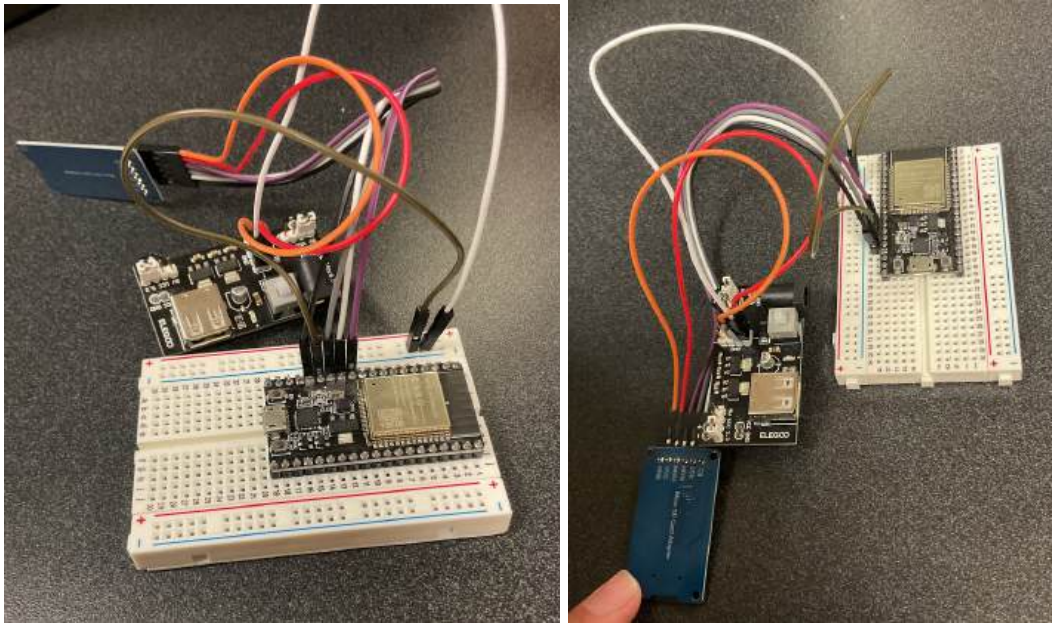


Figure 8.1.2.1.1: Integrated Micro SD Card Adapter

8.1.2.2 Integrating the nRF24L01 Transceiver

Since the nRF24L01 Transceiver also runs on the Serial Peripheral Interface, SPI, communication protocol it requires master in slave out, MISO, master out slave in, MOSI, and serial clock, SCK, pins. In addition to needing pins for its CS, GND, and VCC pins, this device requires a pin for its chip enable, CE. For implementation with the ESP32, pins 1, 26, 29-31, 37, and 38 were used. Again, due to the non-compliance from the micro SD card adapter, the transceiver was connected to operate on its own SPI bus. This device only required 3.3V to function so the board alone is able to power it through pin one. The separate SPI bus is handled in the code that controls this device. Below is an image of the integration of the transceiver:

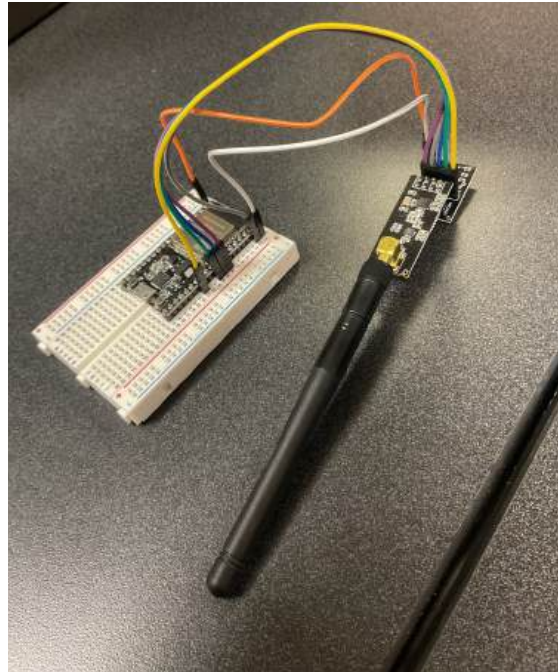


Figure 8.1.2.1.2: Integrated nRF24L01 Transceiver

8.1.2.3 Integrating Buttons

Integration of buttons is very simple. The button must be connected to a GND connected to the board. Then any GPIO button may be used to interact with the button. For the image below the button is connected to pin 23. This allows for the calling of specific functions in the code at a moment's notice without modifying the uploaded code.

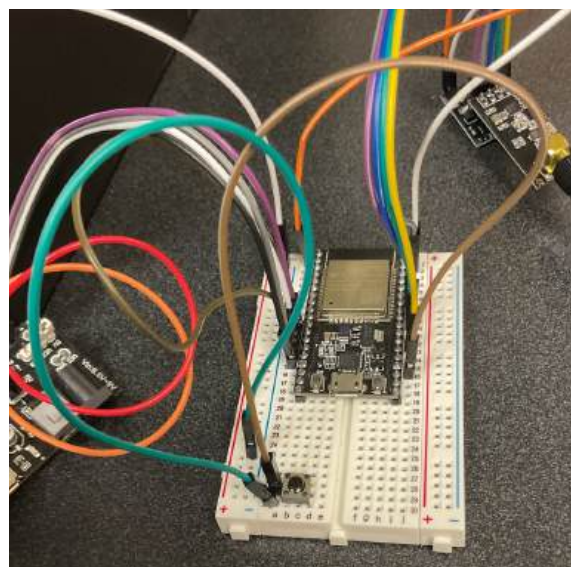


Figure 8.1.2.1.1: Integrated Button

8.1.2.4 Housing the Completed Circuit

The housing has yet to be drafted. However, the housing will use measurements taken of the Adafruit proto-board, transceiver, and micro SD card adapter to size and orient the housing design to properly support, protect, and simplify the ground station. This design will then be 3D printed and hand assembled.

8.1.3 Manufacturing Process

The manufacturing process will involve the use of pre-manufactured COTS components that will be assembled into a cohesive system. The communication hardware will be configured and integrated to ensure seamless interaction with the cubesat, providing reliable data transmission and reception. Once the assembly is complete, performance validation will be conducted through a series of tests to confirm the system's functionality under the anticipated mission parameters. This process ensures that the ground station meets all requirements for mission success.

8.1.4 Facility/Location

The ground station will be assembled in the Senior Design Lab, a well-equipped facility capable of supporting the necessary assembly and testing activities. This lab provides the space and tools required for the fabrication and verification of the communication system.

8.1.5 Equipment, Tools, and Machines Required

No specialized machinery or tools are required for this process, as the ground station will primarily be assembled using hand tools and COTS components. The assembly will involve basic tools for wiring, configuration, and testing, with no major manufacturing equipment necessary.

8.2 Payload Manufacturing

8.2.1 Objective

The payload manufacturing objective is to assemble each critical component in as compact of a manner as possible to allow for easy integration into the CubeSat. The final assembled subsystem should also perform all the design objectives of the payload.

8.2.2 Fabrication Procedure

The prototype payload will mostly be constructed through the modular threaded design of the COTS RF components and wired either with a breadboard and then soldered into a perma-proto board after initial verification testing. Besides these simple tasks, the bulk of the hardware fabrication will revolve around mounting equipment and the antennas.

8.2.2.1 Modulator Circuit Assembly

For initial testing, the modulator circuit, missing only two components below, will be wired through a breadboard. The center component is the black integrated circuit, the monolithic function generator, which is used to produce the voltage ramp needed by the VCO to generate the signal chirp for the SAR. The components surrounding the MFG are all simple resistors, capacitors, and potentiometers. For testing, a DC power supply will be connected to the appropriate lanes to power and provide a reference voltage of +8.2 Volts to this circuit. After initial testing, this circuit was soldered onto a perma-proto board for durability and compactness.

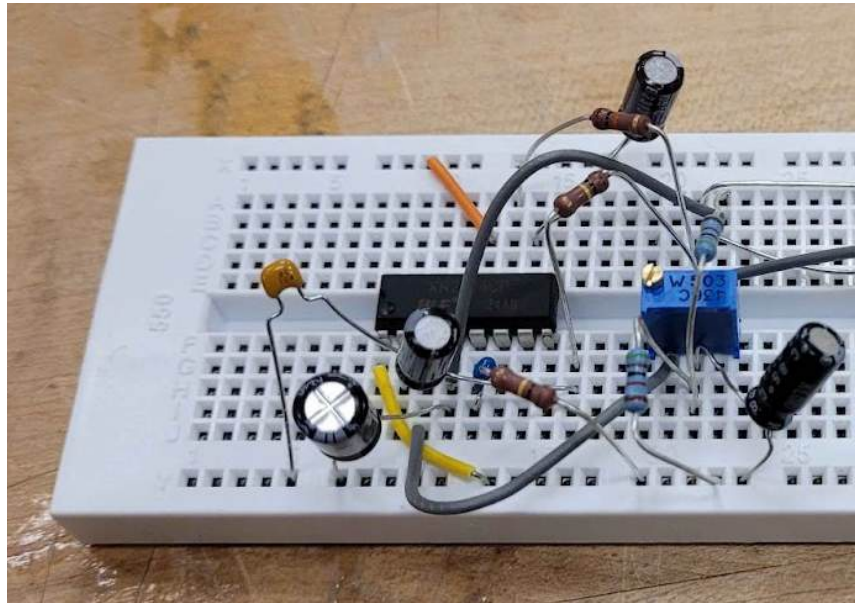


Figure 8.2.2.1.1: Early Modulator Circuit

8.2.2.2 Video Amplifier Circuit Assembly

Exactly the same as the previous circuit, the video amplifier circuit was initially wired using a breadboard. The central component for this assembly is the black integrated circuit, the quad operational amplifier used to process the mixed signal collected by the RF circuit into readable audio data for the ground station. This circuit is less complete in the provided image, but was fully assembled before the end of the week. After testing, this circuit was also soldered.

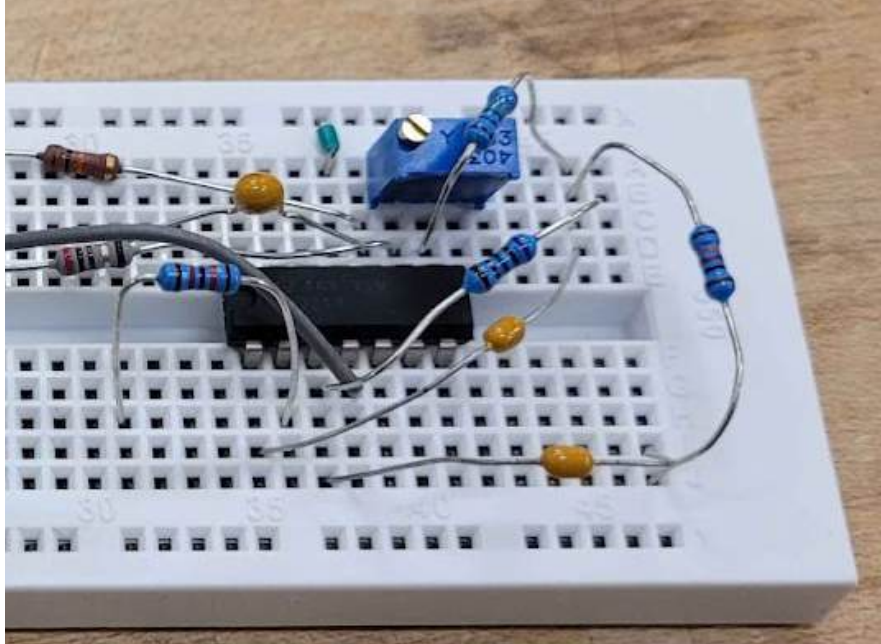


Figure 8.2.2.2.1: Early Video Amplifier Circuit

8.2.2.3 RF Circuit Assembly

The radio frequency circuit is entirely fabricated through modular threading on each component and male-to-male adapters in case of connector miss-match. This process is simple and rudimentary as the combination of adapters and components can only fit a couple ways. Unfortunately, the footprint of this section of the payload is large, but the overall volume it fills is quite small. A clean image of the assembled system is unavailable as of writing this document, but a reference image for visualizing the assembly method using an image taken much earlier this semester is placed below.



Figure 8.2.2.3.1: RF Splitter (left) and RF mixer (right) connected with a male-to-male SMA adapter and with coaxial SMA male-to-male wires attached on either end.

8.2.2.4 Cantenna Assembly

Starting with simple soup cans, they will be cleaned, dried, and measured to ensure appropriate sizing. A $\frac{1}{4}$ inch hole will then be drilled with its center 1.8 inches away from the back wall of the can. The hole will be deburred and sanded to ensure safe handling and a smooth interior surface. Any lip on the inside of the can will be cut to allow maximum return and transmission area. If necessary for a given can, the interior surface will be thoroughly sanded to remove any coat or film that could inhibit radio wave reflection. Then, an SMA bulkhead will be inserted into the hole and a nut will be screwed on to hold it in place. Finally, a short, straight, wire will be soldered into the cup on the interior of the can. This wire was initially made of aluminum, but will be upgraded to copper for further iterations of the cantenna design. Below is an unmodified can and a completed cantenna.



Figures 8.2.2.4.1 and 6.2.2.4.2: Freshly Cleaned Can (left) and Fabricated Cantenna (right)

8.2.3 Manufacturing Process

Nearly all the components of the payload are COTS and will simply be hand manipulated into an assembled state. After initial testing, a soldering iron will be required to fabricate the final circuits. The cantenna assembly requires simple hand tools and an electric drill to make the hole. The cantennas also require extremely simple soldering of the monopole wire into the SMA bulkhead cup. Once each section of the payload is manufactured, they will be mated together through either wiring into the breadboard, soldering onto the perma-proto board, or simply screwing them into each other using the SMA threading. The modulator circuit and the cantennas each need to be calibrated after fabrication to ensure they are tuned for use in the payload.

8.2.4 Facility/Location

The payload can be entirely constructed in the Senior Design labs. The workspace and tools necessary for manufacturing are easily accessible there. Tuning of the modulator circuit can also be done in the Senior Design Labs, but the cantennas will need to be tested in the ECE MakerSpace to have access to the appropriate RF equipment.

8.2.5 Equipment, Tools, and Machines Required

For fabrication, a hand drill, deburring tool, flat-head screwdriver, wire strippers, needle-nose pliers, a wire cutter, metal sanding rod, soldering iron, and some solder/flux are

required. No specialty machines or tools are required for initial assembly of any part of the payload.

8.2 Propulsion Manufacturing

8.2.1 Objective

The objective of the propulsion manufacturing process is to design and assemble a propulsion system capable of providing consistent propulsion for the cubesat. This propulsion system should be capable of deorbiting the CubeSat.

8.2.2 Fabrication Procedure

The prototype was built with a simple assembly method. Since all of the parts are threaded, they were simply screwed together. Plumbers tape and metal glue were used to seal leaks.

8.2.3 Manufacturing Process

The manufacturing process will involve the use of a wrench and clamp. Using these, pipes will be fastened together. This process is simple and there are no exceptions.



Figure 8.2.3: Propulsion System

8.2.4 Facility/Location

The propulsion system will be assembled in the Senior Design Lab, a well-equipped facility capable of supporting the necessary assembly and testing activities. This lab provides the space and tools required for the fabrication and verification of the propulsion system.

8.3 GNC Manufacturing

8.3.1 Objective

The objective of the guidance, navigation, and control system is to be able to orient the differing CubeSat faces in their desired direction. The manufacturing part of this process includes putting together the reaction wheels, making a flywheel with a high moment of inertia,

and assembling the correct circuitry to the arduino. The only other manufacturing for this section comes in creating the script for the arduino control system.

8.3.2 Fabrication Procedure

The prototype was built on a breadboard which interfaced the batteries and arduino. This allowed for an easy to debug and build setup that will provide a reference for the final prototype, which will be built on a PCB. The similar design of the printed circuit board compared to the breadboard will allow for direct transfer of the completed circuit. This circuit includes 4 photosensitive resistors, 1 accelerometer, 3 electronic speed controllers, and the power supply.

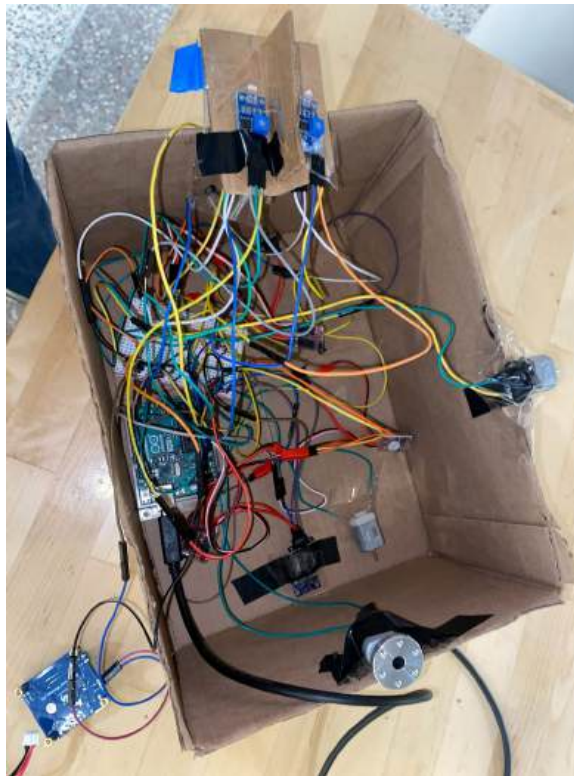


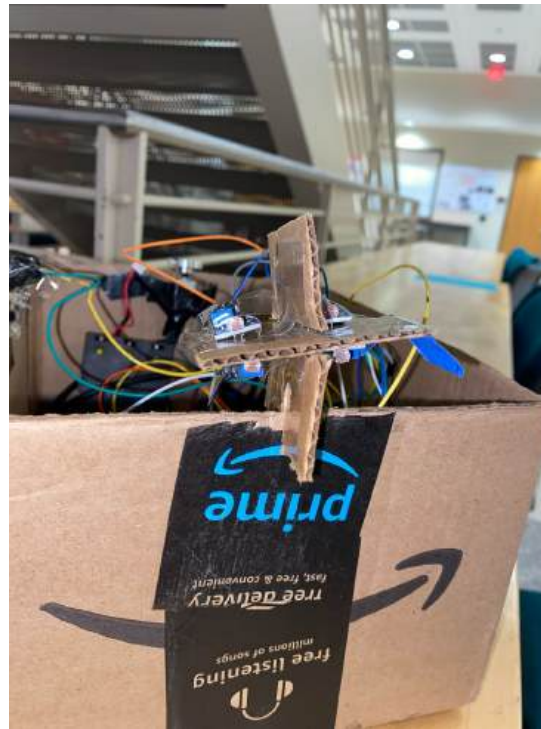
Figure 8.3.2.1:Initial Breadboard Configuration

As the frame is still being manufactured, a cardboard box about the size of the SMARTSat design was used for a visualization. This also allows for the sensors to be tested as the box can be oriented in different positions to verify correct reaction from the wheels.

8.3.2.1 Integrating Photoresistors

Since the photosensitive resistors read as a high/low output only, the manufactured design will need to be more complex than simply one sensor. At this stage in the design process, the SMART team has decided to include a 4 way light-blocking design. This design will allow for the most precise pointing accuracy. With this setup, when a light is offset either up/down or side to side, the cardboard blinders will create a shade on the adjacent sensor. Depending on which sensor is pushing the positive output, the position of the light relative to that face can be

computed. For instance, if the top left and bottom left are reading a positive output, the satellite will be told to move to the left without going up or down, and will not stop until all four signals are positive. In other words, when all 4 sensors are sensing the light, the system will be satisfied. This design also allows for 2 modes of calibration. First, the sensors have a screw that changes the sensitivity to the amount of light that causes them to switch on/off. Second, moving the sensors further back on the blinders will increase the accuracy of the system. This is because a smaller displacement of the light source will cause the shadow to cover one of the sensors. The preliminary design as currently constructed is shown as below, along with final design.



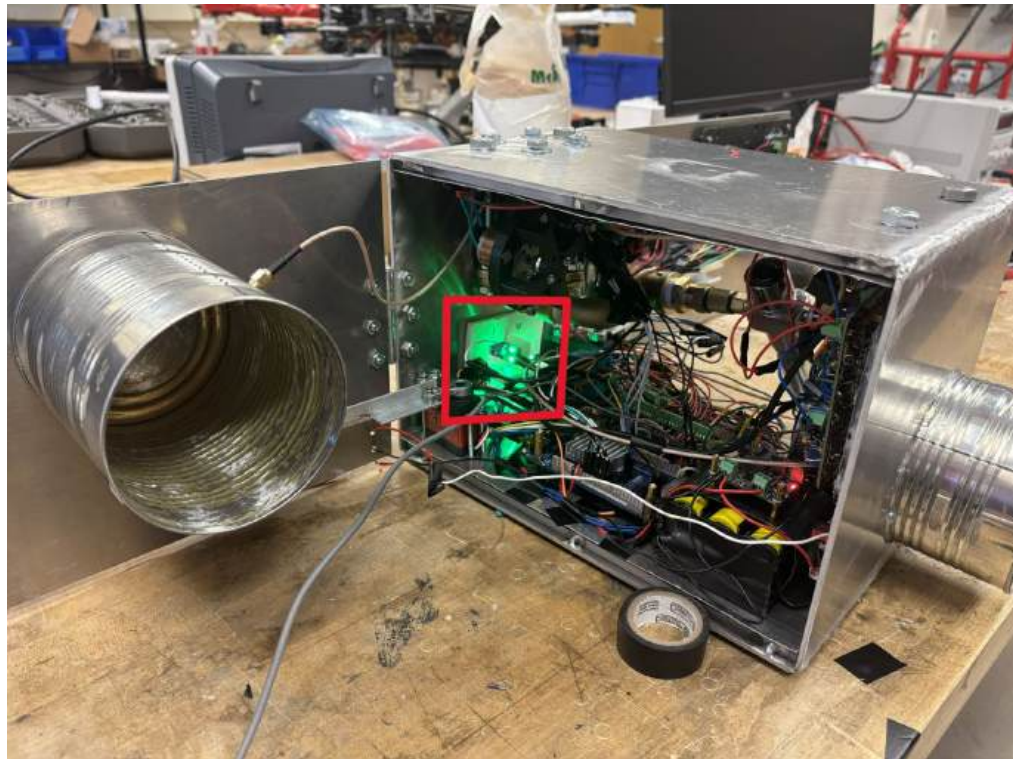


Figure 8.3.2.1.1: Integrated Photoresistor

8.3.2.2 Integrating Accelerometer

Since the photoresistors are for sensing capabilities during the Sun pointing orbits, the rest of the time while in orbit, the satellite shall be engaged in Earth pointing. The accelerometer is the sensor that allows for the CubeSat to do this most efficiently. When static, it will detect gravitational acceleration from Earth along the vertical z axis. Most accelerometers use software algorithms such as high and low pass filters to separate gravitational acceleration and actual motion. This allows for as the sensor is being tilted, this gravitational acceleration is no longer being expressed only in the z axis, but spreads to the x and y respectively. The key part of the manufacturing on this system comes from mounting. This accelerometer must be perfectly flat in order to read gravitational force in only the z axis. This sensor is highlighted in the below image, seen working in conjunction with the rest of the control system.

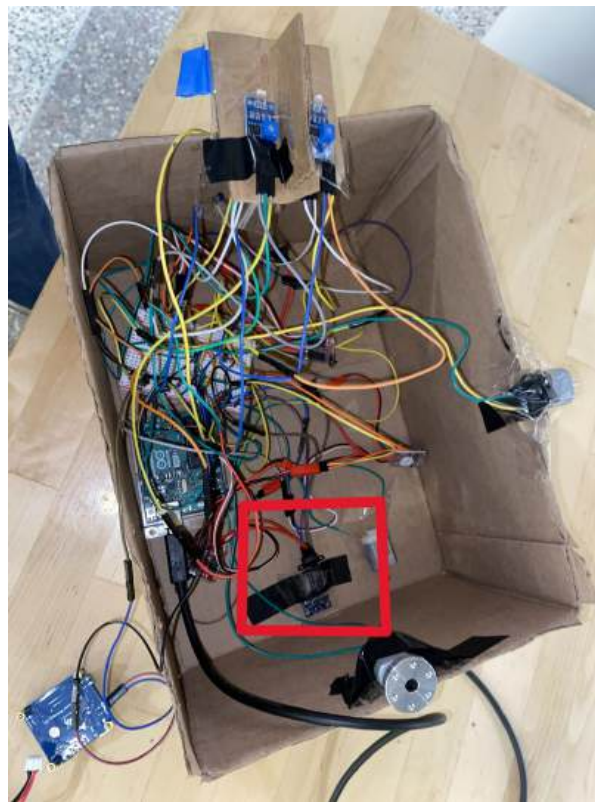


Figure 8.3.2.2.1: Integrated Accelerometer

8.3.2.3 Integrating motors

Integration of the motors is relatively simple. There must be wires that are soldered to the motor terminals that go directly to the speed controllers. These controllers have a power line running directly from the batteries, along with a signal line that connects to the arduino. It is important to include this intermediary component to avoid connecting motors directly to the arduino pins. This could result in overvolting the arduino, if the motors are big enough. These speed controllers are rated for 20 Amps, and 9 Volts. They allow for forward, backwards, and braking functions as well. The layout of how the speed controllers are connected is shown below.

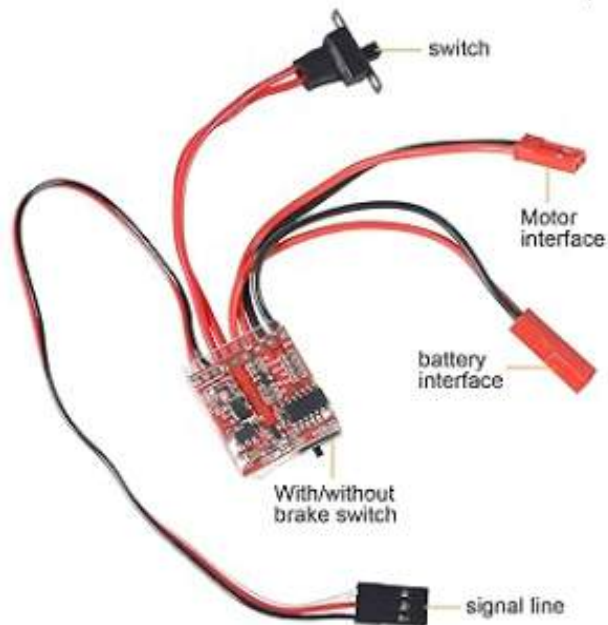
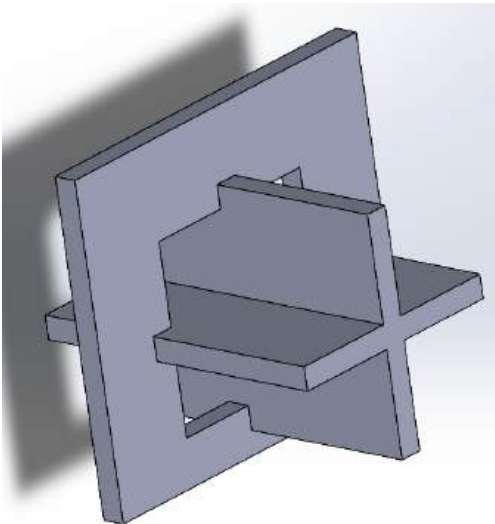


Figure 8.3.2.1.1: ESC Wire Connections

8.3.2.4 Housing the Completed System

The housing for the photoresistors and accelerometer were created via a 3-D printed part. The motors were mounted to the ceiling plate of the satellite, using long screws and pipe hanger straps. Pictures of these are provided below. In the first motor image, the orange arrow points to the long screws attached to the satellite roof, while the blue arrow points to the pipe hanger that is holding the motor in place.



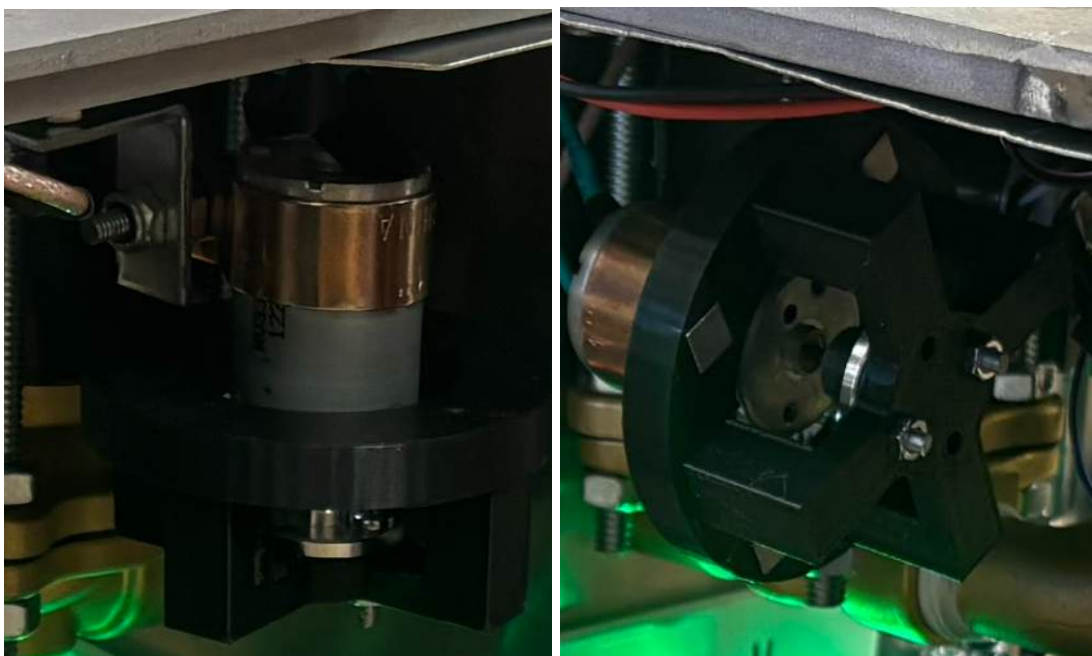


Figure 8.3.2.4.1: Final Motor Integration

8.3.3 Manufacturing Process

The manufacturing process will involve the use of pre-manufactured COTS components that will be assembled into a cohesive system. The reaction wheels will be configured to produce torque on three different axes, and allow for full attitude control. The wiring used will be soldered onto a printed circuit board for more room allowed inside the satellite. The arduino will be used in accordance with communication and data handling requirements.

8.3.4 Facility/Location

The GNC system was assembled in the Senior Design Lab, a well-equipped facility capable of supporting the necessary assembly and testing activities. This lab provides the space and tools required for the fabrication and verification of this system. It is also important to note that the photoresistors require a dimly lit room as the ambient lighting will affect their readings.

8.3.5 Equipment, Tools, and Machines Required

No specialized machinery or tools were required for this process, as the GNC system will primarily be assembled using hand tools and COTS components. The assembly will involve basic tools for wiring, configuration, and testing, with no major manufacturing equipment necessary.

8.6 Thermal Manufacturing

8.6.1 Objective

The objective of thermal manufacturing is to design a thermal management system that will be able to heat and cool the components. The thermal system should be able to do this based on temperature.

8.6.2 Fabrication Procedure

The prototype was fabricated by connecting wires onto a breadboard that connected to the heater, thermocouple, arduino, and relay. The cooling fans will be connected to this setup once they arrive, but they will essentially do the opposite effect of the heater to constantly cool the system to prevent overheating.

8.6.3 Manufacturing Process

The manufacturing process will involve the use of a breadboard and wires in order to connect the thermocouple, heater, and cooling fans together in order to fully develop the thermal management system. The cooling fans have been removed and replaced with heat sinks and critical components.



Figure 8.6.3: Basic thermocouple setup without heater and relay

8.6.4 Facility/Location

The thermal system will be assembled in the Senior Design Lab, a well-equipped facility capable of supporting the necessary assembly and testing activities. This lab provides the space and tools required for the fabrication and verification of the thermal system.

8.7 Structures Manufacturing

8.7.1 Objective

The key objective of manufacturing the structure's subsystem is to design a frame that can maintain its shape without deformation due to loads. Additionally, the frame needs to be able to protect/house the different subsystem components from the outside environment.

8.7.2 Fabrication Procedure

8.7.2.1 Cutting of Parts

Six 30 cm struts (11.81") are cut from an aluminum alloy 6061 sheet with dimensions of 15x15" and 1/4" thickness. The struts have a square cross sectional area, resulting in a width of 1/4" as well. Additionally, six aluminum alloy 6061 sheets of 1/8" thickness are used to cut four side panels with dimensions of 20x30 cm and the top and bottom panels with dimensions 20x20 cm. The bottom panel has a circular cutout with a diameter of 10mm for the nozzle from the propulsion subsystem and a square cutout of 1.97 inches for the sun sensor to go for the GNC subsystem. Additionally, the top panel will have a hole with a diameter of 10 cm in the bottom right for one of the cantennas to be placed for the payload subsystem.



Figure 8.7.2.1.1: Welded cubesat frame with top and bottom cutouts for the three different subsystems

8.7.2.2 Interior skeleton attachment

Aluminum alloy that is significantly thinner than side panels will be cut/trimmed to proper dimensions and then attached via standoff screws to the insides of the side panels. This interior skeleton will serve as a mounting surface for the components of the other subsystems, without compromising the structural integrity of the frame.



Figure 8.7.2.2.1: Interior skeleton material

8.7.2.3 Connection of Parts

The final step in this fabrication process is to connect the struts as well as the side panels in the shape of a 20x20x30 cm cubesat frame. This connection process will be accomplished by welding the different pieces together.

8.7.3 Manufacturing Process

The construction of the S.M.A.R.T frame consisted of performing various cuts/trimms on aluminum sheets, in an effort to get the proper dimensions specified for this project. Additionally, welding served as the method for connecting the different struts, as well as the side panels consisting of the body.

8.7.4 Facility/Location

- Senior design shop, located in EB3 on the ground floor

8.7.5 Equipment, Tools, & Machines Required

- Water Jet
- Welding tool

8.8 Power Manufacturing

8.8.1 Objective

Many components must be constructed to make the power system. Solar panels, additional battery storage, power delivery boards, 12V high power rail, and the panel deployment solution. The power system is nearing completion. The remaining tasks are to connect the solar panels to the frame, attach servos to the panels and the frame for deployment, and solder all power connections to the interior. All these tasks are contingent on getting the frame welded together and mounting components.

8.8.2 Fabrication Procedure

8.8.2.1 Constructing Solar Panels

Solar panel construction is mainly completed by soldering all panels together and ensuring a consistent connection. The panels were soldered together using 14 gauge solid core wire for structural support as well as current delivery. Once the panels were soldered together, they were tested with a multimeter to ensure that the connections between each cell were solid. The main concern with the solar panels at this point is not power delivery, but structural stability

as the solder joints and copper solder pads are incredibly fragile.



Figure 8.8.2.1.1: Solar Panel Front

The first iteration of the solar panels had roughly a few centimeters between each cell to keep them equidistant. However, this configuration was altered to allow for more flexibility once mounted to the exterior panels. The final iteration has each cell touching each other, minimizing the overall footprint of the panel.

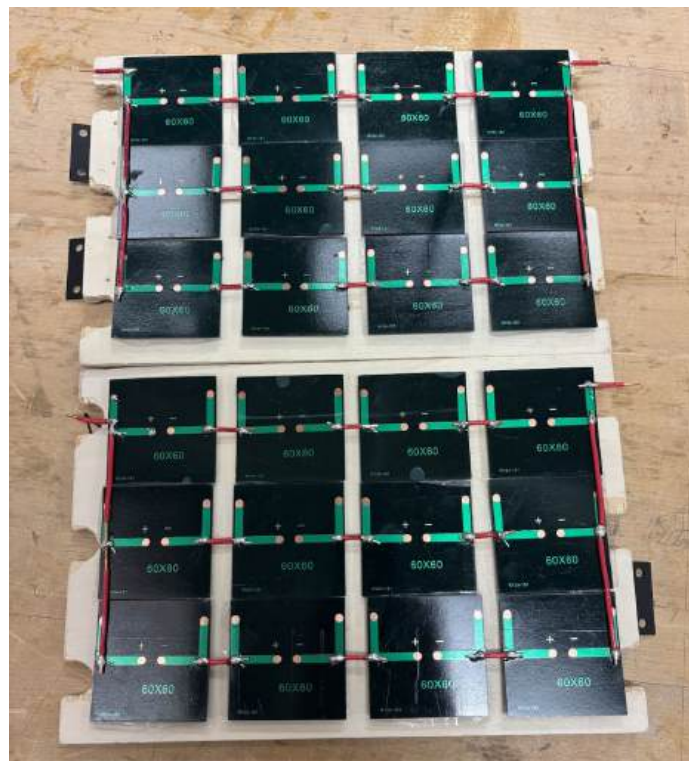


Figure 8.8.2.1.2: Solar Panel Rear



Figure 8.8.2.1.3: Solar Panel final configuration

Initial testing of the solar panels has voltages ranging from 10-14V under ambient shop light. In outdoor conditions, this range was greatly extended to 22V+. This shows that not only are the solar panels within the required 6-24V range, but they outperform the expected 22V maximum.



Figure 8.8.2.1.4: Solar Panel Testing 1



Figure 8.8.2.1.4: Solar Panel Testing 2

8.8.2.2 Making the 12V rail solution

The 12V rail is a necessary component for driving both the solenoid and the heater pad. These are by far the most power demanding components within the system. Both components require 12V from a source capable of delivering high power continuously. Attempting to power the system using the arduino results in either the solenoid not actuating or the heater not getting warm enough.

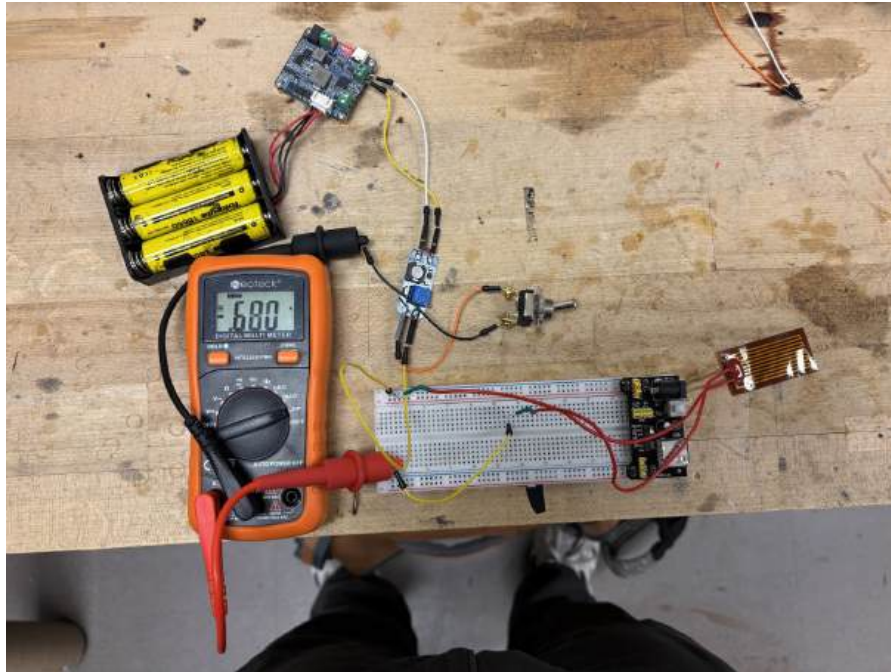


Figure 8.8.2.2.1: Powering Heater Pad

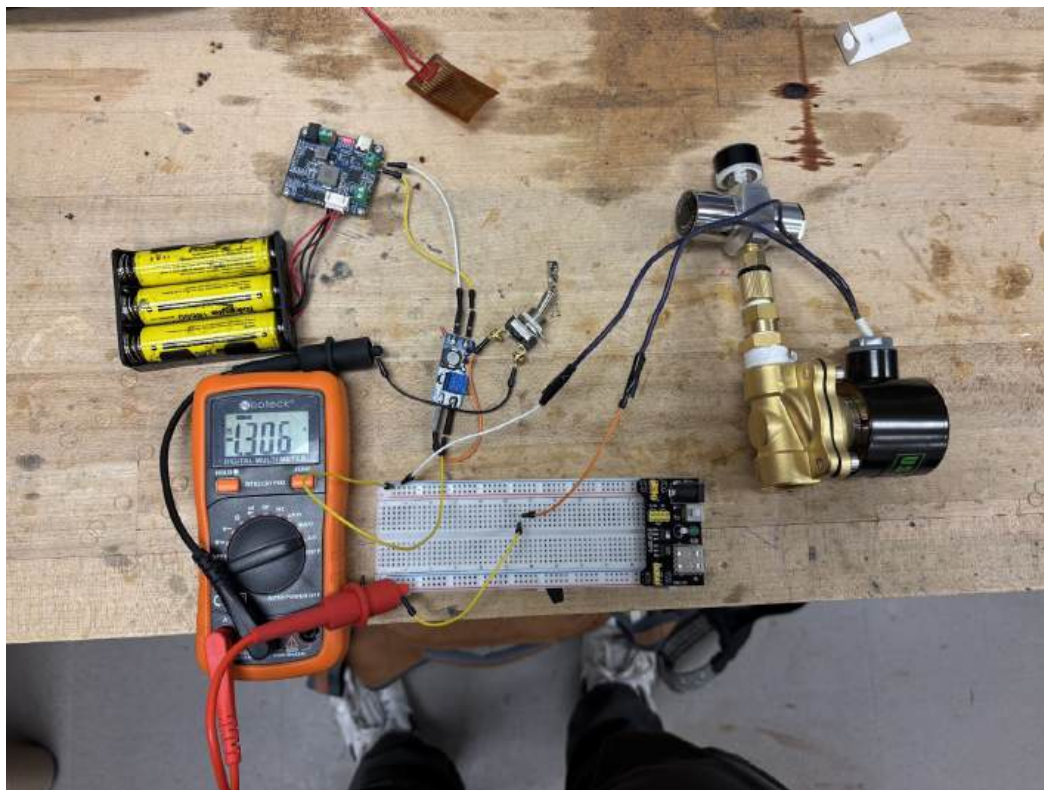


Figure 8.8.2.2.2: Powering Solenoid

The above photo shows the solenoid being driven at 12V and drawing 1.3A of current. This places the solenoid wattage at 15.672 W when driven in the above configuration. The heater being driven at 12V requires 0.68A of current in the following configuration. This brings the heater power usage to 8.16 W, differing from the projected value. Thankfully, both components are not meant to be driven continuously throughout the satellite operation. The following test shows both components driven in parallel.

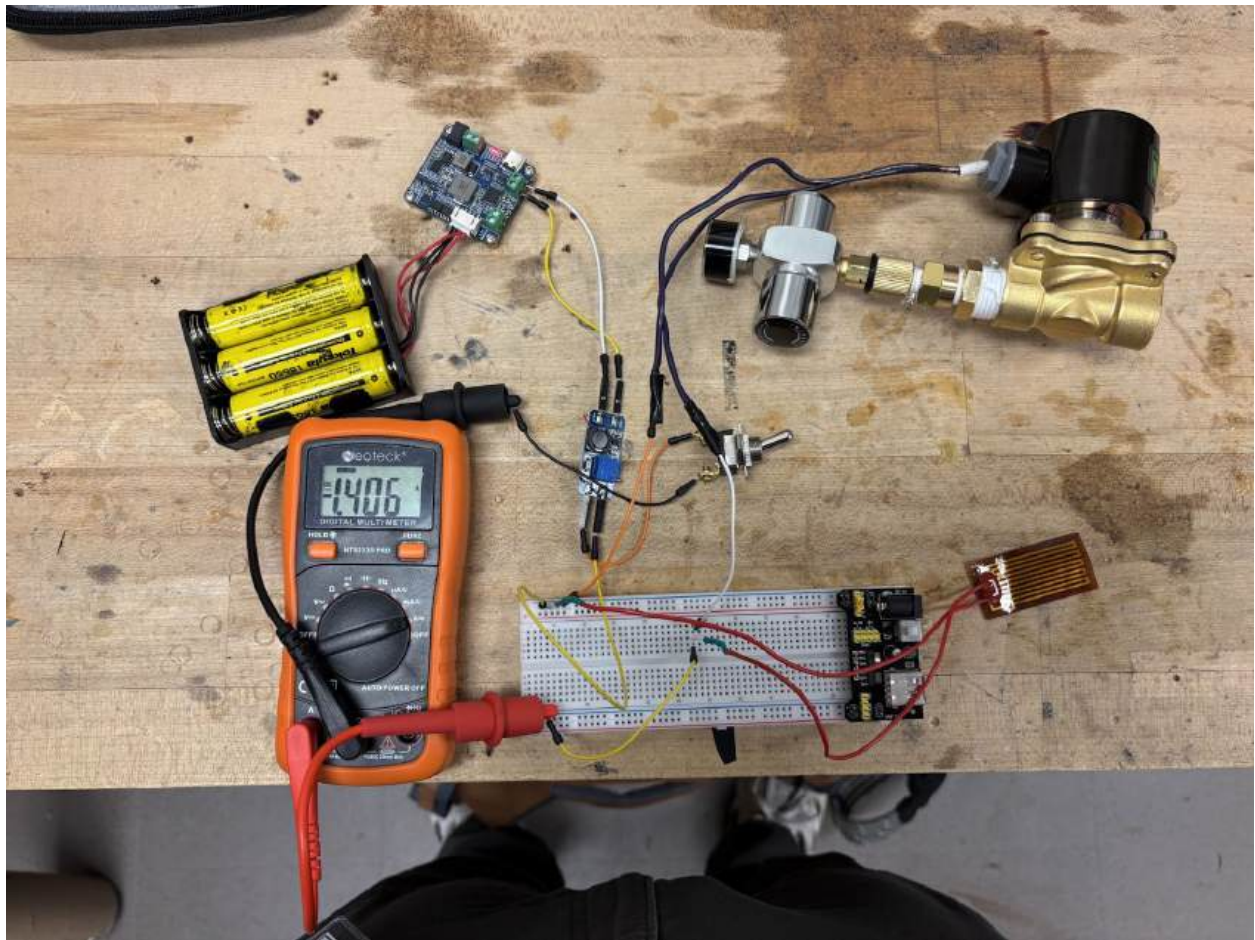


Figure 8.8.2.2.3: Powering Heater Pad and Solenoid

Both the heater and solenoid valve driven in parallel pull 1.406A, placing the combined wattage at 16.872W. This isn't much higher than driving the solenoid alone, but this is due to the heater losing temp when the solenoid is switched on. Because of this power drop in the heater, it may be worth experimenting with higher voltages.

8.8.2.3 Constructing Thermal System

The thermal system wiring consists of a thermocouple, a relay, and the heater pad. The heater pad is initiated when the thermocouple detects a critically low temperature. In testing, it was shown that the relay could reliably control the heater by switching on and off when

necessary to the system. One challenge with this system construction was learning how the relay functions, and how to safely drive the relay without compromising the delicate circuitry of the arduino. Once the wiring configuration was learned, driving the solenoid was a simple process.

The thermal system has plans for expansion using a thermal louver assembly and small raspberry pi fans. The louver is meant to be a mock design of what the full scale satellite will require for venting heat in space, while the fans are necessitated by the running of these electronics in the Earth's environment. These two systems will switch on simultaneously when the thermocouple detects a critically high temperature. The louvers and fans will not require high power delivery and thus can be driven by the arduino and power delivery boards.

8.8.2.4 Wiring other components

The last step in constructing the power delivery system is finalizing all wiring. To do this, components will be soldered to protoboards to ensure stable and reliable connections, power will be delivered in 5V, 12V, and -12V sources, all digital signals will be mapped out, and a separate module will be constructed for the 12V transformer. Before this step can be taken, the frame must be fully constructed so all components can be fixed to their respective locations. Frame construction will allow each subsystem lead to ensure their components have enough space to operate without interference. Parts to construct these boards have been ordered and will begin construction within the next week.

8.8.3 Manufacturing Process

The manufacture of the power system has consisted of iterative testing before settling on the final design. For the solar panels, they were first constructed and soldered to ensure the functionality of the system. While the panels did work as intended, they were altered in favor of smaller and higher quality design. The high power relay was initially not powerful enough to drive the heater or solenoid. While a successful iteration was eventually reached, the transformers in use were found to be too inconsistent and were instead swapped for a higher quality transformer capable of handling the required load. Lastly, the power bank was originally going to consist of 3 lithium ion batteries, differing some from the projected design. However, after testing the high powered components, it was determined that 6 lithium ion batteries would be necessary to drive each system reliably.

8.8.4 Facility/Location

- Senior Design Lab
- ECE Maker Space
- Courtyard

8.8.5 Equipment, Tools, and Machines Required

- Soldering Iron
- Multimeter
- Water Jet

8.9 Communications and Data Handling Manufacturing

8.9.1 Objective

The primary objective of the communications and data handling subsystem is to collect data from the satellite and deliver that data to the ground station via the 2.4Ghz radio connection. This is accomplished by first storing the data to a micro SD card in flash memory. Next, the data is sent through the radio to the ground station. While this task appears simple at the surface, it was much more difficult due to interference with the devices.

8.9.2 Fabrication Procedure

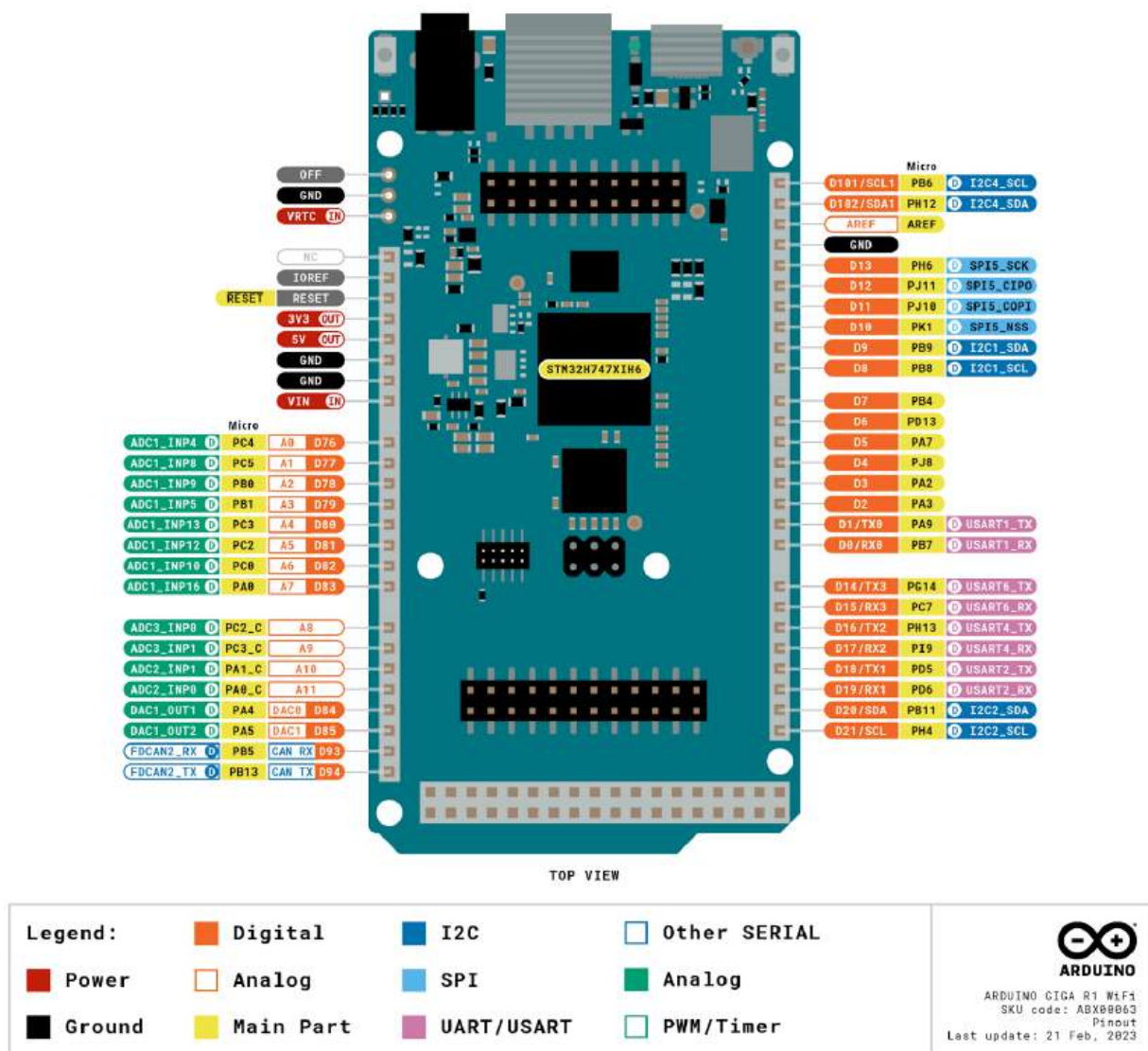


Figure 6.1.2.1: Arduino Giga R1 Pinout Diagram

8.9.2.1 Integrating the Micro SD Card Adapter

Integrating the micro SD card reader was a simple procedure. The SD card reader utilizes the SPI protocol to read and write data. After some trial and error, it was determined that the SD card prefers to use the 6 pin header located at the middle of the board for data communication. In this configuration, the SD card could easily read and write data. The next step was to integrate the communications system.

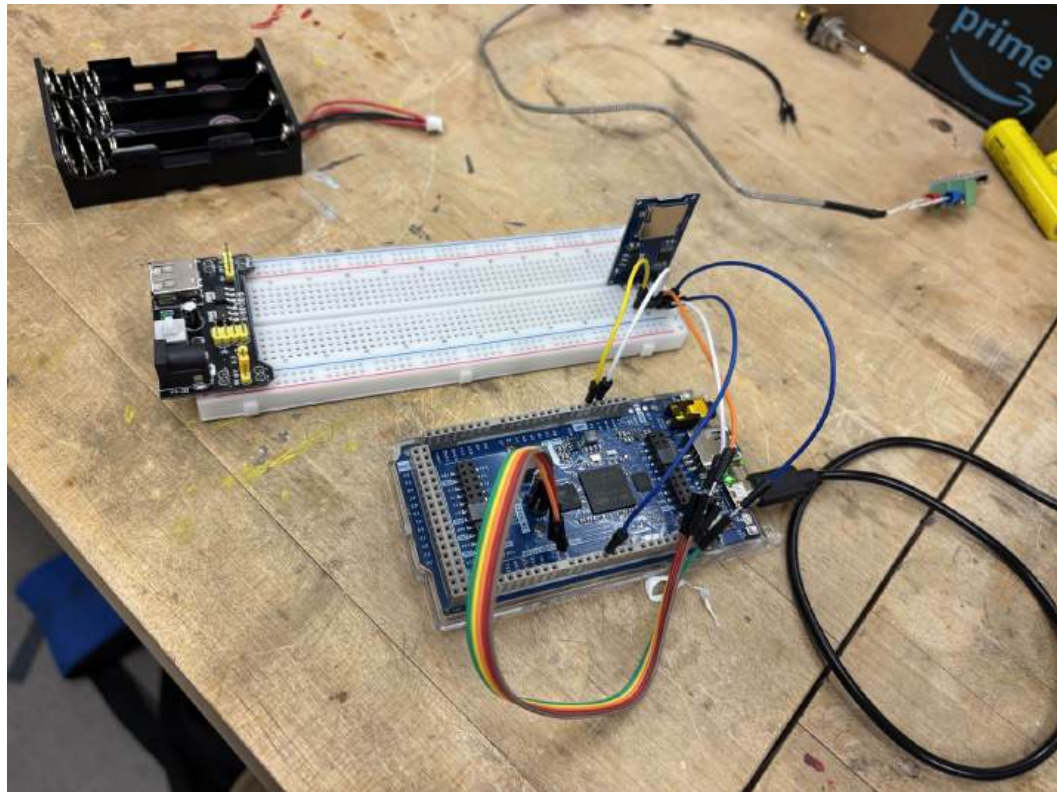


Figure 8.9.2.1.1: Integrated Micro SD Card Adapter

8.9.2.2 Integrating the nRF24L01 Transceiver

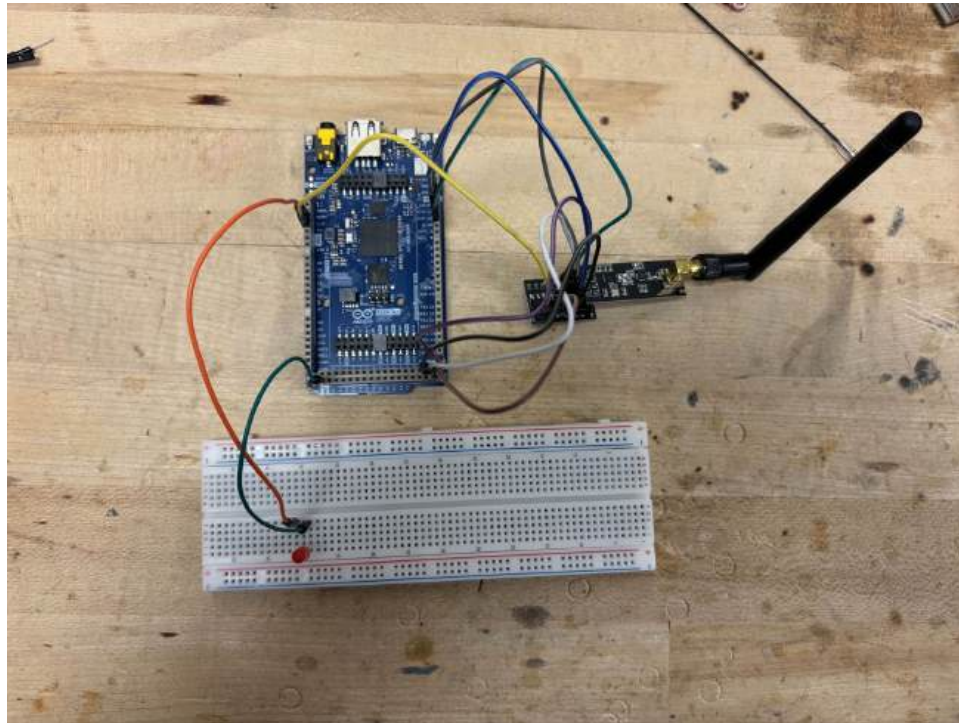


Figure 8.9.2.1.2: Integrated nRF24L01 Transceiver

Wiring the communications system was as simple as wiring the SD card. However, the difficulty with this system came to light with getting the radio to work with the Arduino Giga. Because the Giga is a newer micro controller, there are not as many resources for getting certain libraries to work with it. One major hang up came when trying to install the radio to the secondary SPI bus. Typically, Arduinos will utilize the GPIO pins from 13-10 as the default SPI interface. The Giga on the other hand, uses the 6 pin header as default, requiring the initialization of the secondary SPI bus for the radio to function. Because of the manual and the programming not matching with regards to the naming convention of these SPI busses, wiring this component proved to be quite difficult.

8.9.2.3 Failing to add a third SPI device

The Arduino Giga R1 has two SPI busses in total, while the design settled on requiring three SPI devices. The final device to install was the thermocouple, which had one SPK and one MISO pin. Wiring this component in parallel with the SD card reader resulted in the reader failing to initialize. This issue was never resolved. When trying to wire the thermocouple in parallel with the radio, the signal became corrupted and failed to send or receive any data. This was resolved when it was determined that the radio requires all wires to be the same length to ensure a stable connection. Unfortunately, once the lengths of wire were all the same, the radio functioned while the thermocouple couldn't initialize.

The solution to this problem was found in not initializing the thermocouple on the SPI pins. It was discovered that the thermocouple only needs three data pins rather than two SPI pins and a data pin. This unfortunately means that the satellite has maxed out on SPI devices due to the radio and SD cards incompatibility with other devices. Regardless, this allowed the communication system to retrieve thermal data from the thermocouple without error.



Figure 8.9.2.3.1: SD card, Radio, and Thermocouple connected to arduino.

8.9.2.4 Final Step

The final step in the fabrication of the command and data system will have each component added to the master script and deliver telemetry data to the SD card. Each subsystem will have its own .txt file containing system diagnostics. Angle of rotation for servos, temp for thermocouple, sensor reading for photoresistors, state for fans, heater, propulsion, relays, etc. With each system delivering telemetry data, the final component is to configure SAR data collection. This will require that the SAR is fully operational and delivering reliable outputs. The current plan for this is to collect the raw SAR data to a .txt file and deliver this to the ground station. The ground station will then construct the SAR image.

8.9.3 Manufacturing Process

Manufacturing the C&DH system consisted of wiring and programming components. The wiring for the communication system was a small part of creating the subsystem. Programming took up the majority of the development time for the communication and telemetry system. This was accomplished in no small part by the ground station lead. With his background in computer science and programming, the GS lead made quick work of configuring the SD card reader and radio, enabling both to function together. With this step complete, reading and storing data from other subsystems is a simple process.

One major concern is packet loss due to other subsystems operating in tandem with the communication system. Because of this, the possibility of migrating the communication system to an independent microcontroller is being explored. This would have the SD card connected to two microcontrollers at the same time, enabling one to write and store data while the other reads and sends data. No decision is final for integrating this into the final design.

8.9.4 Facility/Location

- Senior Design Lab

8.9.5 Equipment, Tools, and Machines Required

- Soldering Iron
- Multimeter
- Arduino IDE

8.10 Systems Integration

8.10.1 Mounting

A crucial step in systems integration is a solid mounting solution for each subsystem. For SMART, this can be broken up into three main categories. First is the main skeleton, next is the sub-skeleton or small electronics plate, and lastly is the case mounted components. This section will break down the specifics as well as the chosen method for mounting each component.

8.10.1.1 Skeleton

The skeleton refers to two 2mm thick aluminum plates mounted to the top and the bottom of the main chassis. These plates house most of the electronic components that make up the satellites subsystems. The skeleton is necessary to simplify assembly and maintenance on the prototype system and to ensure any and all component malfunctions are easily accessible in such a tight space. Pictured below is the bottom plate.

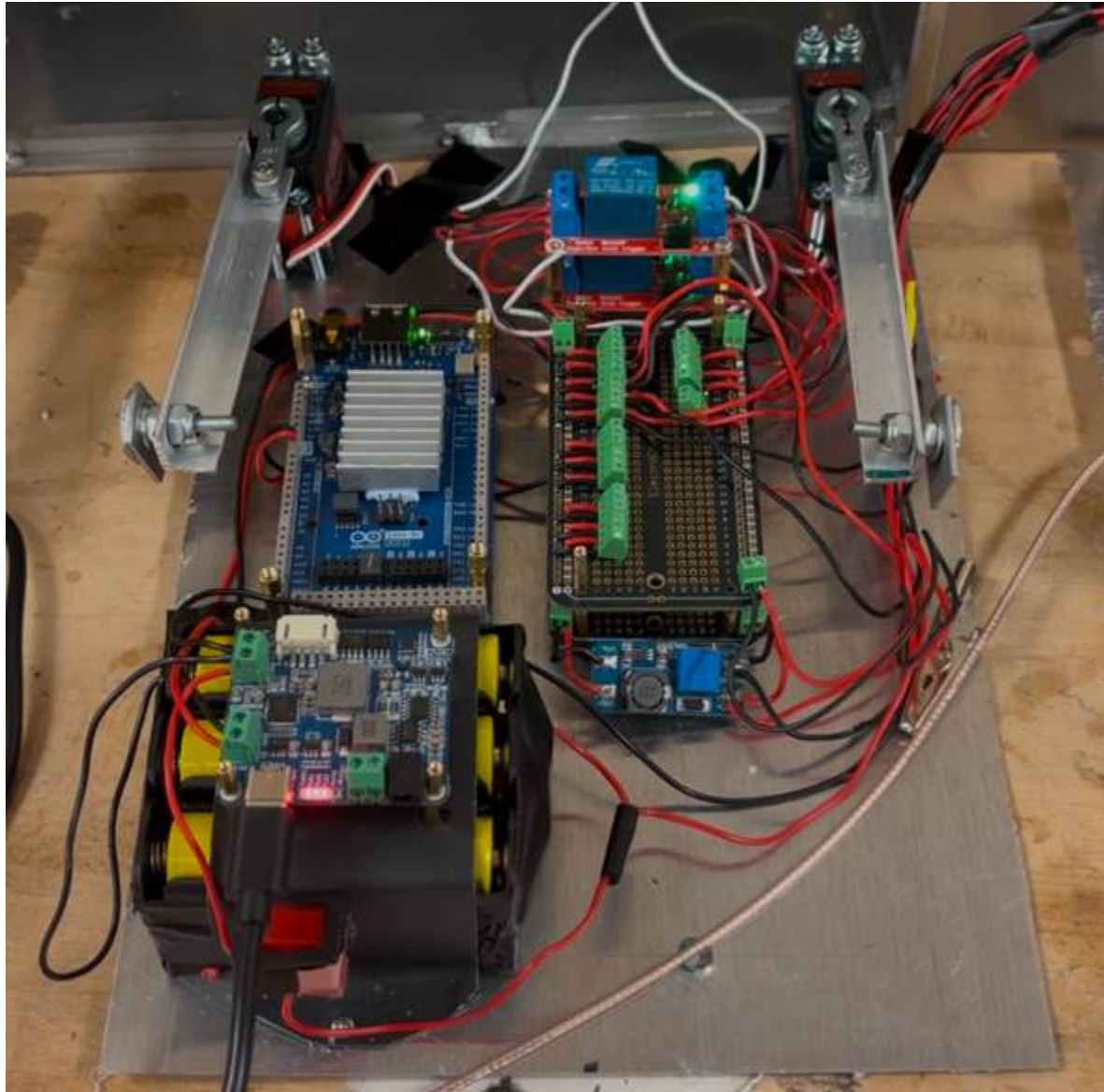


Figure 8.10.1.1.1: Bottom skeleton plate mounts

The bottom plate houses most of the electronic components such as the batteries, MPPT board, arduino, power delivery board, relays, and servos. While not pictured here, the sun sensor is mounted between the sun sensor and It should be noted that in future designs, the servos should be mounted directly to the case and secured using a custom 3D printed housing to ensure no flex during operation. This setup allows the heaviest electronic components to be situated on the ‘floor’ of the satellite to minimize any possible vibrational damages/ dismounts to them. Pictured next is the top plate.

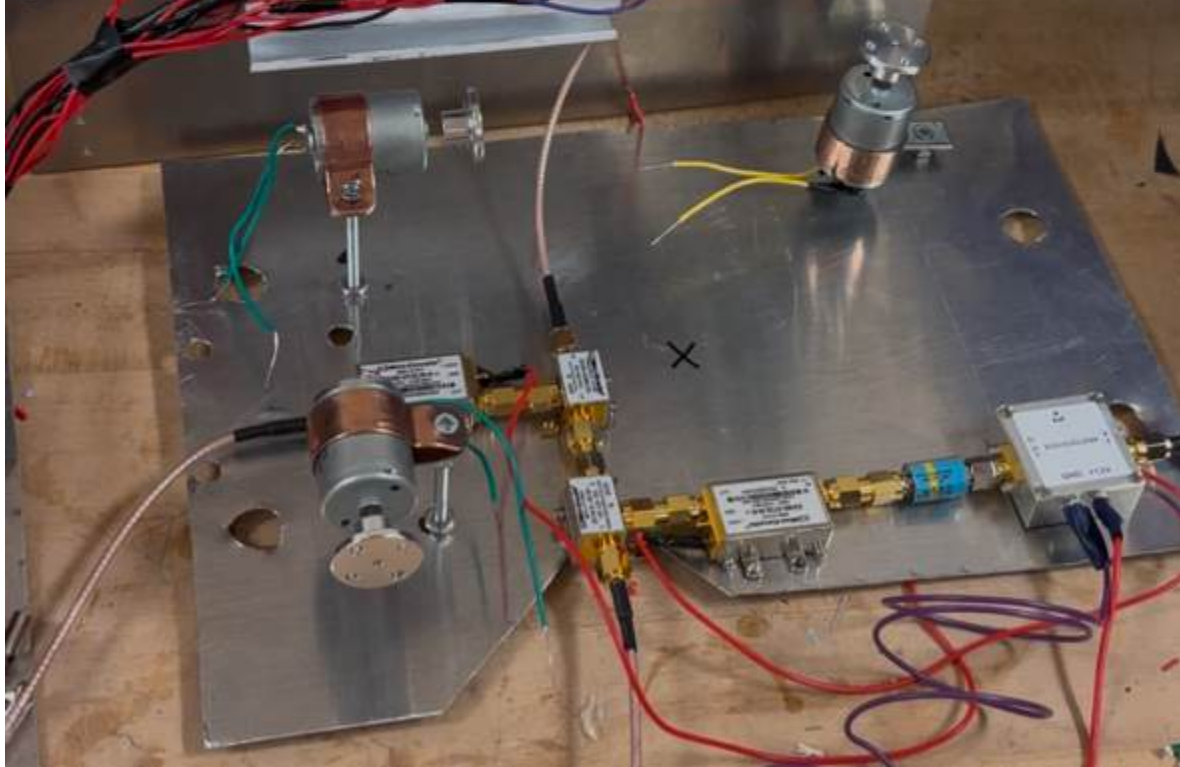


Figure 8.10.1.1.2: Top skeleton plate mounts

The top plate houses the radar components and GNC motors. This location was chosen as opposed to direct case mounts to allow maintenance on these components should the need arise. Another driving factor behind placing the radar components on the top plate is their inherent low-profile mounting, which allows them to be situated close to the GNC wheels without contact. This also allowed for ease of access for wiring the radar components (2x coaxial cables) once in place. The two skeleton plates were mounted to the top and bottom surfaces of the chassis with four nuts and bolts respectively.

8.10.1.2 Electronics Sub-Skeleton

The electronics sub-skeleton is a smaller sheet of metal intended to house the driver circuit for the propulsion system, two transformers (5v-12v and 12v-8.2v respectively), the SD card reader, and thermocouple. This plate is intended to be easily removable to allow for maintenance on the sensitive components mounted to it. This feature proved to be crucial when each component housed on this board had different issues associated with them. Due to the modularity of this board, each damaged component was easily swapped out to allow for a rapid redesign of the power delivery system. Pictured below is the electronics sub-skeleton.

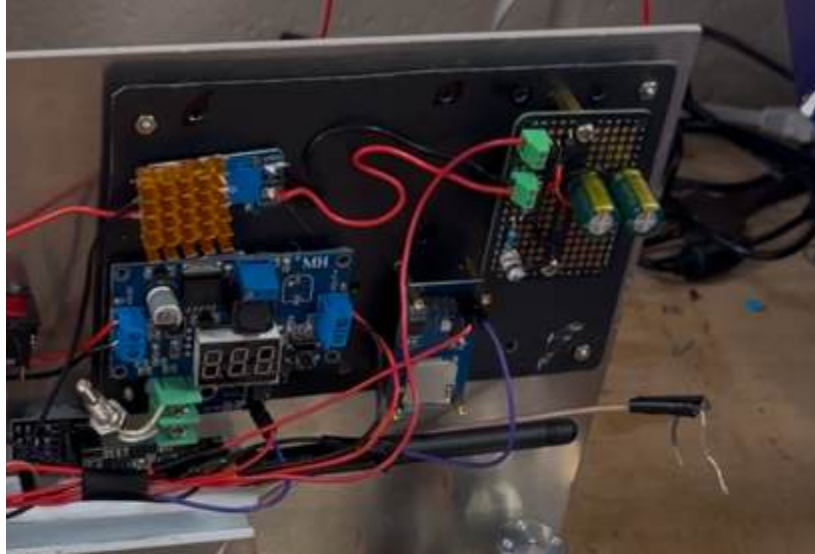


Figure 8.10.1.2.1: Electronics sub-skeleton.

8.10.1.3 Case mounted components

Not all components were easily mounted to the skeleton plates. Certain components had to be mounted directly to the case either out of necessity or convenience. These components include the propulsion solenoid, payload antennas, payload board, communications antenna, deployment rail, heater, and various switches for controlling each power branch. Pictured below are the mounting solutions for the solenoid and antennas.



Figure 8.10.1.3.1: Solenoid Mounting

The solenoid is directly mounted to the case using $\frac{1}{4}$ inch hex bolts. The entire valve needs to be disassembled and reassembled for mounting.

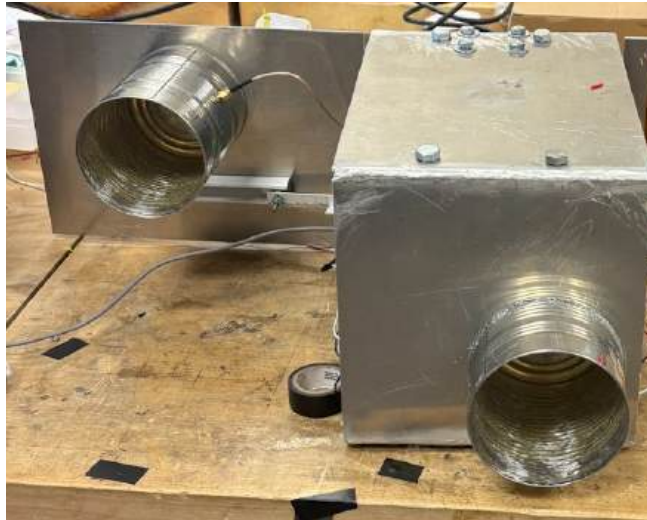


Figure 8.10.1.3.2: Cantenna Mounting

The transceiver antenna is fixed to the front of the satellite using hot glue, while the receiving antenna is fixed to the right deployed panel.

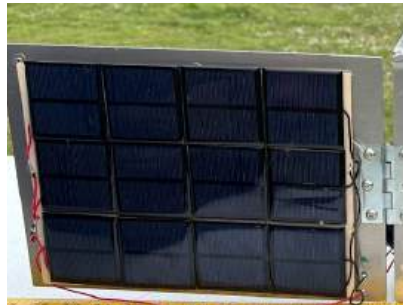


Figure 8.10.1.3.3: Solar Panel Mounting

The solar panels are fixed to a thin sheet of wood and then hot glued to the exterior deployed panels.

8.10.2 Final Weight

The final weight of the satellite was measured to be 6757.5 grams or approximately 15 lbs. This weight is compliant with the 12U form factor by weighing in at less than 24 kg or 2kg per U.



Figure 8.10.2.1: Final Weight

This completes the comprehensive breakdown of the systems integration of the SMART satellite prototype.

8.11 Summary Table

Part/Component	Objective	Fabrication Procedure	Manufacturing Process	Facility/Location	Equipment Required	Schedule
Chassis	Create durable frame that can withstand the forces of launch and translational movement during the mission life of the cubesat	Solidworks/CAD	Cutting, Drilling	Senior Design Lab	Saw, Drill Press	January 8-12
Solar Panels	Fabricate a solar panel power delivery system complete with MPPT.	Solidworks/CAD, Assembly	Cutting, Drilling, Soldering	Senior Design Lab	Soldering Iron, Drill, Saw	January 4-8
Propulsion	Create functional pressurized air feed. Outlet must be accelerated to above mach one.	Solidworks/CAD, Assembly	3D Printing, Hand Assembly	Senior Design Lab	3D Printer, Wrench	January 18-27
Power System	Model and fabricate a reliable circuit capable of	Solidworks/CAD, Assembly, Third	Cutting, Drilling, Soldering	Senior Design Lab	Soldering Iron, Drill, Saw	January 4-8

	delivering proper voltage to subsystems.	party PCB manufacturer.				
C&DH	Connect and mount microcontroller and communications systems.	Solidworks/CAD, Assembly	Cutting, Drilling, Soldering	Senior Design Lab	Soldering Iron, Drill, Saw	January 4-8
Ground Station	Create a location separate from the satellite to relay information to	Hand Assembly	COTS	Senior Design Lab	N/A	January 6-8
GNC	Create flywheels to attach on motors in order to create reaction wheels	Solidworks	3D Printing, Hand Assembly	Senior Design Lab	3D printer	January 12-15
Thermal	Create functional thermal management system	Hand Assembly	COTS	Senior Design Lab	N/A	January 17-31

9. Validation, Verification, and Testing

This section outlines the verification tests required to validate the design requirements of the satellite and ground station system, focusing on the communication capabilities over both S-Band and UHF Band, as well as the ability to develop a soil moisture map. Each test is linked to specific driving requirements, outlining the methodology and necessary resources.

9.1 Power VV&T

Design Req.	Verification	Objective	Method	Resources & Schedule	Success Criteria
Dr.7.1	Solar energy is stored in the power bank.	Ensure that solar panels can charge lithium cells	Exp.	Resources: Solar Charging System Schedule: 02/2025	Cells are charged by solar panels alone.
DR7.1.3	Power reaction wheels to do attitude adjustments with power bank supply.	Ensure reaction wheels can be powered.	Exp.	Resources: Reaction Wheels Schedule: 03/2025	Reaction wheels have enough power supplied to rotate satellite.
DR7.1.4	Perform maneuvers using a propulsion system.	Ensure the propulsion	Exp.	Resources: Propulsion	Propulsion actuates electronically.

		system can be powered.		system. Schedule: 02/2025	
DR7.1.6	Set a design temperature to trigger the thermal system electronically.	Ensure thermal electrical systems work as intended.	Exp.	Resources: Thermal System. Arduino Code Schedule: 02/2025	Heaters activate when design temperature is reached.

9.1.1 Test 1: Solar Panel Charging Test

Verification Test Objective: Validate that the satellite can be charged with solar energy.

Driving Requirement:

- **DR7.1:** The power demand shall be met using solar panels.

Method:

- **Experiment:** Set solar panels in the sun to charge for several hours

Facility/Location: Courtyard

Equipment/Machines required:

- Batteries
- Solar Panels
- Multimeter
- MPPT charging board

Schedule: Testing scheduled for February 2025, after solar panel construction.

Success Criteria: Solar cells increase in charge from solar energy alone.

Results

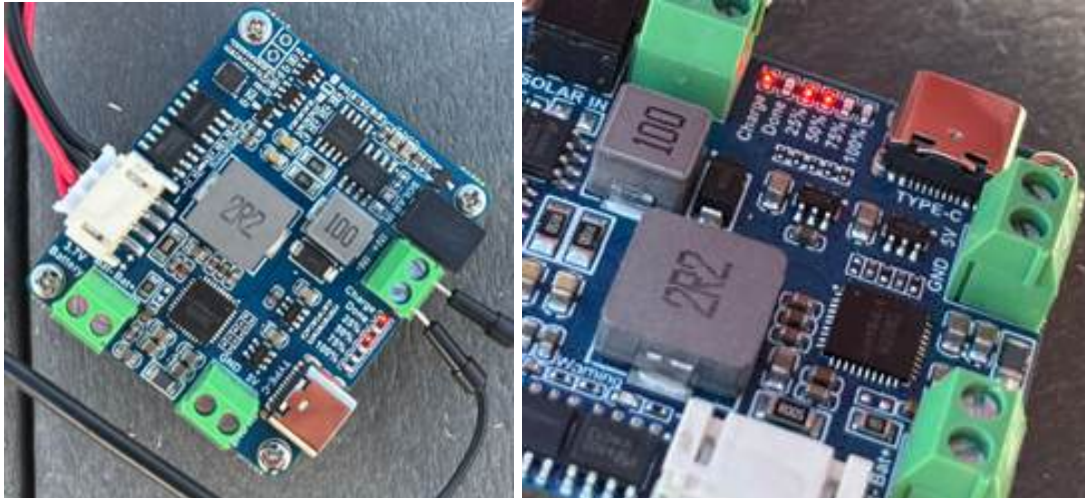


Figure 9.1.1.1: Solar Charging Indicator Increase

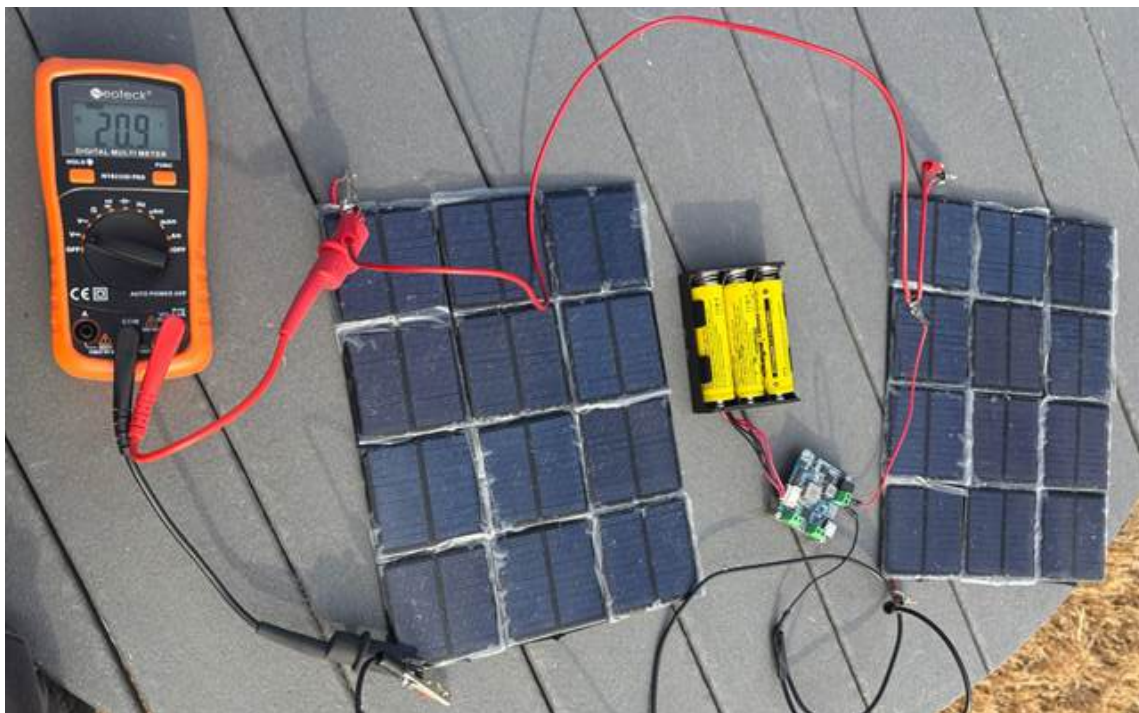


Figure 9.1.1.2: Solar Charging Setup and Regulated Voltage Input

Solar panels charged power cells successfully! Panels sat outside from 3:23 pm to 3:48 and went from 25% charged to 75% charged according to the MPPT board. Panels reached a maximum voltage of 24V unregulated and remained at a stable 20.9V while charging. Current could not be obtained while charging. Interestingly, a frequency of 7.2kHz was measured from the panels even though they should only output DC power.

Success Level

This test was incredibly successful, as the cells were charged far quicker than anticipated. However, they seem to follow a distinct curve where the cell's charging speed dramatically slows down to protect the batteries from overheating and popping. Cells never reached 100%. Also, the MPPT board should not be trusted for its accuracy in power capacity measurements. It is not likely that the batteries increased in charge by 50% in only 20 minutes. However, they could have increased by ~25% in this time but due to the board rounding to 25%, this was received as a 50% increase.

Conclusion

Panels are successful and warrant the remaining six extra solar panels to be integrated into the solar charging system due to input voltage regulation. In addition to measured success, the protective film on the solar panels will remain on until final integration with the satellite. Despite this film, the solar panels are still highly successful. System is ready for full integration.

9.1.2 Test 2: High Powered Relay Test

Verification Test Objective: Validate that has the power requirements to deliver power to the thermal system are met.

Driving Requirement

- **DR7.1.6:** The power bank shall supply enough power to the thermostat to the thermal system to trigger an active thermal management system.

Method

- **Experiment:**
- The thermocouple will be programmed to trigger the high power relay when a critical (low) temperature is reached.
- Once the critical temperature is met by the thermocouple (read through the serial out on Arduino IDE), the temperature of the heating pad will be measured.
- The thermocouple will then be heated up to show the relay turning off when heated beyond the critical temperature.

Facility/Location: Senior Design Lab

Equipment/Machines required:

- MAE AE Senior Design Room
- Laptop
- Arduino IDE
- Thermometer

Schedule: Testing scheduled for February 2025

Success Criteria: If the heater can be thermally switched on and off, the test is successful.

Results:

The thermal switching test was successful. The thermocouple was able to detect an accurate temperature change and switch the heater on. The heater was then able to get up to 200+ degrees

fahrenheit until being switched off when the thermocouple was heated back up. The system was powered by a fixed voltage power supply rather than the satellite power supply.

Success Level

Moderately successful. The satellite power supply system could not meet the power requirements necessary to meaningfully heat the heater pad. Instead, the fixed voltage power supply was used to ensure that the rest of the system was functional.

Conclusion

Due to the moderate success of the power relay switching system, a redesign of the power delivery system was necessary. After consultation, two possibilities were considered for meeting the power requirements. The first was tweaking the 12V rail until a functional configuration was achieved. The second avenue was having a serial lithium ion battery supply to meet the high voltage low current requirements. Wiring lithium batteries in series has the serious concern of possibly popping one of the batteries due to overcharging. Because of this, the 12V rail was adjusted until a successful iteration was found. The system is fully functional now. System is ready for full integration.

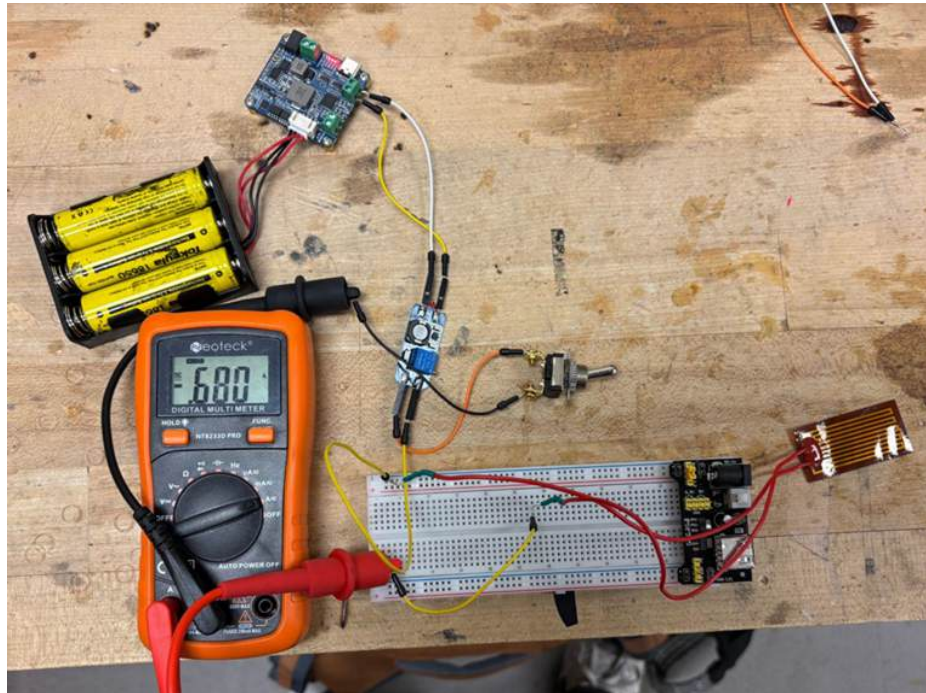


Figure 9.1.2.2: Successful Heater Test

9.1.3 Test 3: Propulsion Actuation Test

Verification Test Objective: Validate that the power requirements to deliver power to the thermal system are met.

Driving Requirement

- **DR7.1.6:** The power bank shall supply enough power to the thermostat to the thermal system to trigger an active thermal management system.

Method

- **Experiment:** The propulsion system will be switched on and off while being driven by the 12V rail from the power delivery system.

Facility/Location: Senior Design Lab

Equipment/Machines required:

- MAE AE Senior Design Room
- Power Delivery System
- Propulsion System

Schedule: Testing scheduled for February 2025

Success Criteria: Propulsion actuates electronically.

Results:

The test was a success. The solenoid actuated as expected while driven by 12V high power supply. Current was measured at 1.3A and input voltage was set to 12V. Total power consumed by the solenoid was 15.6W.

Success Level

Completely Successful. No issues in actuating the system whatsoever.

Conclusion

Power draw is unfortunately high with this system, driving the requirement for more power storage. Due to the success of the solar charging system, adding more power should be no issue. System is ready for full integration.

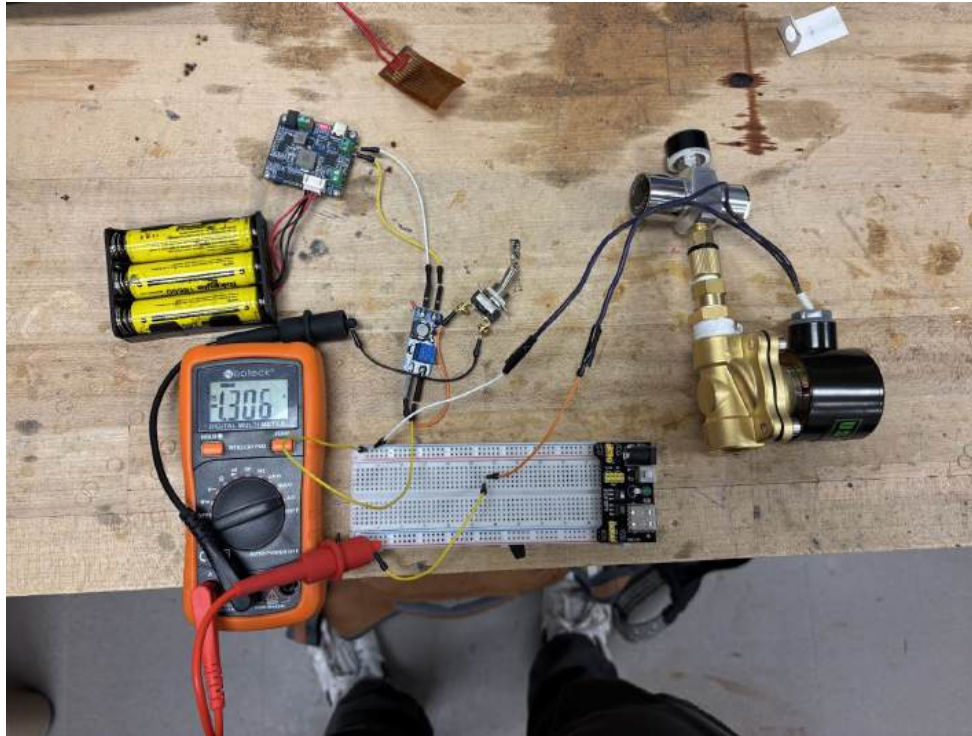


Figure 9.1.3.1: Successful Solenoid Test

9.2 Payload VV&T

Design Requirement	Verification	Objective	Method	Resources & Schedule	Success Criteria
DR4.4	Verify the efficient functionality of the cantennas at the design frequency of 2.4-2.5 GHz	Demonstrate the capability of the cantennas to output an RF signal at the design frequency efficiently.	Experiment	Resources: Cantenna Vector Network Analyzer Wire Cutters SMA Male-to-Male Cable Schedule: 02/2025	Average measured cantenna reflection coefficient is below -10 dB.
DR4.1	Verify the modulator circuit's ability to output a voltage ramp from 3.3 to 4.5 V with a ramp time of 20 microseconds	Demonstrate the ability of the modulator circuit to output a voltage signal in the right range and timing.	Experiment	Resources: Modulator Circuit DC Power Supply Oscilloscope Screwdriver Schedule: 03/2025	The modulator circuit outputs a voltage ramp from 3.3 to 4.5 V with a 20 microsecond leading edge.
DR4.4	Verify the cantennas can	Demonstrate the ability of	Experiment	Resources:	The measured

	be paired with a power coupling near the expected - 24 dB	the antennas to transmit and receive RF signals from each other efficiently.		Cantennas Vector Network Analyzer SMA Male-to-Male Cable Schedule: 02/2025	power coupling between the two antennas is near the expected value of -24 dB.
DR4.1 DR4.4	Verify the SAR's ability to detect small and large objects with a maximum range of 1 km	Demonstrate the ability of the SAR to scan objects of varying size and fidelity and to detect large objects at 1 km range.	Experiment/ Analysis	Resources: SAR Power Supply Ground Station Various Small Scanning Targets Rangefinder or Range Marked Test Area Schedule: 03/2025	All small objects are visibly detected in the produced .wav file and the general shape of the large target 1 km away is visible in the .wav file produced during the long range scan.

9.2.1 Test 1: Cantenna Calibration

Verification Test Objective: Verify that the cantennas can output and receive RF signals at the design frequency efficiently.

Driving Requirement:

- **DR4.1:** The radar system shall have a maximum range of twice the altitude of the CubeSat.

Method:

- **Experiment:** One completed cantenna will be connected to the VNA via an SMA cable and will have its reflection coefficient recorded before trimming its monopole wire and recording the new reflection coefficient until an average reading below -10 dB is recorded.

Facility/Location: ECE MakerSpace

Equipment/Machines required: Cantenna, VNA, SMA Male-to-Male Cable, Wire Cutters

Schedule: Testing scheduled for February 2025 after cantenna fabrication.

Success Criteria: The average measured reflection coefficient of the tested cantenna is below -10 dB.

Results

Both current iteration cantennas have been tested and trimmed, their average reflection coefficients are ~-12 dB and ~-13 dB respectively. Their minimums are -15 dB and -17 dB.

Success Level

The success level of this test is very high, without severely trimming the monopole, both cantennas had measured reflection coefficients well below the desired -10 dB.

Conclusion

The current iteration of cantennas can easily be trimmed to acceptable reflection coefficients. In order to improve the sharpness of the curve, some changes will be made to further antenna iterations such as using copper for the monopole and sanding the inner surface of the cans to remove films or coatings.

9.2.2 Test 2: Modulator Circuit Tuning

Verification Test Objective: Verify that the modulator circuit can output a voltage ramp from 3.3 to 4.5 V with a leading edge time of 20 microseconds.

Driving Requirement:

- **DR4.1:** The radar system shall scan a target area of 10 km².

Method:

- **Experiment:** The modulator circuit will be powered and given a +8.2 V reference voltage and its output will be connected to an oscilloscope to measure the voltage range and ramp time. The two potentiometers governing the voltage range and ramp time will be tuned until the desired values are reached

Facility/Location: Senior Design Lab

Equipment/Machines required: Modulator Circuit, DC Power Supply, Oscilloscope, Flat-Head Screwdriver

Schedule: Testing scheduled for March 2025, after modulator circuit assembly.

Success Criteria: The oscilloscope reads a voltage range of 3.3 to 4.5 V and a ramp time of 20 milliseconds.

Results

The circuit was able to be perfectly tuned using the two potentiometers and was set to the designed voltage range and ramp time with the appropriate reference voltage.

Success Level

The success level of this test is perfect, the tuning was successful at producing the exact voltage range and ramp time required for the rest of the subsystem.

Conclusion

No changes need be made to the modulator circuit or this testing procedure, should the circuit become out of tune, this procedure will be followed to re-tune it quickly.

9.2.3 Test 3: Transmitter/Receiver Pairing

Objective: Verify the antenna's ability to be paired with a power coupling near the expected - 24 dB.

Driving Requirements:

- **DR4.4:** The radar system shall have a maximum range of twice the altitude of the CubeSat.

Method:

- **Experiment:** Both completed antennas will be connected to the VNA and their power coupling will be measured when they are placed exactly 1 meter apart pointing at each other.

Facility/Location: ECE MakerSpace

Equipment/Machines Required: Antennas, VNA, SMA Male-to-Male Cable

Schedule: Testing scheduled for February 2025, following antenna fabrication.

Success Criteria: The power coupling of the two antennas can be reliably measured and is near the -24 dB expected value.

Results

The antennas were arranged as described in the test and their power coupling was measured to be ~-36 dB at 1 meter separation.

Success Level

Not successful. Although it was possible to reliably measure the power coupling in order to compute the antenna's gain, the measured value was far below the expected value.

Conclusion

Improvements must be made to the antenna design to increase their power coupling closer to the expected value. These improvements will include changing the monopole to copper wire, sanding the interior surface of the cans, attempting to procure longer cans, removing any lip from the inside of the opening of the cans to allow for maximum radio wave capture and transmit area, and fabricating cans with a more precisely placed bulkhead to ensure maximum signal boosting at the design frequency.

9.2.4 Test 4: SAR Object Detection and Maximum Range

Objective: Verify the SAR's ability to detect small and large objects with a maximum range of 1 km.

Driving Requirements:

- **DR4.1:** The radar system shall scan a target area of 10 km².
- **DR4.4:** The radar system shall have a maximum range of twice the altitude of the CubeSat.

Method:

- **Experiment/Analysis:** The payload will be tested on various objects of differing size, complexity and range in its SAR mode. This will be accomplished by taking readings every few centimeters over a 1 meter line perpendicular to the target and compiling the data at the ground station into a .wav file that can be viewed and analysed to determine if the target was detected.

Facility/Location: Centennial Campus/Lake Raleigh

Equipment/Machines Required: Payload, Communications System, Power System, Ground Station, Ruler, Laptop, Rangefinder, Small Targets

Schedule: Testing scheduled for March 2025, following each subsystem's full functionality tests.

Success Criteria: Each target is visibly detected in the produced .wav files and the large target placed at 1 km is both visibly detected and its shape can be discerned from the produced .wav file.

Results

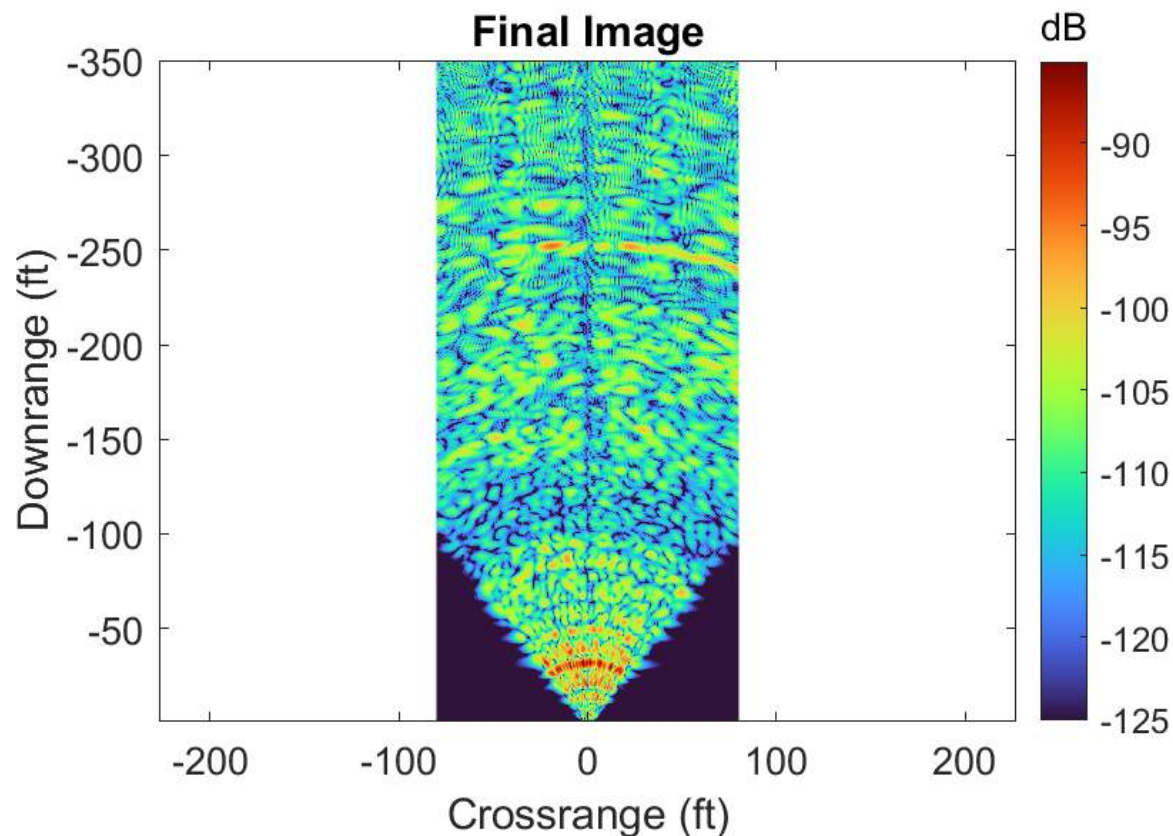


Figure 9.2.4.1: Produced SAR Image of a Treeline Approximately 250 ft Away.

A treeline was scanned and returned a recognizable image with two prominent clusters of foliage easily discernible in the image. The MATLAB code used to produce this image is located in the appendix. The software used to obtain the .wav file required for MATLAB processing was Audacity. The MATLAB

code was originally written by Gregory L. Charvat and a team from the 2011 MIT radar course. This code was modified for use with our radar system and updated to work with modern MATLAB processing software.

Success Level

Middlingly successful, although the maximum range was not confirmed, both a large object, the treeline, and a smaller object, individual tree clusters, were detected and visualized in the resultant image.

Conclusion

Improvements could be made to the existing payload subsystem via small changes to the modulator circuit or antennas, but the overall effectiveness of the payload cannot be improved significantly without considerable extra funding to procure components capable of producing and withstanding more powerful RF signals..

9.3 GNC VV&T

Design Requirement	Verification	Objective	Method	Resources & Schedule	Success Criteria
DR7.2	Validate that the satellite can maintain in Sun pointing condition.	Demonstrate the ability of the satellite to track light sources.	Experiment/ Analytical	Resources: Computer Senior Design Lab Schedule: 03/2025	If the SMARTSat maintains orientation at the light source.
DR9.1	Verify that the satellite mock reaction wheels will spin at differing rates, imposing a torque onto the system.	Demonstrate the ability of the satellite to maneuver autonomously using the reaction wheels.	Experiment	Resources: Senior Design Lab Schedule: 02/2025	If the controllers can spin different wheels at differing RPMs.
DR.9.3	Confirm that CubeSat can switch from using the Sun sensor to using an Earth sensor for Earth pointing.	Demonstrate the ability of the control system to switch from the Sun sensor to Earth pointing and maintain stability.	Experiment	Resources: Computer Senior Design Lab Schedule: 02/2025	Proving SMARTSat wheels will change RPMs when in different operational modes.

9.3.1 Test 1: Reaction Wheel Test

Verification Test Objective: Validate that the satellite can be controlled via reaction wheels. This includes validation that torques applied will be large enough to move a full weight satellite.

Driving Requirement:

- **DR9.1:** CubeSat shall maneuver via 3 axis reaction wheels

Method:

- **Experiment:** The satellite will be connected to a power source while the reaction wheels are connected to the frame. If the wheels spin fast enough, a small torque should be applied to the system.

Facility/Location: Senior Design Lab

Equipment/Machines required: Batteries, Suspension Device

Schedule: Testing scheduled for March 2025, after system integration and satellite calibration.

Success Criteria: If the system has torque applied that will be able to move the satellite while suspended, the satellite should orient itself to face towards the light.

Results

The reaction wheels do not produce enough torque to move the CubeSat as of this test. Increasing motor speed and increasing flywheel moment of inertia for the final prototype design.

Success Level

The success level is low, as the main goal was not achieved. The wheels were able to be controlled via the arduino and electric speed controllers, but the sizing is wrong.

Conclusion

The control subsystem will need to have upgrades to specific components. This includes higher voltage motors, heavier flywheels, and a transformer to step up voltage from 5 V to 9 V.

9.3.2 Test 2: Sun Tracking Test

Verification Test Objective: Validate that the satellite can track a light source via the on board sensors.

Driving Requirement:

- **DR7.2:** The GNC system shall be able to maintain sun pointing for charging orbits.

Method:

- **Experiment:** The satellite will be built with microcontrollers and sensors in place. A light source will be placed in front of the satellite. Check that the correct wheels spin when light is moving around the satellite.

Facility/Location: Senior Design Lab

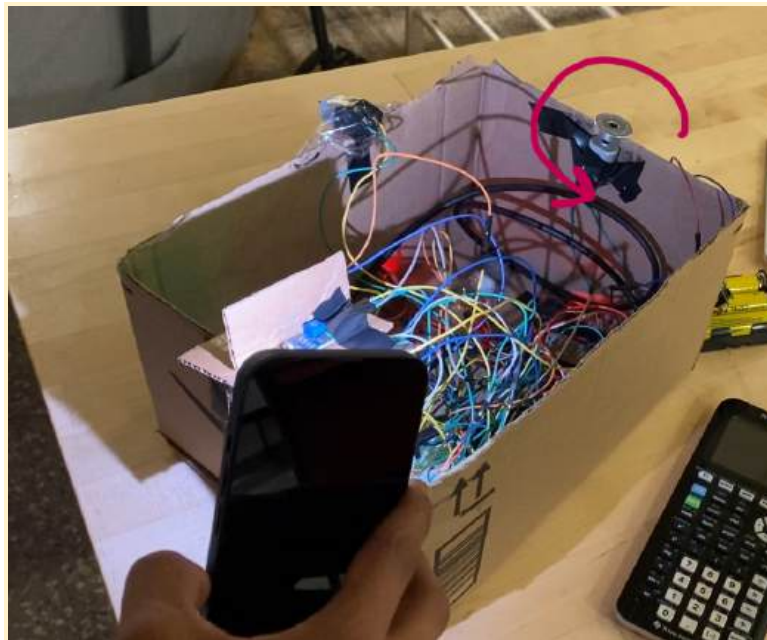
Equipment/Machines required: Light Source

Schedule: Testing happened in February 2025

Success Criteria: If the system has the correct wheels spinning when the light source is at different orientations, then the experiment was successful. If the wheels spin at a higher angular velocity the further out of frame the light gets, the experiment was successful.

Results

The light reception test was deemed successful, as the wheels reacted to a light source in the correct orientation. The expected response was for the reaction wheel to spin opposite to the torque needed for attitude correction. With the sensors properly wired, sending signals to the arduino, a simple script is uploaded to read light intensity as wheel speed commands. A light shined while the system remained stationary, the reaction wheel spun as expected. This test supported the attitude control for the yaw and pitch axes. With these results, we have validated the control logic and sensor integration demonstrating reaction wheel setup functions according to the test conditions. The next steps will involve refining the system accuracy and permanent mounting to the satellite frame. An image taken during the verification testing is shown below.



Success Level

The success level is moderate as all intended tasks, including reaction wheel spinning in response to light exposure was successful. The results could have been better, as the torque from these wheels were not large enough to shift the box, and the light sensing has a limited range.

Conclusion

The control system is deemed as autonomously capable of light pointing capabilities with limited range. Future improvements will focus on optimizing performance, refining control algorithms, and conducting more advanced tests under simulated space conditions.

9.3.3 Test 3: Earth Pointing Test

Verification Test Objective: Validate that the satellite can switch between two operational modes, light tracking and Earth tracking.

Driving Requirement:

- **DR9.3:** CubeSat shall point at desired area for SAR scan.

Method:

- **Experiment:** The satellite will be placed in Sun pointing mode with the light source in view. The satellite will first be placed in Sun pointing mode. When the light source is turned off it should go back to Earth pointing mode. The wheels should change velocity in order to point the correct face at the Earth.

Facility/Location: Senior Design Lab

Equipment/Machines required: Light Source

Schedule: Testing happened in February 2025

Success Criteria: When the operational mode switches, the RPMs of the mock reaction wheels should also change, proving that it switched modes.

Results

The control system code was updated to include the readings from the accelerometer. The algorithm was then set to maintain Earth pointing unless an outside light source was introduced into the environment. This allowed for testing of both systems functionality, along with demonstrating the ability to switch between modes. This test was exactly as expected, and results were very encouraging. When in Earth pointing mode, since the accelerometer outputs a vector quantity, reaction wheel speed is directly correlated to how far from “flat” the sensor is reading. This allows for rotor speed to be linearly mapped to maximum gravitational value, making this system robust.

Success Level

The success level is high, as all intended tasks, including switching to Earth pointing when no light was present, were completed successfully. The system is functioning as expected.

Conclusion

The control system is deemed capable of both light and Earth pointing modes, with capabilities of switching completely autonomously. These verification tests prove an encouraging step towards final design implementation.

9.4 Communications VV&T

Design Req.	Verification	Objective	Method	Resources & Schedule	Success Criteria
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DR11.3	Validate that the satellite can control GNC system remotely.	Demonstrate the ability for the satellite to be controlled via the GNC system.	Experimentally	Resources: Laptop Senior Design Lab Schedule: 03/2025	GNC can be controlled from the ground station.
DR11.2	Read SAR data from satellite and generate an image.	Demonstrate that the on board SAR will yield readable images.	Experimentally	Resources: Radar Laptop Schedule: 03/2025	SAR image is generated.
FR11	CubeSat will collect data using an On-Board Data Handling system (OBDH).	Demonstrate that on board communication systems will be able to save data	Experimentally	Resources: Laptop Schedule: 03/2025	File is generated and saved.
DR10.1	The radio communication system shall transmit data to the ground station.	Demonstrate that the communication system can transmit telemetry data to ground station	Experimentally	Resources: Radar Laptop Schedule: 02/2025	Telemetry is received by ground station.

9.4.1 Test 1: Save Data to SD Card

Verification Test Objective: Validate that the satellite can record data.

Driving Requirement:

- **FR11:** CubeSat will collect data using an On-Board Data Handling system (OBDH).

Method:

- **Experiment:** The satellite will record a test file to the SD card.

Facility/Location: Senior Design Lab

Equipment/Machines required:

- Laptop
- Arduino IDE

Schedule: Testing scheduled for February 2025.

Success Criteria: File is saved successfully without corruption.

Results

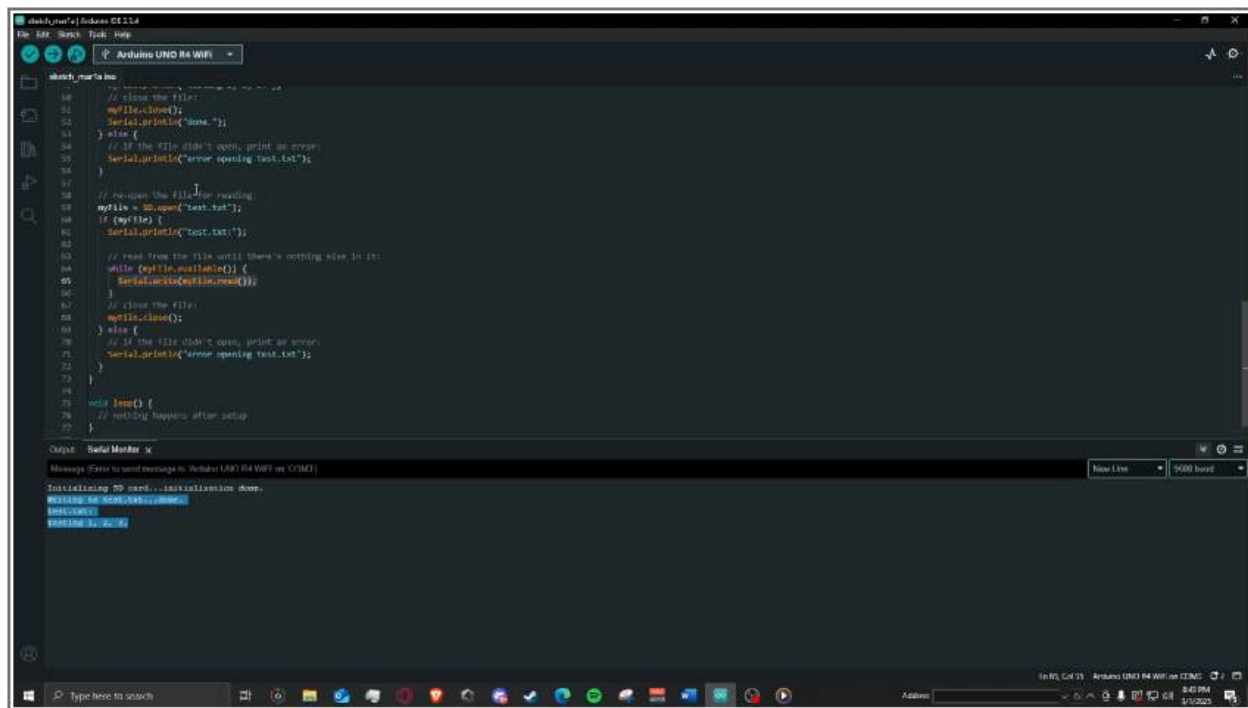


Figure 9.4.1.1: Successful Test 1.

File was generated and saved successfully. No errors or corruption visible on read.

Success Level

Incredibly successful. The system had no hang ups when generating the file. Wiring proved some slight challenges with familiarization with the Giga manual.

Conclusion

System is ready for integration.

9.4.2 Test 2: Telemetry Data Test

Verification Test Objective: Validate that the satellite can actively record telemetry data and send it to the ground station.

Driving Requirement:

- **DR10.1:** The radio communication system shall transmit data to the ground station.

Method:

- **Experiment:**
- A thermocouple will be wired to the arduino as well as the comms system.
- The temperature will be continuously recorded to a .txt file to act as the telemetry data.

- Upon request from the ground station, the telemetry .txt file will be sent to the ground station for viewing.
- To confirm the data was received, the ground station SD card will then be read and compared to the temp recorded by the satellite.

Facility/Location: Senior Design Lab

Equipment/Machines required: Laptop

Schedule: Testing scheduled for February 2025.

Success Criteria: If the file is received free of corruptions reliably, then the system is successful in verification.

Results

System sent file to ground station successfully and ground station received file successfully. System also functioned at multiple distances ensuring long distance communication is possible.

Success Level

Almost completely Successful. Some systems need to be tweaked for increasing reliability in signal transmission. Occasionally the system hangs up and requires a full reset. Issues must be resolved before full integration.

Conclusion

Due to minor issues in signal transmission, a possible redesign is being explored. An additional microcontroller for just communication may be necessary for seamless communication between the satellite and the ground station. System is very close to full integration.

9.5 Ground Station VV&T

Design Requirement	Verification	Objective	Method	Resources & Schedule	Success Criteria
DR6.1	Validate that the satellite can communicate data to the ground station for the creation of a soil moisture map.	Demonstrate the ability of the satellite to communicate data to the ground station to develop a soil moisture map.	Experiment/ Analysis	Resources: Radar Laptop Moisture Meter Schedule: 03/2025	A generally accurate soil moisture map is able to be constructed.
DR6.1.1	Verify that the satellite can communicate raw SAR data over S-Band to the ground station.	Demonstrate the ability of the satellite to communicate the raw SAR data over S-Band.	Experiment	Resources: Laptop Schedule: 03/2025	Raw data is received from the satellite using S-Band.
DR.11.*	Confirm that all telemetry data from	Demonstrate the ability of all telemetry data to be	Experiment/ Analysis	Resources: Laptop	All subsystems providing telemetry

	subsystems is communicated to the ground station over UHF Band.	communicated to the ground station over UHF Band.		Schedule: 03/2025	data are accounted for in data collection using UHF Band.
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9.5.1 Test 1: Soil Moisture Map Development

Verification Test Objective: Validate that the satellite is capable of communicating data to the ground station for constructing a generally accurate soil moisture map.

Driving Requirement:

- **DR6.1:** Demonstrate the ability of the satellite to communicate data to the ground station to develop a soil moisture map.

Method:

- **Experiment:** A field experiment will be conducted where data from the satellite's soil moisture sensors is transmitted to the ground station. The data will then be processed to construct the soil moisture map.

Facility/Location: Senior Design Lab, Public Park

Equipment/Machines required: Radar, Laptop, Moisture Meter

Schedule: Testing scheduled for March 2025 after satellite construction and initial system setup.

Success Criteria: A generally accurate soil moisture map must be generated from the data transmitted by the satellite. The accuracy of the map will be verified by cross-referencing with known soil moisture values from ground-based sensors.

Results

After performing the simulation the prototype code is not performing as expected. This code is very rudimentary and assumes certain types of data files. Once the payload system is complete further progress will be able to be made. Currently the code is overlaying a mock soil moisture data plot onto an image of North Carolina. This will be wildly different then the map that will be generated in the final stage of prototyping. This is the output of the current code structure:

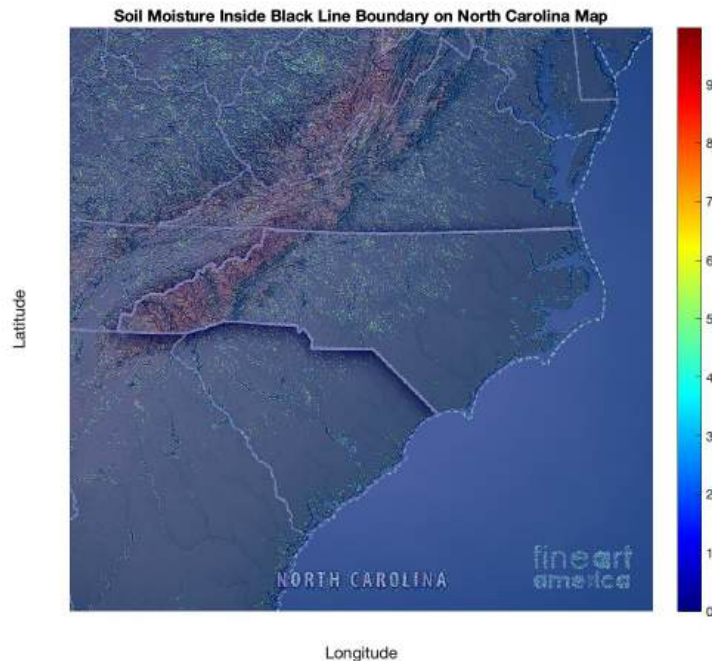


Figure 9.7.5.1.1: Soil Moisture Map

Success Level

This is not correct SAR data and the current code is generally following water veins rather than live data. However, the code is able to generally replicate a map that shows where the soil is highly saturated. Sensitivity must be increased to produce a more useful map.

Conclusion

Although the source code is in its beginning stages, the generated map seems to follow areas with high soil moisture levels. The data file types and other aspects of the data analysis are subject to change. Due to these changes, success may alter in the future iterations of the source code.

9.5.2 Test 2: Communication & Data Storage Test

Verification Test Objective: Validate that the satellite can transmit raw Synthetic Aperture Radar (SAR) data over S-Band to the ground station.

Driving Requirement:

- **DR6.1.1:** Demonstrate the ability of the satellite to communicate the raw SAR data over S-Band.

Method:

- **Experiment:** The satellite will be activated to transmit SAR data over the S-Band frequency. The ground station will receive and log the raw data to verify its integrity.

Facility/Location: Senior Design Lab, Public Park

Equipment/Machines required: Laptop

Schedule: Testing scheduled for February 2025, after system integration and satellite calibration.

Success Criteria: Raw SAR data should be received clearly with no significant data loss, and it should match the expected format for analysis. The integrity of the transmitted data will be verified through comparison with pre-flight data samples.

Results

The ground station successfully received and saved all data from the CubeSat's OBDH System. The system performed without any errors, and the data was correctly stored in the designated data storage system.

Success Level

The success level is high, as all intended tasks, including receiving and saving the data, were completed successfully. The system is functioning as expected.

Conclusion

The ground station has proven capable of handling the CubeSat's data transmission. The system is now prepared for future expansions, such as requesting additional data from the CubeSat as needed.

9.5.3 Test 3: CubeSat Initiation & Telemetry Return Test

Objective: Verify that telemetry data from all subsystems is communicated from the satellite to the ground station over the UHF Band.

Driving Requirements:

- **DR.11.*:** Demonstrate the ability of all telemetry data to be communicated to the ground station over UHF Band.

Method:

- **Experiment/Analysis:** Telemetry data from all satellite subsystems will be transmitted to the ground station over UHF Band. The data will be analyzed to ensure that all subsystems are accounted for and their data is accurately captured.

Facility/Location: Senior Design Lab, Public Park

Equipment/Machines Required: Laptop

Schedule: Testing scheduled for January 2025, following the satellite's construction and system verification.

Success Criteria: All telemetry data from subsystems should be successfully received at the ground station with no significant loss of information. Data consistency will be verified by

cross-checking with subsystems onboard the satellite. If possible over S-Band, it is reasonable to assume it is possible over UHF band.

Results

The ground station successfully turned on and off the required systems, including the LED, which was properly initialized. The system performed the necessary initialization without issues.

Success Level

The success level is high, as the system met its goal of controlling the activation and deactivation of the LED and related subsystems.

Conclusion

The ground station is functioning as expected and is ready for further extension to support turning on/off other subsystems as needed. The system code is prepared for future expansions.

9.6 Thermal and Survivability

Design Requirement	Verification	Objective	Method	Resources & Schedule	Success Criteria
DR8.4 DR8.5	Verify that the heater system will be able to add heat in order to maintain internal temperatures	Demonstrate the ability of the satellite to determine actions relating to temperature regulation	Experiment	Resources: Computer Senior Design Lab Schedule: 02/2025	If the heater can have a high enough temperature, the test is successful
DR8.4	Verify that the thermocouple circuit can read temperatures	Demonstrate the ability of the satellite to read temperature	Experiment	Resources: Computer Senior Design Lab Schedule: 02/2025	If the thermocouple can read temperature and convert to standard temperature units, test is successful

9.6.1 Test 1: Heat Addition Test

Verification Test Objective: Verify that the heater system will be able to add heat at certain temperatures

Driving Requirement:

- **DR8.3:** CubeSat should survive temperatures ranging from -150°C to 150°C.

- **DR8.5:** CubeSat shall maintain a temperature range of 0°C to 50°C to maintain operating temperature range

Method:

- **Experiment:** The heater will be connected to a voltage source. The voltage source will be turned on and the temperature will be measured using a IR camera

Facility/Location: Senior Design Lab

Equipment/Machines required: IR Camera, power supply

Success Criteria: If the heater system can add heat to the system when hitting a certain temperature range and try to keep that temperature

Results: Temperature for heater exceeds expectations and shows we can use lower voltage or the current voltage is fine since the temperature of the heater reaches 300 F. Heater is adequate heat flux source

9.6.2 Test 2: Thermocouple Circuit Test

Verification Test Objective: Verify that the thermocouple is able to read temperatures and can convert voltage difference to F or C

Driving Requirement:

- **DR8.4:** CubeSat shall be able to read temperatures internally to determine what actions to take to cool off or heat up the CubeSat.

Method:

- **Experiment:** The thermocouple will be attached to a circuit that connects thermocouple and arduino. The arduino will be hooked up to a computer to show its reading temperature

Facility/Location: Senior Design Lab

Equipment/Machines required: Laptop, Arduino, Thermocouple

Success Criteria: If the thermocouple circuit can read temperatures to a certain accuracy, the test is successful.

Results: The thermocouple and arduino can read temperatures and convert them to standard units successfully

9.9 Propulsion VV&T

Design Requirement	Verification	Objective	Method	Resources & Schedule	Success Criteria
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DR1.1 DR1.2 DR2.2	Validate that it is possible to fit enough propellant into the CubeSat in order to deorbit (move 100 km)	Prove that the propulsion system both fits physically and weight wise into the requirements while still maintaining deorbit capability.	Experiment/ Analysis	Resources: - A computer that can run ANSYS - ANSYS - An anemometer or pitot probe - EB3. Schedule: 02/2025	Propellant required fits and is underweight.
DR7.1.4	Validate that the power supply is enough for the propulsion system to actuate.	Prove that the propulsion system fits under the power budget of the CubeSat.	Experiment	Resources: - Power meter Schedule: 02/18/2025	Power consumption of the propulsion system doesn't exceed supply (accounting for other systems)
DR1.1 DR1.2 DR2.2 DR7.1.4 (If it fails no requirements met)	Validate structural integrity of the high pressure propulsion system.	Prove that the propulsion system can function under the loads of launch and its own pressure.	Experiment/ Analysis/ Qualitative	Resources: - Functional - Electrical System - ANSYS - All of the measurement tools from other propulsion tests Schedule: 02/18/2025	System does not fail. No structural failure occurs.

9.9.1 Test 1: Propulsion Feasibility Test

The CubeSat shall maintain operational status for 4 years, but the total combination of units shall be 12U, or 20x20x30 cm, and the cubesat mass shall be under 24 kg. These goals can conflict with each other. A test is needed in order to verify that it is in fact possible to meet both requirements at the same time. The CubeSat shall contain enough propellant to reduce the mission life to 4 years. The amount of propellant shall not exceed an amount of space that would make the system too large. The mass of propellant shall not make the CubeSat exceed its weight maximum.

9.9.1.1 Test Design

- Test 1 will be conducted using experimental and analytical testing methods.

9.9.1.2 Procedure

1. While the parts are being delivered, the orbit of the CubeSat shall be simulated in ANSYS. This will show the mission lifetime at the chosen altitude along with the height trajectory of the CubeSat.

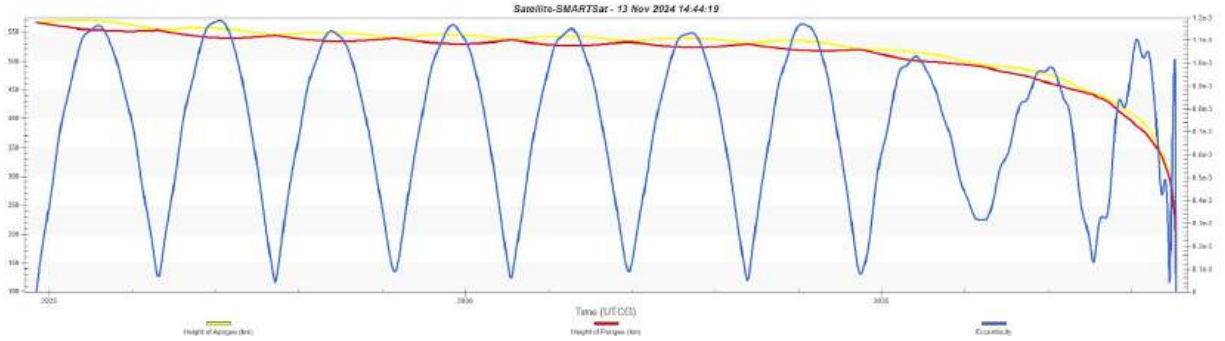


Figure 9.7.9.1.2.1 Orbital Simulation Graph

2. Using the information from the simulation, a height change for a 4 years lifetime shall be acquired.
3. Using calculations, a required deltaV shall be acquired.
4. The mass of all ordered parts shall be measured.
5. The assembled propulsion system shall be tested by reading the dial on the pressure regulator. These findings will be used to determine ISP.
6. Using previously acquired information, the required mass of propellant shall be calculated.
7. The assembled CubeSat dimensions shall be measured (minus the pressure tank).

9.9.1.3 Required Facilities/Equipment/Tools/Software

- A computer that can run ANSYS will be required.
- ANSYS will be required.
- An anemometer or pitot probe will be required.
- This test will be performed in EB3.

9.9.1.4 Expected Results

- The propulsion system will be feasible

9.9.1.5 Success Criteria

- The required propellant can fit inside the prototype CubeSat.

9.9.1.6 Results

- Test has been removed from the team's plans due to other more pressing systems and deadlines.

9.9.2 Test 2: Systems Interaction Test

The CubeSat shall be powered using solar energy. The power demand shall be met using solar panels to provide 100Wh. The power bank shall supply enough power to the propulsion system to perform maneuvers in orbit. This means that the propulsion system needs to be efficient enough to not eat into other systems' power demand.

9.9.2.1 Test Design

- Test 2 will be conducted using experimental testing methods.

9.9.2.2 Procedure

- After the propulsion system is constructed, the power demand of operation shall be measured.
- This shall be done by measuring the power demand of pulse operation and steady state operation, as outlined in the literature review section.
- The power demands will be compared to the power supply and other systems power demands.

9.9.2.3 Required Facilities/Equipment/Tools/Software

- A power supply will be used to supply and measure demands.
- This test will be performed in EB3, most likely inside of the aerospace senior design lab.

9.9.2.4 Expected Results

- The propulsion system will meet power consumption requirements.

9.9.2.5 Success Criteria

- The power consumption of the propulsion system is low enough that it doesn't eat into other systems power demands.

9.9.2.6 Results

- Power requirements are 15 W, 12 V, 1.4 A. This is much too high for what we would like, but it is acceptable given operating conditions (these will only be used on their own).

9.9.3 Test 3: Structural Integrity Test

While the design requirement tests are important in their own right, it is also important to ensure that the system will not fail under the stresses of the mission. This test is designed in order to ensure that the propulsion design solution can function under the stresses required. A failure would imply an inability to perform, thus causing failure of design requirements.

9.9.3.1 Test Design

- Test 2 will be conducted using experimental, analytical, and qualitative testing methods.

9.9.3.2 Procedure

- Cycling tests require the system to be placed under a cyclical load. The team is unable to perform this test as proposed in literature due to the single use nature of CO₂ cartridges. Instead, the solenoid will be actuated and unactuated to mimic this test as closely as possible. This will continue until the CO₂ cartridge is depleted.
- After this test the system will be turned on and will be inspected for leaks.
- A visual inspection will follow this, looking for potential cracks and failure points.
- After these tests, the system will be run normally to ensure no damage has been inflicted.

- This will be followed by final measurements and inspections.

9.9.3.3 Required Facilities/Equipment/Tools/Software

- A working electrical system is required to actuate the propulsion system.
- ANSYS will be required for any simulations.
- Any equipment from other propulsion tests will be required for final damage testing.
 - A power supply is needed.

9.9.3.4 Expected Results

- The propulsion system design solution survives stress testing.

9.9.3 Success Criteria

- The propulsion system functions at similar ISP or pre-nozzle pressure as before testing.
- No material failure occurs.

9.9.4 Results

- The system originally had leaks, though these have been dealt with. Metal glue is being used to seal the leaking pipe fittings.
- No damage has been incurred.

9.10 Structures VV&T

Design Req.	Verification	Method	Resources & Schedule	Success Criteria
DR5.1	Conduct bending moment test on 30 cm strut at a strain rate of 10mm/min to determine the force that causes failure	Experimental/& analytical	Resources: Structures lab with bending moment machine Schedule: 02/2025	If the component doesn't show indications of bending from 511.74 N force (equivalent 5g force) then test is successful

	due to bending			
DR5.1	Conduct bending moment test on 20 cm strut at a strain rate of 1mm/min to determine the force that causes failure due to bending	Experimental/& analytical	Resources: Structures lab with bending moment machine Schedule: 02/2025	If the component doesn't show indications of bending from 511.74 N force (equivalent 5g force) then test is successful
DR5.1	Run ansys program to simulate bending moment forces on frame to ensure that previous tests are accurate	Simulation	Resources: Computer with Ansys Schedule: 02/2025	If simulation shows similar force value of bending moment failure as experimental tests, then simulation is successful

9.10.1 Test 1: 20 cm strut bending moment test

The cubesat frame shall be able withstand loads of at least 5G's and the cubesat frame shall maintain its structural integrity for the expected orbital period of 4 years. Tables describing the various parameters of aluminum alloy 6061 have already been provided. Due to this, it is of the utmost importance to find the force that causes bending moment failure for the 20cm component and ensure it is higher than the 5G force.

9.10.1.1 Test Design

- Test 1 will be conducted using experimental, analytical, and qualitative testing methods

9.10.1.2 Procedure

- Two 20 cm aluminum alloy 6061 ¼” struts will be tested for bending moment failure at a **strain rate of 1mm/min**
- Data will be recorded from both tests to ensure that a similar pattern occurs

9.10.1.3 Required Facilities/Equipment/Tools/Software

- Structures testing lab
- Bending moment testing machine

9.10.1.4 Expected Results

- The 20 cm strut will be able to withstand the 5G force and not fail due to bending

9.10.1.5 Success Criteria

- 20 cm strut survives the 511.74 N force (5G equivalent with estimated cubesat weight of 23 lb)

9.10.1.6 Results

- Both trials showed that the 20cm strut can survive the calculated **511.74N** force during launch. More specifically, in **trial 1**, bending started to occur at **527.3 N** and in **trial 2**, bending started to occur at **513.4 N**. Furthermore, **trial 1** displayed a flexure stress value of **431.469 MPa** and **trial 2** displayed a value of **439.436 MPa** at the first indication of bending, which serves as supplemental data for future calculations.

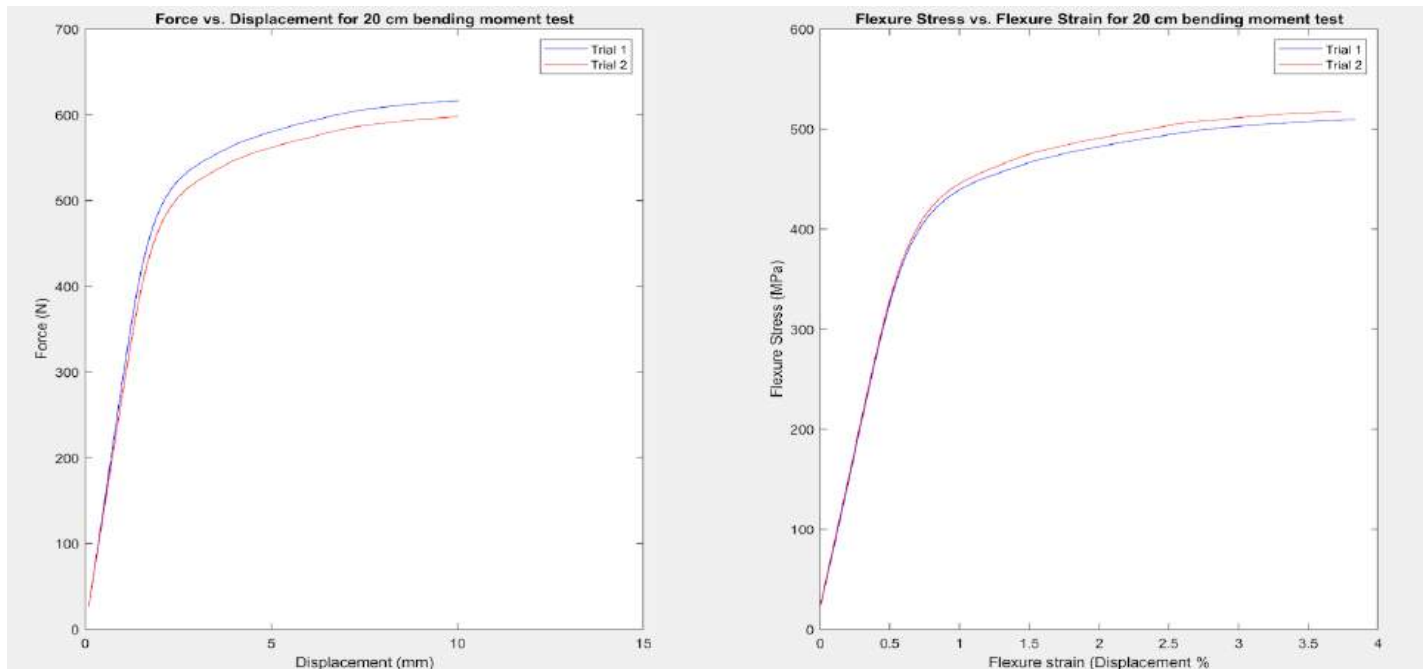


Figure 9.10.1.6.1: Test 1 results

9.10.1.7 Conclusion

- Test was successful and frame is currently being welded, allowing for other subsystems to attach or place components inside frame

9.10.2 Test 2: 30 cm strut bending moment test

The cubesat frame shall be able withstand loads of at least 5G's and the cubesat frame shall maintain its structural integrity for the expected orbital period of 4 years. Tables describing the various parameters of aluminum alloy 6061 have already been provided. Due to this, it is of the utmost importance to find the force that causes bending moment failure for the 20cm component and ensure it is higher than the 5G force.

9.10.2.1 Test Design

- Test 2 will be conducted using experimental, analytical, and qualitative testing methods

9.10.1.2 Procedure

- Two 30 cm aluminum alloy 6061 1/4" struts will be tested for bending moment failure at a **strain rate of 10mm/min**
- Data will be recorded from both tests to ensure that a similar pattern occurs

9.10.1.3 Required Facilities/Equipment/Tools/Software

- Structures testing lab
- Bending moment testing machine

9.10.1.4 Expected Results

- The 30 cm strut will be able to withstand the 5G force and not fail due to bending

9.10.1.5 Success Criteria

- 30 cm strut survives the 511.74 N force (5G equivalent with estimated cubesat weight of 23 lb)

9.10.1.6 Results

- Both trials showed that the 30cm strut can't survive the calculated **511.74N** force during launch. More specifically, in **trial 1**, bending started to occur at **330 N** and in **trial 2**, bending started to occur at **334.4 N**. Furthermore, **trial 1** displayed a flexure stress value of **397.751 MPa** and **trial 2** displayed a value of **427.829 MPa** at the first indication of bending, which serves as supplemental data for future calculations.

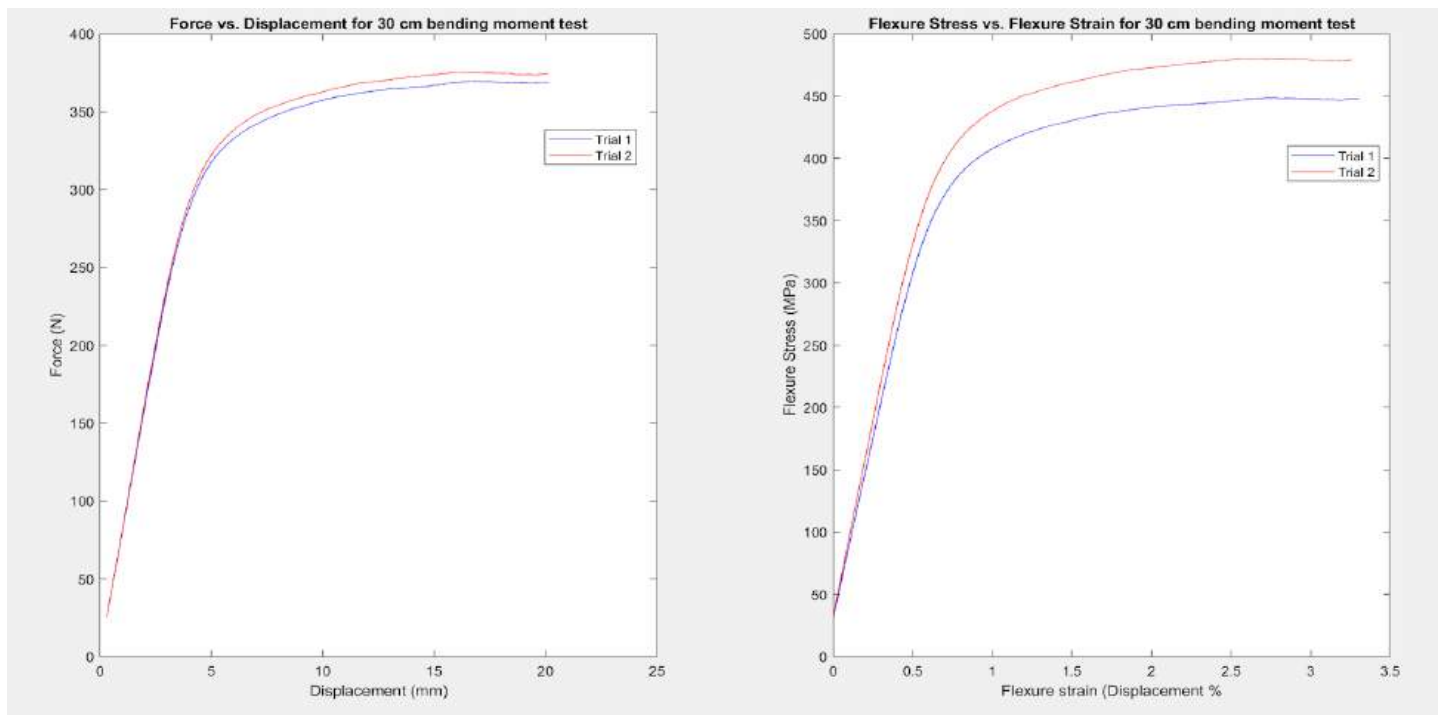


Figure 9.10.1.6.1: Test 2 results

9.11 VV&T Summary Table

Test Name	Testers	System	Status
Soil Moisture Map Development	Zumi Cuji	Ground Station	Complete: Almost Successful
S-Band Raw Data Transmission	Zumi Cuji	Ground Station	Complete: Successful
UHF Band Telemetry Data Communication	Zumi Cuji	Ground Station	Complete: Successful
Propulsion Feasibility Test	Zachariah Moore	Propulsion	Removed from team plan. (Proved inconsequential)
Propulsion System Interactions Test	Zachariah Moore, Waylon Ponder	Propulsion	Complete: Successful
Propulsion Structural Integrity Test	Zachariah Moore, Waylon Ponder	Propulsion	Complete: Successful
Heat Addition Test	Dylan Perdue, Zach Moore, Waylon Ponder	Thermal	Complete: Successful
Thermocouple Circuit Test	Dylan Perdue, Zach Moore, Waylon Ponder	Thermal	Complete: Successful
20 cm strut bending moment Test (strain rate 1mm/min)	Jamison Fuller, Sam Martinez, Dr. Henry Ware	Structures	Complete: Successful
30 cm strut bending moment Test (strain rate 10mm/min)	Jamison Fuller, Sam Martinez, Dr. Henry Ware	Structures	Complete: Almost successful
Ansys bending moment simulation Test	Jamison Fuller	Structures	Incomplete: To be completed by 3/18/2025
Photoresistor Light Reception test	Jack Lamont	GNC	Complete: Moderately Successful
Multi-pointing	Jack Lamont	GNC	Complete: Successful

Earth/light control system			
SD Card Basic Read Write Test	Waylon Ponder	Command and Data Handling	Complete: Successful
Telemetry Data Transmission Test	Waylon Ponder, Zumi Cuji	Command and Data Handling	Complete: Successful
High Powered Relay Test	Waylon Ponder, Dylan Perdue	Power/Thermal	Complete: Moderately Successful (required redesign, now fully functional).
Solar Charging Test	Waylon Ponder	Power	Complete: Successful
Propulsion Actuation Test	Waylon Ponder, Zach Moore	Power/Propulsion	Complete: Successful
Cantenna Calibration	Samuel Martinez	Payload	Complete: Successful
Transmitter/Receiver Pairing	Samuel Martinez	Payload	Complete: Unsuccessful
Modulator Circuit Tuning	Samuel Martinez	Payload	Complete: Successful
SAR Detecting and Maximum Range	Samuel Martinez, Waylon Ponder, Zumi Cuji	Payload, Communications, Power, Ground Station	Complete: Moderately Successful

10. Risk Management

Risk management is a crucial aspect of ensuring project success, particularly in aerospace and technical missions, where the stakes are high, and uncertainties can have significant consequences. Effective risk management involves identifying potential risks, evaluating their likelihood and severity, and implementing strategies to mitigate or control them. By proactively addressing risks, the project can reduce the impact of unforeseen events and maintain progress toward mission objectives.

This section outlines the major risks associated with the project, categorized into schedule, cost, safety, and technical risks. Each risk is evaluated based on its likelihood of occurrence and the severity of its potential consequences. A mitigation plan is proposed for each

risk to either minimize the likelihood of occurrence, reduce the severity of the consequences, or both. Finally, a risk matrix is presented to visually assess the risk landscape and prioritize mitigation efforts.

10.1 Risk Description

The identified risks range from delays in project timelines and budget overruns to technical failures and manufacturing hazards. Each risk is carefully described to ensure a clear understanding of its nature and impact on the project. For example, schedule risks might delay mission readiness, while technical failures could jeopardize the satellite's functionality during or after launch.

10.2 Likelihood/Frequency & Severity/Consequence

Risks are rated on two scales:

1. Likelihood/Frequency - The probability of a risk occurring, rated from 1 (low likelihood) to 3 (high likelihood).
2. Severity/Consequence - The impact of the risk if it occurs, rated from 1 (low impact) to 5 (catastrophic impact).

By evaluating risks on these scales, the project team can prioritize mitigation efforts and focus resources on the most critical threats.

Table 10.2.1: Risk Identification Information

Risk	Risk Type	Call Sign	Likelihood	Consequences
Failure to meet project/mission deadlines	Schedule	S-1	3	3
Project exceeds provided budget	Cost	C-1	3	4
Non-life threatening dangers during manufacturing process	Danger	D-1	2	5
Catastrophic structural failure during launch	Technical	T-1	3	5
Space debris destroys satellite	Technical	T-2	2	5
Power system fails prematurely	Technical	T-3	3	5
Thermal system control failure	Technical	T-4	3	5
Single event latchup	Technical	T-5	2	5

Pressurized feed system failure	Technical	T-6	1	3
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10.3 Mitigation Plan

To effectively address the identified risks, a comprehensive mitigation plan has been developed. This plan outlines specific actions to reduce the likelihood of risks occurring, minimize the severity of their consequences, or both. Below are the detailed mitigation strategies for each risk:

- 1. Failure to meet project/mission deadlines (S-1)**

To ensure timely completion of the project, team-specific deadlines will be set for each phase. These internal milestones will help maintain consistent progress and avoid delays.

- 2. Project exceeds provided budget (C-1)**

A price-performance optimization process will be performed to identify areas where resources can be allocated most efficiently. By prioritizing high-value tasks and mitigating overspending, the consequences of budget overruns will be minimized.

- 3. Non-life-threatening dangers during manufacturing (D-1)**

To ensure a safe manufacturing process, all team members will wear proper personal protective equipment (PPE). Additionally, comprehensive training on the safe use of machinery and tools will be provided to reduce risks further.

- 4. Catastrophic structural failure during launch (T-1)**

Extensive testing of the chassis will be conducted to determine its maximum force-handling capabilities. By pushing the structure to its limits during controlled tests, the risk of failure during launch can be mitigated, and design adjustments can be made to improve resilience.

- 5. Space debris destroys satellite (T-2)**

The target altitude for the satellite will be carefully selected based on mission requirements and an analysis of space debris at that altitude. This approach ensures the satellite operates in a safer orbital environment, reducing the likelihood of debris impact.

- 6. Power system fails prematurely (T-3)**

Multiple safeguards will be implemented to ensure power system reliability. These include watchdog circuits, surge protection, and redundant power delivery mechanisms. If the budget allows, a backup power source will be integrated to provide an additional layer of security.

- 7. Thermal system control failure (T-4)**

The thermal control system will undergo rigorous testing in extreme environmental conditions to simulate worst-case scenarios. A high factor of safety will be incorporated into the temperature control system design to mitigate risks of failure.

- 8. Single event latch-up (SEL) (T-5)**

To address potential SEL issues in the electronic systems, guard rings will be integrated around sensitive circuit areas, and the circuit layout will be optimized. Additionally, dedicated latch-up current limiting circuits will be included to detect and respond to excessive current draw caused by SEL events.

- 9. Pressurized feed system failure (T-6)**

The pressurized feed system will be tested extensively before use in the actual

CubeSat. By conducting thorough pre-use testing, potential failures can be identified and mitigated in a controlled environment, ensuring they do not impact other systems.

This mitigation plan combines proactive measures and rigorous testing to ensure that risks are effectively managed throughout the project lifecycle. This mitigation plan is subject to change as testing is performed and additional risks are identified.

Table 10.3.1: Risk Mitigation Plan

Call Sign	Risk Mitigation Plan	Likelihood	Consequences
S-1	Team specific deadlines will be set to maintain project progress goals.	1	3
C-1	Price-performance optimization will be performed to determine where most of the budget should be allocated to. Over budget amount will be mitigated, less consequential	1	3
D-1	Proper PPE will be worn. Proper and safe use of machines will be practiced.	1	3
T-1	Chassis will be thoroughly tested and pushed to its limits to determine max capabilities. Structural maximum force capability will mitigate degree of failure.	2	3
T-2	Target altitude will be determined with mission requirements as well as space debris collection at that altitude.	1	5
T-3	Implementation of watchdog circuits, surge protection, and redundant power delivery for the system. Possible inclusion of backup power source should the budget allow it.	1	4
T-4	Extremely rigorous testing of the system in extreme environments. Implement a very rigid and high factor of safety temperature control regiment.	1	5
T-5	Possible integration of guard rings around sensitive circuit areas, optimizing circuit design with proper layout techniques, and incorporating dedicated latch-up current limiting circuits to detect and react to excessive current draw caused by SEL events	1	4
T-6	Pressure feed system will be tested before use in the actual CubeSat. This ensures that failures do not happen near other systems. The actual effect of a failure stays the same.	1	2

10.4 Risk Matrix

The following matrix provides the determined likelihood and severity of each risk to provide visualization of the current risks. This allows for easier understanding of what risks are to be considered more thoroughly. The colors are directly correlated with how much of a threat the risk is to the team and mission, green being the least and red being the most threatening. The first figure is the risk matrix before mitigation plans are in place. Which is then followed by the risk matrix after mitigation plans are in place.

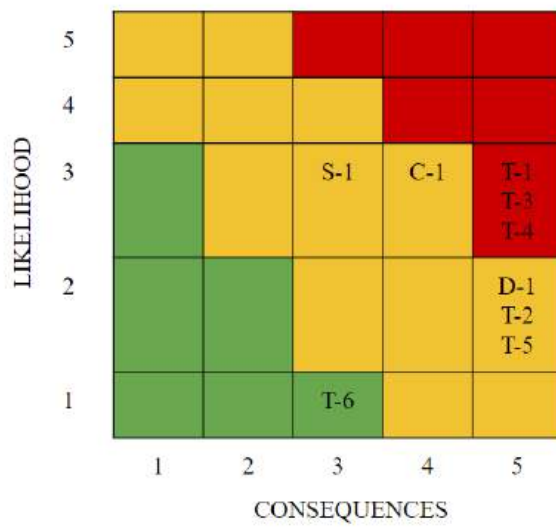


Figure 10.4.1: Pre-Mitigation Risk Matrix

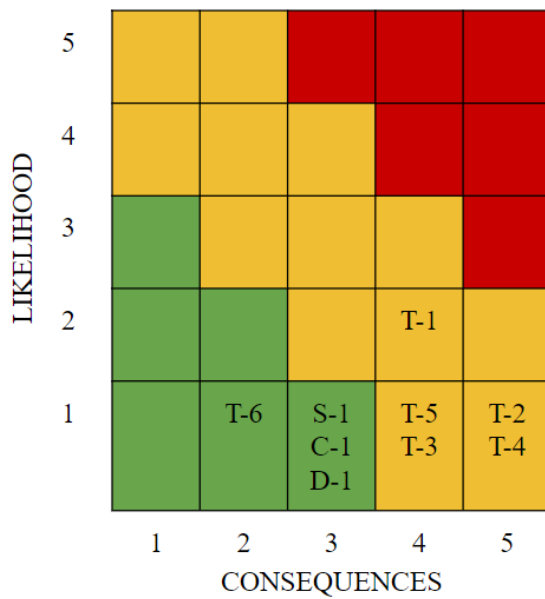


Figure 10.4.2: Post-Mitigation Risk Matrix

This allows for the easy identification of risks moving from dangerous to the mission to considerably safe or inconsequential. This is shown by the migration of risks from red/yellow areas of the matrix to the green/yellow areas.

10.5 Risk Realization

A lot of the risks were prominent in the fabrication stage of our project, more specifically, the power, propulsion, and payload subsystems. We realized immediately that if wires were not properly connected or if the GIGA was blown, then none of the internal components would function, which would ultimately not allow the cubesat to carry out its mission. To mitigate this, our Power lead was able to do extensive research to ensure that this setback would not happen. The structures test had another risk associated with it because if the struts failed significantly below the 5G load, new ones would have to be made, which could push the welding of the frame back further than anticipated. To mitigate this, more sheets of aluminum alloy 6061 were purchased so that new struts could be cut immediately to help save time. One final prominent risk associated with our project was falling behind on deadlines. To prevent this from happening, our team manager drew up deadlines each week so that the team could stay on track as a whole and not fall behind in the fabrication stage of this project. For thermal, one component would overheat very quickly and needed to be cooled down in order to prevent failure of the propulsion system.

A good number of the expected risks never made any problems for the team, while unexpected risks found ways to cause problems. In our project, there were a large number of electrical shorts. These caused damage to parts that were not anticipated to due the team's overall lack of electrical experience, along with some of the demanding circuits causing issues. The propulsion system had unexpected issues with size constraints which made the team make an executive decision to curve the system to ensure it could fit. This issue was caused by a last minute change in parts, but ended without causing irreversible damage to the project itself. Another issue was actually testing the SAR. The initial testing site had too much wifi interference which caused the SAR to create a faulty image. Another test location was selected and the test was completed successfully.

11. Project Management

Table 11.1: Organizational Chart

Team Members	
Dylan Perdue Thermal Lead Financial Lead	Jamison Fuller Structures & Mechanism Lead Test/Safety Lead
Samuel Martinez Payload Lead	Zachariah Moore Propulsion Lead

CAD/Simulation Lead	Manufacturing Lead
Zumi Cuji Project Manager Ground Station Lead	Jack Lamont Guidance Navigation and Control Lead Systems Engineering Lead
Waylon Ponder Power/Electronics lead Data and Communication lead	

11.1 Gantt Chart

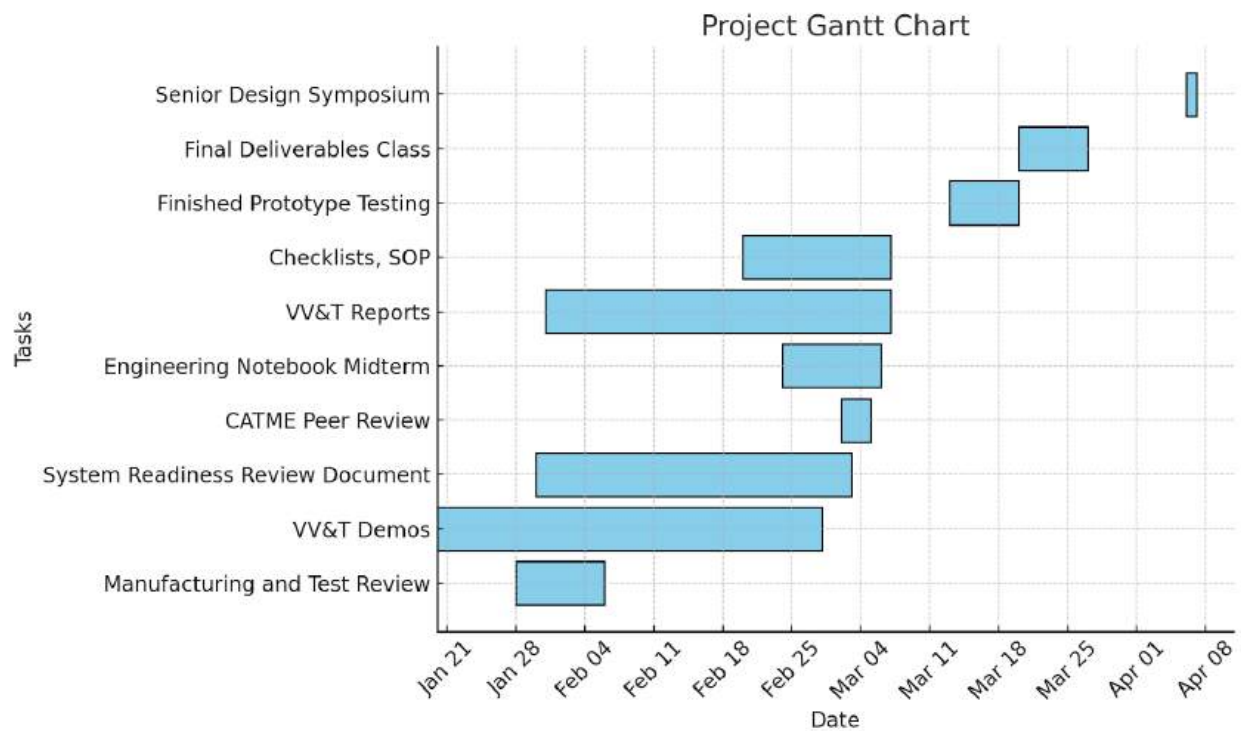


Table 11.1.1: Current Gantt Chart

11.2 Timeline

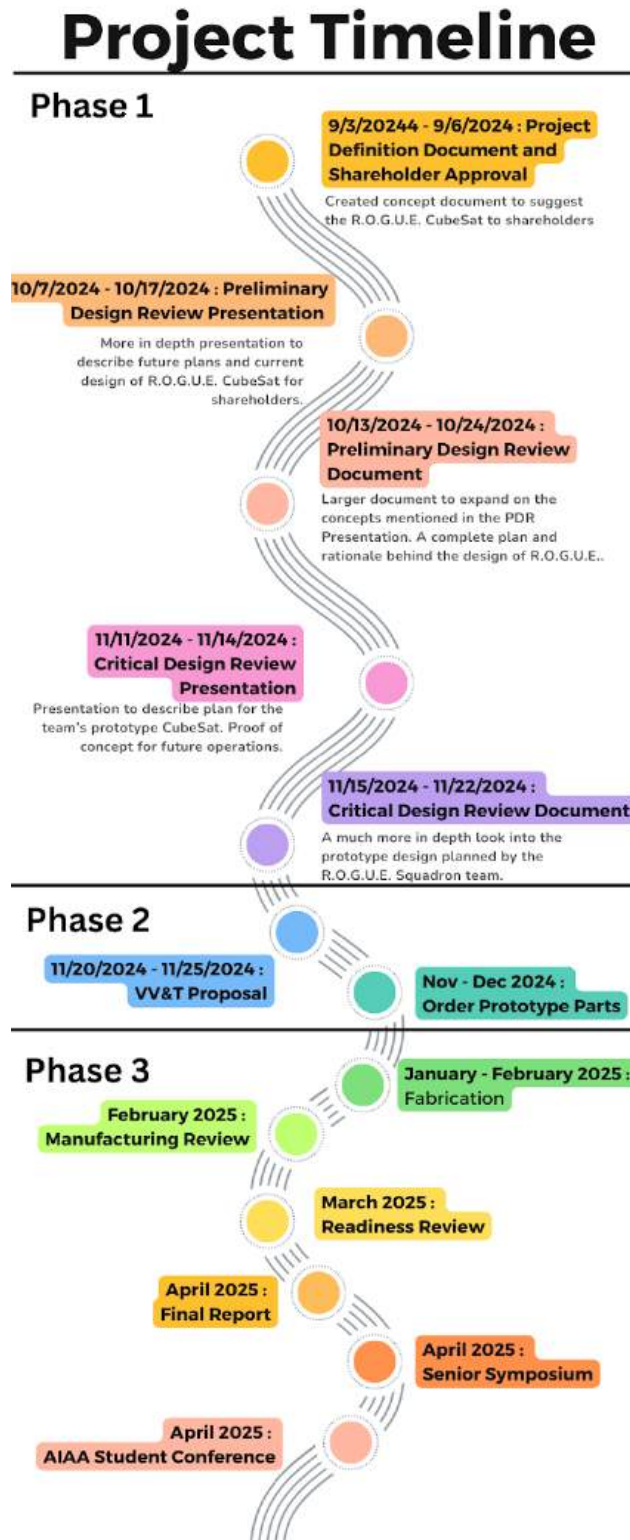


Figure 11.2.4: Project Timeline

11.3 Tasks/Activities

11.3.1 Manufacturing and Test Review

The manufacturing review describes preliminary design solutions on a subsystem level, as well as outlining any changes that have been made during the beginning stages of manufacturing. This presentation gives an update on manufacturing progress as well as verification test status.

11.3.2 System Readiness Review Report

The SRR examines, tests, demonstrations, analyses, and audits that determine the system's readiness for a safe and successful flight or launch and for subsequent flight operations. It also ensures that all flight and ground hardware, software, personnel, and procedures are operationally ready.

11.3.3 VV&T Report

Outlines the verification tests for the official “for the record” testing performed on a system or element to show that it meets its allocated requirements or specifications including physical and functional interfaces. Verification tests use instrumentation and measurements and are generally accomplished by engineers, technicians, or operator-maintainer test personnel in a controlled environment to facilitate failure analysis.

11.3.4 Vehicle Inspection Checklist, SOP

The vehicle inspection checklist outlines the standard operational procedures along with pre-flight requirements for the whole system. This includes breaking down each specific subsystem's pre flight routine.

11.3.5 Final Report

Comprehensive document with all phases of the spring semester capstone project.

11.4 Project Phases

11.4.1 Phase 1 (Fall 2024)

Table 11.5.1.1: Phase 1

Task	Timeline
Project Definition Document and Shareholder Approval	9/3/2024 - 9/6/2024

Preliminary Design Review Presentation	10/7/2024 - 10/17/2024
Preliminary Design Review Document	10/13/2024 - 10/24/2024
Critical Design Review Presentation	11/11/2024 - 11/14/2024
Critical Design Review Document	11/15/2024 - 11/22/2024

11.4.2 Phase 2: (Fall 2024)

Table 11.5.2.1: Phase 2

Task	Timeline
VV&T Proposal	11/20/2024 - 11/25/2024
Order Prototype Parts	Nov - Dec 2024

11.4.3 Phase 3: (Spring 2025)

Table 11.5.3.1: Phase 3

Task	Timeline
Fabrication	January - February 2025
Manufacturing Review	February 6, 2025

Readiness Review	March 3, 2025
Final Report	April 18, 2025
Senior Symposium	April 7, 2025
AIAA Student Conference	April 18, 2025

11.5 Work Breakdown Structure

The work breakdown was a mutually understood distribution of work and assignments relating to each member's subsystems and administrative roles. The following breakdown reflects the general distribution of work among the team. Some assignments required collaboration of the entire team while others required solo or small group collaboration. Assignments where collaboration of the entire team was required indicates each member completed respective sections relating to their technical and administrative responsibilities.

Table 11.5.1: Work Breakdown Summary

Assignment	Members	Finish Date
Parts Pickup & Organization	Zumi Cuji & Waylon Ponder	1/13/2025
UNP Proposal Draft	SMART Squadron	1/17/2025
Manufacturing & Test Review Presentation	SMART Squadron	2/6/2025
Subsystem Manufacturing	SMART Squadron	N/A
VV&T Demonstrations	SMART Squadron	2/28/2025
System Readiness Review	SMART Squadron	3/3/2025
Flight Test	SMART Squadron	3/31/2025
Poster	SMART Squadron	4/2/2025
Senior Symposium	SMART Squadron	4/7/2025
Final Report	SMART Squadron	4/17/2025

AIAA Report	SMART Squadron	4/18/2025
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11.6 Test Plan

Test	Date Conducted	Facility Used	Securral of use
Bending moment Test on 30 cm strut	February 2025	Dr. Ware's Structures lab	Emailing Dr. Ware for permission
Heat Addition Test	February 2025	SD space lab	N/A
Thermocouple Circuit Test	February 2025	SD space lab	N/A
Systems Integration Test	February 2025	SD space lab	N/A
Structural Integrity Test	February 2025 (partially) April 2025 (completely)	SD space lab	N/A
Solar Panel Charging Test	February 2025	SD space lab	N/A
High Powered Relay Test	February 2025	SD space lab	N/A
Cantenna Reflection Coefficient Test	February 2025	ECE Makerspace	Via ECE Makerspace Training
Modulator Circuit Tuning	February 2025	ECE Makerspace	Via ECE Makerspace Training
Cantenna Power Coupling Test	February 2025 (partially) March-early April 2025 (completely)	ECE Makerspace	Via ECE Makerspace Training
SAR imaging Test	April 2025	Outside	N/A
Data & Communication Storage Test	February 2025	SD space lab	N/A

CubeSat Initiation Test	February 2025	SD space lab	N/A
Sun Pointing Test	February 2025	SD space lab	N/A
Multi-Pointing Test	February 2025 (on rough model) March-April 2025 (on finalized cubesat model)	SD space lab	N/A
Data Storage Test	February 2025	SD space lab	N/A
Telemetry Data Test	February 2025	SD space lab	N/A

12. Budget Breakdown

12.1 Real World Cost

Costs are important when looking over any engineering projects. The cost breakdown of each subsystem is essential to see if we are over budget or if there is any room for improvement or adjustment to give the costs to other systems. It would be good practice to have about 5% of the budget free in order to give us room for other unforeseen problems. Table 10-1 shows the different parts' generic costs based on current price estimates.

Table 12.1.1: Table of Cost Estimates

Component	Price (\$)
Cold Gas Propulsion	45,000
Aluminum 6061	190.77
Lithium Ion Battery Pack	44,300

Payload	21750
Manufactured Solar Panel	20,000
Thermal Coating	2000
Electrical Heater	100
Thermostat	1500
Sensors	11800
Reaction Wheel (x4)	28000
Hybrid Ground Station	25000
UHF & S-band Transceiver	17400
Antenna	6100
On-Board Computer	5400
Total	228540.77

12.2 Budget Overview

It is also important to have a breakdown of the costs based on subsystems to see where all of our costs are currently going towards to see if those subsystems have cheaper, effective solutions or to get an idea of where our costs are generally going to go for this project. It is important to note that unless a manufacturer is contacted, the costs are either extreme or very broad. Figure 9-2 shows the percentage of the total costs that go into each subsystem.

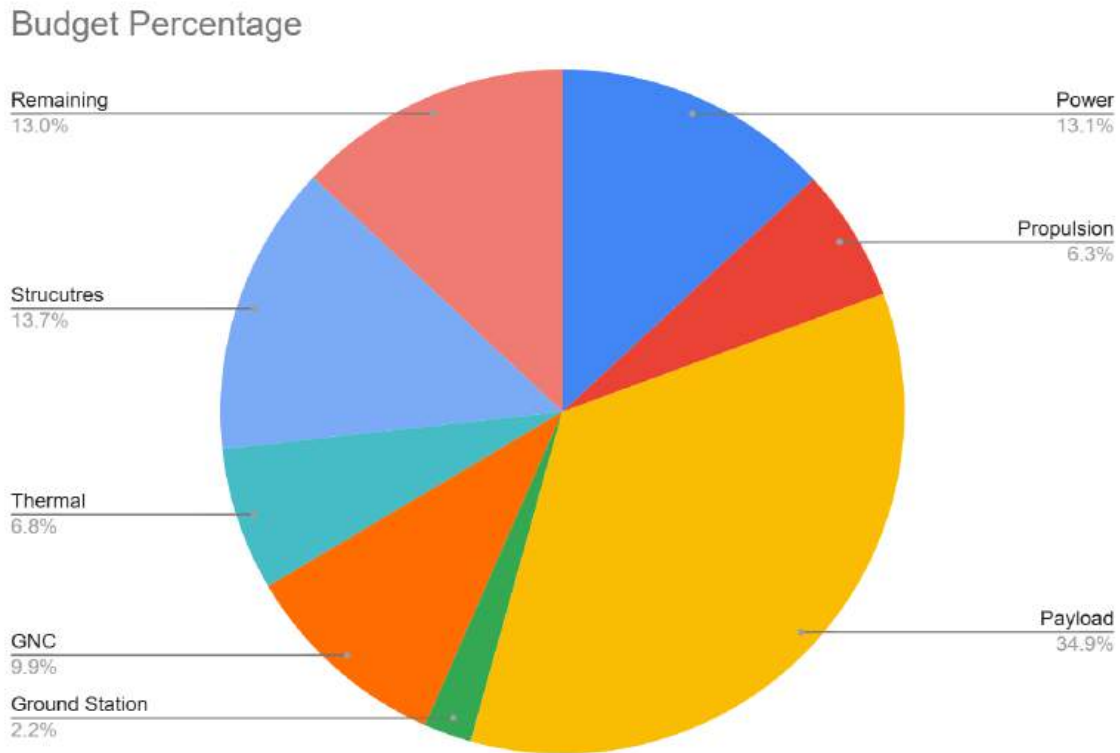


Figure 12.2.1:: S.M.A.R.T. CubeSat Project Budget Pie Chart

12.3 Prototype Budget

The scaling down of costs for prototype testing is essential and we must be careful with spending too much money on the scale downed products. Some subsystems will have the same product from prototype to final design. The cost breakdown of each subsystem is essential to see if we are over budget or if there is any room for improvement or adjustment to give the costs to other systems. It would be good practice to have about 5% of the budget free in order to give us room for other unforeseen problems. Table 10-3 shows the different parts' generic costs based on current price estimates.

Table 12.3.: Table of Prototype Estimates

Component	Price (\$)
Structures	307.74
Power	331.09
Ground Station	74.95
Thermal	127.62
Payload	689.7
Propulsion	151.83
GNC	200.50
Total	1892.43

12.4 Prototype Budget Overview

It is also important to have a breakdown of the costs based on subsystems to see where all of our costs are currently going towards to see if those subsystems have cheaper, effective solutions or to get an idea of where our costs are generally going to go for this project. It is important to note that unless a manufacturer is contacted, the costs are either extreme or very broad. Figure 10-4 shows the percentage of the total costs that go into each subsystem.

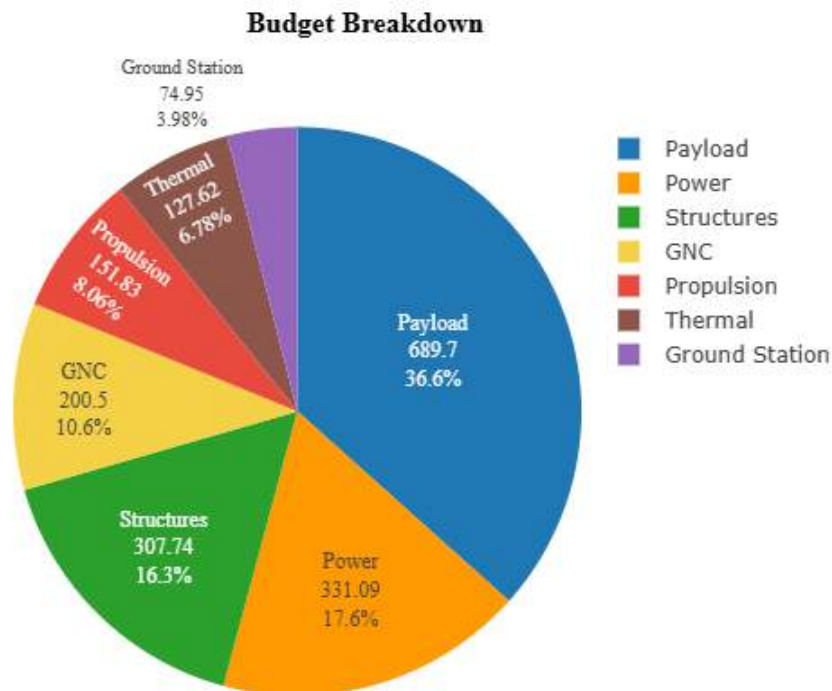


Figure 12.4 Prototype Budget Breakdown

12.5 Benefit to Customer

Compared to cost estimates in the PDR, we are now \$12000 under budget compared to \$3000, so we have greatly improved since then. Our current solution is much cheaper than current solutions when compared to SMAP and Sentinel 1a. When considering the prototype budget, we are currently \$390 under budget, so we have room to provide more materials to do vigorous testing if needed. The CubeSat will measure soil moisture concentration before a disaster in order to prevent flooding and help disaster planners with aiding in reduction of damage. The CubeSat should be able to show flooding after as well in order to help with search and rescue operations. Outside of disaster relief, the DoD can use the CubeSat in order to help them navigate terrain in order to prevent vehicles from becoming stuck and costing manpower and potential losses if the vehicle is damaged enough.

Compared to current solutions, the CubeSat should be able to have higher frequency scanning where SMAP takes about 2-3 days to complete world wide mapping and Sentinel 1a takes about 12 days which caused it to miss hurricane Helene. Also, it would take time for world wide scanning, so we want to have more specific target areas to get data back much quicker than current solutions.

13. Lessons Learned

Zumi Cuji

I learned how to manage and get engineers to collaborate to produce a functional final product. I also learned an incredible amount about electrical development, something that will most definitely aid in my future career as an engineer. I wish I would have known how much work each subsystem lead would generally be taking on. This would allow for more even distribution of work and more efficient development of the product. For example, the power system was an immense amount of work that was hard to collaborate on.

Dylan Perdue

I learned how to take conceptual concepts we have learned throughout our classes and try to apply them for real-world applications. I have learned how important simulations are in the design process since they provide a groundwork that we then compare to actual experiments. It was unfortunate no actual vigorous thermal testing could be done due to lack of thermal vacuum testing. I have also learned how important it is to be well-rounded as an engineer especially today with how the power lead had a lot of work to do due to lack of knowledge of electrical systems but some subsystem leads.

Jamison Fuller

I learned about the engineering process as a whole because we were tasked with brainstorming an idea that could ultimately benefit society and then were tasked to fabricate the design solution. With the numerous due dates associated with senior design, I also learned how to manage my time more effectively because I had to balance this course load with other classes for two semesters. This project has also allowed me to become more comfortable with using Solidworks and making models because I had to replicate the structural design in CAD so that the TA's in the MECH E shop could weld it properly. Lastly, I also gained collaboration experience by working with my group members who were all in charge of different subsystems and relaying information to them so that our assembly process would be effortless.

Zachariah Moore

I learned valuable team planning and scheduling skills. My propulsion subsystem led me to learn about common thread types for piping and how to navigate standards for use in engineering and construction. Integrating the propulsion system taught me some electrical skills that I picked up from the electrical lead. One of the major things that I learned over this project was the importance of an idea and how you pitch it. It was something our team actually struggled with for the early portion of the project, until settling on this concept. I found that it can be very beneficial to make plans such that there is a backup in case something goes wrong. Notably, it was difficult to shift direction on our original design concept.

Samuel Martinez

At the beginning of this semester I knew only the basics of radar engineering and the electrical engineering surrounding the circuits required to make a functioning radar. Over the course of these two semesters I have not only improved my theoretical knowledge of radar,

specifically SAR technology, but also my practical knowledge of the nature of radar systems as a whole. Beyond my subsystem, I learned how important it is to properly communicate with other teams and engineers to ensure successful integration of individual subsystems. I learned how much planning, preparing, and presenting must be done to deliver a complicated product to investors even before prototyping begins. I learned the limitations and difficulties of certain CAD software in presenting ideas and design choices. Most of what I learned and applied to this design process regarding radar was acquired from the MIT OpenCourseWare class on making a coffee can radar system [PAY50].

Jack Lamont

Over the course of the past two semesters, I have learned valuable lessons in team building/communication and technical engineering. Throughout each phase of this assignment, the subsystem I was involved with required extensive collaboration with team members. Specifically, the power lead and I had to stay in touch throughout the whole project in order to confirm the correct voltage and current would be supplied. I learned a lot about electrical engineering and basic circuitry in order to create my subsystem in the prototyping phase. I also learned that engineering design is an iterative process, for multiple reasons. It is impossible to consider every factor before design, therefore it is important that each component can be changed/modified later. From power draw issues to it simply won't fit, things never go to plan, and solving these problems quickly and effectively while under time restrictions is what makes good engineers.

Waylon Ponder

I started my senior year with a limited knowledge of electronic components and circuitry in general. Now, not only do I have a much greater knowledge on these topics, but I have applied experience in designing such systems and a new-found love for electronics. I went from failing to solder strain gauges in my structures lab to hand soldering our entire wiring system from scratch with included voltage transformers, 3 different voltage rails, and a driver circuit for an irrigation solenoid. I've also learned so much about how important it is to have a strong plan going into any project. No matter what you are designing or building, you need a solid foundation before you can make any progress. The more hurdles you can predict, the easier you can avoid them. Back up plans are crucial too. I never tackled a part of the project without several back up designs in mind in case things went wrong. Given that nearly everything did go wrong, I was right to over-prepare. Multiple systems needed complete redesigns on the fly, and I always had a failsafe. This lesson will follow me for the rest of my life. You always need a back-up to your back-up.

14. Individual Contributions

Zumi Cuji

I was in charge of leading the team through the completion of all assignments, presentations, etc. I ensured the team had a solid foundation to develop documents and their subsystems. Like normal, I set up this document and directed the team what needed to be done to reach completion. I aided in the development of everyone's subsystem by guiding them through

absolute necessities and errors. I aided Waylon in the completion of the final project, as this portion required wiring and systems integration. I am solely responsible for the development of the ground station subsystem.

Dylan Perdue

I was in charge of leading the thermal subsystem and finance. I was responsible for making sure the team didn't go over budget and made sure everyone updated the sheet to make sure all costs were accounted for. The team didn't go over budget thanks to the ease of access of the spreadsheet. I also collaborated with power lead multiple times to ensure the electrical components wouldn't overheat and asked questions about what types of thermal management he thought would be appropriate to ensure electronic safety.

Jamison Fuller

I oversaw the mock up and fabrication process of the cubesat frame during the entirety of this project. This included making calculations for our factor of safety, the center of mass, as well as the 5g equivalent load that the cubesat will experience during launch. I also performed tests on the struts to find what force caused bending moment failure to better analyze whether design alterations needed to be made to further strengthen the frame. Furthermore, I created the CAD model of the frame so that it could be used to help the TA's see where the welds needed to go to connect the various pieces as well as serve as a visual representation of what our frame would look like for the other subsystems.

Zachariah Moore

I was in charge of the manufacturing and the propulsion system. This means that I designed all of the testing for the propulsion system, along with the design itself. This process involved CAD modelling the nozzle, selecting pipe threadings that are compatible, and picking a suitable fuel source. Prior to that, it involved sorting through a large selection of propulsion systems and choosing the best one for our project. The manufacturing ended up more hands off than the propulsion due to how much the electrical system was ingrained into every decision for assembling the CubeSat. All of the team members made great effort to buy easy to assemble parts, so not a lot of leading was necessary. The CubeSat frame itself was manufactured out of house, so the job was handled by the structure lead. My main contribution as the manufacturing lead was making an effort to be available to assist in manufacturing if needed. In this respect I made contributions to other subsystems such as the thermal and GNC subsystems (particularly in advising design of CAD models for 3D printing).

Samuel Martinez

I was the payload subsystem lead. This means I was responsible for part selection, designing, building, testing, and integrating the team's chosen payload into the CubeSat. I was responsible for ensuring the final assembled payload met or exceeded the design requirements of our mission. I was also the CAD and simulation lead. While a great deal of individual simulations and some CAD models were created by the other team members for their subsystems, the majority of the modelling was done by me. Many of the simulations not done for verification tests were also done by me using STK. I also assisted in the overall assembly and integration testing of the CubeSat which was primarily done by Waylon Ponder.

Jack Lamont

The subsystems that I was in charge of were the guidance, navigation, and control subsystem, along with the system engineering lead. The primary goal for my system was to control the attitude of the satellite. This is a crucial element for the SMART Sat, because the SAR requires exact pointing during scans. Along with designing this system, I also fulfilled my role as the systems engineering lead. The main challenge from this system came during the prototyping stage when there was very little space inside the satellite. This involved taking apart the GNC system and finding small areas in which it will fit.

Waylon Ponder

My main subsystems were power and communications and data handling. However, most of the assembly fell to me so in a way I had ties to every subsystem. I helped design the sunsensor and reaction wheels for GNC. I developed a driver circuit for the propulsion system as well as redesigned the pipe-fittings to actually fit within the confines of the 12U satellite. I integrated the heat sinks on the at-risk components to avoid system failures. With Sam's help, I soldered his complex radar circuit for the payload. I designed the skeleton-plate mounting solution to allow every component to be easily fixed to the satellite and ease the assembly process. In addition, I hand soldered our power delivery board with four voltage rails (5.5V, 12V, 8.2V, and 18V), assembled a custom 6-cell power-bank, hand wired 24 solar panels, and developed the deployment solution using servo actuation arms built from spare parts and a small microcontroller. Over the course of around 2.5 weeks, I was able to assemble the satellite with the help of Zumi, Sam, and Jack for their respective subsystem integrations.

15. Conclusion

Manufacturing and testing subsystems before integrating them into the cubesat has allowed for easy and efficient development of each system. As discussed, the testing of individual systems shows promising results of full functionality as originally designed. The system integration is as follows:

The S.M.A.R.T. CubeSat prototype has provided a testbed to allow the team to verify:

- Structural stability
- System interaction
- Thermal reliability
- Signal transmission and receiving
- Pressure stress failure
- Power generation
- Thrust generation
- Propellant required
- Weight estimation
- Size and layout estimation
- Electrical systems
- Attitude control
- Light tracking

- Earth tracking
- Control systems

Post testing, the team has verified a fully integrated CubeSat system. Functionalities include working SAR, functional communication and ground station subsystems, custom GNC and propulsion subsystems, and thermal and structural survivability subsystems. The main electrical and communication subsystem has been designed to control these other subsystems, and the overall prototype shows great promise for future work. The SAR proves capable of long distance scanning.

While not selected for competition, the S.M.A.R.T. CubeSat team believes that the current prototype proves a capable testbed and proof of further capabilities and potential, and is proud of the accomplishments over the previous half year of design and development.

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17. Appendix

17.2 Payload Appendix

Modified MATLAB Code:

Initial processing:

```
%MIT IAP Radar Course 2011

%Resource: Build a Small Radar System Capable of Sensing Range, Doppler,
%and Synthetic Aperture Radar Imaging

%

%Gregory L. Charvat

%SAR algorithm from:

%Range Migration Algorithm from ch 10 of Spotlight Synthetic Aperture Radar
%Signal Processing Algorithms, Carrara, Goodman, and Majewski

%NOTE: set up-ramp sweep from 2-3.2V to stay within ISM band
```

```

%change fstart and fstop bellow when in ISM band

%-----%

%Process raw data here

clear all;

close all;

%read the raw data .wave file here

[Y,FS] = audioread('TestOne.wav');
Audinf = audioinfo('TestOne.wav');
NBITS = Audinf.BitsPerSample;

%constants

c = 3E8; %(m/s) speed of light

%radar parameters

Tp = 20E-3; %(s) pulse time

Trp = 0.25; %(s) min range profile time duration

N = Tp*FS; %# of samples per pulse

%fstart = 2260E6; %(Hz) LFM start frequency

%fstop = 2590E6; %(Hz) LFM stop frequency

fstart = 2210E6; %(Hz) LFM start frequency for ISM band

fstop = 2460E6; %(Hz) LFM stop frequency for ISM band

BW = fstop-fstart; %(Hz) transmti bandwidth

f = linspace(fstart, fstop, N/2); %instantaneous transmit frequency

%the input appears to be inverted

trig = -1*Y(:,1);

s = -1*Y(:,2);

clear Y;

%parse data here by position (silence between recorded data)

rpstart = abs(trig)>mean(abs(trig));

count = 0;

Nrp = Trp*FS; %min # samples between range profiles

for ii = Nrp+1:size(rpstart,1)-Nrp

```



```

    if rpstart(ii) == 1 & sum(rpstart(ii-Nrp:ii-1)) == 0
        count = count + 1;
        RP(count,:) = s(ii:ii+Nrp-1);
        RPtrig(count,:) = trig(ii:ii+Nrp-1);
    end
end

%parse data by pulse
count = 0;
thresh = 0.08;
clear ii;
for jj = 1:size(RP,1)
    %clear SIF;
    SIF = zeros(N,1);
    start = (RPtrig(jj,:) > thresh);
    count = 0;
    jj
    for ii = 12:(size(start,2)-2*N)
        [Y I] = max(RPtrig(jj,ii:ii+2*N));
        if mean(start(ii-10:ii-2)) == 0 & I == 1
            count = count + 1;
            SIF = RP(jj,ii:ii+N-1)' + SIF;
        end
    end
end

%hilbert transform
q = ifft(SIF/count);
sif(jj,:) = fft(q(size(q,1)/2+1:size(q,1)));
end

sif(find(isnan(sif))) = 1E-30; %set all Nan values to 0

%SAR data should be ready here

clear s;

```

```

s = sif;

save routsidewarehouse2 s; %for image data

%-----%

%load additional varaibles and setup constants for radar here

clear all;

c = 3E8; %(m/s) speed of light

%load IQ converted data here

load routsidewarehouse2 s; %load variable sif %for image data

for ii = 1:size(s,1)

    s(ii,:) = s(ii,:) - mean(s,1);

end

%sif = s-sif_sub; %perform coherent background subtraction

%sif = sif_sub; %image just the background

sif = s; %image without background subtraction

clear s;

clear sif_sub;

%*****

%radar parameters

fstart = 2210E6; %(Hz) LFM start frequency

fstop = 2460E6; %(Hz) LFM stop frequency

fc = (fstop - fstart)/2 + fstart; %(Hz) center radar frequency

B = (fstop - fstart); %(hz) bandwidth

cr = B/20E-3; %(Hz/sec) chirp rate

Tp = 20E-3; %(sec) pulse width

%VERY IMPORTANT, change Rs to distance to cal target

%Rs = (12+9/12)*.3048; %(m) y coordinate to scene center (down range), make
this value equal to distance to cal target

Rs = 0;

Xa = 0; %(m) beginning of new aperture length

delta_x = 2*(1/12)*0.3048; %(m) 2 inch antenna spacing

L = delta_x*(size(sif,1)); %(m) aperture length

```

```

Xa = linspace(-L/2, L/2, (L/delta_x)); %(m) cross range position of radar on
aperture L

Za = 0;

Ya = Rs; %THIS IS VERY IMPORTANT, SEE GEOMETRY FIGURE 10.6

t = linspace(0, Tp, size(sif,2)); %(s) fast time, CHECK SAMPLE RATE

Kr = linspace(((4*pi/c)*(fc - B/2)), ((4*pi/c)*(fc + B/2)), (size(t,2)));

%Save background subtracted and callibrated data

save sif sif delta_x Rs Kr Xa;

%clear all;

%run IFP

SBAND_RMA_IFP;

```

Code by Gregory L. Charvat:

```

%

%This RMA SAR algorithm was written by Gregory L. Charvat as part of his
dissertation:

%G. L. Charvat, ``A Low-Power Radar Imaging System," Ph.D. dissertation,
%Dept. of Electrical and Computer Engineering, Michigan State University, East
Lansing, MI, 2007.

%

%Please cite appropriately.

%

%This algorithm was based on:

%Range Migration Algorithm from ch 10 of Spotlight Synthetic Aperture Radar
%Signal Processing Algorithms, Carrara, Goodman, and Majewski

%

%*****

%a note on formatting, our convention is sif(Xa,t)

clear all;

load sif;

%apply hanning window to data first

N = size(sif,2);

for ii = 1:N

```

```

    H(ii) = 0.5 + 0.5*cos(2*pi*(ii-N/2)/N);
end
for ii = 1:size(sif,1)
    sif_h(ii,:) = sif(ii,:).*H;
end
sif = sif_h;
figcount = 1;
close_as_you_go = 0;
do_all_plots = 0;

set(0,'defaultaxesfontsize',13); %set font size on plots so we can see it in
the dissertation

% NOTE: the function 'dbv.m' is just dataout = 20*log10(abs(datain));
%*****

if do_all_plots == 1,
    figure(figcount);

    S_image = angle(sif);
    imagesc(Kr, Xa, S_image);
    colormap(gray);

    title('Phase Before Along Track FFT');
    xlabel('K_r (rad/m)');
    ylabel('Synthetic Aperture Position, Xa (m)');
    cbar = colorbar;

    set(get(cbar, 'Title'), 'String', 'radians','fontsize',13);
    print(gcf, '-djpeg100', 'phase_before_along_track_fft.jpg');

    if close_as_you_go == 1,
        close(figcount);
    end

    figcount = figcount + 1;
end

%along track FFT (in the slow time domain)
%first, symetrically cross range zero pad so that the radar can squint

```

```

zpad = 2048; %cross range symmetrical zero pad
szeros = zeros(zpad, size(sif,2));
for ii = 1:size(sif,2)
    index = round((zpad - size(sif,1))/2);
    szeros(index+1:(index + size(sif,1)),ii) = sif(:,ii); %symmetrical zero pad
end
sif = szeros;
clear ii index szeros;
S = fftshift(fft(sif, [], 1), 1);
clear sif;
Kx = linspace((-pi/delta_x), (pi/delta_x), (size(S,1)));
if do_all_plots == 1,
    figure(figcount);
    S_image = dbv(S);
    imagesc(Kr, Kx, S_image, [max(max(S_image))-40, max(max(S_image))]);
    colormap(gray);
    title('Magnitude After Along Track FFT');
    xlabel('K_r (rad/m)');
    ylabel('K_x (rad/m)');
    cbar = colorbar;
    set(get(cbar, 'Title'), 'String', 'dB','fontsize',13);
    print(gcf, '-djpeg100', 'mag_after_along_track_fft.jpg');
    if close_as_you_go == 1,
        close(figcount);
    end
    figcount = figcount + 1;
end
if do_all_plots == 1,
    figure(figcount);
    S_image = angle(S);

```

```

    imagesc(Kr, Kx, S_image);

    colormap(gray);

    title('Phase After Along Track FFT');

    xlabel('K_r (rad/m)');

    ylabel('K_x (rad/m)');

    cbar = colorbar;

    set(get(cbar, 'Title'), 'String', 'radians','fontsize',13);

    print(gcf, '-djpeg100', 'phase_after_along_track_fft.jpg');

    if close_as_you_go == 1,
        close(figcount);
    end

    figcount = figcount + 1;
end

if do_all_plots == 1,
    figure(figcount);

    S_image = dbv(fftshift(fft(S, [], 2), 2));

    imagesc(linspace(-0.5, 0.5, size(S, 2)), Kx, S_image,
[max(max(S_image))-40, max(max(S_image))]);

    colormap(gray);

    title('Magnitude of 2-D FFT of Input Data');

    xlabel('R_{relative} (dimensionless)');

    ylabel('K_x (rad/m)');

    cbar = colorbar;

    set(get(cbar, 'Title'), 'String', 'dB','fontsize',13);

    print(gcf, '-djpeg100', 'mag_after_2D_fft.jpg');

    if close_as_you_go == 1,
        close(figcount);
    end

    figcount = figcount + 1;
end

end

%*****

```

```

%matched filter

%create the matched filter eq 10.8
for ii = 1:size(S,2) %step thru each time step row to find phi_if
    for jj = 1:size(S,1) %step through each cross range in the current time
        step row
            %phi_mf(jj,ii) = -Rs*Kr(ii) + Rs*sqrt((Kr(ii))^2 - (Kx(jj))^2);
            phi_mf(jj,ii) = Rs*sqrt((Kr(ii))^2 - (Kx(jj))^2);
            Krr(jj,ii) = Kr(ii); %generate 2d Kr for plotting purposes
            Kxx(jj,ii) = Kx(jj); %generate 2d Kx for plotting purposes
        end
    end
end

smf = exp(j*phi_mf); %%%%%%%%%%%

%note, we are in the Kx and Kr domain, thus our convention is S_mf(Kx,Kr)

%appply matched filter to S
S_mf = S.*smf;

clear smf phi_mf;

if do_all_plots == 1,
    figure(figcount);

    S_image = angle(S);

    imagesc(Kr, Kx, S_image);

    colormap(gray);

    title('Phase After Matched Filter');

    xlabel('K_r (rad/m)');

    ylabel('K_x (rad/m)');

    cbar = colorbar;

    set(get(cbar, 'Title'), 'String', 'radians','fontsize',13);

    print(gcf, '-djpeg100', 'phase_after_matched_filter.jpg');

    if close_as_you_go == 1,
        close(figcount);
    end

    figcount = figcount + 1;
end

```

```

end

clear S;

if do_all_plots == 1,
    figure(figcount);

    S_image = dbv(fftshift(fft(S_mf, [], 2), 2));

    imagesc(linspace(-0.5, 0.5, size(S_mf, 2)), Kx, S_image,
[max(max(S_image))-40, max(max(S_image))]);

    colormap(gray);

    title('Magnitude of 2-D FFT of Matched Filtered Data');
    xlabel('R_{relative} (dimensionless)');
    ylabel('K_x (rad/m)');

    cbar = colorbar;

    set(get(cbar, 'Title'), 'String', 'dB', 'fontsize', 13);

    print(gcf, '-djpeg100',
'mag_after_downrange_fft_of_matched_filtered_data.jpg');

    if close_as_you_go == 1,
        close(figcount);

    end

    figcount = figcount + 1;

end

%*****

%perform the Stolt interpolation

%FOR DATA ANALYSIS

kstart = 73;

kstop = 108.5;

%kstart = 95;

%kstop = 102;

Ky_even = linspace(kstart, kstop, 1024); %create evenly spaced Ky for interp
for real data

clear Ky S_St;

%for ii = 1:size(Kx,2)

count = 0;

```



```

for ii = 1:zpad;

%for ii = round(.2*zpad):round((1-.2)*zpad)

    count = count + 1;

    Ky(count,:) = sqrt(Kr.^2 - Kx(ii)^2);

    %S_st(ii,:) = (interp1(Ky(ii,:), S_mf(ii,:), Ky_even)).*H;

    S_st(count,:) = (interp1(Ky(count,:), S_mf(ii,:), Ky_even));

end

S_st(find(isnan(S_st))) = 1E-30; %set all Nan values to 0

clear S_mf ii Ky;

if do_all_plots == 1,

    figure(figcount);

    S_image = angle(S_st);

    imagesc(Ky_even, Kx, S_image);

    %imagesc(S_image);

    colormap(gray);

    title('Phase After Stolt Interpolation');

    xlabel('K_y (rad/m)');

    ylabel('K_x (rad/m)');

    cbar = colorbar;

    set(get(cbar, 'Title'), 'String', 'radians','fontsize',13);

    print(gcf, '-djpeg100', 'phase_after_stolt_interpolation.jpg');

    if close_as_you_go == 1,

        close(figcount);

    end

    figcount = figcount + 1;

end

%apply hanning window to data, cleans up data ALOT

N = size(Ky_even,2);

for ii = 1:N

    H(ii) = 0.5 + 0.5*cos(2*pi*(ii-N/2)/N);

```

```

end

for ii = 1:size(S_st,1)
    S_sth(ii,:) = S_st(ii,:).*H;
end

%S_st = S_sth;

%*****

%perform the inverse FFT's

%new notation:  v(x,y), where x is crossrange

%first in the range dimension

clear v Kr Krr Kxx Ky_even;

v = ifft2(S_st,(size(S_st,1)*4),(size(S_st,2)*4));

%bw = (3E8/(4*pi))*(max(xx)-min(xx));

bw = 3E8*(kstop-kstart)/(4*pi);

max_range = (3E8*size(S_st,2)/(2*bw))*1/.3048;

figure(figcount);

S_image = v; %edited to scale range to d^3/2

S_image = fliplr(rot90(S_image));

cr1 = -80; %(ft)

cr2 = 80; %(ft)

dr1 = 1 + Rs/.3048; %(ft)

dr2 = 350 + Rs/.3048; %(ft)

%data truncation

dr_index1 = round((dr1/max_range)*size(S_image,1));

dr_index2 = round((dr2/max_range)*size(S_image,1));

cr_index1 = round(( (cr1+zpad*delta_x/(2*.3048))
/(zpad*delta_x/.3048))*size(S_image,2));

cr_index2 = round((
(cr2+zpad*delta_x/(2*.3048))/(zpad*delta_x/.3048))*size(S_image,2));

trunc_image = S_image(dr_index1:dr_index2,cr_index1:cr_index2);

downrange = linspace(-1*dr1,-1*dr2, size(trunc_image,1)) + Rs/.3048;

crossrange = linspace(cr1, cr2, size(trunc_image, 2));

```

```

%scale down range columns by range^(3/2), delete to make like
%disertation again

clear ii;

for ii = 1:size(trunc_image,2)
    trunc_image(:,ii) =
(trunc_image(:,ii)').*(abs(downrange*.3048)).^(3/2);
end

trunc_image = dbv(trunc_image); %added to scale to d^3/2

imagesc(crossrange, downrange, trunc_image, [max(max(trunc_image))-40,
max(max(trunc_image))-0]);

colormap('turbo');

title('Final Image');

ylabel('Downrange (ft)');

xlabel('Crossrange (ft)');

axis equal;

cbar = colorbar;

set(get(cbar, 'Title'), 'String', 'dB','fontsize',13);

print(gcf, '-djpeg100', 'final_image.jpg');

if close_as_you_go == 1,
    close(figcount);
end

figcount = figcount + 1;

```

17.9 Propulsion Appendix

- $D = .5 * \rho * v^2 * A * Cd$, with $Cd = 2$. [PROP18]. (N)
- $deceleration = D/mass$ (m/s/s)
- $\Delta V_{drag} = deceleration * time$ (m/s)
- $Impulse = Force * time$